

genome is desirable for genomic restriction mapping (McClelland et al., 1987), with clear advantages over frequently cutting endonucleases for the production of fingerprints characteristic of different taxa (Mielenz et al., 1979; Dobritsa, 1985; Sorensen et al., 1985). Application of this principle has resulted in a practical method for constructing and analyzing macrorestriction patterns of 234 strains of different species of aerobic pseudomonads (Grothues and Tümmeler, 1991). The restriction nucleases *AsnI*, *DraI*, *SpeI*, *XbaI*, and *PacI* were found to be most appropriate by being specific for AT-rich regions, or for sites including the extremely uncommon tetranucleotide CTAG (McClelland et al., 1987).

The macrorestriction patterns are compared for the number and position of the bands, and use of appropriate equations often indicates a correlation between the estimated similarities of fingerprints and conventional taxonomic groupings. The results obtained by Grothues and Tümmeler (1991) in general confirm the classification of pseudomonads based on extensive numbers of phenotypic properties and nucleic acid hybridization studies.

Strains sharing the same mol% G + C contents in the chromosomal DNA, the same codon usage, and similar genome size, give similarity coefficients useful for determinative purposes. The genome patterns also have the practical advantage of helping in epidemiological studies. Differences have been found between *Pseudomonas* strains that are susceptible or multiply resistant to antibiotics (Yamashita et al., 1997).

Differences in restriction fragment size distribution may result from various chromosomal rearrangements and/or mutations at the restriction sites, which suggests that the methodology based on restriction fingerprints should be supplemented with other approaches to determine the degree of homology of the fragments. Large genomic rearrangements have been observed in *P. aeruginosa* strains isolated from clinical samples or from the environment (Schmidt et al., 1996c). The former strains came mostly from cystic fibrosis cases. A 95-kb plasmid was detected in environmental strains and it integrated into the chromosome in cystic fibrosis strains. Exchange of DNA blocks and large DNA inversions led to divergence of clones in this species. The presence of inversions only in cystic fibrosis strains suggests that this niche causes or selects for substantial changes in the genome (Römling et al., 1997).

A study of pathovars of fluorescent plant pathogenic pseudomonads (Grothues and Rudolph, 1991) indicated that two strains belonging to the same pathovar, but of different origins, can give almost identical restriction fingerprints. However, this is not always the case. Pathogens of wide host range give more diverse restriction patterns than those of restricted host range. The methodology was also used for the examination of representative strains of *P. stutzeri*, a species notorious for its heterogeneity (Stanier et al., 1966; Palleroni et al., 1970). The results show a marked correlation of genome structure with fatty acid composition, and with data of nucleic acid hybridization experiments (Rainey et al., 1994b). A high degree of heterogeneity in macrorestriction patterns did not correlate with the subdivision of the species in genomovars was also observed by another group of workers, and the marked heterogeneity of *P. stutzeri* was attributed, at least in part, to large chromosomal rearrangements (Ginard et al., 1997). The results obtained by these two groups of workers also indicate that the genome size of *P. stutzeri* ranges from 3.4–4.64 Mb for the strains subjected to their studies.

Rotating field electrophoresis was used to separate fragments of DNA of common and of genetically modified strains of *Pseu-*

domonas species after digestion with rare-cutting restriction endonucleases. The technique was adapted to the identification of *Pseudomonas* strains deliberately released in the environment. The fingerprints were different for strains of the same species, although they were identical for related strains, and the differences were not affected by the presence of natural or genetically modified plasmids (Claus et al., 1992).

Sequence determination of 16S/23S spacer regions A study of the genomic organization of the ribosomal RNA genes in the *P. aeruginosa* has been the basis for the hypothesis that this species carries at least four sets of genes, each containing the genes for 16S, 23S, and 5S (Hartmann et al., 1986). Much information is now available for the 16S rRNA genes, in contrast with the other components of the ribosomal RNA complex, and it has been suggested that the spacer regions between the respective genes may be interesting subjects for sequence studies applicable to identification and typing procedures. Working with *Pseudomonas* strains isolated from different types of soils, Gill et al. (1994) amplified the spacer regions by PCR and subjected them to sequencing studies. The results suggest a limited degree of variability among strains identified as *P. putida* and *P. fluorescens*. However, the data may be useful for the recognition of particular *Pseudomonas* environmental strains. A review with information on studies of the spacer regions in a number of bacterial species, including some aerobic pseudomonads, is now available (Gürtler and Stanisich, 1996).

Distribution of repetitive sequences in the genome These sequences have been taken as the basis for genomic studies based on amplification of the fragments separating the repeated sequences. Work on pseudomonads (Louws et al., 1994) is an extension of studies performed mostly on rhizobia and *Xanthomonas* (Judd et al., 1993), following an original report suggesting the use of the distribution of the repetitive sequences as a tool for the characterization of the genome (Versalovic et al., 1991).

Identification of special genetic markers Saint-Onge et al. (1992) succeeded in cloning and sequencing the lipoprotein I gene (*oprI*) of *P. aeruginosa* PAO. The sequence showed similarity to published sequences from other sources, and oligonucleotide primers were designed to amplify the gene by PCR. The amplified gene or the oligonucleotide primers are convenient identification tools in Southern blot analyses. DNA obtained from small amounts of material taken from colonies could be used for amplification, and when bands of the expected mobility were obtained, they could be further identified by the use of a probe prepared with the amplified gene from *P. aeruginosa*. The gene was identified in species of *Pseudomonas* (rRNA similarity group I) and in strains of the genera *Chryseomonas* and *Flavimonas*, whose genomic DNAs reacted with the probe in Southern blots. Saint-Onge and collaborators concluded that the *oprI* gene was found primarily in strains of group I, but they wisely added that it would be of interest to search for other indications leading to a reclassification of the organisms assigned to these newly created genera. Evidence supporting this prediction was eventually found, and, as mentioned above in the section on DNA–DNA hybridization, supported the conclusion that the genera *Chryseomonas* and *Flavimonas* should be considered as synonyms of *Pseudomonas* (Anzai et al., 1997). The main conclusions of the work of Anzai and collaborators are summarized graphically in Fig. BXII.γ.125. For the time being, the two genera *Chryseomonas* and *Flavimonas* should be considered as junior subjective synonyms of *Pseudomonas*.

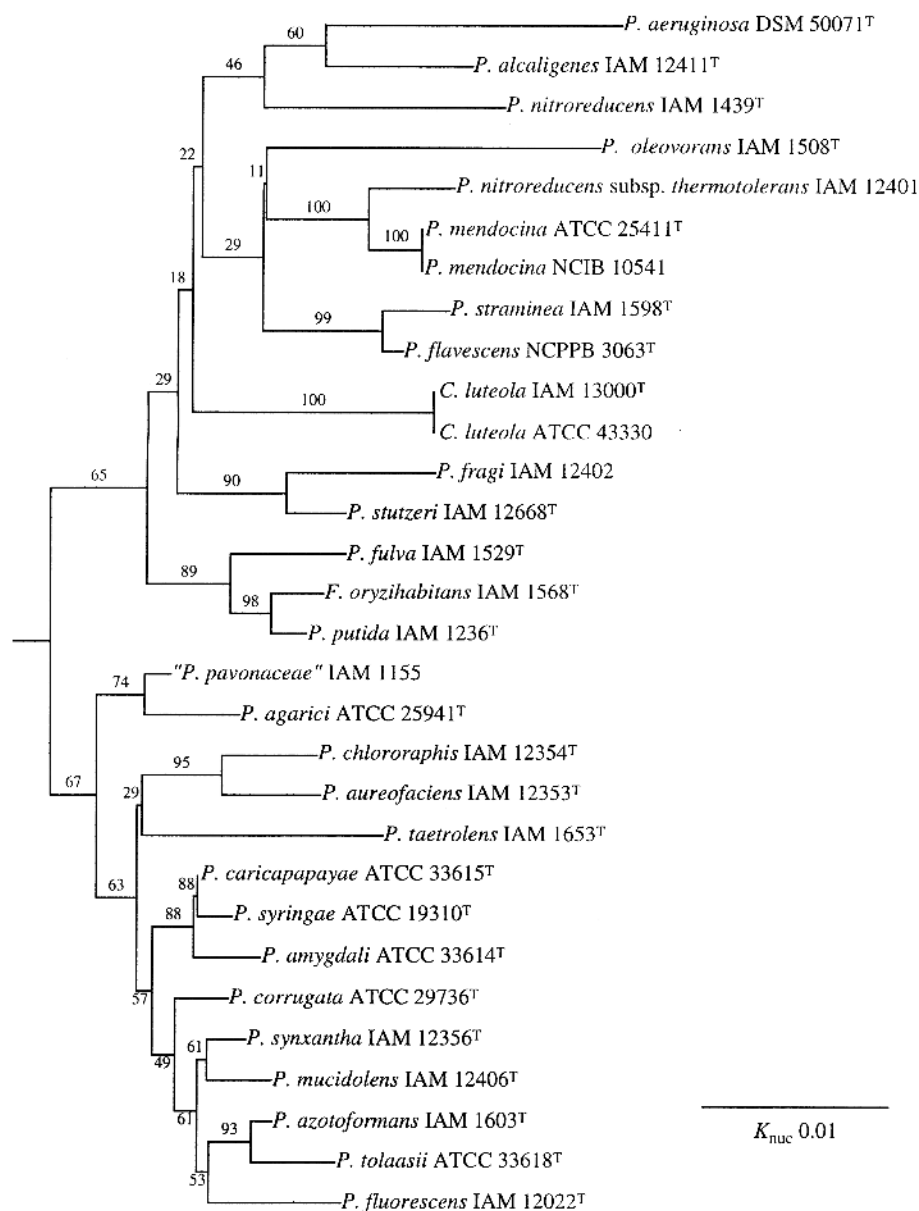


FIGURE BXII.γ.125. Phylogenetic tree for different species of the genus *Pseudomonas* derived from 16S rDNA sequence similarities to show the relative position of *Chryseomonas* and *Flavimonas*. Bootstrap values are indicated at the branch points. The confidence limits for the position of the branches has been estimated using the bootstrap analysis. *Escherichia coli* was chosen as root organism. (Reproduced with permission from Y. Anzai et al., International Journal of Systematic Bacteriology 47: 249–251, 1997, ©International Union of Microbiological Societies.)

Additional confirmation of the relationship among species of *Pseudomonas* came from work demonstrating that four alginate production genes were conserved within the group (Fialho et al., 1990). Conservation also was observed in the *oprP* gene in various species of similarity group I (Siehnel et al., 1990).

rRNA–DNA hybridization This technique has had fundamental importance in *Pseudomonas* taxonomic and phylogenetic studies because its application resulted in the demonstration of the complexity of the aerobic pseudomonads as a bacterial group (Palleroni et al., 1973). Because of these studies, *Pseudomonas* is now restricted to one of five rRNA similarity groups. In these early studies, the competition technique was used (see Kilpper-Bälz, 1991, for a review of the methodology). Later, modifications

of the hybridization technique were used by other workers, confirming and extending the original results (De Vos and De Ley, 1983; De Vos et al., 1985, 1989). Since the development of practical methods for sequencing the components of the rRNA operon, rRNA–DNA hybridization methods are seldom being used for taxonomic purposes.

rRNA gene sequencing Following the initial rRNA–DNA hybridization studies on the aerobic pseudomonads, oligonucleotide catalogs (Uchida et al., 1974) were used for phylogenetic characterization of bacterial species. Comparisons among catalogs were expressed numerically by means of association coefficients (S_{AB}), and the values could be represented graphically in dendrograms (Fox et al., 1977). Cataloging eventually gave way

to a more comprehensive approach directed to the determination of the total sequence of ribosomal RNA components. Initially this was achieved by the isolation of rRNA and DNA synthesis by transcriptase reaction, but the advent of the PCR technique has greatly simplified the experimental methodologies by avoiding RNA isolation, and it is widely used today. An article by Moore et al. (1996) specifically refers to determination and comparison of the 16S rRNA sequences of *Pseudomonas* species. A European compilation of pseudomonad sequences is available (van de Peer et al., 1996).

The results of the contribution by Moore et al. (1996) have been summarized in the creation of "lineages", which are grouped into two "clusters", the *P. aeruginosa* intrageneric cluster and the *P. fluorescens* intrageneric cluster. Within the first cluster there are four lineages: the *P. aeruginosa* lineage (*P. aeruginosa*, *P. citronellolis*, *P. alcaligenes*, *P. stutzeri*), the *P. resinovorans* lineage (*P. resinovorans*, *P. balearica*), the *P. mendocina* lineage (*P. mendocina*, *P. pseudoalcaligenes*, *P. oleovorans*), and the *P. flavescens* lineage (only *P. flavescens*). Within the *P. fluorescens* cluster there are located five lineages: the *P. fluorescens* lineage (*P. fluorescens*, *P. marginalis*, *P. tolaasii*, *P. chlororaphis*, *P. aureofaciens*, *P. viridiflava*), the *P. syringae* lineage (*P. syringae*, *P. amygdali*, "*P. coronafaciens*", *P. ficuserectae*), the *P. cichorii* lineage (*P. cichorii*), the *P. putida* lineage (*P. putida*, *P. asplenii*), and the *P. agarici* lineage (*P. agarici*) (Moore et al., 1996). It is, however, hard to draw any solid conclusions on the proposed classification. In the first place, since only type strains have been included in the published sequences, it is impossible to know the intraspecific variations, and although several strains of some of the species have been included, the selection has not taken into account the degree of heterogeneity in both DNA similarities and in phenotypic features characteristic of some of these species. Thus, while *P. aeruginosa* and *P. mendocina* are notoriously homogeneous in their phenotypic and genotypic properties, they are represented by a total of 12 strains. On the other hand, a very heterogeneous species in both phenotypic features and in DNA similarity such as *P. stutzeri* is represented by only one strain. The authors did not find obvious correlations of their molecular data with the results of standard phenotypic criteria. This is actually information from papers published by other groups; data corresponding to the strains subjected to the 16S rRNA sequence analysis are not reported.

The above-mentioned study included a comparison of sequences of the 16S rRNA variable region (*E. coli* positions 453–479) for the various lineages and for the species within each lineage. With respect to this approach, it may be interesting to mention here the findings on the *P. putida* biovars reported by Yamamoto and Harayama (1998). In a study of fluorescent pseudomonads done on the basis of the sequences of 16S rRNA, gyrase B subunit (*gyrB*) and RNA polymerase $\sigma 70$ factor (*rpoD*), they found that in *P. putida* B the pairwise distances estimated from the variable regions of the 16S rRNA gene correlated rather poorly with the synonymous distances estimated from the *gyrB* and *rpoD* genes. On the other hand, the correlation of the non-variable regions of these genes was highly significant, and a recommendation is formulated that only the nonvariable regions should be used for phylogenetic analysis.

Ribotyping The methodology allows comparisons of restriction patterns of ribosomal RNA genes (Stull et al., 1988), and it became widely used for epidemiological purposes because of its capacity for detection of prokaryotic diversity at the infraspecific level. When applied to the aerobic pseudomonads (*Pseudomonas*

sensu lato), using the restriction enzymes *Sma*I and *Hind*III, followed by hybridization with 16 and 23S rRNA, the method was capable of resolving 169 and 159 unique ribotyping patterns, respectively (Brosch et al., 1996). Of the two enzymes, *Sma*I was the most discriminative, except for some of the species. An analysis of *P. putida* biovar A strains in combination with a phenotypic analysis supported earlier observations, and suggested that a revision of the taxonomic status and internal subdivision of this species might be in order (Elomari et al., 1994). These considerations are mentioned in some detail in the section on taxonomic comments for this species. The methodology has limitations from the phylogenetic point of view, particularly because of significant differences that may be created by single mutations. However, aside from being a useful typing tool (Grimont and Grimont, 1986), it also appears to have taxonomic implications, and some of the observations by Brosch and collaborators confirm the marked heterogeneity of some of the species, particularly *P. fluorescens*.

Nucleic acid probes Based on the capacity of nucleic acid hybridization for the elucidation of genomic relatedness among bacteria, the development of nucleic acid probes allows for high-resolution, rapid, automated identification of specific microorganisms when they are mixed with other members of natural communities. The preparation and use of probes have been described in detail (Stahl and Amann, 1991; Schleifer et al., 1993).

Schleifer et al. (1992) have described a very effective use of the nucleic acid probes for the specific identification of pseudomonads in the environment. More recently, Amann et al. (1996a) have described the application of such probes to the detection of certain *Pseudomonas* species and pseudomonad species now assigned to other genera. In this report, the main goal was to identify the target organisms as either "former" or "genuine" pseudomonads. Unfortunately, the title of the paper carries this distinction, which is most confusing, since all pseudomonads, no matter what genera they have been assigned to, are equally "genuine".

A simplified cell blot technique combined with the use of 16S rRNA-directed probes was used for the identification of environmental isolates. The sequence of a group probe identifying *P. aeruginosa*, *P. mendocina*, *P. fluorescens*, *Comamonas acidovorans*, and "*Flavobacterium lutescens*" is described (Braun-Howland et al., 1993).

Low molecular weight (LMW) RNA profiles As an important addition to phylogenetic methodologies, Höfle (1991) has proposed a "rapid genotypic approach that bridges the gap between the solid systematics of type strains and the everyday need for relating new isolates and old reference strains". His technique is based on the examination of profiles of low-molecular-weight (LMW) RNA components in *Pseudomonas* species using high-resolution gel electrophoresis (Höfle, 1988). The reference strains of this genus, according to the author, showed very specific band patterns with discriminative power at the genus and species level (Höfle, 1990).

Whole-cell protein fingerprinting Protein banding patterns obtained by polyacrylamide gel electrophoresis, particularly when performed under denaturing conditions, are highly characteristic at the strain level, and have been used for many years for the purpose of typing and classification. An interesting application is the verification of strain authenticity, a sensitive and rapid control method that bypasses lengthy and tedious proce-

dures for rechecking exchanged cultures (Kerstens et al., 1994). At present, objective quantification of the band patterns, particularly when a substantial number of strains are analyzed, is being done by computer-assisted methods. Among the variants on the basic theme of protein separation in an electric field, electrophoresis in the presence of a denaturing agent such as sodium dodecyl ("lauryl") sulfate (SDS) (Laemmli, 1970) is the technique of choice. SDS polyacrylamide gel electrophoresis (SDS-PAGE) combines high resolution and good reproducibility, which from a practical point of view compensate for the fact that no specific identification of the bands is usually performed. A description of various methodologies and their evaluation has been conveniently summarized by Jackman (1985, 1987).

Of the various references available for descriptions of whole-cell protein fingerprinting methods, a recent publication of a one-dimensional electrophoretic analysis applied to *Pseudomonas* species is a convenient source of technical and bibliographic information (Vancanneyt et al., 1996a). Over 200 reference strains of various species were included. Aside from the few species for which only the type strain has been analyzed, uniform and distinct patterns were observed for many species of fluorescent and nonfluorescent organisms (*P. aeruginosa*, *P. agarici*, *P. alcaligenes*, *P. amygdali*, *P. caricapapayae*, *P. chlororaphis*, *P. chichorii*, "*P. coronafaciens*", *P. corrugata*, *P. ficuserectae*, *P. fragi*, *P. mendocina*, *P. pertucinogena*, *P. tolaasii*, and *P. viridiflava*). In contrast, *P. fluorescens*, *P. marginalis*, *P. pseudoalcaligenes*, *P. putida*, *P. stutzeri*, and *P. stutzeri* were found to be particularly heterogeneous, in confirmation with earlier data on their phenotypic characteristics.

Ribosomal proteins These proteins, similarly to the ribonucleic acids with which they are associated in the ribosome, are of a highly conservative nature. Obviously, this should be reflected in a comparison of their sequences. In fact, partial amino acid sequences of preparations of protein L30 from species of various ribosomal RNA similarity groups give clusters similar to those that had originally been defined in rRNA-DNA hybridization studies (Ochi, 1995). Despite different levels of conservatism, other ribosomal proteins (S20, S21, L27, L20, L31, L32, and L33 protein families) give comparable results.

Other membrane proteins The discussion on this point will be limited to the identification of lipoprotein I in species of *Pseudomonas*. Work performed on the specificity of the *P. aeruginosa* lipoprotein I gene used as a probe for rapid identification purposes was an extension of the experiments of Saint-Onge et al. (1992). The work included more strains of species of RNA group I (*Pseudomonas*), as well as the related "*Azotomonas*" and *Azomonas* species and other species of Gram-negative organisms. All of the species of *Pseudomonas* that were tested gave positive reactions of variable intensities in PCR and dot blot tests using monoclonal antibodies for the lipoprotein I, whereas species of other similarity groups and of other Gram-negative genera gave negative reactions. These data, still considered to be of a preliminary nature, suggest that *P. aeruginosa* can be differentiated from the other species of the same genus that have been tested, with the exception of *P. mendocina*, *P. pseudoalcaligenes*, and *P. oleovorans*, which have similar restriction patterns in their lipoprotein I gene (De Vos et al., 1993).

Fatty acid analysis Fatty acid composition and the quinone system have been used for groupings of *Pseudomonas* species (Ikemoto et al., 1978; Yamada et al., 1982; Oyaizu and Komagata,

1983). A study of the fatty acid composition of 50 strains of various *Pseudomonas* species revealed the presence of the straight-chain saturated acid of C_{16:0} and straight-chain unsaturated acids C_{16:1} and C_{18:1} in all strains (Ikemoto et al., 1978). Distribution of hydroxy acids, cyclopropane acids, and branched-chain acids follows quite closely the accepted subdivision of the aerobic pseudomonads into RNA similarity groups. These cellular components are very useful for identification purposes.

The results of fatty acid and quinones analyses, performed on a collection of 75 strains that included phytopathogenic pseudomonads, have been reported (Oyaizu and Komagata, 1983). The survey gave a basis for the classification of the collection into groups with particular reference to the presence of 3-OH fatty acids, and some of the groups coincided with the RNA similarity groups already defined (Palleroni et al., 1973).

The analysis of a large number of fatty acid profiles obtained from a collection of 340 strains of pseudomonads that includes plant pathogenic species has been published (Stead, 1992). The results on the distribution of 2-OH and 3-OH fatty acids could serve as one of the bases for a subdivision of the collection into six groups warranting independent generic designations.

The above comments on fatty acid composition studies refer to *Pseudomonas sensu lato* and coincide in supporting the division of these organisms into the various genera that later were to be proposed in various communications from different laboratories. A summary of the results of the above sources is presented in Table BXII.γ.110.

In a more recent communication (Vancanneyt et al., 1996b) the fatty acid composition of whole-cell hydrolysates and phospholipid fractions are reported. The results confirm some of the conclusions of the above-cited references. In general, *Pseudomonas* species are characterized by the presence of C_{10:0 3OH} and C_{12:0 3OH} acids. When the fatty acid contents of phospholipids were taken into account, no clear separation could be found between *Pseudomonas* RNA group I organisms and RNA similarity groups II and III.

Polyamine composition Polyamine patterns appear to be good chemotaxonomic markers for the Gram-negative bacteria in the phylum *Proteobacteria*, allowing ready differentiation of some of the pseudomonad groups (Busse and Auling, 1988). In addition to quinone analysis, polyamines are useful for rapid

TABLE BXII.γ.110. Fatty acid and ubiquinone composition of *Pseudomonas* and other rRNA similarity groups^a

Compound	Ribosomal RNA similarity groups				
	I	II	III	IV	V
<i>Fatty acids:</i>					
C _{10:0 3OH}	+		+		+
C _{11:0 3OH}					+
C _{11:0 iso 3OH}					+
C _{12:0 3OH}	+			+	+
C _{12:0 iso 3OH}					+
C _{13:0 iso 3OH}					+
C _{14:0 3OH}		+		+	
C _{16:1 3OH}		+			
C _{12:0 2OH}	(+)				
C _{16:0 2OH}		(+)			
C _{16:1 2OH}		(+)			
C _{18:1 2OH}		+			
Ubiquinones	Q-9	Q-8	Q-8	Q-10	Q-8

^aData taken from Oyaizu and Komagata (1983) and Stead (1992). Data in parentheses: not all strains of the group are positive.

identification at the genus level (Busse et al., 1989). This last contribution includes a discussion of various other approaches applicable to species and strain identification.

In an attempt to relieve the dearth of simple and reliable tests for the classification of newly isolated strains of the genus *Xanthomonas*, a determination of polyamine composition of strains of this genus and of phytopathogenic *Pseudomonas* species was introduced as a rapid chemotaxonomic identification tool (Auling et al., 1991). Spermidine was found to be the main polyamine of *Xanthomonas*, whereas strains of *Pseudomonas* RNA group I were characterized by the presence of putrescine. The polyamine pattern of *Azotobacter* and *Azomonas* was the same as that of the fluorescent pseudomonads, which are placed in the same branch of *Proteobacteria*. The results obtained by Auling et al. (1991) were confirmed in a more recent study by Goris et al. (1998). Aside from the presence of putrescine and spermidine mentioned above, the species of *Pseudomonas* could be divided into two sublineages, of which only one contained cadaverine. *P. aeruginosa* (as well as *Azotobacter vinelandii*) is in the sublineage that contains this polyamine.

FURTHER READING

- Achouak, W., L. Sutra, T. Heulin, J.-M. Meyer, N. Fromin, S. Degraeve, R. Christen and L. Gardan. 2000. *Pseudomonas brassicacearum* sp. nov. and *Pseudomonas thivervalensis* sp. nov., two root-associated bacteria isolated from *Brassica napus* and *Arabidopsis thaliana*. Int. J. Syst. Evol. Microbiol. 50: 9–18.
- Andersen, S.M., K. Johnsen, J. Sorensen, P. Nielsen and C.S. Jacobsen. 2000. *Pseudomonas frederiksbergensis* sp. nov., isolated from soil at a coal gasification site. Int. J. Syst. Evol. Microbiol. 50: 1957–1964.
- Anzai, Y., H. Kim, J.Y. Park, H. Wakabayashi and H. Oyaizu. 2000. Phylogenetic affiliation of the pseudomonads based on 16S rRNA sequence. Int. J. Syst. Evol. Microbiol. 50: 1563–1589.
- Brown, G.R., I.C. Sutcliffe and S.P. Cummings. 2001. Reclassification of [*Pseudomonas*] *doudoroffii* (Baumann et al. 1983) in the genus *Oceanomonas* gen. nov. as *Oceanomonas doudoroffii* comb. nov., and description of a phenol-degrading bacterium from estuarine water as *Oceanomonas baumannii* sp. nov. Int. J. Syst. Evol. Microbiol. 51: 67–72.
- Guasp, C., E.R.B. Moore, J. Lalucat and A. Bennisar. 2000. Utility of internally transcribed 16S–23S rDNA spacer regions for the definition of *Pseudomonas stutzeri* genomovars and other *Pseudomonas* species. Int. J. Syst. Evol. Microbiol. 50: 1629–1639.
- Nishimori, E., K. Kita-Tsukamoto and H. Wakabayashi. 2000. *Pseudomonas plecoglossicida* sp. nov., the causative agent of bacterial haemorrhagic ascites of ayu, *Plecoglossus altivelis*. Int. J. Syst. Evol. Microbiol. 50: 83–89.
- Rowe, N.J., J. Tunstall, L. Galbraith and S.G. Wilkinson. 2000. Lipid composition and taxonomy of (*Pseudomonas*) *echinoides*: Transfer to the genus *Sphingomonas*. Microbiology (Read.) 146: 3007–3012.
- Uchino, M., Y. Kosako, T. Uchimura and K. Komagata. 2000. Emendation of *Pseudomonas straminea* Iizuka and Komagata 1963. Int. J. Syst. Evol. Microbiol. 50: 1513–1519.
- Yumoto, I., K. Yamazaki, M. Hishinuma, Y. Nodasaka, A. Suemori, K. Nakajima, N. Inoue and K. Kawasaki. 2001. *Pseudomonas alcaliphila* sp. nov., a novel facultatively psychrophilic alkaliphile isolated from seawater. Int. J. Syst. Evol. Microbiol. 51: 349–355.

List of species of the genus *Pseudomonas*

The following list includes those species whose assignment to the genus *Pseudomonas* has been definitely established. Part of the information for some of the species has been condensed in the form of tables representing updated versions of those of the first edition of this *Manual*.

The order is alphabetical, thus avoiding the problem of ordering the species on the basis of properties such as pathogenicity, pigmentation, or certain physiological peculiarities, which only seldom correlate with the order of groups based on 16S rRNA sequence similarities (Figs. BXII.γ.124 and BXII.γ.125). *Pseudomonas* can be subdivided into two big groups, the fluorescent and the nonfluorescent species and, with respect to plant pathogenicity, a subdivision of the genus into saprophytes and plant pathogenic species is possible. These subdivisions do not correlate with each other. Some correlations, however, are apparent. In Fig. BXII.γ.124 the plant pathogens congregate mostly in the so-called *P. syringae* group.

The members of some pairs of species (*P. migulae*–*P. mandelii*; *P. veronii*–*P. gessardii*; *P. cedrella*–*P. orientalis*; *P. marginalis*–*P. libanensis*; *P. fuscovaginae*–*P. asplenii*, and *P. monteilii*–“*P. ayucida*”) were found to have identical 16S rRNA sequences (see Fig. BXII.γ.124) (Anzai et al., 1997). It is to be hoped that the precise taxonomic allocation of these taxa may be further investigated. Some comments are included in the corresponding descriptions below.

A number of species assigned to *Pseudomonas* will not be treated here. As mentioned by Kersters et al. (1996), the phylogenetic position of the following species has not yet been determined. In the list that follows, references to the original descriptions are given in parentheses, and the number in italics refers to the page where the species is described in the first edition of this *Manual* (Palleroni, 1984): *P. antimicrobica* (Attafuah and Bradbury, 1989), *P. cissicola* (Goto and Makino, 1977), *P. flectens* (Johnson, 1956), *P. gelidicola* (Kadota, 1951), *P. halophila*

(Fendrich, 1988), *P. indigofera* (Elazari-Volcani, 1939), *P. iners* (Iizuka and Komagata, 1964b), *P. lanceolata* (Leifson, 1962a), *P. mephitica* (Haynes and Burkholder, 1957), and *P. spinosa* (Leifson, 1962b). Since the descriptions of *P. antimicrobica*, *P. flectens*, and *P. halophila* are not given in the first edition of the *Manual*, they will be added at the end of the list of species, under the subtitle Other Species.

Table BXII.γ.111 lists a number of species previously assigned to *Pseudomonas sensu lato*, which have been transferred to various other genera or groups. The information has been taken mainly from Kersters et al. (1996), and also Behrendt et al. (1999) and Anzai et al. (1997). The paper by Kersters et al. (1996) includes a handy table with the current classification of the species that have been assigned to *Pseudomonas* prior to the subdivision of the genus into rRNA similarity groups, their present assignments, and references to the studies deciding phylogenetic allocations and/or revised taxonomic status for each species.

Recently, three new species have been added to the list of those requiring further confirmation—*Pseudomonas abietaniphila*, *P. multiresinivorans*, and *P. vancouverensis*, which are isolated from resin acids (Mohn et al., 1999a, b). Even though the reported catabolic activities of these organisms are unique, DNA–DNA hybridization data to confirm their genomic relationships vs. related organisms have not been reported. For this reason, the description of these species will be found at the end, in the section Other Species.

Finally, many new species of *Pseudomonas* have been described in the *International Journal of Systematic and Evolutionary Microbiology* since mid-2001, the cut-off point for inclusion of new genera and species in this edition of the *Manual*.

1. ***Pseudomonas aeruginosa*** (Schroeter 1872) Migula 1900, 884^{AL} (*Bacterium aeruginosum* Schroeter 1872, 126.) *ae.ru.gi.no'sa*. L. fem. adj. *aeruginosa* full of copper rust or verdigris, hence green.

Characteristics of the species are summarized in Tables BXII.γ.108, BXII.γ.112, BXII.γ.113, and BXII.γ.114. Rods, motile by means of a single polar flagellum. Two main colony types can be observed on common solid media. One is large, smooth, with flat edges and elevated center ("fried

egg" appearance), and the other is small, rough, convex. On the large colonies, silver-gray metallic shining patches may be observed, and pitting is fairly common. Clinical materials are, in general, good sources of the large colony type, whereas the small type is commonly obtained from

TABLE BXII.γ.111. *Pseudomonas sensu lato* species transferred to other taxa

<i>Pseudomonas</i> species	Previous description in <i>Bergey's Manuals</i>	New allocation	References
<i>P. aminovorans</i>	Palleroni (1984)	<i>Aminobacter</i>	Urakami et al. (1992)
<i>P. beijerinckii</i>	Palleroni (1984)	<i>Halomonas</i> rRNA lineage	De Vos et al. (1989)
<i>P. beteli</i>	Palleroni (1984)	<i>Xanthomonas</i> rRNA lineage	De Vos et al. (1985)
<i>P. boreopolis</i>	Palleroni (1984)	<i>Xanthomonas</i> rRNA lineage	De Vos et al. (1989)
<i>P. carboxydohydrogena</i>	Palleroni (1984)	<i>Bradyrhizobium-Rhodopseudomonas</i> rRNA lineage	Auling et al. (1988)
" <i>P. carboxydovorans</i> "	Palleroni (1984)	<i>Oligotropha</i>	Meyer et al. (1993)
" <i>P. compransoris</i> "	Palleroni (1984)	<i>Zavarzinia</i>	Meyer et al. (1993)
<i>P. doudoroffii</i>	Palleroni (1984)	<i>Aeromonadaceae</i>	De Vos et al. (1989)
<i>P. echinoides</i>	Palleroni (1984)	<i>Sphingomonas</i> rRNA lineage	De Vos et al. (1989)
<i>P. elongata</i>	Palleroni (1984)	<i>Oceanospirillum</i>	De Vos et al. (1989)
" <i>P. extorquens</i> "	Haynes and Burkholder (1957)	<i>Methylobacterium</i>	Bousfield and Green (1985)
<i>P. geniculata</i>	Haynes and Burkholder (1957)	<i>Xanthomonas</i> rRNA lineage	Byng et al. (1983)
<i>P. hibiscicola</i>	Palleroni (1984)	<i>Xanthomonas</i> rRNA lineage	De Vos et al. 1985)
<i>P. huttiensis</i>	Palleroni (1984)	<i>Herbaspirillum</i>	Kerstens et al. (1996)
<i>P. lemoignei</i>	Palleroni (1984)	rRNA group II	De Vos and De Ley (1983), Mergaert et al. (1996)
<i>P. marina</i>	Palleroni (1984)	<i>Halomonas</i> rRNA lineage	De Vos et al. (1989)
<i>P. mesophilica</i>	Palleroni (1984)	<i>Methylobacterium</i>	Green and Bousfield (1983)
<i>P. mixta</i>		<i>Telluria</i>	Bowman et al. (1988, 1993a)
<i>P. nautica</i>	Palleroni (1984)	<i>Oceanospirillum</i>	De Vos et al. (1989)
<i>P. paucimobilis</i>	Palleroni (1984)	<i>Sphingomonas</i>	De Vos et al. (1989)
<i>P. pictorum</i>	Palleroni (1984)	<i>Xanthomonas</i> rRNA lineage	De Vos et al. (1989)
<i>P. radiora</i>	Palleroni (1984)	<i>Methylobacterium</i>	Green and Bousfield (1983)
<i>P. rhodos</i>	Palleroni (1984)	<i>Methylobacterium</i>	Green and Bousfield (1983)
" <i>P. riboflavina</i> "	Haynes and Burkholder (1957)	<i>Devosia</i>	Nakagawa et al. (1996)
" <i>P. rosea</i> "		<i>Methylobacterium</i>	Bousfield and Green (1985)
<i>P. rubrisubalbicans</i>	Palleroni (1984)	<i>Herbaspirillum</i>	Baldani et al. (1996)
<i>P. saccharophila</i>	Palleroni (1984)	<i>Comamonadaceae</i>	Palleroni et al. (1973), Willems et al. (1992)
<i>P. stanieri</i>		<i>Marinomonas</i>	Y. Ansai, personal communication

TABLE BXII.γ.112. General phenotypic characteristics of some fluorescent *Pseudomonas* species^a

Characteristics	<i>P. aeruginosa</i>	<i>P. chlororaphis</i> subsp. <i>chlororaphis</i>	<i>P. chlororaphis</i> subsp. <i>aureofaciens</i>	<i>P. fluorescens</i>	<i>P. fluorescens</i> bv. I	<i>P. fluorescens</i> bv. II	<i>P. fluorescens</i> bv. III	<i>P. fluorescens</i> bv. IV	<i>P. fluorescens</i> bv. V	<i>P. lundensis</i>	<i>P. montelii</i>	<i>P. putida</i> biovar A	<i>P. putida</i> biovar B	<i>P. rhodesiae</i>	<i>P. veronii</i>
Number of flagella	1	>1	>1	1	>1	>1	>1	>1	>1	1	>1	>1	>1	1	1
Nonfluorescent pigment, color:															
Green	—	+	—	—	—	—	—	—	—	—	—	—	—	—	—
Orange	—	—	+	—	—	—	—	—	—	—	—	—	—	—	—
Yellow	—	—	—	+	—	—	—	—	—	—	—	—	—	—	—
Blue	—	—	—	—	—	—	—	+	—	—	—	—	—	—	—
Gelatin liquefaction	+	+	+	—	+	+	+	+	+	+	—	—	—	—	+
Denitrification	+	+	—	—	—	+	+	+	—	—	—	—	—	—	+
Levan formation	—	+	+	—	+	+	—	+	—	—	—	—	—	+	—
Lecithinase	—	+	+	—	+	d	+	+	d	—	—	—	—	+	—
Lipase	+	+	+	—	+	—	d	d	d	—	—	—	—	—	—
Growth at 4°C	—	+	+	—	+	+	+	+	+	+	—	d	+	+	+
Growth at 41°C	+	—	—	—	—	—	—	—	—	—	+	—	—	—	—
Mol% G + C of the DNA	67	63	64	63	60	60	60	60	60	60	60	62	61	59	61

^aFor symbols see standard definitions. Data from Palleroni (1984), Molin et al. (1986), Hildebrand et al. (1994), Elomari et al. (1996, 1997), and Coroler et al. (1996).

TABLE BXII.γ.113. Nutritional characteristics of some fluorescent *Pseudomonas* species^a

Substrates	<i>P. aeruginosa</i>	<i>P. chlororaphis</i> subsp. <i>chlororaphis</i>	<i>P. chlororaphis</i> subsp. <i>aureofaciens</i>	<i>P. fluorescens</i>	<i>P. fluorescens</i> bv. I	<i>P. fluorescens</i> bv. II	<i>P. fluorescens</i> bv. III	<i>P. fluorescens</i> bv. IV	<i>P. fluorescens</i> bv. V	<i>P. lundensis</i>	<i>P. monteilii</i> ^b	<i>P. putida</i> biovar A	<i>P. putida</i> biovar B	<i>P. rhodesiae</i> ^c	<i>P. veronii</i>
Acetamide	+	—	—	—	—	—	—	—	—	—	—	d	d	—	—
Acetate, L-alanine ^d , γ-aminobutyrate, L-asparagine e, L-aspartate, betaine, caprate, caprylate, citrate, fumarate, L-glutamate ^d , glutarate, glycerol, heptanoate, β-hydroxybutyrate, DL-lactate, L-malate ^d , pelargonate ^e , L-proline, putrescine, pyruvate ^d , succinate ^d	+	+	+	+	+	+	+	+	+	+	+	+	+	+	—
cis-Aconitate	+	d	+	+	+	+	d	+	d	—	+	+	+	+	+
Adipate	+	—	—	—	—	—	d	—	—	—	—	—	d	—	d
Adonitol	—	—	—	—	+	—	d	—	d	—	—	—	—	—	—
β-Alanine, L-arginine, spermine, L-tyrosine	+	+	+	—	+	+	+	+	+	+	+	+	+	+	+
D-Alanine	+	+	—	+	+	+	+	+	—	—	+	+	+	+	+
α-Aminobutyrate	—	—	—	—	—	—	—	—	—	—	—	d	—	+	+
α-Aminovalerate	—	—	—	—	—	—	d	—	—	—	—	d	—	—	—
δ-Aminovalerate	+	d	d	—	d	+	+	+	+	—	+	+	+	+	—
α-Amylamine	—	—	+	—	—	d	d	—	d	—	+	d	+	—	—
Anthranilate	+	+	+	—	d	d	d	—	d	—	—	—	+	—	d
D-Arabinose	—	—	—	—	—	—	—	—	—	+	—	—	—	—	—
L-Arabinose	—	—	+	+	+	+	d	+	d	+	+	d	+	+	+
Azelate	+	—	—	—	—	—	d	—	—	—	—	—	—	—	—
Benzoate	+	+	d	—	d	d	d	+	d	+	+	d	+	—	—
Benzoylformate	+	+	+	—	—	—	—	—	—	—	—	d	d	—	—
Benzylamine	—	—	—	—	—	—	d	—	d	—	+	d	+	—	—
Butanol	+	d	—	—	d	d	d	+	d	—	—	+	+	—	—
Butylamine	—	—	d	—	—	—	—	—	d	—	+	+	+	—	—
2,3-Butylene glycol	+	d	d	+	d	+	d	+	d	—	—	d	d	—	—
Butyrate	+	+	d	—	—	d	d	+	d	+	+	+	+	—	+
Caproate	+	d	+	+	+	d	+	+	+	+	+	+	+	+	+
Cellobiose, ethylene glycol ^e , D-fucose, inulin ^f , isopropanol ^{e,f} , lactose, maleate, maltose, methanol ^{e,f} , oxalate, phthalate, poly-β-hydroxybutyrate ^{e,f} , salicin ^f , starch, L-threonine ^{e,f}	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Citraconate	—	—	—	—	d	d	d	+	d	—	—	—	d	—	—
L-Citrulline	d	d	d	+	d	d	d	—	d	—	—	d	d	—	d
Creatine	—	—	—	+	—	—	d	—	—	+	+	d	d	—	—
Dodecane, hexadecane	d	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Erythritol	—	—	—	—	d	d	+	—	d	—	—	—	—	—	+
Ethanol, propanol	+	d	—	—	—	+	d	—	d	—	—	d	d	—	—
Ethanolamine	d	d	+	+	+	d	d	+	d	+	+	d	d	+	+
D-Fructose	+	d	d	+	+	d	+	+	+	+	+	+	+	+	+
D-Galactose	—	d	+	+	+	+	d	+	d	—	—	—	d	+	+
Geraniol	+	—	—	—	—	—	—	—	—	—	—	—	—	—	—
D-Glucose	+	+	d	+	+	+	+	+	+	+	+	+	+	+	+
Gluconate	+	+	d	+	+	+	+	+	+	+	+	+	+	+	+
Glycerate	+	+	+	+	+	+	d	+	d	—	+	d	+	+	+
Glycine	d	—	—	—	—	—	d	—	d	—	—	d	—	—	—
Glycolate	—	—	—	+	—	—	—	—	—	—	—	d	—	—	d
Hippurate	—	—	—	—	—	—	d	—	—	+	—	d	d	—	—
Histamine	+	d	d	—	d	—	d	—	d	—	d	d	+	—	—
L-Histidine	+	+	+	+	+	d	+	+	d	+	+	+	+	+	d
o-Hydroxybenzoate	—	d	—	—	—	—	—	—	—	—	—	d	d	—	—
m-Hydroxybenzoate	—	d	d	—	—	—	—	—	—	—	—	d	d	—	—
p-Hydroxybenzoate	+	+	+	+	+	+	d	+	d	—	+	+	+	+	+
Hydroxymethylglutarate	—	—	—	+	d	d	—	—	d	—	—	—	—	—	—

^aFor symbols see standard definitions. Data from Palleroni (1984), Molin et al. (1986), Hildebrand et al. (1994), Elomari et al. (1996, 1997), and Coroler et al. (1996).

^b*P. monteilii* was also reported to be unable to use the following carbon sources: *N*-acetylglucosamine, esculin, *m*-aminobenzoate, *p*-aminobenzoate, 3-aminobutyrate, amygdalin, D-arabitol, L-arabitol, arbutin, L-cysteine, dulcitol, ethylamine, L-fucose, β-gentiobiose, glucosamine, glycogen, isophthalate, 5-ketogluconate, D-lyxose, L-lyxose, melezitose, melibiose, L-methionine, α-methylglucoside, α-methyl-D-mannoside, α-methyl-xyloside, raffinose, L-sorbose, terephthalate, raffinose, tagatose, D-tryptophan, D-turanose, xylitol, and L-xylose.

^c*P. rhodesiae* was also found to use acetylglucosamine and D-arabitol, and was unable to grow on esculin, 2-aminobenzoate, 3-aminobenzoate, 4-aminobenzoate, 3-aminobutyrate, amygdalin, L-arabitol, arbutin, L-cysteine, dulcitol, ethylamine, L-fucose, gentiobiose, glucosamine, glycogen, norvaline, raffinose, salicin, sorbose, D-tagatose, terephthalate, D-turanose, urea, xylitol, and L-xylose.

^dPositive for *P. veronii*; results on other substrates not reported.

^e*P. monteilii* not tested.

^fNo information on *P. veronii*.

(continued)

TABLE BXII.γ.113. (cont.)

Substrates	<i>P. aeruginosa</i>	<i>P. chlororaphis</i> subsp. <i>chlororaphis</i>	<i>P. chlororaphis</i> subsp. <i>aureofaciens</i>	<i>P. flavescens</i>	<i>P. fluorescens</i> bv. I	<i>P. fluorescens</i> bv. II	<i>P. fluorescens</i> bv. III	<i>P. fluorescens</i> bv. IV	<i>P. fluorescens</i> bv. V	<i>P. lundensis</i>	<i>P. monteilii</i> ^b	<i>P. putida</i> biovar A	<i>P. putida</i> biovar B	<i>P. rhodesiae</i> ^c	<i>P. veronii</i>
<i>m</i> -Inositol	—	+	+	—	d	+	d	+	d	—	+	—	—	+	+
Isobutanol	+	—	—	—	—	d	d	—	d	—	—	d	d	—	+
Isobutyrate	+	—	d	—	—	d	d	—	d	—	+	d	d	—	+
L-Isoleucine, L-valine	d	+	+	—	+	+	+	+	+	+	+	+	+	+	+
Isovalerate	+	+	+	—	d	d	d	—	d	—	+	+	+	d	—
Itaconate, mesaconate	+	+	+	+	+	d	d	—	d	—	—	d	—	—	—
2-Ketogluconate	+	+	d	—	+	+	+	d	+	—	+	d	+	+	+
α-Ketoglutarate	+	+	+	+	+	+	+	+	+	+	—	+	+	—	—
L-Kynurenine	+	+	+	—	d	d	d	—	d	—	—	—	+	—	—
Kynurenate	d	d	d	—	d	—	d	—	—	—	—	—	d	—	—
L-Leucine	+	+	+	+	+	d	+	+	+	+	+	+	+	+	d
Levulinate	+	d	d	—	d	—	d	—	d	—	—	d	d	—	d
L-Lysine	+	d	d	—	+	d	d	+	d	+	+	+	d	—	—
D-Malate	d	d	—	+	—	d	d	+	d	—	d	d	d	—	—
Malonate	+	+	+	+	+	+	d	+	d	—	d	d	+	+	+
D-Mandelate	—	—	—	—	—	—	—	—	—	—	—	d	d	—	—
L-Mandelate	+	—	—	—	—	—	—	—	d	—	—	—	d	—	—
Mannitol	+	+	+	+	+	+	d	+	d	—	—	d	d	+	+
D-Mannose	—	+	+	+	+	+	+	+	d	+	—	d	d	+	+
Mucate	—	+	+	+	+	+	d	+	+	—	—	d	+	+	d
Naphthalene	—	—	—	—	—	—	—	—	—	—	—	—	d	—	—
Nicotinate	—	—	—	—	—	—	—	—	d	—	—	d	+	—	—
DL-Norleucine	—	—	—	—	—	—	—	—	—	—	+	—	—	—	—
L-Ornithine	+	d	+	—	+	d	d	d	d	+	+	+	+	—	d
Pantothenate	—	—	—	—	—	—	d	—	—	—	—	—	—	—	—
Phenol	—	—	—	—	—	—	d	—	—	—	—	d	d	—	—
Phenylacetate	—	d	+	—	—	—	d	—	d	—	+	d	+	—	—
L-Phenylalanine	d	d	+	—	d	d	d	+	d	+	—	+	+	+	+
Phenylethanediol	—	—	—	—	—	—	—	—	—	—	—	d	—	—	—
Pimelate, suberate	d	—	—	—	—	—	d	—	—	—	—	—	—	—	—
Propionate	+	+	+	—	+	+	d	+	+	+	+	+	+	+	+
Propylene glycol	d	—	—	—	—	—	d	—	—	—	—	—	—	—	—
Quinate	d	+	+	—	+	+	+	+	d	—	—	+	+	—	—
L-Rhamnose	—	—	—	—	—	d	d	—	d	—	—	—	—	—	—
D-Ribose	+	+	+	+	+	+	d	+	d	+	+	d	d	+	+
Saccharate	—	+	+	+	+	+	d	+	d	—	—	+	+	—	—
Sarcosine	d	d	+	—	+	d	d	+	d	+	+	+	+	+	+
Sebacate	+	—	—	—	—	—	d	—	—	—	—	—	—	—	—
L-Serine	d	d	+	+	+	d	+	+	d	—	d	d	d	+	+
Sorbitol	—	—	—	—	+	+	d	+	d	—	—	—	d	+	+
Sucrose	—	+	d	+	+	+	—	+	d	—	—	—	d	—	+
D(–)-Tartrate	—	—	—	—	—	—	d	—	—	—	—	d	d	—	—
L(+)-Tartrate	—	d	—	—	—	—	—	+	—	—	—	d	d	—	—
<i>m</i> -Tartrate	—	—	—	+	—	—	d	—	d	—	—	d	—	—	—
Testosterone	—	—	—	—	—	—	—	—	—	—	—	—	+	—	—
Trehalose	—	+	d	+	+	+	d	+	d	—	—	—	—	+	d
Trigonelline	—	—	—	+	d	d	d	—	d	—	d	d	+	—	+
Tryptamine	—	—	—	—	—	d	d	—	—	—	—	d	+	—	—
D-Tryptophan	—	—	—	—	—	—	—	—	—	—	—	—	d	—	—
L-Tryptophan	d	+	+	—	+	d	d	—	d	—	—	—	+	—	+
Valerate	+	+	+	—	d	d	d	—	d	—	+	+	+	+	+
D-Xylose	—	—	—	+	+	d	d	d	d	—	—	d	d	+	+

^aFor symbols see standard definitions. Data from Palleroni (1984), Molin et al. (1986), Hildebrand et al. (1994), Elomari et al. (1996, 1997), and Coroler et al. (1996).

^b*P. monteilii* was also reported to be unable to use the following carbon sources: *N*-acetylglucosamine, esculin, *m*-aminobenzoate, *p*-aminobenzoate, 3-aminobutyrate, amygdalin, D-arabitol, L-arabitol, arbutin, L-cysteine, dulcitol, ethylamine, L-fucose, β-gentiobiose, glucosamine, glycogen, isophthalate, 5-ketogluconate, D-lyxose, L-lyxose, melezitose, melibiose, L-methionine, α-methylglucoside, α-methyl-D-mannoside, α-methyl-xyloside, raffinose, L-sorbose, terephthalate, raffinose, tagatose, D-tryptophan, D-turanose, xylitol, and L-xylose.

^c*P. rhodesiae* was also found to use acetylglucosamine and D-arabitol, and was unable to grow on esculin, 2-aminobenzoate, 3-aminobenzoate, 4-aminobenzoate, 3-aminobutyrate, amygdalin, L-arabitol, arbutin, L-cysteine, dulcitol, ethylamine, L-fucose, gentiobiose, glucosamine, glycogen, norvaline, raffinose, salicin, sorbose, D-tagatose, terephthalate, D-turanose, urea, xylitol, and L-xylose.

^dPositive for *P. veronii*; results on other substrates not reported.

^e*P. monteilii* not tested.

^fNo information on *P. veronii*.

TABLE BXII.γ.114. Characteristics differentiating *Pseudomonas aeruginosa*, *P. balearica*, *P. stutzeri*, and *P. putida**

Characteristics	<i>P. aeruginosa</i>	<i>P. balearica</i>	<i>P. putida</i>	<i>P. stutzeri</i>
<i>Type of colony:</i>				
Smooth	+		+	
Wrinkled		+		+
Number of flagella	1	1	>1	1
<i>Hydrolysis of:</i>				
Gelatin	+	—	—	—
Starch	—	+	—	+
<i>Utilization of:</i>				
Maltose	—	+	d	+
Xylose	—	+	d	—
γ-Aminobutyrate	—	—	d	d
Malate	d	+	—	+
Suberate	d	—	—	d
Mannitol	+	—	—	d
Ethylene glycol	—	—	—	+
Denitrification	+	+	—	+
<i>Growth at:</i>				
42°C	+	+	—	d
46°C	—	+	—	d
Growth in media with 8.5% NaCl	—	+	—	—
<i>Fatty acid content (%):</i>				
C _{17:0} cyclo	0.8	4.71	>5	0.28–1.72
C _{19:0} cyclo	1.2	3.8	Traces	0.32–1.45
Mol% G + C of the DNA	67	64.1–64.4	60.7–62.5	60.9–64.9

*For symbols see standard definitions. Data from Bennisar et al. (1996) and Stanier et al. (1966).

natural sources (Véron and Berche, 1976). Variation of the large type to the small is rather common but the reverse variation is extremely rare. A third colony type (mucoid) often can be obtained from respiratory and urinary tract secretions and was first observed by Sonnenshein (1927). Mucoid mutants can be divided into two groups according to the capacity of forming mucus (alginate) in chemically defined media (Fyfe and Govan, 1980). Aside from pyoverdine and pyocyanin, other pigments may be produced by some strains, including a dark red pigment (see above section on pigments).

Denitrification and gelatin liquefaction is present in the great majority of strains. The hydrolysis of Tween 80 is weak and the egg-yolk reaction is negative. Optimum temperature, 37°C. *P. aeruginosa* is probably the most widespread of all bacterial species. It can be isolated from soil and water, particularly from enrichment cultures for denitrifying bacteria. Commonly isolated from clinical specimens (wound, burn, and urinary tract infections). Causative agent of “blue pus”, which accounts for the origin of the synonym pyocyanus. Occasionally pathogenic for plants. Strains isolated from leaf spot of tobacco, identical with or similar to *P. aeruginosa* have been named “*P. polycolor*” (Clara, 1930). The species can be internally divided into a number of subgroups (types) useful for epidemiological purposes.

The mol% G + C of the DNA is: 67.2 (Bd).

Type strain: ATCC 10145, DSM 50071, NCIB 8395, NCTC 10332.

GenBank accession number (16S rRNA): X06684, Z76651.

2. *Pseudomonas agarici* Young 1970, 985^{AL}

a.gar'i.ci. M.L. n. *Agaricus* a genus of fungi; M.L. gen. n. *agarici* of *Agaricus*.

The following description is a summary of the one presented by Young (1970).

Short rods (no dimensions given), motile by one or, rarely, two polar flagella. Green diffusible pigment with

weak fluorescence under ultraviolet light of unspecified wavelength (in our experience, the maximum intensity of fluorescence is observed at low wavelength UV, ~257 nm). Acid is produced from arabinose, glucose, and mannitol; little acid from fructose, galactose, and ribose, and no acid from rhamnose, xylose, mannose, lactose, sucrose, maltose, trehalose, melibiose, cellobiose, raffinose, starch, inulin, dextrin, glycogen, adonitol, sorbitol, inositol, and salicin.

Acetate, benzoate, citrate, formate, fumarate, gluconate, lactate, propionate, and succinate are utilized. Galacturonate, oxalate, and tartrate are not. Oxidase and catalase reactions positive; nitrate reduction, pectate liquefaction, starch hydrolysis, esculin hydrolysis, levan production, and growth factor requirements are all negative. Further extensive phenotypic studies on strains of this species in comparison with other fluorescent organisms have been published (Fahy, 1981). Some of the reported properties are summarized in Table BXII.γ.115. Details of its fatty acid composition are known (Stead, 1992).

The organism causes drippy gill of mushrooms, and one of the main differences with another mushroom pathogen, *P. tolaasii*, is in the utilization of benzoate. Catechol is oxidized to a black pigment diffusing into the medium, a character also present in *P. agarici*. The species was tentatively assigned to RNA group I by Byng et al. (1980), and this position was further confirmed by rRNA–DNA hybridization (De Vos et al., 1985) and by rDNA sequencing (Moore et al., 1996).

The mol% G + C of the DNA is: 58.8–61.1 (*T_m*).

Type strain: ATCC 25941, DSM 11810, LMG 2289.

GenBank accession number (16S rRNA): Z76652.

3. *Pseudomonas alcaligenes* Monias 1928, 332^{AL}

al.ca.li'ge.nes. M.L. adj. *alcaligenes* alkali-producing.

Characteristics useful to differentiate the species from other *Pseudomonas* species are given in Tables BXII.γ.116 and BXII.γ.117. For further descriptive information see Ral-

TABLE BXII.γ.115. Characters distinguishing some fluorescent *Pseudomonas* species associated with mushroom culture^a

Characteristics	<i>P. agarici</i>	<i>P. cichorii</i>	<i>P. fluorescens</i> biovar II	<i>P. tolaasii</i>
Levan formation from sucrose	—	—	+	—
Arginine dihydrolase	—	—	+	+
Denitrification	—	—	+	—
Gelatin hydrolysis	—	—	+	+
Egg yolk reaction	—	+	—	+
Growth at the expense of:				
Trehalose	—	—	+	d
2-Ketogluconate	—	—	+	d
meso-Inositol	—	+	+	+
L-Valine	d	—	+	+
β-Alanine	+	—	+	+
L-Arabinose	—	+	+	d
Sucrose	—	+	+	—
Sorbitol	—	—	+	+
Adonitol	—	—	d	d
Ethanol	—	—	+	d
meso-Tartrate	—	+	—	+
Nicotinate	—	—	—	+
Staining of mushroom caps	d			+
Pitting of mushroom caps	—			+

^aFor symbols see standard definitions. Data from Fahy (1981) and Stanier et al. (1966).**TABLE BXII.γ.116.** General characteristics of some nonfluorescent *Pseudomonas* species^a

Characteristics	<i>P. alcaligenes</i>	<i>P. corrugata</i>	<i>P. luteola</i>	<i>P. mendocina</i>	<i>P. oryzae</i> habitans	<i>P. pseudoalcaligenes</i>	<i>P. stutzeri</i>
Cell diameter, μm	0.5		0.8	0.7–0.8	0.8	0.7–0.8	0.7–0.8
Cell length, μm	2.0–3.0		2.5	1.4–2.8	2	1.2–2.5	1.4–2.8
Number of flagella	1	>1	>1	1b	1	1	1 ^b
Yellow or orange pigments	d	+	+	+	+	—	—
Oxidase reaction	+	+	—	+	—	+	+
PHB accumulation	—	+	+	—	—	d	—
Gelatin liquefaction	d	d	+	—	—	d	—
Starch hydrolysis	—	—	—	—	—	—	+
Lecithinase	—	+	—	—	—	—	—
Lipase	d	d	—	+	—	—	+
Growth at 41°C	+	—	+	+	+	+	+
Denitrification	+	—	—	+	—	d	+
Arginine dihydrolase	+	d	+	+	—	d	—
Mol% G + C of the DNA	64–68	58–61	55.4	63–64	65.1	62–64	61–66

^aFor symbols see standard definitions. Data from Palleroni (1984), Kodama et al. (1985), and Sutra et al. (1997).^bLateral flagella of short wavelength may be produced under certain conditions.

ston-Barrett et al. (1976) and Stanier et al. (1966). The nutritional spectrum is very narrow, resembling that of highly mutated fluorescent organisms. The gelatinase reaction is negative. The type strain was isolated from swimming pool water (Hugh and Ikari, 1964).

The mol% G + C of the DNA is: 64–68 (Bd).

Type strain: Stanier 142, ATCC 14909, LMG 1224NCIB 9945, NCTC 10367.

GenBank accession number (16S rRNA): Z76653.

4. *Pseudomonas amygdali* Psallidas and Panagopoulos 1975, 105^{AL}

a.myg'da.li. L. n. *amygdalum* almond; L. gen. n. *amygdali* of the almond.

The following description is taken from Psallidas and Panagopoulos (1975).

Rods, 0.7 × 1.7 μm or much longer (filaments can be 10–15 times the length of normal cells). Motile by means of one to six polar flagella. No PHB accumulated. Grows better in potato-dextrose medium than in nutrient agar. Growth range, 3–32°C. No growth below pH 5. No fluo-

rescent pigment produced. Acid is formed from D-ribose, L-arabinose, glucose, mannose, galactose, fructose, sucrose, mannitol, and sorbitol. No utilization of xylose, L-rhamnose, L-sorbose, cellobiose, lactose, maltose, melibiose, trehalose, raffinose, inulin, esculin, amygdalin, arbutin, salicin, dulcitol, erythritol, glycerol, inositol, dextrin and α-methyl-D-glucoside. Malate, citrate, succinate, and fumarate are utilized. Gluconate is slowly assimilated. Acetate, propionate, oxalate, maleate, malonate, tartrate, lactate, sulfanilic acid, picrate, hippurate, and benzoate are not utilized. Among the natural amino acids, serine, aspartate, glutamate, arginine, asparagine, proline, and histidine are utilized. Not used as carbon and/or nitrogen sources are glycine, β-alanine, leucine, isoleucine, valine, lysine, ornithine, tyrosine, phenylalanine, tryptophan, cystine, cysteine, methionine, and creatine.

Some isolates are urease positive. Tween 80 and tributyrin are rapidly hydrolyzed. Lecithinase and arginine dihydrolase negative. Gelatin, casein, esculin, arbutin, and starch are not hydrolyzed. Nitrates are not reduced. Rotting of potato slices does not occur, but the organism is positive

TABLE BXII.γ.117. Nutritional characteristics of some nonfluorescent species of *Pseudomonas*^a

Substrates ^b	<i>P. alcaligenes</i>	<i>P. corrugata</i> ^c	<i>P. mendocina</i>	<i>P. pseudoalcaligenes</i>	<i>P. stutzeri</i>
Acetate, L-alanine, caprate, caprylate, fumarate, L-glutamate, α-ketoglutarate, lactate, L-proline, succinate	+	+	+	+	+
Aconitate, caproate, heptanoate, L-tyrosine, valerate	d	+	+	d	d
Adipate	—	—	—	—	d
β-Alanine	d	—	+	d	—
D-Alanine, citrate	d	+	+	d	+
α-Aminobutyrate	—	—	+	—	d
γ-Aminobutyrate	+	+	—	+	—
L-Arabinose, D-galactose, <i>m</i> -inositol, D-mannose, D-ribose, sucrose, trehalose, trigonelline, D-xylose	—	+	—	—	—
δ-Aminovalerate	d	+	—	d	d
α-Amylamine	d	—	—	—	—
L-Arginine	+	+	+	+	—
L-Aspartate, L-isoleucine, malonate, L-valine	—	+	+	—	d
Azelate, maltose, sebacate, starch	—	—	—	—	+
Benzoate	—	—	d	—	d
Betaine	—	+	+	+	—
Butanol, propanol, putrescine	d	—	+	+	d
Butylamine	d	—	—	—	—
2,3-Butylene glycol	—	—	—	—	d
Butyrate	d	d	+	+	+
Isobutyrate, citraconate	—	d	d	—	—
Creatine, sorbitol	—	—	—	d	—
Dodecane	d	—	—	—	—
Ethanolamine	—	d	—	+	d
Ethylene glycol	—	—	+	d	+
D-Fructose	—	+	d	+	d
Gluconate, glutarate, L-serine	—	+	+	d	d
D-Glucose	—	+	+	—	+
Glycerate	—	d	+	+	d
Glycerol	—	+	d	d	+
Glycine, D-malate	—	—	d	d	d
Glycolate	—	—	+	—	+
Histamine	d	d	—	d	—
L-Histidine	d	+	+	d	—
<i>p</i> -Hydroxybenzoate, mannitol	—	+	—	—	d
β-Hydroxybutyrate	—	+	+	+	+
Hydroxymethylglutarate	—	—	d	—	d
Isobutanol	d	—	+	—	d
Isovalerate, mucate, saccharate	—	—	+	—	d
Itaconate	—	—	+	d	+
L-Leucine	+	+	+	d	+
Levulinate	—	—	+	—	—
L-Lysine, DL-norleucine, L-ornithine, D(—)-tartrate, m-tartrate	—	d	—	—	—
L-Malate	+	+	—	+	+
Mesaconate	—	—	+	+	+
Pelargonate, propionate, spermine	+	+	+	+	d
L-Phenylalanine	—	+	d	d	d
Pyruvate	+	d	d	+	+
Sarcosine	—	+	+	d	—
L(+) -Tartrate, tryptamine	—	—	d	—	—

^aFor symbols see standard definitions. Data from Palleroni (1984) and Sutra et al. (1997).

^bThe following compounds were not used by any of the species: adonitol, anthranilate^d, D-arabinose, benzoyl-formate^d, benzylamine, cellobiose, citrulline, erythritol, hexadecane^d, hippurate^d, *o*-hydroxybenzoate, *m*-hydroxybenzoate, isopropanol, 2-ketogluconate, kynurenate^d, lactose, maleate, mandelate, D- and L-mandelate, naphthalene^d, nicotinate^d, oxalate, panthothenate^d, phenol^d, phenyl-acetate, phenyl-ethanediol^d, phthalate, pimelate, L-rhamnose, salicin, suberate, testosterone^d, and L-threonine.

^cN-acetylglucosamine, D-arabitol, diaminobutane are all positive for *P. corrugata* (other species untested). D-lyxose was variable for *P. corrugata* (other species untested). Substrates not used by *P. corrugata* (other species not tested) are adonitol, 2, 3, and 4-aminobenzoates, amygdalin, L-arabitol, arbutin, creatine, cysteine, dulcitol, ethylamine, D- and L-fucose, gentiobiose, glycogen, inulin, 5-ketogluconate, melezitose, melibiose, methionine, methylglucoside, methylmannoside, methylxyloside, norvaline, isophthalate, terephthalate, raffinose, tagatose, turanose, urea, and xylitol.

^dCompounds not been tested for *P. corrugata*.

in the hypersensitivity test on tobacco leaves. Further details are given in the original paper. Assignment to rRNA group I has been decided based on rRNA–DNA hybridization studies (De Vos et al., 1985), and later confirmed by rDNA sequence analysis (Moore et al., 1996; Anzai et al., 1997). The fatty acid composition places the species in one of the subgroups of group I (Stead, 1992). Pathogenic for the almond tree (*Prunus dulcis*, family Rosaceae), in which it

produces a hyperplastic bacterial canker. Not pathogenic for other fruit trees.

The mol% G + C of the DNA is: 57.7–58.5 (T_m); strain NCPPB 2610, 52.2.

Type strain: ATCC 33614, DSM 7298, LMG 2123, NCPPB 2607.

GenBank accession number (16S rRNA): D84007, Z76654.

Additional Remarks: The DNA similarity studies of Gardan

et al. (1999) have included the type strains of this species as well as those of *P. ficuserectae*, *P. meliae*, and *P. savastanoi*. They indicate that these names should be considered as synonymous, and that *P. amygdali* is the correct one. Unfortunately, no decision on this point seems advisable in view of the similarity in known phenotypic properties among these organisms.

5. ***Pseudomonas anguilliseptica*** Wakabayashi and Egusa 1972, 584^{AL}
an.guil.li.sep' ti.ca. L. n. *anguilla* eel; Gr. adj. *septica* putrefactive; M.L. adj. *anguilliseptica* pertaining to diseased eels.

The description below is taken from the original paper.

Rods, $0.4 \times 2.0 \mu\text{m}$, with a tendency to become filamentous. Motile by a single polar flagellum; motility is better at 15°C than at 25°C. Catalase and oxidase reactions positive. Gelatin is liquefied. Nitrate reduction, urease, fluorescent pigment production, and starch hydrolysis are all negative. There is no acid production from arabinose, xylose, rhamnose, fructose, galactose, glucose, mannose, sorbose, lactose, maltose, raffinose, sucrose, starch, dextrin, salicin, glycerol, mannitol, or inositol. rDNA sequencing studies place this species close to *P. aeruginosa* (see Fig. BXII.γ.124). Isolated from diseased pond-cultured eels (*Anguilla japonica*).

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 33660, DSM 12111, NCMB 1949.

GenBank accession number (16S rRNA): X99540.

6. ***Pseudomonas asplenii*** (Ark and Tompkins 1946) Săvulescu 1947, 11^{AL} (*Phytomonas asplenii* Ark and Tompkins 1946, 760.)
a.spl'e ni.i. M.L. neut. n. *Asplenium* genus of ferns, spleenworts; M.L. gen. n. *asplenii* of *Asplenium*.

A description is given by Haynes and Burkholder (1957) in the seventh edition of the *Manual*.

The species was tentatively placed in RNA group I by Byng et al. (1980), an assignment confirmed by nucleic acid hybridization studies (De Vos et al., 1985; Kersters et al., 1996). Pathogenic for the bird's-nest fern (*Asplenium nidus*), which has been the source of isolation.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 23835, LMG 2137.

GenBank accession number (16S rRNA): Z76655.

7. ***Pseudomonas aurantiaca*** Nakhimovskaya 1948, 64^{AL}
au.ran.ti' a.ca. M.L. adj. *aurantiaca* orange colored.

The description is taken from the original paper.

Rods, $0.3\text{--}0.5 \times 0.8\text{--}2.0 \mu\text{m}$; motile, lophotrichous, with four to six flagella. Two main diffusible pigments, green and orange, are produced. The colony may remain orange, while the medium is stained green and fluoresces. Colorless colonies are also produced, presumably by mutation. Gelatin liquefaction is rapid. Starch is not hydrolyzed. Growth is good in complex and synthetic media. Nitrogen sources assimilated include ammonia, nitrate, amino acids, and peptone. The organism has been described as "oligonitrophilic", that is, its nitrogen requirements are satisfied by the traces of nitrogenous compounds that are contained in the "nitrogen-free" medium used in the experiments (Nakhimovskaya, 1948). Optimum temperature 25°C. Capable of growth and acid formation from arabinose, xylose, glucose, galactose, sucrose, raffinose, glycerol, and mannitol. Lactose, maltose, and starch are not utilized.

Studies on the type strain of this species have shown that it actually corresponds to *P. chlororaphis* subsp. *aureofaciens* (Kiprianova et al., 1985). These authors propose to retain the name *P. aurantiaca* for pigmented strains of biovar II of *P. fluorescens*.

The mol% G + C of the DNA is: unknown.

Type strain: NCIB 10068.

8. ***Pseudomonas avellanae*** Janse, Rossi, Angelucci, Scortichini, Derks, Akkermans, De Vrijer and Psallidas 1997, 601^{VP} (Effective publication: Janse, Rossi, Angelucci, Scortichini, Derks, Akkermans, De Vrijer and Psallidas 1996, 594.)
a'vel.la.nae. L. adj. *avellana* the species name of the host, *Corylus avellana*.

The following description is taken from Janse et al. (1996), who transcribed the information from the original paper by Psallidas (1993).

Rods occurring single or in pairs. Motile by means of one to four polar flagella. PHB is not accumulated. Slow growth in nutrient agar. On sucrose-containing media they form circular, dome-shaped, glistening, semitranslucent, butyrous, radially striated, cream-white or pearl-white colonies, 2–2.5 mm and 3.5–4 mm in diameter after 3 and 7 d, respectively. A levan is formed on nutrient medium plus 5% sucrose. Weak, blue-green fluorescent diffusible pigment formed in King B medium. Pantothenate is required for growth. 4% NaCl in the medium is tolerated. Catalase and lipase (Tween 80 hydrolysis) are positive. Urease, oxidase reaction, tyrosinase, arginine and ornithine dihydrolyase, amylase, gelatin liquefaction, hydrolysis of casein, and protopectinase reaction (rotting of potato slices) are all negative. Nitrates are not reduced. Hydrolysis of esculin and arbutin are negative. H₂S is not produced from peptone, cysteine, or thiosulfate. Ammonia and 2-ketogluconate are not produced.

The following carbon compounds are used for growth: L-arabinose, D-ribose, D-xylose, D-glucose, D-fructose, D-galactose, D-mannose, sucrose, glycerol, mannitol, meso-inositol, citrate, formate, fumarate, maleate, malate, malonate, mucate, nonanoate, propionate, and succinate. Some strains use D-alanine, L-alanine, L- and DL-aspartate, L-glutamate, glutamine, L-proline, and L- and DL-serine. Organic compounds not used for growth include adonitol, esculin, amygdalin, arbutin, cellobiose, dulcitol, inulin, lactose, maltose, melezitose, melibiose, meso-erythritol, α-methylglucoside, L-rhamnose, salicin, L-sorbose, sorbitol, trehalose, adipate, benzoate, cinnamate, hippurate, itaconate, lactate, oxalate, picrate, sulfanilate, D(–)-tartrate, L(+)-tartrate, DL-tartrate, L-aspartate, β-alanine, L-arginine, L- or DL-asparagine, L-citrulline, L-cysteine, L-cystine, glycine, L-histidine, DL-homoserine, L-hydroxyproline, L-leucine, L-lysine, L-methionine, DL-ornithine, DL-phenylalanine, L-sarcosine, DL-tyrosine, L-threonine, DL-tryptophan, and L-valine. Further details are mentioned in the original paper (Janse et al., 1996). Optimum growth temperature, 23–25°C. Maximum, 30°C. Strains contain plasmids in the range of 33–100 MDa. It causes hypersensitive reaction in tobacco leaves. Isolated from hazelnut trees (*Corylus avellana* L.), where it causes cankers and dieback on stems and branches.

The mol% G + C of the DNA is: unknown.

Type strain: F11, DSM 11809, ICMP 9746, NCPPB 3487.

Additional Remarks: Interesting additional information

on this species is found in the literature (Janse et al., 1996). Fourteen strains were analyzed for plasmid DNA, fatty acids of whole cells, whole-cell proteins, and restriction fragment patterns of 16S rRNA. In comparison with other fluorescent species of the genus *Pseudomonas*, *P. avellanae* strains are homogeneous in their properties, and the molecular approaches indicate that the species, until recently considered to be a pathovar of *P. syringae* (*Pseudomonas syringae* pathovar *avellanae*, Psallidas, 1993), warrants independent species allocation.

9. ***Pseudomonas azotoformans*** Iizuka and Komagata 1963b, 137^{AL}

a.zo.to.for' mans. Fr. n. *azote* nitrogen; L. v. *formo* to fashion, form; M.L. part. adj. *azotoformans* nitrogen forming (by denitrification).

The description is taken from the paper by Iizuka and Komagata (1963b).

Rods, 0.6–0.8 × 1.4–2.0 µm, motile with polar flagella. Fluorescent pigment is produced. Oxidase, gelatin liquefaction, nitrate reduction, and denitrification are all positive. H₂S is not produced and starch is not hydrolyzed. Acid is produced from glucose, glycerol, xylose, sucrose, but not from lactose or starch. Glucose, gluconate, 2-ketogluconate, citrate, succinate, ethanol, *p*-hydroxybenzoate, protocatechuate, and anthranilate are assimilate, but not phenol, benzoate, salicylate, *m*-hydroxybenzoate, gentisate, *p*-aminobenzoate, or 5-ketogluconate. Additional details can be found in the original paper. Optimum growth temperature, 25–30°C. No growth at 37°C. The description suggests that this species is similar to some of the denitrifying biovars of *P. fluorescens*, but no comparative studies have been performed. Assignment to the genus *Pseudomonas* has been confirmed by 16S rDNA sequence analysis studies (Anzai et al., 1997). Isolated from Japanese rice paddies.

The mol% G + C of the DNA is: unknown.

Type strain: IAM 1603.

GenBank accession number (16S rRNA): D84009.

10. ***Pseudomonas balearica*** Bennasar, Rosselló-Mora, Lalucat and Moore 1996, 204^{VP}

ba.le.a'ri.ca. L. fem. adj. *balearica* of the Balearic Islands, where the organism was isolated.

Description taken from the paper by Bennasar et al. (1996).

Short, straight rods, 0.3–0.5 × 1.5–3.0 µm. Motile by means of a single polar flagellum. It grows aerobically, or under anaerobic conditions in the presence of nitrate. Vigorous denitrifier, liberating copious amounts of nitrogen gas from nitrate. Nonpigmented. It shares with *P. stutzeri* some of the phenotypic and morphological characteristics, including starch hydrolysis, growth with maltose, negative arginine dihydrolase, and gelatinase reactions, and a typical wrinkled colony morphology when freshly isolated. Differences with *P. stutzeri* include the ability to grow at 46°C and to tolerate 8.5% NaCl in the medium. It is also able to grow on xylose, but it does not utilize ethylene glycol, mannitol, 4-aminobutyrate, and suberate. Table BXII.γ.114 summarizes differences with other species of *Pseudomonas*. Some characteristics that differentiate the species from other denitrifying pseudomonads are presented in the chapter on *Burkholderia*. The two strains that have been isolated are

able to degrade 2-methylnaphthalene at 40°C in wastewater or marine sediment samples.

The mol% G + C of the DNA is: 64.1–64.4 (HPLC).

Type strain: SP1402, DSM 6083.

GenBank accession number (16S rRNA): U26418.

11. ***Pseudomonas caricapapayae*** Robbs 1956, 74^{AL}

ca.ri.ca.pa'pay.ae. M.L. gen. n. *caricapapayae* of *Carica papaya*, pawpaw.

The description is taken from Robbs (1956).

Straight or slightly curved rods, 0.9–1.1 × 1.3–3.0 µm. Motile by three to six polar flagella. Green fluorescent pigment produced. Gelatin is liquefied. Starch is not hydrolyzed. Nitrates are not reduced. Acid is produced from glucose, mannose, sucrose, glycerol, and mannitol, but not from lactose, maltose, salicin, or starch. Citrate and tartrate are utilized; lactate is not. rRNA–DNA hybridization studies have confirmed the allocation of this species among the fluorescent members of the genus *Pseudomonas* (De Vos et al., 1985). Optimum growth temperature: 23–29°C; no growth below 7°C and above 45°C. Isolated from water-soaked, angular spots on leaves of pawpaw.

The mol% G + C of the DNA is: 58.9–59.4 (*T_m*).

Type strain: ATCC 33615, NCPPB 1873.

GenBank accession number (16S rRNA): D84010.

12. ***Pseudomonas cedrella*** Dabboussi, Hamze, Elomari, Verhille, Baida, Izard and Leclerc 1999b, 303

ced.rel'la. M.L. fem. gen. n. *cedrella* of the cedar tree.

Rods, motile by means of a single polar flagellum. Colonies on nutrient agar are circular and smooth. On blood agar there are no signs of hemolysis. Fluorescent pigment is produced on King B agar. Oxidase, catalase, and arginine dihydrolase reactions are all negative. Some strains (2/7) liquefy gelatin. Levan is produced from sucrose. Two out of seven strains liquefied gelatin. Growth occurs at 4°C but not at 41°C. No growth occurs with 5% NaCl in the medium. Various enzymatic reactions are described in the paper (Dabboussi et al., 1999b), but gelatinase is not reported. Denitrification is a variable character. Growth at the expense of various carbon sources, particularly carbohydrates and derivatives, and various enzymatic activities are reported in the description. A comparison of growth characteristics and some general phenotypic characters, with those of other fluorescent species, is presented in the paper.

The results of DNA–DNA hybridization experiments suggest that perhaps this species should be considered as a genomovar of some of the biovars of *P. fluorescens*. Isolated from Lebanese spring water.

The mol% G + C of the DNA is: 59–60.

Type strain: CFML 96-198; CIP 105541.

GenBank accession number (16S rRNA): AF064461.

13. ***Pseudomonas chlororaphis*** (Guignard and Sauvageau 1894) Bergey, Harrison, Breed, Hammer and Huntton 1930, 166^{AL} (*Bacillus chlororaphis* Guignard and Sauvageau 1894, 841.)

chlo.ro.ra'phis. Gr. adj. *chlorus* green; Gr. n. *raphis* a needle; M.L. fem. n. *chlororaphis* a green needle.

Fluorescent pigment is produced. Even though the species name refers to the production of the green insoluble phenazine pigment chlororaphin, which often is excreted and crystallizes around the colonies, *P. chlororaphis* has been

found to be closely related to *P. aureofaciens* Kluyver (which does not produce this pigment, but instead makes another phenazine pigment that is orange and freely diffuses into the medium. The relationship between the two species was demonstrated by a high DNA similarity as determined by DNA–DNA hybridization (Palleroni et al., 1972; Johnson and Palleroni, 1989). For this reason, *P. chlororaphis* may include two subspecies: subsp. *chlororaphis*, producing chlororaphin, and subsp. *aureofaciens*, which produces the water-soluble phenazine monocarboxylic acid. Other phenotypic properties that differentiate the two subspecies are the utilization of L-arabinose and α -amylamine by subsp. *aureofaciens*, and of D-alanine by subsp. *chlororaphis*. Optimum temperature, about 30°C. The type strain of *P. chlororaphis* subsp. *chlororaphis* was isolated from dead larvae of the cockchafer (a large European beetle). Strains of this subspecies and of subsp. *aureofaciens* have been isolated from various sources, including water. The first strain of subsp. *aureofaciens* (ATCC 13985, NCIB 9030) was isolated from clay suspended in kerosene for 3 weeks (Kluyver, 1956).

The mol% G + C of the DNA is: 63.5 (Bd).

Type strain: ATCC 9446, DSM 50083, IFO 3904, NRRL B-560, NCIB 9392.

GenBank accession number (16S rRNA): D86004, Z76657.

14. ***Pseudomonas cichorii*** (Swingle 1925) Stapp 1928, 291^{AL} (*Phytomonas cichorii* Swingle 1925, 730.)
ci.ch'o'ri.i. Gr. pl. *cichora* succory, chicory; L. n. *cichorium* chicory; M.L. gen. n. *cichorii* of chicory.

Selected characteristics of the species are given in Tables BXII.γ.115 and BXII.γ.118. A fluorescent pigment is produced. The oxidase reaction is weak and slow and the gelatinase reaction is negative. Optimum temperature, ~30°C. Isolated from *Cichorium intybus* and *C. endivia*, for which it is pathogenic.

The mol% G + C of the DNA is: 59 (Bd).

Type strain: ATCC 10857, DSM 50259, LMG 2162, NCPPB 943, PDDCC 5707.

GenBank accession number (16S rRNA): Z76658.

15. ***Pseudomonas citronellolis*** Seubert 1960, 428^{AL}
cit.ro.nel'lo.lis. M.L. gen. n. *citronellolis* of citronellol.

The following description is summarized from the original paper by Seubert (1960).

Rods, 0.5 × 1.0–1.5 µm. Motile by means of a single polar flagellum. Optimum growth temperature, approximately 31°C. Gelatin is not liquefied. Nitrates are reduced to nitrites. Growth can occur anaerobically in the presence of nitrate. H₂S is not produced. Besides citronellol, glucose, acetate, farnesol, and ionone can support growth. No growth was observed with squalene or camphoric acid. Ammonium salts, nitrate, peptone, or yeast extract can be used as nitrogen sources. Acid is produced from glycerol, but not from glucose, galactose, arabinose, fructose, sucrose, maltose, lactose, dulcitol, inositol, mannitol, inulin, or dextrin.

16S rDNA sequence analysis allocates this species in the genus *Pseudomonas* (Moore et al., 1996). Isolated from soil by enrichment with citronellol as carbon source.

The mol% G + C of the DNA is: unknown.

Type strain: ATCC 13674, DSM 50332.

GenBank accession number (16S rRNA): Z76659.

16. ***Pseudomonas corrugata*** Roberts and Scarlett 1981, 216^{VP} (Effective publication: Roberts and Scarlett in Scarlett, Fletcher, Roberts and Lelliott 1978, 109) emend. Sutra, Siverio, Lopez, Hunault, Bollet and Gardan 1997, 1032.
cor.ru'ga.ta. L. v. *corrugare* to wrinkle up; part adj. *corrugatus* wrinkled up.

The following description is summarized from the original paper by Scarlett et al. (1978).

Rods motile by multitrichous polar flagella. According to the original description, the cells are reported to accumulate PHB as carbon reserve material; however, the granules are in fact poly-hydroxyalkanoates of different monomer composition (Kessler and Palleroni, 2000). The colonies are wrinkled (the reason for the species name), yellowish, sometimes with a green center. With age, the color may change to khaki or fawn, depending on the medium. A yellow to yellow-green diffusible, nonfluorescent pigment is produced. Gelatin is hydrolyzed, but starch is not. Egg yolk (lecithinase) reaction positive. Levan production negative. There is growth at 37°C but not at 41°C. Among the characteristics that differentiate this species from *Burkholderia cepacia* and *B. gladioli* (*"Pseudomonas alliicola"*) are the absence of pectate hydrolysis and rot of onion slices, and the lack of utilization of D-arabinose, cellobiose, adipate, meso-tartrate, and citraconate. All these characteristics were positive for the strains of the last two species included in the study.

The above description has been emended recently (Sutra et al., 1997). This detailed study included a large number of strains of the species as well as strains of fluorescent pseudomonads also isolated from tomato pith necrosis, and the results of the analysis were studied by numerical methods, which showed that *P. corrugata* constituted a single phenon. The general phenotypic characters and nutritional properties of the strains, taken from this comprehensive report, are summarized in Tables BXII.γ.116 and BXII.γ.117. The cellular fatty acid composition places this species in a subgroup within the genus (Stead, 1992). The results of rRNA–DNA hybridization experiments allocate this species as a member of the genus *Pseudomonas* (De Vos et al., 1985), which was later confirmed by 16S rDNA sequence analysis (Anzai et al., 1997). Isolated from tomato pith necrosis.

The mol% G + C of the DNA is: 58.4–60.8 (*T_m*).

Type strain: ATCC 29736, DSM 7228, NCPPB 2445.

GenBank accession number (16S rRNA): D84012.

17. ***Pseudomonas ficuserectae*** Goto 1983a, 547^{VP}
fi.cus.e.rec'tae. L. n. *ficuserectae* of *Ficus erecta*, the name of the host species.

The following description is taken from the paper by Goto (1983a)

Nonencapsulated straight rods, 0.5 × 2.0 µm. Motile by means of one to five polar flagella. The original description reports the accumulation of PHB granules by the cells. However, an examination of the type strain has shown that these granules are of poly-β-hydroxyalkanoates other than PHB (Kessler and Palleroni, 2000). Colonies on yeast extract-peptone agar at 28°C are white, transparent, circular, convex, 0.2–3.0 mm in diameter after 6 d. Growth on yeast extract-peptone agar slants becomes very viscid after several days. Denitrification, nitrate reduction, growth factor re-

TABLE BXII.γ.118. Nutritional properties of some of the fluorescent phytopathogenic *Pseudomonas* species^a

Characteristics	<i>P. cichorii</i> ^b	<i>P. syringae</i> pathovars ^c	<i>P. viridiflava</i> ^d	<i>P. viridiflava</i> ^e
<i>Utilization of^f:</i>				
Glucose, mucate, succinate, glycerol, L-aspartate, L-glutamate, L-glutamine, γ-aminobutyrate	+	+	+	+
D-Ribose, D-xylose, acetate, propionate, β-hydroxybutyrate	+	d	d	d
L-Arabinose, gluconate, L-malate, citrate, aconitate	+	d	d	+
D-Mannose, D-galactose, caproate, L-arginine, betaine	+	d	+	d
D-Fructose, caprylate, pelargonate, lactate, mannitol, <i>m</i> -inositol, <i>p</i> -hydroxybenzoate, quinate, L-serine, L-proline	+	d	+	+
Raffinose	d	d	—	—
Fumarate	+	+	+	—
Sucrose, glutarate	+	d	+	—
Saccharate	+	d	d	+
Valerate	d	d	d	—
Caprate	d	+	+	d
Malonate, <i>m</i> -tartrate	+	d	d	—
D-Malate, glycerate, trigonelline	+	d	+	—
D(—)-Tartrate	—	—	+	+
L(+)-Tartrate, α-ketoglutarate	+	d	—	—
Hydroxymethylglutarate	—	d	d	—
Pyruvate	+	+	+	d
Erythritol, sorbitol	—	d	+	+
L-Alanine	+	+	d	+
D-Alanine	d	d	+	—
L-Leucine	—	—	—	+
L-Histidine	d	d	+	d
L-Tyrosine	+	—	d	—
L-Tryptophan	—	—	—	d
Putrescine	—	d	d	—
Sarcosine	d	d	—	d
Laurate	—	—	—	d
L-Sorbose, melezitose, amygdalin, dextrin, formate, dulcitol, isophthalate, L-methionine, <i>m</i> -aminobenzoate, <i>p</i> -aminobenzoate, methylamine	—	—	—	—
Linolenate	d	d	+	—
Triacetin	+	d	+	—
Tripropionin	+	d	+	—
Tricaproin	d	d	+	—
Ascorbate	—	d	—	—
Isoascorbate	—	d	+	—
Lecithin	+	d	d	—
L-Asparagine	+	d	+	—

^aFor symbols see standard definitions.^bData from Sands et al., 1970.^cData from Sands et al., 1970.^dData from Sands et al., 1970.^eData from Billing, 1970b.

^fThe following compounds were not utilized by any strains of the species in the table: D-arabinose, D-fucose^g, L-rhamnose, trehalose, maltose, cellobiose, lactose, melibiose, methylglucoside^g, starch, inulin, 2-ketogluconate^g, salicin^g, N-acetylglucosamine^g, isobutyrate, isovalerate, linoleate^g, laurylsulfate^g, tannate^g, oxalate, maleate, adipate, pimelate, suberate, azelate, sebacate, glycolate, thioglycolate^g, levulinate, citraconate, itaconate, mesaconate, 3-phosphoglycerate^g, hydroxymethylbutyrate^g, adonitol, ethylene glycol, propylene glycol, 2,3-butyleneglycol, methanol^g, ethanol, n-propanol, isopropanol, n-butanol, isobutanol, geraniol^g, D-mandelate, L-mandelate^g, benzoylformate^g, benzoate, *o*-hydroxybenzoate, *m*-hydroxybenzoate, phthalate, phenylacetate, phenylethanediol^g, eicosenedioate^g, naphthalene, phenol, testosterone, glycine, β-alanine, L-threonine, L-isoleucine, L-norleucine, L-valine, L-lysine, L-ornithine, L-citrulline, α-aminobutyrate, δ-aminovalerate^g, L-phenylalanine, L-hydroxyproline^g, D-tryptophan^g, indoleacetic acid^g, L-kynurenine^g, kynurenate^g, anthranilate, methylamine, ethanalamine, benzylamine, spermine, histamine^g, tryptamine^g, butylamine, α-amylamine, creatine, choline^g, hippurate, urate^g, pantothenate, acetamide, nicotinate, dodecane, hexadecane, poly-β-hydroxybutyrate^g, pectate^g, chlorogenate^g, and uridine^g.

^gCompounds not been tried by Billing, 1970b.

quirement, arginine dihydrolase reaction, starch hydrolysis, decarboxylase reaction with lysine and ornithine, phenylalanine deaminase, growth in 5% NaCl, gelatin liquefaction, hydrolysis of Tween 80, urease, and growth at 41°C are all negative. The strains grow with L-arabinose, glucose, mannose, sucrose, raffinose, glycerol, and malonate, but do not grow at the expense of galactose, maltose, trehalose, erythritol, dulcitol, xylan, inulin, glycogen, salicin, carboxymethyl cellulose, α-methylglucoside, esculin, oxalate, glycolate, butyrate, sebacate, hippurate, arginine, betaine, valine, or ethylene glycol. Additional characteristics can be found in the original paper by Goto (1983a).

A numerical analysis that included strains of this species as well as many others of aerobic pseudomonads gave results showing that *P. ficuserectae* and *P. meliae* formed a single cluster, and suggested that they may be pathovars of *P. syringae* (Hu et al. 1991). This comment hinges on the observation that the strains under study did not accumulate PHB. For the moment, the position of *P. ficuserectae* remains unsettled, and more authentic strains will be needed for examination to confirm the correct taxonomic status. However, rDNA sequence analysis confirms its allocation to *Pseudomonas* (Moore et al., 1996). See also Additional Remarks at the end of the description of *P. amygdali* in this list. Path-

ogenic for *Ficus erecta*, in which it produces leaf spots that develop beside the thick veins.

The mol% G + C of the DNA is: 59 (T_m).

Type strain: ATCC 35104, DSM 6929, LMG 5694, PDDCC 7848.

GenBank accession number (16S rRNA): Z76661.

18. ***Pseudomonas flavescens*** Hildebrand, Palleroni, Henderson, Toth and Johnson 1994, 413^{VP}

fla.ves' cens. L. v. *flavesco* to become golden yellow; L. part. adj. *flavescens* becoming golden yellow.

The information for the following description is taken from Hildebrand et al. (1994).

Characteristics useful for differentiation from some of *Pseudomonas* species are given in Tables BXII.γ.112 and BXII.γ.113. Slightly curved rods, $0.6\text{--}0.7 \times 1.6\text{--}2.3 \mu\text{m}$. Motile by means of a single polar flagellum. No PHB accumulated. The following characteristics are negative: arginine dihydrolase, ice nucleation, hypersensitivity reaction, denitrification, and hydrolysis of gelatin, starch, or Tween 80. It grows well in nutrient agar and other common complex media. The colonies are yellow on King medium B and other complex media. In King B, a diffusible fluorescent pigment is produced. The original report gives a list of more than 50 carbon compounds that can be used individually as carbon and energy sources, and a list of more than 80 compounds that do not support growth. Optimum temperature 28°C. No growth occurs at 37°C. Isolated from walnut blight cankers.

The mol% G + C of the DNA is: 63 (T_m).

Type strain: B62, DSM 12071, NCPPB 3063.

GenBank accession number (16S rRNA): U01916.

19. ***Pseudomonas fluorescens*** (Trevisan 1889b) Migula 1895a, 29^{AL} (*Bacillus fluorescens* Trevisan 1889b, 18.)

flu.o.res' cens. L. n. *fluor* a flux; M.L. v. *fluoresce*; M.L. part. adj. *fluorescens* fluorescing.

Characteristics of the species are given in Tables BXII.γ.108, BXII.γ.112, BXII.γ.113, BXII.γ.115, and BXII.γ.119. Optimum temperature, 25–30°C. Found in soil and water, from which it can be isolated after enrichment in media containing various carbon sources, incubated aerobically; strains of the denitrifying biovars can be enriched in similar media containing nitrate, incubated under anaerobic conditions. Commonly associated with spoilage of foods (eggs, cured meats, fish, and milk). Often isolated from clinical specimens. Some strains assigned to this species (biovar II) have been isolated from diseased plants (e.g., lettuce), and identified as *Pseudomonas marginalis* (Brown, 1918) Stevens 1925^{AL}. For more details on this species see the description below. Biovar I (biotype A of Stanier et al., 1966) is considered to be typical of *P. fluorescens*, and the type strain of the species belongs to this group. Aside from *P. marginalis* strains, biovar II (previously biotype B) also includes saprophytic organisms. In their studies on the species *P. aurantiaca* Nakhimovskaya 1948, Kiprianova et al. (1985) found that the type strain was actually *P. aureofaciens* (now a subspecies of *P. chlororaphis*), but they propose to retain the name *P. aurantiaca* for the pigmented strains of biovar II of *P. fluorescens*. Within biovar III (biotype C of Stanier et al., 1966) at least two subgroups, which differ from each other in their capacity for utilization of dicarboxylic acids, can be defined. Biovar IV (biotype F of Stanier

et al., 1966) contains the type strain of "*P. lemonnieri*" (Laseur 1913) Breed 1948, 178. Several strains are known at present, and these can be grouped into at least two clusters.

The group of miscellaneous strains assigned to *P. fluorescens* biotype G by Stanier et al. (1966) constitutes the last (V) biovar of the species. The biovar is very heterogeneous in its nutritional properties, and it may consist of strains that have lost one or more of the properties considered to be of diagnostic importance in differentiating among the better characterized biovars. Among the non-authentic strains assigned to this biovar are strains labeled "*P. schuylkilliensis*" and *P. geniculata* (Wright 1895) Chester 1901, 313^{AL} (see Stanier et al., 1966). The group of strains isolated from meat that eventually was described as a new species under the name *P. lundensis* (Molin et al., 1986) corresponded to a cluster identified within biotype G (Barrett et al., 1986) (see below). Strains of biovar V are common in various materials, and have been isolated from soils (Sands and Rovira, 1971). It is quite possible that isolation of more strains of each of the different biovars may result in the future in the proposal of new species with the general properties of *P. fluorescens*, but circumscribed based on genomic differences.

The mol% G + C of the DNA is: 59.4–61.3 (Bd).

Type strain: ATCC 13525, DSM 50090, NCIB 9046, NCTC 10038.

GenBank accession number (16S rRNA): Z76662.

20. ***Pseudomonas fragi*** (Eicholz 1902) Gruber 1905, 122^{AL} (*Bacterium fragi* Eicholz 1902, 425.)

fra' gi. L. neut. n. *fragum* strawberry; L. gen. n. *fragi* of the strawberry.

The description of this species is given in the seventh edition of the *Manual* (Haynes and Burkholder, 1957).

Rods motile with a polar flagellum. The gelatinase reaction is positive. Nitrate is not reduced to nitrite. Ammonia is produced from peptone. Description of growth in various media and acid production from some sugars and sugar-alcohols are given by Haynes and Burkholder (1957). Growth occurs from 10–30°C. No growth at 37°C. The organism is very sensitive to heat.

This species was tentatively placed in rRNA group I by Byng et al. (1980), a conclusion supported by rRNA–DNA hybridization studies (De Vos et al., 1989) and by 16S rDNA sequence analysis (Moore et al., 1996; Anzai et al., 1997). Isolated from milk, dairy products, water.

The mol% G + C of the DNA is: 60.6 (T_m).

Type strain: ATCC 4973, DSM 3456, LMG 2191.

21. ***Pseudomonas fulva*** Iizuka and Komagata 1963b, 138^{AL} *ful' va.* L. adj. *fulva* tawny, yellowish brown.

Rods, $0.6\text{--}0.8 \times 1.2\text{--}1.8 \mu\text{m}$. Motile with one to three polar flagella. Fluorescent pigment is produced. Oxidase reaction feebly positive. Gelatin is not liquefied. Nitrate not reduced to nitrite or denitrified. H_2S not produced. Starch is not hydrolyzed. Acid is produced from glucose. No acid from glycerol, xylose, sucrose, lactose, or starch. Glucose, gluconate, 2-ketogluconate, citrate, succinate, ethanol, *p*-hydroxybenzoate, and protocatechuate are assimilated. Phenol, benzoate, *m*-hydroxybenzoate, gentisate, *p*-amino-benzoate, and 5-ketogluconate are not assimilated.

Optimum temperature for growth, 25–30°C. Poor growth at 37°C and no growth at 42°C. Allocation to the

TABLE BXII.γ.119. Characteristics of *Pseudomonas chlororaphis* and *P. fluorescens* biovars^a

Characteristics	<i>P. chlororaphis</i> subsp. <i>chlororaphis</i>	<i>P. chlororaphis</i> subsp. <i>aureofaciens</i>	<i>P. fluorescens</i> bv. I	<i>P. fluorescens</i> bv. II	<i>P. fluorescens</i> bv. III	<i>P. fluorescens</i> bv. IV	<i>P. fluorescens</i> bv. V
<i>P. fluorescens</i> biovars as designated by Stanier et al. (1966)	D	E	A	B	C	F	G
<i>Nonfluorescent pigments, color:</i>							
Green (chlororaphin)	+	—	—	—	—	—	—
Orange (phenazine-1-carboxylate)	—	+	—	—	—	—	—
Blue, nondiffusible	—	—	—	—	—	+	—
Levan formation from sucrose	+	+	+	+	—	+	—
Denitrification	+	d	—	+	+	+	—
<i>Substrates used for growth:</i>							
L-Arabinose	—	+	+	+	d	+	d
Sucrose	+	d	+	+	—	+	d
Saccharate	+	+	+	+	d	+	d
Propionate	+	+	+	—	d	+	+
Butyrate	+	d	—	d	d	+	d
Sorbitol	—	—	+	+	d	+	d
Adonitol	—	—	+	—	d	—	d
Propylene glycol	—	—	—	+	d	—	d
Ethanol	d	—	—	+	d	—	d

^aFor symbols see standard definitions. Data from Palleroni (1984).

genus *Pseudomonas* has been confirmed by rRNA sequence analysis (Anzai et al., 1997). Isolated from Japanese rice paddies.

The mol% G + C of the DNA is: 60.6 (T_m).

Type strain: IAM 1529.

GenBank accession number (16S rRNA): D84015.

22. ***Pseudomonas fuscovaginae*** (ex Tanii, Miyajima and Akita 1976) Miyajima, Tanii and Akita 1983, 656^{VP}
fus.co.va.gi'nae. L. adj. *fuscus* fuscous; L. fem. n. *vagina* vagina, sheath; M.L. fem. n. *fuscovaginae* of a fuscous vagina.

The following description is taken from the original paper.

Rods with round ends, 0.5–0.8 × 2.0–3.5 μm. Cells occur singly or in pairs. Motile by one to four polar flagella. Moderate growth on nutrient agar after 4 or 5 d at 28°C; white to light brown colonies, smooth, glistening, raised, translucent, circular, butyrous. A green fluorescent, diffusible pigment is produced on King's medium B. No slime is produced on nutrient agar plus 5% sucrose. Catalase and Kovács's oxidase reaction are positive. Denitrification and nitrate reduction are negative. Hydrolysis of Tween 80 and of gelatin is positive; esculin and arbutin are not hydrolyzed. Arginine dihydrolase reaction positive, but phenylalanine deaminase and formation of 2-ketogluconate and H₂S are negative. No organic growth factors are required. Citrate, malonate, succinate, urate, acetate, β-alanine, L-valine, and L-lysine are utilized for growth, but not tartrate, hippurate, 2-ketogluconate, or polygalacturonic acid. Optimal growth temperature is approximately 28°C. No growth occurs at 37°C. Additional characteristics may be found in the paper by Miyajima et al. (1983).

Pathogenic for rice (*Oryza sativa*) and various other plants of the family Gramineae. In rice it produces first water-soaked, dark green spots, which later become brown to dark brown. Characteristics differentiating this species from other fluorescent pseudomonads are mentioned in the original paper. Assignment to the genus *Pseudomonas* has been confirmed by rRNA–DNA hybridization studies (De Vos et al., 1985). Isolated from diseased leaf sheaths of *O. sativa* in Japan.

The mol% G + C of the DNA is: unknown.

Type strain: 6801, DSM 7231, NCPPB 3085, PDDCC 5940.

23. ***Pseudomonas gessardii*** Verhille, Baïda, Dabboussi, Hamze, Izard and Leclerc 1999a, 1566^{VP}
ges.sar' di.i. gessardii of C. Gessard, French chemist who isolated "*Bacterium aeruginosum*" for the first time in 1882 and studied its pigment.

The following description is taken from the original paper (Verhille et al., 1999a).

Rods, motile by means of a single flagellum. Fluorescent pigment is produced on King B; no phenazine pigments are observed on King A. Oxidase, catalase, and production of levan from sucrose are all positive. Many compounds were tested as carbon and energy sources for growth and a number of enzymatic reactions were determined with the cultures. The results are listed in the original paper. According to the authors, the utilization of L(–)-arabitol, xy-litol, *myo*-inositol, adonitol, and *i*-inositol and the absence of growth in the presence of L-arabinose, D-xylose, D-saccharate, *meso*-tartrate, tricarballic acid, glucuronate, galacturonate phenylacetate, and histamine, differentiate this species from *P. migulae*, a closely related species. Differentiation from other *Pseudomonas* species is given in tables in the original report. A minority of the strains (1/13) had gelatinase activity. No growth is observed in the presence of 5% NaCl in the medium. The range of growth temperatures goes from 4°C to 35°C, and the optimum is around 30°C. Isolated from natural mineral waters.

The mol% G + C of the DNA is: 58.

Type strain: CIP 105469.

GenBank accession number (16S rRNA): AF074384.

24. ***Pseudomonas graminis*** Behrendt, Ulrich, Schumann, Erler, Burghardt and Seyfarth 1999, 306^{VP}
gra' mi.nis. L. n. *gramen* grass; *graminis* of grass, the source of the organism.

Rods, 0.5–1.0 × 3.5–5.0 μm, motile by one polar flagellum. Colonies are yellow, glistening, moderately convex, circular, with smooth edges. Fluorescent pigment production, levan production from sucrose, starch hydrolysis, ox-

idase, nitrate reduction to nitrite, denitrification, arginine dihydrolase, lysine and ornithine decarboxylase, tryptophan deaminase, DNase, urease, β -hemolysis of sheep blood, and β -galactosidase reactions are all negative. Catalase and hydrolysis of Tweens 40 and 80 are positive. Utilization of a number of carbon compounds is listed in the original paper by Behrendt et al. (1999). Unfortunately, few of the substrates tested are the same as the ones that appear in the tables of this chapter, and the species has not been included for comparison. However, the paper gives a selection of properties that allow a differentiation of this species from others of similar characteristics. The strains use monosaccharides but not the disaccharides tested.

A comparison of rRNA sequences with those of other species of *Pseudomonas* and species of other RNA similarity groups indicates that the assignment to the genus *Pseudomonas* is correct. The creation of a new species name for the strains is supported by low DNA similarity with other species of the genus. The paper is an excellent source of taxonomic information on the allocation of this species within the group of fluorescent pseudomonads. The original description includes the presence of ubiquinone Q-9 and the hydroxy fatty acids 3-hydroxydodecanoic and 2-hydroxydodecanoic. Isolated from the phyllosphere of grasses.

The mol% G + C of the DNA is: 60–61.

Type strain: P 294/08, DSM 11363.

GenBank accession number (16S rRNA): Y11150.

25. ***Pseudomonas jessenii*** Verhille, Baïda, Dabboussi, Izard and Leclerc 1999b, 54^{VP}
jes.sen' ni.i. M.L. masc. gen. n. *jessenii* of Jessen, named after O. Jessen, eminent Danish bacteriologist who contributed substantially to the knowledge of fluorescent pseudomonads.

Rods, motile by means of a single polar flagellum. Colonies on nutrient agar are circular, smooth, nonpigmented. Nonhemolytic when grown on blood agar. Fluorescent pigment is produced on King B agar medium. Catalase, oxidase, and arginine dihydrolase reactions are positive. Denitrification, gelatinase, and lecithinase are negative. Nitrate is reduced to nitrite. The strains grow at 4°C but not at 41°C (range: 4–35°C). Optimal: 30°C. No growth occurs in the presence of 5% NaCl. Other properties as well as the results of nutritional and enzymatic screenings are listed in the original paper. Isolated from mineral water.

The mol% G + C of the DNA is: 57–58.

Type strain: CIP 105274.

GenBank accession number (16S rRNA): AF068259.

26. ***Pseudomonas libanensis*** Dabboussi, Hamze, Elomari, Verhille, Baïda, Izard and Leclerc 1999a, 1099^{VP}
li.ba.nen' sis. L. n. *Libanus* Lebanon, in southern Syria; L. adj. *libanensis* from or of Lebanon.

The description is taken from the original paper.

Rods, motile by means of a single polar flagellum. No poly- β -hydroxybutyrate is accumulated. Colonies on nutrient agar are smooth, circular, and nonpigmented. No hemolysis is observed on blood agar. A fluorescent pigment is produced on King B medium, and no phenazine pigments are produced on King A medium. Lipase, elastase, and tetrathionate reductase are negative. Arginine dihy-

drolase, lecithinase, catalase, levan formation from sucrose, and oxidase reaction are all positive. The strains do not denitrify. The majority of the strains (6/7) liquefy gelatin. A number of enzymatic reactions and the results of a nutritional screening on a number of compounds (mostly carbohydrates and derivatives) are reported in the original paper by Dabboussi et al. (1999a). The cultures tolerate 3% but not 7% of NaCl in the medium. Growth occurs between 4 and 36°C, with optimal growth at 30°C. The strains were isolated from Lebanese spring waters.

The authors of the original description (Dabboussi et al., 1999a) discuss the relationship of this species to other fluorescent organisms. *P. libanensis* seem to be closer to *P. fluorescens* biovar A, as suggested by the results of DNA–DNA hybridization experiments. This similarity extends to the main characteristics of biovar, namely, a negative denitrification test and the capacity of formation of levan from sucrose.

The mol% G + C of the DNA is: 58.

Type strain: CFML 96-195; CIP 105460.

GenBank accession number (16S rRNA): AF057645.

27. ***Pseudomonas lundensis*** Molin, Ternström and Ursing 1986, 339^{VP}
lund.en' sis. M.L. adj. *lundensis* referring to the city of Lund, Sweden.

The following description is taken from the original paper.

Rods, 0.5–1.0 $\times$ 1.0–3.0 μ m, motile by means of a single polar flagellum. Colonies on nutrient agar are 1–5 mm in diameter after 3 d at 25°C, circular, and smooth. No organic growth factors are required. Fluorescent pigment is produced in King B medium. Denitrification is negative. PHB is not accumulated in the cells. Arginine dihydrolase, catalase, and oxidase are positive. Gelatin liquefaction is variable. Only one strain has been observed to produce lipase against Tween 80.

All strains produce acid but not gas from L-arabinose, D-galactose, D-glucose, D-mannose, D-ribose, cellobiose, maltose, melibiose, and D-xylose. In assimilation tests, the last four sugars do not serve as carbon and energy sources. The following carbon compounds are assimilated (number of negative strains from a collection of 60 strains of the species, are given in parentheses): acetate, L-alanine, β -alanine, 4-aminobutyrate, 2-aminoethanol (1), 5-aminopentanoate, L-arabinose (1), L-arginine, L-asparagine, L-aspartate, betaine (2), citrate (1), fructose, fumarate, D-gluconate (4), D-glucose, L-glutamate, L-glutamine, glycerate (1), glycerol (1), heptanoate, L-histidine, *meso*-inositol (6), L-lactate (2), L-leucine (1), L-malate (2), L-ornithine, 2-oxoglutarate, pelargonate, L-phenylalanine, L-proline, propionate, putrescine, pyruvate, succinate, DL-tyrosine and DL-valine (1). Less than 90% of the strains can use D-arabinose, benzoate, carnosine, creatine, deoxycholate, D-galactose, DL-hydroxybutyrate, L-isoleucine, D-malate, D-ribose, sarcosine, and tauracholate. Slow utilization of butyrate, caproate, L-lysine, D-mannose, and trehalose. The main characteristics of the species are summarized in Tables BXII.γ.112 and BXII.γ.113. They refer to the type strain.

Growth occurs between 0 and 33°C, and one-third of the strains are able to grow at 37°C. Optimal growth is at 25°C. No growth at pH below 4.5. Isolated from meat. The strains

correspond to cluster 2 of a previous numerical taxonomic study of psychrotrophic pseudomonads that had been isolated from meat, soil, and water (Molin and Ternström, 1982, 1986). Interestingly, results of numerical taxonomic studies performed independently at about the same time on the fluorescent pseudomonad collection at Berkeley, which included the biotype G of *P. fluorescens*, identified a tight cluster with the same properties as *P. lundensis* (Barrett et al., 1986).

The mol% G + C of the DNA is: 58–60.

Type strain: ATCC 49968, CCM 3503, DSM 6252.

28. ***Pseudomonas luteola*** Kodama, Kimura and Komagata 1985, 473^{VP} (*Chryseomonas luteola* (Kodama, Kimura and Komagata 1985) Holmes, Steigerwalt, Weaver and Brenner 1987, 246.)

lu.te' o.la. L. dim. adj. *luteola* yellowish.

The following description is taken from the original paper.

Rods, $0.8 \times 2.5 \mu\text{m}$, with rounded ends, occurring singly, rarely in pairs. Motile by means of polar multitrichous flagella. A small number of PHB granules accumulate in the cells. Colonies on 0.5% glucose nutrient agar are smooth or wrinkled, entire or erose, flat or convex, light yellow or pale yellow, 3 mm in diameter after 2 d of growth at 30°C. A pellicle is formed on the surface of 0.5% glucose nutrient broth. A water-insoluble yellow pigment is produced. No water-soluble fluorescent pigment is produced. Catalase positive but oxidase negative. Nitrate is reduced to nitrite, but no denitrification occurs. Production of indole, hydrogen sulfide, and hydrolysis of starch, gelatin, Tween 80, esculin, and ONPG are all negative. No organic growth factors are required. Growth does not occur in a medium with 6.5% NaCl. Growth occurs at 42°C. Cleavage of protocatechuate is of the *ortho* type.

Acid is produced from L-arabinose, D-xylose, D-ribose, D-glucose, D-fructose, D-mannose, D-galactose, L-rhamnose, maltose, trehalose, mannitol, inositol, and salicin, but not from sucrose, lactose, cellobiose, adonitol, sorbitol, or inulin. L-arabinose, D-xylose, D-ribose, D-glucose, D-fructose, D-mannose, D-galactose, maltose, trehalose, mannitol, glycerol, acetate, pyruvate, malonate, DL- β -hydroxybutyrate, fumarate, 2-ketogluconate, gluconate, succinate, *p*-hydroxybenzoate, and glutamate are utilized, but sucrose, lactose, raffinose, inulin, starch, phenol, *o*-hydroxybenzoate, and *m*-hydroxybenzoate are not. Some properties are summarized in Table BXII.γ.116. The fatty acid and ubiquinone composition is typical of that of the genus *Pseudomonas*. Isolated from clinical specimens.

The mol% G + C of the DNA is: 55.4 (T_m).

Type strain: KS0921, DSM 6975, IAM 13000, JCM 3352.

GenBank accession number (16S rRNA): D84002.

Additional Remarks: This species, originally assigned to the genus *Pseudomonas* (Kodama et al., 1985), was subsequently considered a member of a newly proposed genus, *Chryseomonas* (Holmes et al., 1987). Later studies (Anzai et al., 1997) demonstrated that *Pseudomonas* should be its correct generic allocation.

29. ***Pseudomonas mandelii*** Verhille, Baïda, Dabboussi, Izard and Leclerc 1999b, 56^{VP}
man.de' li.i. M.L. gen. n. *mandelii* of Mandel, dedicated to M. Mandel, American bacteriologist.

Colonies on nutrient agar are smooth, circular, and non-pigmented. Fluorescent pigment produced on King B agar medium. Catalase, oxidase, denitrification, and levan formation are positive. Gelatin and starch are not hydrolyzed. Sodium chloride inhibits growth at 5% concentration. Growth occurs at 4°C but not at 41°C. Optimal growth at 30°C. Growth at the expense of many organic compounds, as well as the result of many enzymatic reactions, are listed in the original paper (Verhille et al., 1999b).

DNA–DNA hybridization experiments indicate a relative high DNA similarity with that of biovar IV (*"P. lemonnieri"*) of *P. fluorescens*. Isolated from mineral water.

The mol% G + C of the DNA is: 57.

Type strain: CFML 95-303, CIP 105273.

GenBank accession number (16S rRNA): AF058286.

30. ***Pseudomonas marginalis*** (Brown 1918) Stevens 1925, 30^{AL} (*Bacterium marginale* Brown 1918, 386; *Phytomonas marginalis* (Brown 1918) Bergey, Harrison, Breed, Hammer and Hinton 1923, 182; *Phytomonas intybi* Swingle 1925, 730; *Chlorobacter marginale* (Brown 1918) Patel and Kulkarni 1951b, 80.)

mar.gi.na' lis. L. *margo*, *marginis* edge, margin; M.L. adj. *marginalis* marginal.

Description taken from Haynes and Burkholder (1957), who cite the original sources.

Rods, motile with one to three polar flagella. Green fluorescent pigment produced in culture. Agar colonies cream to yellowish. Gelatin is liquefied. There is a discrepancy among different authors on nitrate reduction. Feeble hydrolysis of starch. Acid but no gas from glucose, galactose, fructose, mannose, arabinose, xylose, rhamnose, mannitol, and glycerol. Alkali from (indicating utilization of) acetate, citrate, malate, formate, lactate, succinate, and tartrate. Sucrose, maltose, raffinose, and salicin are not degraded. Optimum growth temperature between 25 and 26°C. Minimum, 0°C; maximum, 38°C.

Isolated from marginal lesion on lettuce from Kansas; pathogenic on lettuce and related plants. Strains of this species have been included in biovar II of *P. fluorescens* (Stanier et al., 1966), and it appears under such designation in Tables BXII.γ.112, BXII.γ.113, and BXII.γ.119. Later, the taxon was subdivided by Young et al. (1978) into three pathovars. *P. marginalis* pathovar *marginalis*, which includes the type strain of the species, ATCC 10844; *P. marginalis* pathovar *alfalfa* Shinde and Lukezic 1974b, associated with discolored alfalfa (reference strain PDDCC 5708, NCPPB 2644), and *P. marginalis* pathovar *pastinacae* (Burkholder 1960a), a pathogen of cultivated parsnip (*Pastinaca sativa*) (reference strain ATCC 13889, PDDCC 5709, NCPPB 806). Fatty acid analysis data were taken as the basis for placing this species in a subgroup with other fluorescent phytopathogenic and saprophytic *Pseudomonas* (Stead, 1992). This subgroup also contains the nonfluorescent *P. meliae*.

The mol% G + C of the DNA is: 60 (Bd).

Type strain: ATCC 10844, LMG 2210, NCPPB 667, PDDCC 3553.

GenBank accession number (16S rRNA): Z76663.

31. ***Pseudomonas meliae*** Ogimi 1981, 382^{VP} (Effective publication: Ogimi 1977, 547.)
me' li.ae. Gr. n. *Melia* Chinaberry tree, *Melia azedarach*; M.L. gen. n. *meliae* of *Melia*.

The following description is taken from the original paper.

Rods, $0.4\text{--}0.5 \times 1.4\text{--}2.0 \mu\text{m}$ (average $0.5 \times 1.8 \mu\text{m}$). Motile by means of one to two polar flagella. Noncapsulated. Do not accumulate PHB as carbon reserve material. Colonies on nutrient agar very small after 48 h, and $0.3\text{--}1.0 \text{ mm}$ in 72 h. Circular, smooth, convex, translucent, and white when viewed with reflected light. On potato-sucrose agar, the colonies have a rough surface and irregular shape. The rough colonies revert to the smooth type after transfer to nutrient agar. The following properties are negative: nitrate reduction, gelatin liquefaction, H_2S and indole formation, production of pyocyanine, fluorescein, and other pigments, levan formation on nutrient-sucrose agar, arginine and lysine dihydrase reactions, starch and esculin hydrolysis, production of gluconate, tyrosinase and urease reactions, egg-yolk and lipase (margarine hydrolysis), malonate utilization, phenylalanine deaminase, methyl red, and Voges-Proskauer reactions. The catalase and oxidase reactions are positive. Nicotinate is required as an organic growth factor.

The following carbon compounds are used for growth: ribose, glucose, mannose, galactose, fructose, sucrose, glycerol, citrate, succinate, and malate. No growth is observed at the expense of xylose, rhamnose, arabinose, lactose, maltose, cellobiose, melibiose, trehalose, dextrin, glycogen, starch, inulin, mannitol, sorbitol, inositol, adonitol, dulcitol, salicin, or tartrate. Growth occurs between 4 and 37°C (optimum, 27°C) and between pH 5 and 8 (optimum, pH 7–8). rRNA–DNA hybridization studies have confirmed the allocation of this species in the genus *Pseudomonas* (De Vos et al., 1985). Based on fatty acid analysis, *P. meliae* was placed in the same subgroup constituted by phytopathogenic and saprophytic fluorescent *Pseudomonas* species (Stead, 1992). Isolated from Chinaberry tree (*Melia azedarach* L.) (family Meliaceae), where this pathogen produces the bacterial galls. The galls are formed in the leaf petioles and in branches and trunk.

The mol% G + C of the DNA is: 57.9 (T_m).

Type strain: ATCC 33050, DSM 6759, LMG 2220, NCPPB 3033.

Additional Remarks: See the Additional Remarks at the end of the description of *P. amygdali* in this list.

32. ***Pseudomonas mendocina*** Palleroni *in* Palleroni, Doudoroff, Stanier, Solanes and Mandel 1970, 220^{AL}
men.do.ci' na. Sp. fem. n. *mendocina* native of Mendoza (Argentina).

Characteristics of the species useful for differentiation from other species are given in Tables BXII.γ.116 and BXII.γ.117. Properties useful for distinguishing the species from some denitrifying pseudomonads are summarized in the chapter on the genus *Burkholderia*. Colonies are yellowish because of production of carotenoid pigment; not adherent or wrinkled in appearance. Optimum temperature, approximately 35°C . For further descriptive information, see Palleroni et al. (1970). Found in soil and water; isolated by enrichment in media with nitrate under anaerobic conditions, especially at 40°C . Ethanol and L(+)-tartrate can be used as carbon sources in the enrichments. It has also been isolated from clinical specimens (Hugh and Gilardi, 1980).

The mol% G + C of the DNA is: 62.8–64.3 (Bd).

Type strain: ATCC 25411, DSM 50017, LMG 1223.

GenBank accession number (16S rRNA): M59154, Z76664.

33. ***Pseudomonas migulae*** Verhille, Baïda, Dabboussi, Hamze, Izard and Leclerc 1999a, 1570^{VP}
mi' gu.lae. M.L. gen. n. *migulae* of W. Migula, who created the generic name *Pseudomonas*.

The following description is taken from the original paper.

Rods, motile by means of a single polar flagellum. Colonies on nutrient agar are smooth and circular. Fluorescent pigment is produced on King B agar. No phenazine pigments are synthesized. Starch and gelatin are not hydrolyzed. Levan is produced from sucrose and the arginine dihydrolase system is present in the cells. Low concentrations of NaCl (up to 0.8%) are tolerated, and 5% concentration is inhibitory. Growth occurs at 4°C but not at 41°C . The original paper reports the results of a number of enzymatic tests, as well as a list of compounds that can serve as sources of carbon and energy for growth, and a selection of these characters is used for differentiation from other fluorescent species.

DNA–DNA hybridization experiments suggest that this species may be considered as a genomovar of some of the biovars of *P. fluorescens*. Isolated from natural mineral water.

The mol% G + C of the DNA is: unknown.

Type strain: CFML 95-321, CIP 105470.

GenBank accession number (16S rRNA): AF074383.

34. ***Pseudomonas monteilii*** Elomari, Coroler, Verhille, Izard and Leclerc 1997, 849^{VP}
mon.tei' li.i. M.L. masc. gen. n. *monteilii* of Monteil, in honor of Henri Monteil, a French microbiologist.

The following description is taken from the original paper.

Motile rods. Colonies on nutrient agar are circular and nonpigmented. Fluorescent pigment is produced. No hemolysis on blood agar. The strains are lipase, elastase, lecithinase, and tetrathionate reductase negative. Arginine dihydrolase, catalase, and cytochrome oxidase are produced. Denitrification, formation of levan from sucrose, gelatinase, starch hydrolysis, hydrolysis of esculin and starch, deamination of phenylalanine, exonuclease activity, lysine and ornithine decarboxylase, and the tributyrin and fibrinolysis tests are all negative. Temperature range for growth is $10\text{--}36^\circ\text{C}$; optimal growth at 30°C . Growth occurs in the presence of 3% NaCl but not with 5% NaCl. The original paper by Elomari et al. (1997) includes an extensive nutritional and enzymatic characterization of the species. Part of the available information has been summarized in Tables BXII.γ.112 and BXII.γ.113. The strains have been isolated from clinical specimens.

The mol% G + C of the DNA is: 60 (T_m).

Type strain: CFML 90-60, CIP 104883.

GenBank accession number (16S rRNA): AF064458.

35. ***Pseudomonas mucidolens*** Levine and Anderson 1932, 344^{AL}
mu.ci' do.lens. L. adj. *mucidus* musty; L. v. *oleo* to smell of; L. part. adj. *mucidolens* musty smelling.

The description below is taken from the original paper. Rods, rounded ends, occurring singly and in pairs, ac-

tively motile. Filamentous cells are frequently observed. Fluorescent pigment is produced. Gelatin is liquefied. Nitrates reduced to nitrites and gas. Optimum growth temperature 23–25°C; slight growth at 37°C and at 10°C. Acid produced from glucose, rhamnose, arabinose, erythritol, sorbitol, trehalose, galactose, fructose, and mannose. Moderate amount of acid from glycerol, mannitol, dulcitol, glycogen, inulin, maltose, melezitose, pectin, raffinose, salicin, starch, sucrose, and xylan. Lactate, citrate, and urate can be used for growth. Acetate, oxalate, sulfanilate, tartrate, salicylate, and formate do not support growth. A physiologically less active form of this species is described in the same paper under the name "*P. mucidolens* biovar *tarda*".

P. mucidolens is one of the species causing mustiness in eggs, which were the source for its isolation. The general characteristics as described above roughly conform to those of one of the denitrifying biovars of *P. fluorescens*, but extensive comparative studies of the phenotypic properties have not been done. However, DNA–rRNA hybridization data (De Vos et al., 1989) and rRNA sequence analysis (Anzai et al., 1997) confirm the allocation to the genus *Pseudomonas*.

The mol% G + C of the DNA is: 61.0 (T_m).

Type strain: ATCC 4685, LMG 2223.

36. ***Pseudomonas nitroreducens*** Iizuka and Komagata 1964a, 214^{AL}

ni.tro.re.du' cens. L. n. *nitrum* nitrate; L. v. *reduco* to draw backwards, bring back to a state or condition; M.L. part. adj. *nitroreducens* nitrate reducing.

The following description is taken from the original paper.

Rods, 0.4–0.6 × 1.4–1.8 µm, occurring singly, rarely in pairs; motile with polar flagella. Fluorescent pigment produced. Gelatin is not liquefied. Nitrates are reduced to nitrites and to gas. Oxidase reaction positive. H₂S not produced. Starch is not hydrolyzed. Acid is produced from glucose. Glucose, gluconate, 2-ketogluconate, citrate, succinate, ethanol, *p*-hydroxybenzoate, and protocatechuate can be utilized as sole carbon sources. 5-Ketogluconate, benzoate, salicylate, gentisate, anthranilate, and *p*-aminobenzoate are not utilized. The fresh isolate could utilize kerosene, but the activity was lost after subcultivation in the laboratory. Optimum growth temperature, 25–30°C. No growth at 37°C. The source of isolation was oil brine in Japan.

The species has two characteristics never found in association in any of the fluorescent species described in RNA group I, namely, the capacity for denitrification and the inability to utilize gelatin. Its assignment to the genus *Pseudomonas* has been confirmed by molecular studies (Anzai et al., 1997).

The mol% G + C of the DNA is: unknown.

Type strain: IAM 1439.

GenBank accession number (16S rRNA): D84021.

37. ***Pseudomonas oleovorans*** Lee and Chandler 1941, 377^{AL}

o.le.o'vo.rans. L. n. *oleum* oil; L. v. *voro* to destroy, consume; M.L. part. adj. *oleovorans* oil consuming.

Description taken from the original paper.

When grown on agar, the cells are almost coccoid (0.5 × 0.8 µm), but the length increases to about 1.5 µm during

the exponential phase in broth. The colonies have a typical fluorescence that is not imparted to the medium. Nitrate is reduced to nitrite. Gelatin not liquefied. Starch is hydrolyzed. Isolated from oil-water emulsions used as lubricants and cooling agents in the cutting and grinding of metals. Apparently the organism lives on some normal constituent of the cutting compound, probably the naphthenic acids, which act as emulsifying agents. Consequently, the name *oleovorans* may not be appropriate. Allocation to the genus *Pseudomonas* was confirmed by DNA–rRNA hybridization studies (De Vos et al., 1989) and by 16S rRNA sequence analysis (Moore et al., 1996; Anzai et al., 1997).

The mol% G + C of the DNA is: 63.5 (T_m).

Type strain: ATCC 8062, DSM 1045.

GenBank accession number (16S rRNA): Z76665.

38. ***Pseudomonas orientalis*** Dabboussi, Hamze, Elomari, Verhille, Baida, Izard and Leclerc 1999b, 303^{VP}
ori.en.ta'l is. M.L. masc. adj. *orientalis* pertaining to the Orient.

Rods, motile by means of a single polar flagellum. Colonies on nutrient agar are circular, smooth, and nonpigmented. Fluorescent pigment is produced on King B agar. Catalase, oxidase, gelatinase, esterase-lipase, urease, and arginine dihydrolase reactions are all positive. Levan is produced from sucrose. Reduction of nitrate to nitrite is positive, but there is no denitrification. All strains liquefied gelatin. Various enzymatic reactions are described in the original paper, but not the ability to liquefy gelatin. Assimilation of various organic compounds, particularly carbohydrates, is described.

DNA–DNA hybridization experiments give low reannealing values with other fluorescent species and biovars. Growth occurs at 4°C. Isolated from Lebanese spring water.

The mol% G + C of the DNA is: 60.

Type strain: CFML 96-170, CIP 105540.

GenBank accession number (16S rRNA): AF064457.

39. ***Pseudomonas oryzae*** Kodama, Kimura and Komagata 1985, 472^{VP} (*Flavimonas oryzae* (Kodama, Kimura and Komagata 1985) Holmes, Steigerwalt, Weaver and Brenner 1987, 245.)

o.ry.za'i ha.bi.tans. L. fem. n. *oryza* rice; L. fem. adj. *habitans* inhabiting; M.L. fem. adj. *oryzohabitans* rice.

The description is taken from the original paper.

Rods, 0.8 × 2.0 µm, with rounded ends, occurring singly, rarely in pairs. Motile by means of a monotrichous flagellum. Granules of PHB do not accumulate. Colonies on 0.5% glucose nutrient agar are smooth or wrinkled, entire or erose, flat to convex, light yellow, 2 mm in diameter after 2 d incubation at 30°C. A pellicle is formed on the surface of 0.5% glucose nutrient broth. A water-insoluble yellow pigment is produced. No water-soluble fluorescent pigment is produced. Catalase is produced; the oxidase reaction is negative. Nitrate reduced to nitrite, but no denitrification occurs. Urease is produced. Production of indole, hydrogen sulfide, and hydrolysis of starch, Tween 80, esculin, and ONPG are all negative.

Growth occurs in medium containing 6.5% NaCl. No organic growth factors are required. Cleavage of protocatechuate is of the *ortho* type. Acid is produced from L-arabinose, D-xylose, D-glucose, D-fructose, D-mannose, D-galactose, maltose, trehalose, mannitol, sorbitol, and inositol, but

not from L-rhamnose, sucrose, lactose, cellobiose, adonitol, salicin or inulin. L-arabinose, D-xylose, D-ribose, D-glucose, D-mannose, D-galactose, maltose, trehalose, mannitol, glycerol, acetate, pyruvate, malonate, DL- β -n-hydroxybutyrate, fumarate, gluconate, 2-ketogluconate, succinate, *p*-hydroxybenzoate, and glutamate are utilized, but not sucrose, lactose, raffinose, inulin, starch, phenol, *o*-hydroxybenzoate, or *m*-hydroxybenzoate. Hydrogen is not utilized. Some properties are given in Table BXII.γ.116. The cellular fatty acid and ubiquinone composition is the typical type of the genus *Pseudomonas*. Isolated from a Japanese rice paddy.

The mol% G + C of the DNA is: 65.1 (T_m).

Type strain: KS0036, ATCC 43272, DSM 6835, IAM 1568, JCM 2592.

GenBank accession number (16S rRNA): D84004.

Additional Remarks: This species was originally described as a member of the genus *Pseudomonas* (Kodama et al., 1985), but was later assigned to the proposed new genus name *Flavimonas* (Holmes et al., 1987). Further studies (Anzai et al., 1997) demonstrated that the latter should be considered a synonym of *Pseudomonas*.

40. ***Pseudomonas pertucinogena*** Kawai and Yabuuchi 1975, 318^{AL}

per.tu.ci.no'ge.na. pertucin (coined word), a bacteriocin active against *Bordetella pertussis*; L. v. *gigno* to produce; M.L. fem. adj. *pertucinogena* intended to mean pertucin producing.

The description is from the original paper, referring to two strains of the species.

Rods, 0.4×1.1 μ m, motile by means of single polar flagellum. Do not accumulate PHB. Oxidase reaction positive. No pigments are produced. Hydrolysis of gelatin, starch, or Tween 80, and arginine dihydrolase reaction are all negative. Acid production from sugars is in general weak or negative (glucose, D-arabinose, and galactose are positive). Pyruvate, succinate, oxaloacetate, β -hydroxybutyrate, and L-alanine are utilized for growth. A number of amino acids, carbohydrates, and alcohols are not utilized. Further details are given in the original paper. The two known strains produce pertucin, a bacteriocin active against *Bordetella pertussis*, and were kept for many years in the American Type Culture Collection as members of this species.

The mol% G + C of the DNA is: ~60.

Type strain: ATCC 190.

41. ***Pseudomonas pseudoalcaligenes*** Stanier in Stanier, Palleroni and Doudoroff 1966, 247^{AL}

pseu.do.al.ca.li'ge.nes. Gr. adj. *pseudes* false; M.L. adj. *alcaligenes* alkali-producing; M.L. adj. *pseudoalcaligenes* false alkali-producing.

Characteristics useful for differentiation from other *Pseudomonas* species are given in Tables BXII.γ.116 and BXII.γ.117, and the chapter on the genus *Burkholderia*, which summarizes information on denitrifying pseudomonads. Optimum temperature, 35°C. Further descriptive information can be found in Ralston-Barrett et al. (1976) and Stanier et al. (1966). The nutritional spectrum is quite narrow (see *P. alcaligenes*). The gelatinase reaction is negative. The species is rather heterogeneous. The collection examined by Stanier et al. (1966) included at least two groups. Strains of one of the groups were capable of PHB accumulation and gave a positive arginine dihydrolase re-

action, while the strains of the second group were negative for the two properties and also differed in some nutritional characteristics (Doudoroff and Palleroni, 1974).

Three subspecies were thought to integrate this species, namely, *P. pseudoalcaligenes* subsp. *pseudoalcaligenes* (corresponding to the original description by Stanier et al., 1966), *P. pseudoalcaligenes* subsp. *citrulli* Schaad et al. 1978 (a name given to strains isolated from water soaked lesions on cotyledons of watermelon, *Citrullus lanatus*, and also pathogenic for other plants of the same family) (Schaad et al., 1978), and *P. pseudoalcaligenes* subsp. *konjaci* (the causal agent of bacterial blight of konjac, *Amorphophallus konjac* Koch) (Goto, 1983b). Nutritional characteristics that were found useful for rapid identification were the utilization of L-arabinose and galactose by the subspecies *citrullus* (subsp. *pseudoalcaligenes* and subsp. *konjaci*, and *P. alcaligenes* are negative), and the utilization of mannitol by subsp. *konjaci* but not by subsp. *pseudoalcaligenes* and subsp. *citrulli*, and *P. alcaligenes* (Goto, 1983b).

The fatty acid composition of *P. pseudoalcaligenes* and its subspecies placed this taxon in one of the subgroups of *Pseudomonas* together with other nonfluorescent species (Stead, 1992). In spite of the above considerations, the results of a numerical analysis supplemented by DNA-DNA reassociation experiments (Hu et al., 1991) were taken as the basis for the conclusion that subsp. *citrulli* and subsp. *konjaci* should be transferred to a different species, *P. avenae*, which had been allocated to rRNA similarity group III (Palleroni, 1984). Consequently, this species should include three subspecies: *P. avenae* subsp. *avenae* Manns 1909 (type strain ICMP 3183), *P. avenae* subsp. *citrulli* (Schaad et al., 1978) comb. nov. (type strain ICMP 7500), and *P. avenae* subsp. *konjaci* (Goto, 1983a) comb. nov. (type strain ICMP 7733) (Hu et al., 1991). Studies by Willems et al. (1992) support the conclusion that these various phytopathogenic pseudomonad taxa should be transferred to the genus *Acidovorax* and the new combinations *Acidovorax avenae* subsp. *citrulli* and *A. avenae* subsp. *cattleyae* as well as the new name *Acidovorax konjaci* have been proposed. *P. pseudoalcaligenes* has been isolated from various natural materials and from clinical specimens.

The mol% G + C of the DNA is: 62.2–63.2 (Bd).

Type strain: ATCC 17440, LMG 1225, NCIB 9946.

GenBank accession number (16S rRNA): Z76666.

42. ***Pseudomonas putida*** (Trevisan 1889b) Migula 1895a, 29^{AL}

(*Bacillus putidus* Trevisan 1889b, 18.)
pu'ti.da. L. fem. adj. *putida* stinking, fetid.

Characteristics of the species (103 strains of biovar A and 9 strains of biovar B) are given in Tables BXII.γ.112 and BXII.γ.113. Other properties of interest are summarized in Tables BXII.γ.108 and BXII.γ.114. The characteristics that are useful for a separation from *P. aeruginosa* and *P. fluorescens* include the inability to liquefy gelatin, to produce any phenazine pigments, to denitrify or to give an egg-yolk reaction, and to grow at 41°C. In this constellation of negative properties, the incapacity to hydrolyze gelatin is the one that has classically defined *P. putida*. Optimum temperature, 25–30°C.

Isolated from soil and water after enrichment in mineral media with various carbon sources. The majority of the strains have been assigned to biovar A (biotype A of Stanier

et al., 1966), which is considered typical. Biovar B differs from biovar A only in a few phenotypic properties: all known strains of this biovar utilize L-tryptophan, kynurenine, and anthranilate, and most use D-galactose as carbon sources. None of the strains of biovar B uses nicotinate. Recent studies on many strains of this species, combining a numerical analysis of phenotypic data with rRNA gene restriction patterns (ribotyping), have indicated the convenience of a revision of its present internal subdivision into biovars (Elomari et al., 1994), as previously suggested by Barrett et al. (1986) and by Stenström et al. (1990) (see also the sections on taxonomic comments and on ribotyping, in particular the contribution by Brosch et al., 1996).

The mol% G + C of the DNA is: 62.5 (biovar A; thermal denaturation); 60.7 (biovar B).

Type strain: ATCC 12633, DSM 291, NCIB 9494.

GenBank accession number (16S rRNA): D37923.

43. ***Pseudomonas resinovorans*** Delaporte, Raynaud and Daste 1965, 1075^{AL}

re.si.no' vor.ans. L. n. *resina* resin; L. v. *voro* to devour, digest; M.L. part. adj. *resinovorans* resin digesting.

The description is taken from the original paper.

Rods, 0.6–0.7 × 2.0–2.5 µm. Motile by means of a polar flagellum. Fluorescent pigment is produced. Gelatin is not liquefied. Nitrate reduction is weak, and denitrification is negative. Oxidase reaction positive. Optimum growth temperature, 28–30°C; no growth at 5°C or 42°C. No acid is produced from arabinose, xylose, rhamnose, glucose, fructose, galactose, mannose, lactose, maltose, sucrose, raffinose, inulin, salicin, dextrin, glycerol, mannitol, inositol, or dulcitol. Starch hydrolysis very weak. Growth occurs at the expense of colophony, Canada balsam, or abietic acid. Phenol, phenanthrene, salicylic acid, *m*-cresol, and naphthalene can also be used as carbon and energy sources for growth. Further information may be found in the original paper by Delaporte et al. (1965). Assignment to the genus *Pseudomonas* is confirmed by results on nucleic acid hybridization (De Vos et al., 1989) and by 16S rDNA sequence analysis (Moore et al., 1996).

The mol% G + C of the DNA is: 63.7 (T_m).

Type strain: ATCC 14235, CCUG 4439, LMG 2274.

GenBank accession number (16S rRNA): Z76668.

44. ***Pseudomonas rhodesiae*** Coroler, Elomari, Hoster, Izard and Leclerc 1996, 603^{VP}

rho.de' si.ae. M.L. fem. gen. n. *rhodesiae* of Rhodes, in honor of M.R. Rhodes, an English microbiologist.

Rods, motile by means of single polar flagella. Colonies on nutrient agar are smooth, circular, and nonpigmented. No hemolysis observed on blood agar. Fluorescent pigment is produced in King B medium. Arginine dihydrolase, catalase, cytochrome oxidase, lecithinase, nitrate reduction to nitrite, formation of levan from sucrose, growth on cetrimide agar, lecithinase reaction, and decomposition of L-tyrosine are all positive. PHB is not accumulated. Phenazine pigments are not produced. Gelatin liquefaction, lipase reaction, esculin and starch hydrolysis, deamination of phenylalanine, and exonuclease production are all negative. A large number of enzymatic properties are described in the original paper. A summary of properties of this species is presented in Tables BXII.γ.112 and BXII.γ.113. A number

of additional characteristics can be found in the original paper and elsewhere (Elomari et al., 1995). Isolated from natural mineral water.

The mol% G + C of the DNA is: 59 ± 1 (T_m).

Type strain: CIP 104664.

GenBank accession number (16S rRNA): AF064459.

45. ***Pseudomonas savastanoi*** (Janse 1982) Gardan, Bollet, Abu Ghorrah, Grimont and Grimont 1992, 611^{VP} (*Pseudomonas syringae* subsp. *savastanoi* Janse 1982, 168.)

sa.vas.ta' no.i. L. gen. n. *savastanoi* of Savastano, the first worker who studied olive knot disease.

The description is taken from the above-mentioned paper.

Rods, 0.4–0.8 × 1.0–3.0 µm. Motile by means of one to four polar flagella. Rather slow growing. Colonies are white or cream, smooth, flat, and glistening with entire or erose margins. Strains produce an hypersensitive reaction on tobacco leaves. Other properties of interest are summarized in Table BXII.γ.109. Metabolism is respiratory. Oxidase negative. Nitrates are not reduced. Blue fluorescent pigment is produced on King B medium and can be observed under UV light. Arginine dihydrolase negative. Esculin, gelatin, and starch are not hydrolyzed. Strains assimilate sucrose, L-arabinose, gluconate, caprylate, fumarate, DL-glycerate, L-malate, pyruvate, citrate, D-alanine, and L-proline. They do not assimilate lactose, L-xylose, adonitol, 2-aminobutyrate, DL-lactate, D-β-hydroxybutyrate, D(–)-tartrate, L-cysteine, L-methionine, and L-valine.

Strains have been isolated for many years from members of the family Oleaceae (olive, oleander). These do not produce levan from sucrose. Strains isolated from another species of the same family (*Fraxinus excelsior* L.), *Phaseolus vulgaris*, and *Glycine max* (Leguminosae) do produce levan. The species is now divided into three pathovars. *P. savastanoi* pathovar *savastanoi* causes knots, galls, and cankers on plants of various genera of the family Oleaceae; *P. savastanoi* pathovar *glycinea* causes bacterial blight of soybean; and *P. savastanoi* pathovar *phaseolicola* causes halo blight of bean.

The mol% G + C of the DNA is: 60 (T_m).

Type strain: ATCC 13522, CFBP 1670, ICMP 4352, NCPPB 639.

Additional Remarks: The paper by Gardan et al. (1992) essentially is a proposal to revive the name *P. savastanoi*, which had been created by Stevens in 1913 in a valid publication. This name was omitted from the 1980 approved list of bacterial species, an omission that had its origin in the fact that the name had been included in a list of pathovars of the species *P. syringae*.

The restoration of *P. syringae* pathovar *savastanoi* to its original status as independent species (*P. savastanoi*) after two demotions (one to the subspecies category, and the other to pathovar), has been formulated based on a phenotypic analysis and DNA–DNA hybridization studies (Gardan et al., 1992). In fact the phenotypic properties that differentiate this taxon from other related fluorescent, oxidase negative plant pathogens, are very few. In addition, the conditions of the nucleic acid hybridization used by Gardan et al. (1992) apparently were those used in the original description of the adopted methodology (Crosa et al., 1973b), even to the point of the incubation at 60°C, which is low for organisms of mol% G + C content higher

than *E. coli*. As a consequence, the estimated similarities reported in the paper tend to be high, although the figures can be grouped into blocks that reflect differences between *P. savastanoi* and other *P. syringae* pathovars, with the exception of the type strains of pathovars *phaseolicola* and *glycinea*. Consequently, the proposal subdivides *P. savastanoi* into pathovar *savastanoi*, pathovar *phaseolicola*, and pathovar *glycinea*. These last two names had also been used until now to designate pathovars of *P. syringae*.

Recent studies on DNA similarity by hybridization methods indicate that *P. savastanoi*, *P. ficusecetae*, *P. meliae*, and *P. amygdali* should be considered as synonyms, in which case the latter is the correct name of the species (Gardan et al., 1999).

46. ***Pseudomonas straminiae*** Iizuka and Komagata 1963b, 139^{AL} emend. Uchino, Kosako, Uchimura and Komagata 2000, 1518.

stra.mi' na.e. L. adj. *straminiae* made of straw.

Slender rods, 0.3×3.0 μm . Motile with single polar flagellum. Colonies are yellow. Fluorescent pigment is produced. Oxidase reaction positive. Gelatin is liquefied. Nitrates are not reduced to nitrites. Denitrification is negative. H_2S is produced. Starch is not hydrolyzed. Acid produced from glucose. Glucose, gluconate, citrate, succinate, and ethanol are assimilated. Phenol, benzoate, salicylate, *m*-hydroxybenzoate, *p*-hydroxybenzoate, protocatechuate, gentisate, anthranilate, *p*-aminobenzoate, 2-ketogluconate, and 5-ketogluconate are not assimilated. Optimum growth temperature, 25–30°C. No growth at 37°C. Assignment to the genus *Pseudomonas* has been confirmed by 16S rDNA sequence analysis (Anzai et al., 1997). The organism has been isolated from Japanese rice paddies.

The mol% G + C of the DNA is: unknown.

Type strain: IAM 1598.

GenBank accession number (16S rRNA): D84023.

47. ***Pseudomonas stutzeri*** (Lehmann and Neumann 1896) Sijderius 1946, 115^{AL} (*Bacterium stutzeri* Lehmann and Neumann 1896, 237.)

stut' ze.ri. *stutzeri* Stutzer patronymic, of Stutzer.

Characteristics differentiating the species from other species of *Pseudomonas* are given in Tables BXII.γ.116 and BXII.γ.117. Differentiation from a number of denitrifying pseudomonads can be found in Table BXII.γ.114 and in the chapter on the genus *Burkholderia* appearing in Volume 2 Part C in this *Manual*. Recently isolated colonies are adherent, dark brown, and have a characteristic wrinkled appearance, which usually is lost after repeated transfers in laboratory media. The colonies may eventually become smooth, butyrous, and pale in color. The gelatinase reaction is negative. Among the characteristics useful for its identification are the vigorous denitrification, the appearance of the colonies, and the use of starch as a carbon and energy source. Variations in some of the typical characteristics are observed in some populations, depending of the enrichment conditions. Some strains grow at 43°C. Optimum temperature, ~35°C.

The species is markedly heterogeneous in nutritional properties and in DNA base composition. Further descriptive information: van Niel and Allen (1952), Sijderius (1946), and Palleroni et al. (1970). Found in soil and water,

from which it can be isolated after enrichment in media with nitrate under anaerobic conditions at 30°C using various carbon sources. L(+)–Tartrate gives excellent results in the enrichments (van Niel and Allen, 1952) although, paradoxically, strains obtained in this manner may not grow with tartrate in pure culture. Many strains have been isolated from clinical specimens. Recently, the results of studies on the genomic structure and organization, as well as on the genotypic and phenotypic diversity of *P. stutzeri*, have been published (Rosselló et al., 1991, Rosselló–Mora et al., 1994b; Rainey et al., 1994b; Ginard et al., 1997). Seven DNA–DNA similarity groups (genomovars) have been defined in this species (Rosselló–Mora et al., 1994b). This subdivision correlated with clusters obtained by comparison of 16S rDNA gene sequences. One of the genomovars was raised to the category of species, for which the name *P. balearica* was created (see below). The name *P. stanieri* (Mandel 1966) has been proposed for strains with a mol% G + C of ~62. However, this species is not clearly differentiated from *P. stutzeri* based on phenotypic characteristics (Palleroni et al., 1970).

The mol% G + C of the DNA is: from 60.6–66.3 (Bd).

Type strain: AB 201, ATCC 17588, CCUG 11256, DSM 5190.

GenBank accession number (16S rRNA): U26262.

48. ***Pseudomonas synxantha*** (Ehrenberg 1840) Holland 1920, 220^{AL} (*Vibrio synxanthus* Ehrenberg 1840, 202.)

syn.xan' tha. Gr. pref. *syn* along with, together; Gr. adj. *xanthus* yellow; M.L. adj. *synxanthus* with yellow.

The description of this species is given by Haynes and Burkholder (1957) in the seventh edition of the *Manual*.

Gelatinase positive. Growth characteristics in various solid and liquid media are described by Haynes and Burkholder (1957). A distinctive characteristic is the production of an intense, diffusible, yellow to orange pigment in cream or in the cream layer in milk.

rRNA–DNA hybridization studies indicate that this is a species of the genus *Pseudomonas* (De Vos et al., 1989), which was later confirmed by 16S rDNA sequencing analysis (Anzai et al., 1997). Isolated from bitter milk.

The mol% G + C of the DNA is: 61. (T_m).

Type strain: ATCC 9890, LMG 2190.

49. ***Pseudomonas syringae*** Van Hall 1902141^{AL} (*Phytomonas syringae* (van Hall 1902) Bergey, Harrison, Breed, Hammer and Huntoon 1930, 257; *Pseudomonas barkeri* (Berridge 1924) Clara 1934, 11; *Pseudomonas citrarefaciens* (Lee 1917) Stapp 1928, 190; *Pseudomonas citriputealis* (Smith 1913) Stapp 1928, 190; “*Pseudomonas hibisci*” (Nakada and Takimoto 1923) Stapp 1928; *Pseudomonas prunicola* Wormald 1930, 742; *Pseudomonas punctulans* (Bryan 1933) Săvulescu 1947, 12; *Pseudomonas rimaefaciens* Koning 1938, 11; *Pseudomonas spongiosa* (Leifson 1962b); *Pseudomonas tonelliana* (Ferraris 1926) Burkholder 1948b, 132; *Pseudomonas trifoliorum* (Jones, Williamson, Wolf and McCulloch 1923) Stapp 1928; *Pseudomonas utiformica* Clara 1932, 111; *Pseudomonas vignae* Gardner and Kendrick 1923, 275; *Pseudomonas viridifaciens* Tisdale and Williamson 1923, 150.) *sy.rin' gae.* M.L. fem. n. *Syringa* generic name of lilac; M.L. fem. gen. n. *syringae* of the lilac.

Characteristics of the species and of some of its pathovars are described in Tables BXII.γ.118 and BXII.γ.120. The

TABLE BXII.γ.120. Characteristics of some *Pseudomonas syringae* pathovars and of *Pseudomonas savastanoi* and its two biovars, biovar glycinea and biovar phaseolicola^a

Characteristics	<i>P. savastanoi</i>	<i>P. savastanoi</i> pathovar <i>glycinea</i>	<i>P. savastanoi</i> pathovar <i>phaseolicola</i>	<i>P. syringae</i> pathovar <i>antirrhini</i>	<i>P. syringae</i> pathovar <i>aptata</i>	<i>P. syringae</i> pathovar <i>cannabina</i>	<i>P. syringae</i> pathovar <i>coronafaciens</i>	<i>P. syringae</i> pathovar <i>delphinii</i>	<i>P. syringae</i> pathovar <i>eribiotryae</i>	<i>P. syringae</i> pathovar <i>lachrymans</i>	<i>P. syringae</i> pathovar <i>mori</i>	<i>P. syringae</i> pathovar <i>morsprunorum</i>	<i>P. syringae</i> pathovar <i>passiflorae</i>	<i>P. syringae</i> pathovar <i>persicae</i>	<i>P. syringae</i> pathovar <i>pisii</i>	<i>P. syringae</i> pathovar <i>sesami</i>	<i>P. syringae</i> pathovar <i>strigifaciens</i>	<i>P. syringae</i> pathovar <i>syringae</i>	<i>P. syringae</i> pathovar <i>tabaci</i>	<i>P. syringae</i> pathovar <i>tomato</i>
Levan formation	—	+	+	+	+	+	+	d	+	+	+	+	—	+	+	+	+	+	+	+
Pectate gel pitting ^b	4	4	4	—	—	4	4,8	4	4	4,8	—	—	4	4	—	4	—	—	4,8	4
β-Glucosidase	—	—	—	+	+	—	+	+	+	+	—	—	+	—	d	—	+	+	+	+
Growth on:																				
Mannitol	+	d	—	+	+	—	+	+	+	+	—	+	+	+	+	—	+	+	+	+
Betaine	+	—	d	+	+	+	+	+	+	—	—	+	+	—	+	+	+	+	+	+
Inositol	d	+	—	+	+	—	+	+	—	+	+	+	+	—	+	+	+	+	+	+
Sorbitol	d	—	—	+	+	—	+	+	+	+	d	+	+	+	+	—	+	+	+	+
Trigonelline	d	+	+	+	+	+	—	+	+	+	+	+	+	—	+	+	—	+	+	+
Quinate	—	+	+	+	+	—	+	+	+	+	—	+	+	—	+	+	—	+	+	+
Erythritol	—	—	—	—	+	—	+	+	+	+	—	d	+	—	d	+	+	+	d	—
L-Tartrate	d	—	—	—	—	—	—	—	+	+	—	+	—	—	—	+	—	—	+	—
D-Tartrate	—	—	—	+	+	—	—	—	—	—	—	—	—	—	—	—	—	d	—	+
L-Lactate	—	—	—	—	+	—	—	—	—	—	—	—	—	—	d	—	—	+	—	—
Anthranilate	d	—	—	—	—	—	—	—	—	—	—	+	—	—	—	—	—	—	—	—
Homoserine	—	—	—	+	—	—	—	—	—	—	—	—	—	—	+	—	—	—	—	—

^aFor symbols see standard definitions. Data from Sands et al. (1970) and from Gardan et al. (1992).^bMethod of Hildebrand (1971); numbers represent pH values at which pitting occurs.

pattern of acid production from sugars will be found in Haynes and Burkholder (1957) and in the original papers describing the nomenclatures here included as pathovars.

Cytochrome *c* is not detectable. Fluorescent pigment is produced. The gelatinase reaction is positive. Rare strains may require organic growth factors. Growth of most strains is slow in mineral media with a single carbon source and is relatively slow in complex media. Nutritional spectrum is less extensive and more heterogeneous than that of the saprophytic fluorescent pseudomonads. Optimum temperature, 25–30°C. The original strain was isolated from lilac (*Syringa vulgaris*, family Oleaceae) but strains conforming to the original description are pathogenic for many unrelated plants.

The mol% G + C of the DNA is: 60.5.

Type strain: ATCC 19310, DSM 6693, LMG 1247, NCPPB 281, PDDCC 3023.

Additional Remarks: The following is a list of pathovars of *P. syringae* and of the respective pathotypes proposed by Dye et al. (1980). The names of the most important host plants and of their botanical families are also included. The list includes pathovar *mori*, although this name has priority over *syringae*. As recommended by Young et al. (1978), *P. syringae* is widely known and it should be conserved over "*P. mori*". Many of the pathovars in the list may not be distinguishable from *P. syringae* except for their host range. Few phenotypic characters are taxonomically useful.

Studies on the internal subdivision of *P. syringae* into pathovars using various molecular techniques, in particular DNA–DNA hybridization, have suggested that some of the pathovars should be elevated to independent species status.

At present, however, attempts in this direction have materialized only in a few instances. The difficulty is in finding sets of phenotypic properties that could help in the ready identification of such species without the need of nucleic acid similarity experiments. A recent study centered mainly on this approach and in ribotyping (Gardan et al., 1999) extended early investigations that defined clusters of genomically related strains within *P. syringae* (Pecknold and Grogan, 1973).

The studies of Gardan and collaborators (1999) defined nine genomospecies. The conclusions included the proposal for synonymy of the names *P. savastanoi*, *P. ficusectae*, *P. meliae*, and *P. amygdali*, with *P. amygdali* as the earliest name in this group. In addition, genomospecies 3 and 7 may be named in the future "*P. tomato*" and "*P. tagetis*", respectively, once phenotypic data for differentiation become available. Finally, genomospecies 5 and 9, for which such data are available, are given the species names *P. tremae* and *P. cannabina*, respectively.

The genomospecies 5 (*P. tremae*) is represented by a single strain (Gardan et al., 1999). According to the description given by Gardan and collaborators, *P. tremae* is a non-fluorescent organism, but evaluation of its relationships with other species or pathovars or with nonfluorescent species is not clear at this stage. The description includes a nutritional analysis where the strain appears to be unable to use L-alanine, fructose, glycerol, or gluconate, and it is stated that it did not assimilate the 91 other carbon sources of the Biotype-100 strips (BioMérieux). Since there is only one strain available at this moment, it is not possible to ascertain the intraspecific diversity of the species. Experi-

ments on the DNA relatedness of *P. tremae* and *P. cannabina* to other taxa have been limited mainly to *P. syringae* pathovars but have not included most other species of the genus. Descriptions of *P. tremae* and *P. cannabina*, as given in the paper by Gardan et al. (1999), are transcribed below in the section Other Species.

As a general comment, the taxonomic situation of the members of the conglomerate known under the name *P. syringae* still remains largely unsettled, but it is reasonable to expect that future studies will contribute to reach a satisfactory internal subdivision of the species and a more precise circumscription of pathovars segregated as individual species on the basis of genomic and phenotypic properties.

- a. ***Pseudomonas syringae* pathovar *syringae*** van Hall 1902.
Hosts: lilac (*Syringa vulgaris*, family Oleaceae) and several unrelated plants.
Deposited strain: ATCC 19310, NCPPB 281, PDDCC 3023 (neopathotype strain).
- b. ***Pseudomonas syringae* pathovar *aceris*** (Ark 1939) Young et al. 1978.
Hosts: *Acer macrophyllum*, *Acer* spp. (family Aceraceae).
The mol% G + C of the DNA is: 61.1.
Deposited strain: ATCC 10853, NCPPB 958, PDDCC 2802 (Sneath and Skerman, 1966) (neopathotype strain).
- c. ***Pseudomonas syringae* pathovar *antirrhini*** (Takimoto 1920) Young et al. 1978.
Host: *Antirrhinum majus* (family Scrophulariaceae).
The mol% G + C of the DNA is: 60.4.
Deposited strain: NCPPB 1817, PDDCC 4303 (neopathotype strain).
- d. ***Pseudomonas syringae* pathovar *apii*** (Jagger 1921) Young et al. 1978.
Host: celery (*Apium graveolens*, family Umbelliferae).
The mol% G + C of the DNA is: 60.8.
Deposited strain: ATCC 9654, NCPPB 1626, PDDCC 2814 (neopathotype strain).
- e. ***Pseudomonas syringae* pathovar *aptata*** (Brown and Jamieson 1913) Young et al. 1978.
Hosts: sugar beet (*Beta vulgaris*, family Chenopodiaceae), *Nasturtium* sp. (family Cruciferae); lettuce (*Lactuca sativa*, family Compositae).
Deposited strain: NCPPB 871, PDDCC 459 (neopathotype strain).
- f. ***Pseudomonas syringae* pathovar *atrofaciens*** (McCulloch 1920) Young et al. 1978.
Hosts: wheat (*Triticum* sp.) and other plants of the family Gramineae.
Deposited strain: NCPPB 2612, PDDCC 4394 (neopathotype strain).
- g. ***Pseudomonas syringae* pathovar *atropurpurea*** (Reddy and Godkin 1923) Young et al. 1978.
Hosts: *Bromus inermis* and many other *Bromus* spp., *Agropyron repens* and many other plants of the family Gramineae.
Deposited strain: NCPPB 2397, PDDCC 4457 (neopathotype strain).
- h. ***Pseudomonas syringae* pathovar *berberidis*** (Thornberry and Anderson 1931a) Young et al. 1978.
Hosts: barberry (*Berberis thunbergii*, *B. inermis*, family Berberidaceae).
The mol% G + C of the DNA is: 59.7–60.1.
Deposited strain: NCPPB 2724, PDDCC 4116 (neopathotype strain).
- i. ***Pseudomonas syringae* pathovar *cannabina*** (Šutić and Dowson 1959) Young et al. 1978.
Hosts: *Cannabis sativa* (family Moraceae); *Phaseolus vulgaris*, *Vicia sativa* (family Leguminosae).
The mol% G + C of the DNA is: 59.3–60.3.
Deposited strain: PDDCC 2823 (NCPPB 1437) (neopathotype strain).
- j. ***Pseudomonas syringae* pathovar *ciccaronei*** (Ercolani and Caldarola 1972) Young et al. 1978.
Host: carob (*Ceratonia siliqua*, family Leguminosae).
Deposited strain: NCPPB 2355, PDDCC 5710 (pathotype strain).
- k. ***Pseudomonas syringae* pathovar *coronafaciens*** (Elliott 1920) Young et al. 1978.
Hosts: *Avena sativa*, *Bromus inermis*, *Agropyron repens* and various other wild and cultivated plants of the family Gramineae. Description as “*P. coronafaciens*” is transcribed by Haynes and Burkholder (1957) in the seventh edition of *Bergey’s Manual of Determinative Bacteriology*.
The mol% G + C of the DNA is: 59.3.
Deposited strain: NCPPB 600, PDDCC 3113 (neopathotype strain).
- l. ***Pseudomonas syringae* pathovar *delphinii*** (Smith 1904) Young et al. 1978.
Host: *Delphinium* sp. (family Ranunculaceae).
Deposited strain: NCPPB 1879, PDDCC 529 (neopathotype strain).
- m. ***Pseudomonas syringae* pathovar *dysoxyli*** (Hutchinson 1949) Young et al. 1978.
Host: *Dysoxylum spectabile* (family Meliaceae).
Deposited strain: : ATCC 19863, NCPPB 225, PDDCC 545 (neopathotype strain).
- n. ***Pseudomonas syringae* pathovar *erobotryae*** (Takimoto 1931) Young et al. 1978.
Host: loquat (*Eriobotrya japonica*, family Rosaceae).
The mol% G + C of the DNA is: 58.3.
Deposited strain: NCPPB 2331, PDDCC 4455 (neopathotype strain).
- o. ***Pseudomonas syringae* pathovar *garcae*** (Amaral, Teixeira and Pinheiro 1956) Young et al. 1978.
Host: coffee (*Coffea arabica*, family Rubiaceae) and many other unrelated plants.
Deposited strain: ATCC 19864, NCPPB 588, PDDCC 4323 (pathotype strain).
- p. ***Pseudomonas syringae* pathovar *helianthi*** (Kawamura 1934) Young et al. 1978.
Host: sunflower (*Helianthus debilis*, family Compositae).
Deposited strain: NCPPB 2640, PDDCC 4531 (neopathotype strain).

- q. *Pseudomonas syringae* pathovar **japonica** (Mukoo 1955) Dye et al. 1980 (*Pseudomonas striafaciens* pathovar *japonica* Mukoo 1955.)
Hosts: rye (*Secale cereale*), barley (*Hordeum sativum*), wheat (*Triticum vulgare*), rice (*Oryza sativa*) and many other Gramineae of the genera *Setaria*, *Panicum*, *Bromus*, *Lolium*, *Andropogon*, *Alopecurus*, etc., as well as plants of other botanical families (Solanaceae, Chenopodiaceae, Oxalidaceae, etc.).
Deposited strain: NCPPB 3093, PDDCC 6305 (neopathotype strain).
- r. *Pseudomonas syringae* pathovar **lachrymans** (Smith and Bryan 1915) Young et al. 1978.
Host: cucumber (*Cucumis sativus*, family Cucurbitaceae).
Deposited strain: ATCC 7386, NCPPB 537, PDDCC 3988 (neopathotype strain).
- s. *Pseudomonas syringae* pathovar **lapsa** (Ark 1940) Young et al. 1978.
Hosts: corn (*Zea mays*), sugarcane (*Saccharum officinarum*) (family Gramineae).
The mol% G + C of the DNA is: 59.4.
Deposited strain: NCPPB 2096, PDDCC 3947 (neopathotype strain).
- t. *Pseudomonas syringae* pathovar **maculicola** (McCulloch 1911) Young et al. 1978.
Hosts: cabbage, cauliflower (*Brassica oleracea*, family Cruciferae).
The mol% G + C of the DNA is: 61.3.
Deposited strain: NCPPB 2039, PDDCC 3935 (neopathotype strain).
- u. *Pseudomonas syringae* pathovar **melles** (Johnson 1923) Young et al. 1978.
Host: tobacco (*Nicotiana tabacum*, family Solanaceae).
Deposited strain: NCPPB 2356, PDDCC 5711 (neopathotype strain).
- v. *Pseudomonas syringae* pathovar **mori** (Boyer and Lambert 1893) Young et al. 1978.
Host: mulberry (*Morus* spp., family Moraceae).
Deposited strain: ATCC 19873, NCPPB 1034, PDDCC 4331 (neopathotype strain).
- w. *Pseudomonas syringae* pathovar **morsprunorum** (Wormald 1931) Young et al. 1978.
Host: *Prunus* spp. (family Rosaceae).
Deposited strain: ATCC 19322, NCPPB 2995, PDDCC 5795 (Sneath and Skerman, 1966) (neopathotype strain).
- x. *Pseudomonas syringae* pathovar **panici** (Elliott 1923) Young et al. 1978.
Host: proso or broom-corn millet (*Panicum miliaceum*, family Gramineae).
The mol% G + C of the DNA is: 60.7.
Deposited strain: ATCC 19875, NCPPB 1498, PDDCC 3955 (neopathotype strain).
- y. *Pseudomonas syringae* pathovar **papulans** (Rose 1917) Dhanvantari 1977.
Host: apple (*Pyrus malus*, family Rosaceae).
Deposited strain: NCPPB 2848, PDDCC 4048 (neopathotype strain).
- z. *Pseudomonas syringae* pathovar **passiflorae** (Reid 1938) Young et al. 1978.
Host: *Passiflora edulis* (family Passifloraceae).
The mol% G + C of the DNA is: 59.4–60.3.
Deposited strain: NCPPB 1387, PDDCC 129 (neopathotype strain).
- aa. *Pseudomonas syringae* pathovar **persicae** (Prunier et al. 1970) Young et al. 1978.
Host: peach (*Prunus persicae*, family Rosaceae).
Deposited strain: NCPPB 2761, PDDCC 5846 (neopathotype strain).
- ab. *Pseudomonas syringae* pathovar **pisi** (Sackett 1916) Young et al. 1978.
Host: pea (*Pisum sativum*, family Leguminosae).
Deposited strain: NCPPB 2585, PDDCC 2452 (neopathotype strain).
- ac. *Pseudomonas syringae* pathovar **primulae** (Ark and Gardner 1936) Young et al. 1978.
Host: *Primula polyantha* (family Primulaceae).
The mol% G + C of the DNA is: 60.4.
Deposited strain: ATCC 19306, NCPPB 133, PDDCC 3956 (neopathotype strain).
- ad. *Pseudomonas syringae* pathovar **ribicola** (Bohn and Maloit 1946) Young et al. 1978.
Host: golden currant (*Ribes aureum*, family Saxifragaceae).
The mol% G + C of the DNA is: 60.6.
Deposited strain: ATCC 13456, NCPPB 963, PDDCC 3882 (neopathotype strain).
- ae. *Pseudomonas syringae* pathovar **sesami** (Malkoff 1906) Young et al. 1978.
Host: sesame (*Sesamum indicum*, family Pedaliaceae).
The mol% G + C of the DNA is: 59.4.
Deposited strain: ATCC 19879, NCPPB 1016, PDDCC 763 (neopathotype strain).
- af. *Pseudomonas syringae* pathovar **striafaciens** (Elliott 1927) Young et al. 1978.
Hosts: oats (*Avena sativa*), barley (*Hordeum sativum*) (family Gramineae).
The mol% G + C of the DNA is: 59.1.
Deposited strain: ATCC 10730, NCPPB 1898, PDDCC 3961 (pathotype strain); NCPPB 2394, PDDCC 4483 (pathogenic reference strain).
- ag. *Pseudomonas syringae* pathovar **tabaci** (Wolf and Foster 1917) Young et al. 1978.
Host: tobacco (*Nicotiana tabacum*, family Solanaceae).
Deposited strain: NCPPB 1427, PDDCC 2835 (neopathotype strain).
- ah. *Pseudomonas syringae* pathovar **theae** (Hori 1915) Young et al. 1978.
Host: tea plant (*Thea sinensis*, family Theaceae).
Deposited strain: NCPPB 2598, PDDCC 3923 (neopathotype strain).
- ai. *Pseudomonas syringae* pathovar **tomato** (Okabe 1933) Young et al. 1978.
Host: tomato (*Lycopersicon esculentum*, family Solanaceae).
Deposited strain: NCPPB 1106, PDDCC 2844 (neopathotype strain).

- aj. ***Pseudomonas syringae* pathovar *ulmi*** (Šutić and Tešić 1958) Young et al. 1978.

Host: elm (*Ulmus* sp., family Ulmaceae).

The mol% G + C of the DNA is: 58.7.

Deposited strain: ATCC 19883, NCPPB 632, PDDCC 3962 (neopathotype strain).

- ak. ***Pseudomonas syringae* pathovar *viburni*** (Thornberry and Anderson 1931b) Young et al. 1978.

Host: elm (*Ulmus* sp., family Ulmaceae).

The mol% G + C of the DNA is: 60.1.

Deposited strain: ATCC 13458, NCPPB 1921, PDDCC 3963 (neopathotype strain).

50. ***Pseudomonas taetrolens*** Haynes 1957, 108^{AL}.

taet'ro.lens. L. adj. *taeter* offensive; L. part. adj. *olens* having an odor; M.L. part. adj. *taetrolens* foul smelling.

The description of this species is given by Haynes and Burkholder (1957) in the seventh edition of the *Manual*.

The cells are short rods with rounded ends, motile by means of one to five polar flagella. Gelatin and starch are not hydrolyzed. Nitrate is not reduced to nitrite. Acid production from a number of substrates is described by Haynes and Burkholder (1957). A decision to allocate the species in the genus *Pseudomonas* has been made based on rRNA–DNA hybridization studies (De Vos et al., 1989). Isolated from foods that have a musty odor.

The mol% G + C of the DNA is: 59.8 (T_m).

Type strain: ATCC 4683, IAM 1653, LMG 2336.

GenBank accession number (16S rRNA): D84027.

51. ***Pseudomonas tolaasii*** Paine 1919, 210^{AL}.

to.laa'si.i. *Tolaas* patronymic; M.L. gen. n. *tolaasii* of Tolaas.

Description is given by Haynes and Burkholder (1957) in the seventh edition of the *Manual*.

Rapid identification tests have been described (Wong and Preece, 1979). Results of a detailed phenotypic study of a collection of strains of this species have been published (Fahy, 1981). Some of the reported characteristics are summarized in Table BXII.γ.115. A gene cluster encoding proteins required for the synthesis of the toxin tolaasin has been identified and characterized (Rainey et al., 1993). Detailed composition of fatty acids is known (Stead, 1992). rRNA–DNA hybridization studies indicate allocation of this species to the genus *Pseudomonas* (De Vos et al., 1985). Fluorescent pigment is produced. Gelatin is hydrolyzed (Haynes and Burkholder, 1957). Pathogenic for cultivated mushrooms. Isolated from brown-spot of cultivated mushrooms.

The mol% G + C of the DNA is: 60.8–61.3 (T_m).

Type strain: ATCC 33618, LMG 2342, NCPPB 1873.

GenBank accession number (16S rRNA): D84028, Z76670.

52. ***Pseudomonas veronii*** Elomari, Coroler, Hoste, Gillis, Izard and Leclerc 1996, 1142.^{VP}

ve.ro'ni.i. M.L. masc. gen. n. *veronii* of Véron, in honor of Professor M.M. Véron, a distinguished French microbiologist.

The following description is taken from the original paper.

Rods, motile by means of single polar flagella. PHB is not accumulated by the cells. The cultures produce a fluorescent pigment on King B medium. The colonies on nutrient agar are smooth, circular, and nonpigmented. They

are nonhemolytic on blood agar. The oxidase, catalase, and arginine dihydrolase reactions are positive. Most strains liquefy gelatin. All the strains grow on α-aminobutyrate, D-xylene, L-arabinose, D-mannose, D-galactose, sucrose, butyrate, isobutyrate, erythritol, sorbitol, inositol, D-alanine, L-tryptophan, and trigonelline as the sole source of carbon and energy. No strain is able to use isovalerate, sebacate, azelate, L-mandelate, benzoate, L-kynurenine, histamine, or acetamide. The original paper (Elomari et al., 1996) includes extensive information on a variety of enzymatic activities. Additional properties are summarized in Tables BXII.γ.112 and BXII.γ.113 and in a previous communication by the same group of workers (Elomari et al., 1995).

All strains have been isolated from natural mineral waters.

The mol% G + C of the DNA is: 61–62.

Type strain: CFML 92-134, CIP 104663, DSM 11331.

GenBank accession number (16S rRNA): AF064460.

53. ***Pseudomonas viridiflava*** (Burkholder 1930) Dowson 1939, 177^{AL} (*Phytomonas viridiflava* Burkholder 1930, 63.)

vi.ri.di.fl'a.va. L. *viridis* green; L. *flavus* yellow; M.L. adj. *viridiflavus* greenish yellow.

Characteristics that are useful for comparison with related taxa are given in Table BXII.γ.118. Further information is given by Haynes and Burkholder (1957), Clara (1934), and Billing (1970b). The following characteristics are mentioned in this last paper. The bacterial mass usually has a yellow tinge in media with 5% sucrose and olive to golden brown in media with yeast extract and glycerol. A blue-green insoluble pigment is produced by some strains. Aside from the characteristics described in the tables, the potato rot and esculin reactions are positive. Fluorescent pigment is produced, and gelatin is liquefied (Haynes and Burkholder, 1957).

From her extensive nutritional screening of strains of the species, Billing (1970b) has concluded that few substrates have diagnostic value. The only ones distinguishing *P. viridiflava* from most other oxidase-negative plant pathogens are the inability to use sucrose and the capacity for use of D(–)-tartrate. This substrate is used by *P. viridiflava*, *P. syringae* pathovar *tomato*, and rarely by other species and pathovars. In addition to these characteristics, Billing (1970b) mentions that the reaction in beans was similar to that produced by *P. syringae* and both were different from the water-soaked lesions produced by "*P. phaseolicola*".

Pathogenic on bean (*Phaseolus vulgaris*).

The mol% G + C of the DNA is: 59.9 (NCPPB 1810).

Type strain: ATCC 13223, DSM 6694, LMG 2352, NCPPB 635, PDDCC 2848.

GenBank accession number (16S rRNA): Z76671.

Other Species

Some putative *Pseudomonas* species have an uncertain phylogenetic position. The following descriptions are those that are not included in the first edition of the *Manual* (Palleroni, 1984) or in previous editions of *Bergey's Manual of Determinative Bacteriology*.

1. ***Pseudomonas abietaniphila*** Mohn, Wilson, Bicho and Moore 1999b, 935^{VP} (Effective publication: Mohn, Wilson, Bicho and Moore 1999a, 76.)

a.bie.ta.ni'phi.la. M.L. neut. n. *abietanum* abietane; Gr. adj. *philos* loving, friendly to; M.L. fem. adj. *abietaniphila* abietane-loving.

The following description is taken from the paper by Mohn et al. (1999a).

Rods. When grown on dehydroabietic acid, the cells are $0.7 \times 1.2\text{--}2.5 \mu\text{m}$. Motile. No information is given on the flagellar number and insertion. Colonies are clear, translucent, smooth, circular, and convex. Catalase positive and oxidase negative. Abietic, dehydroabietic, linoleic and pyruvic acids, as well as L-arabinose, D-galactose, D-glucose, D-xylose, and glycerol are used for growth; 12- plus 14-chlorodehydroabietic and palmitic acids are used poorly. Pimaric, isopimaric, and acetic acids are not used. Main cellular fatty acids are $C_{16:1 \omega 7c}$, $C_{16:0}$, and $C_{18:0}$; minor ones include $C_{10:0 \ 3OH}$, $C_{12:0 \ 2OH}$, and $C_{12:0 \ 3OH}$. Isolated from bleached Kraft pulp mill effluent treatment system near Kamloops, British Columbia, Canada.

The mol% G + C of the DNA is: unknown.

Type strain: Strain BKME-9, ATCC 700689.

2. ***Pseudomonas antimicrobica*** Attafuah and Bradbury 1990, 320^{VP} (Effective publication: Attafuah and Bradbury 1989, 571.)

an.ti.mi.cro'bi.ca. Gr. pref. *anti* against; Gr. adj. *micrus* small; Gr. n. *bios* life; *antimicrobica* against microbes (referring to a wide antimicrobial activity).

Description taken from the original paper (Attafuah and Bradbury, 1989).

Rods, $0.5 \times 0.8\text{--}1.5 \mu\text{m}$, occurring singly, in groups, and occasionally in pairs. Motile by means of one or two polar flagella. Two types of colonies were observed on nutrient agar. Those of one variant (NCIB 9897) were circular, raised, creamy-white, and opaque, rugose and butyrous when raised, with microundulate margins; the other type (NCIB 9898) was moist, smooth, glistening, and slightly viscid, with entire margins and occasionally an orange tinge. A yellowish-orange, water-soluble, nonfluorescent pigment was produced in King A and B media, tyrosine medium and others. No pigment was produced on potato dextrose agar. Gelatinase activity, levan formation, accumulation of PHB, starch hydrolysis, pectolytic activity, and rotting of plant tissue were all negative. Growth occurs with NaCl up to 4%, but it decreases to a trace with 6%.

Acid was produced from arabinose, dulcitol, fructose, galactose, glucose, glycerol, lactose, mannitol, *meso*-inositol, sorbitol, and xylose, but not from maltose, raffinose, salicin, or sucrose. Carbon sources for growth were N-acetylglucosamine, D-alanine, L-arabinose, caprate, citrate, gluconate, glucose, glutamate, *p*-hydroxybenzoate, *meso*-inositol, α -ketoglutarate, malonate, mannitol, D-mannose, phenylacetate, L-proline, propionate, and raffinose. No growth on adipate, D-iso-ascorbate, malate, *meso*-tartrate, starch, or ethanol. No organic growth factors were required. A wide spectrum of antibiotic activity was present. Good growth occurred between 15 and 37°C; optimum around 30°C. No growth at 41°C. Additional information may be found in the original paper (Attafuah and Bradbury, 1989) and some comments on the relationship of this species to those of other groups of aerobic pseudomonads. Isolated from the mealybug *Planococcoides njalensis*.

The mol% G + C of the DNA is: unknown.

Type strain: DSM 8361, NCIB 9898.

3. ***Pseudomonas cannabina*** (ex Šutić and Dowson 1959) Gardan, Shafik, Belouin, Broch, Grimont and Grimont 1999, 477^{VP}

can.na'bi.na. L. fem. adj. *cannabina* pertaining to *Cannabis*, the generic name of the host plant, *Cannabis sativa* L.

Rods, $1.1\text{--}3.0 \times 3.0\text{--}4.0 \mu\text{m}$, motile by means of one to four polar flagella. Colonies are gray color and are slightly convex. Production of fluorescent pigment in King B medium. The results of LOPAT (levan production, oxidase test, potato rotting, arginine dihydrolase, and tomato hypersensitivity) are +, −, −, −, and +. Nitrate is not reduced. Negative for hydrolysis of starch, esculin, and gelatin. The strains assimilate D-glucose, glycerol, D-saccharate, mucate, citrate, D-gluconate, L-histidine, L-aspartate, L-glutamate, L-proline, L-alanine, and L-serine, but not the other 83 carbon sources of the Biotype-100 strips (BioMérieux). Pathogenic on *Cannabis sativa* L.

The mol% G + C of the DNA is: 60.2 (T_m).

Type strain: CFBP 2341; ICMP 2823; NCPPB 1437.

4. ***Pseudomonas flectens*** Johnson 1956, 144^{AL}
flec'tens. L. v. *flectere* to bend; *flectens* that bends.

The following description is summarized from that in the original paper.

Rods, $0.5\text{--}0.75 \times 1.4\text{--}2.0 \mu\text{m}$, with rounded ends. Motile by means of one to two polar flagella. Colonies on meat infusion agar, after 5 d at 27.5°C, circular, 1–1.5 mm in diameter, convex, amorphous, smooth, glistening, with an entire edge, grayish white. No fluorescent pigment is formed. Gelatin liquefaction does not occur. Acid is produced from glucose, mannose, and sucrose, but not from maltose, lactose, starch, glycerol, mannitol, sorbitol, or esculin. Starch hydrolysis is very slight. H₂S is not produced and nitrate is not reduced. The organism is the agent of the pod twist disease of French beans (*Phaseolus vulgaris* L.), where it produces large, diffuse water-soaked lesions on twisted young pods.

The mol% G + C of the DNA is: 31.8 (type strain).

Type strain: ATCC 12775.

Additional Remarks: Strain NCPPB 539, presumed to be the type strain, is not phylogenetically related to the genus *Pseudomonas*, as shown by rRNA–DNA hybridization experiments (De Vos et al., 1989). However, the authenticity of this strain may be questioned because of its unexpectedly low mol% G + C.

5. ***Pseudomonas halophila*** Fendrich 1989, 205^{VP} (Effective publication: Fendrich 1988, 42.)

ha.lo'phi.la. Gr. n. *hals*, *halos* salt; Gr. v. *philein* to love; M.L. adj. *halophila* salt-loving.

The description is taken from the original paper (Fendrich, 1988).

Rod-shaped cells, $0.8\text{--}1.0 \times 1.5\text{--}5.0 \mu\text{m}$, single or in pairs. Motile by means of a polar flagellum. Colonies are circular with entire margins and smooth surfaces, slimy, raised, and reddish-brown colored. Chemoorganotrophic growth on arabinose, cellobiose, fructose, galactose, glucose, lactose, maltose, mannitol, sorbitol, sucrose, trehalose, and xylose with no acid production, and on acetate, caproate, citrate, lactate, pelargonate, propionate, pyruvate,

succinate, alanine, glutamine, esculin, ethanol, and glycerol as carbon and energy sources. Catalase and cytochrome oxidase positive. Gelatinase and urease positive. Hydrolysis of starch or cellulose, production of indole or sulfide, arginine dihydrolase reaction, and lysine and ornithine decarboxylase activities are all negative. Strictly aerobic. Growth occurs between 0.02 and 3.3 ml/l NaCl, with optimum at 0.8 mol/l. The pH range is 4.5–9.6; optimum at 7. Temperature range, 4–37°C; optimum at 28°C. Susceptible to ampicillin, chloramphenicol, erythromycin, nalidixic acid, and penicillin G. Isolated from the north arm of Great Salt Lake, Utah, USA.

The mol% G + C of the DNA is: 57.

Type strain: DSM 3050.

6. ***Pseudomonas multiresinivorans*** Mohn, Wilson, Bicho and Moore 1999b, 935^{VP} (Effective publication: Mohn, Wilson, Bicho and Moore 1999a, 77.)

mul.ti.re.si.ni.vo'rans. L. adj. *multi* many; L. fem. n. *resina* resin; L. part. adj. *vorans* devouring; M.L. fem. adj. *multiresinivorans* devouring many resins.

Rods. When grown on isopimaric acid, cells are 1×1.5 – $2 \mu\text{m}$, motile. No information is given about insertion and number of flagella. The cells form clumps. Colonies are clear, translucent, smooth, asymmetrical, and flat. Catalase and oxidase reactions are positive. Can live anaerobically in the presence of nitrate. Pimaric, isopimaric, palmitic, linoleic, benzoic, and acetic acids, as well as L-arabinose, D-glucose, citronellol, ethanol, glycerol, and n-hexadecane are used for growth. Poor utilization of abietic, dehydroabietic, and pyruvic acids, 12- plus 14-chlorodehydroabietic acids, D-galactose, or D-xylose. Main cellular fatty acids are C_{18:1}, C_{16:0}, and C_{16:1 ω7c}; minor ones include C_{10:0 3OH}, C_{12:0 2OH}, and C_{12:0 3OH}. Isolated from a laboratory sequencing batch reactor in Vancouver, British Columbia, Canada.

The mol% G + C of the DNA is: unknown.

Type strain: Strain IpA-1, ATCC 700690.

7. ***Pseudomonas tremae*** Gardan, Shafik, Belouin, Broch, Grimont and Grimont 1999, 477^{VP}
tre'ma.e. M.L. gen. fem. n. *tremae* of *Trema*; generic name of the host plant, *Trema orientalis* BL.

Rods, motile by means of one to four polar flagella. No fluorescent pigment is produced on King B medium. Of the LOPT tests (levan formation from sucrose, oxidase reaction, potato rotting ability, arginine dihydrolase reaction, and tobacco hypersensitive test), the first four are negative. Nitrate is not reduced. Negative for hydrolysis of starch, esculin, gelatin, and Tween 80. The strain assimilates D-saccharate, D- and L-malate, citrate, succinate, fumarate, L-aspartate, and L-glutamate, but not any of the 91 other carbon sources of the Biotype-100 strips (BioMérieux). Pathogenic for *Trema orientalis*.

The mol% G + C of the DNA is: 60.5 (*T_m*).

Type strain: CFBP3229; ICMP 9151; NCPPB 3465.

8. ***Pseudomonas vancouverensis*** Mohn, Wilson, Bicho and Moore 1999b, 935^{VP} (Effective publication: Mohn, Wilson, Bicho and Moore 1999a, 76.)

van.cou.ver'en'sis. M.L. adj. *vancouverensis* pertaining to the city of Vancouver, Canada.

Rods. When grown on dehydroabietic acid, the cells are 0.9×0.9 – $3.0 \mu\text{m}$. Motile. No information is given about number and insertion of flagella. The cells form clumps. Colonies pale yellow, translucent, smooth, circular, and convex. Catalase and oxidase reactions are positive. Abietic, acetic, benzoic, dehydroabietic, linoleic, palmitic, and pyruvic acids, as well as β-citronellol, D-glucose, and glycerol, are used for growth. Acetic, isopimaric, and pimaric acids, and L-arabinose, D-galactose, and D-xylose are not used. Main cellular fatty acids are C_{16:1 ω7c}, C_{16:0}, C_{17:0 cyc}, and C_{18:1}; minor ones include C_{10:0 3OH}, C_{12:0 2OH}, and C_{12:0 3OH}. Isolated from forest soil in Vancouver, British Columbia, Canada.

The mol% G + C of the DNA is: unknown.

Type strain: Strain DhA-51, ATCC 700688.

Genus II. *Azomonas* Winogradsky 1938, 391^{AL}

CHRISTINA KENNEDY AND PAUL RUDNICK

A.zo.mo'nas. Gr. adj. *a* not; Gr. n. *zoë* life; Gr. n. *azoë* not sustaining life, nitrogen Gr. n. *monas* a unit, monad; M.L. fem. n. *Azomonas* nitrogen monad.

Cells are $2.0 \mu\text{m}$ or more in diameter, of variable length and shape, although usually 2.5 – $3.5 \mu\text{m}$ in length and generally ellipsoidal- to rod-shaped. They cannot easily be distinguished from cells of *Azotobacter*. Cells may occur singly, in pairs, or in clumps. Motile by peritrichous, lophotrichous, or polar flagella. Gram negative, but sometimes Gram variable, depending on culture age. Cysts are not formed in aging cultures. All species fix atmospheric nitrogen under aerobic conditions. Alternative nitrogenases containing vanadium (nitrogenase-2) or iron (nitrogenase-3) only may be synthesized in Mo-deficient media. Cultures can grow both aerobically and microaerobically. Chemoorganotrophic. Sugars, alcohols, and organic acids are used as carbon sources. Ammonium salts and sometimes nitrate (*A. insignis* only) are used as nitrogen sources; amino acids are not used. Water-soluble and fluorescent pigments are produced by nearly all strains. Species are catalase-positive. The optimum pH

for nitrogen fixation is close to neutrality, but certain strains can also fix nitrogen at a pH of 4.6–4.8. Species are isolated from water or soil.

The mol% G + C of the DNA is: 52–58.6.

Type species: *Azomonas agilis* (Beijerinck 1901) Winogradsky 1938, 400 (*Azotobacter agilis* Beijerinck 1901, 577.)

FURTHER DESCRIPTIVE INFORMATION

Please note that the large body of data collected by Thompson and Skerman (1979) forms the current basis for the description of the genus *Azomonas* and its species. Most of the information below is provided from a summary of the work of those authors. Exceptions and new data are noted.

The cell shape of *Azomonas* species is similar to that for *Azotobacter* species (Fig. BXII.γ.126) but may vary in aging cultures of *A. agilis*. In *A. insignis*, the cell shape is retained but the size

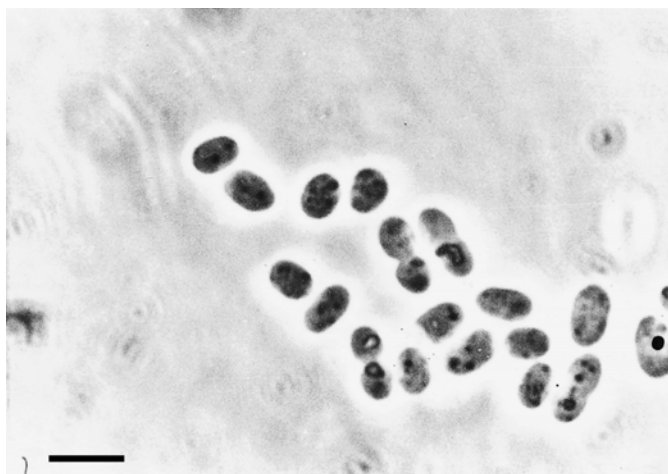


FIGURE BXII.γ.126. Cells of *Azomonas agilis* (phase-contrast microscopy). Bar = 10 μm.

can change. *A. macrocytogenes* produces very large cells and sometimes spindle-shaped or filamentous forms, especially when ethanol is used as the carbon source (Fig. BXII.γ.127). On peptone-yeast extract medium, *A. macrocytogenes* may show distortion of cells, with irregular, filamentous forms predominating. *A. insignis* does not show any distortion, but the cells appear thinner than when grown in glucose medium. *A. agilis* is less affected and does not show distortion.

No cysts are formed by any species; however, in *A. macrocytogenes* cells emerging from saclike capsular structures may occur and be mistakenly identified as germinating cysts (Jensen, 1955a; Tchan, 1968). Page and Collinson (1987) described five *A. macrocytogenes* isolates from Alberta soils, none of which formed cysts after prolonged growth on glucose or with butanol—an encystment inducer for *Azotobacter* species. Moreover, no 5-*n*-alkyl resorcinols—characteristic components of *Azotobacter vinelandii* cysts—could be identified in lipid extracts of the Alberta *A. mac-*

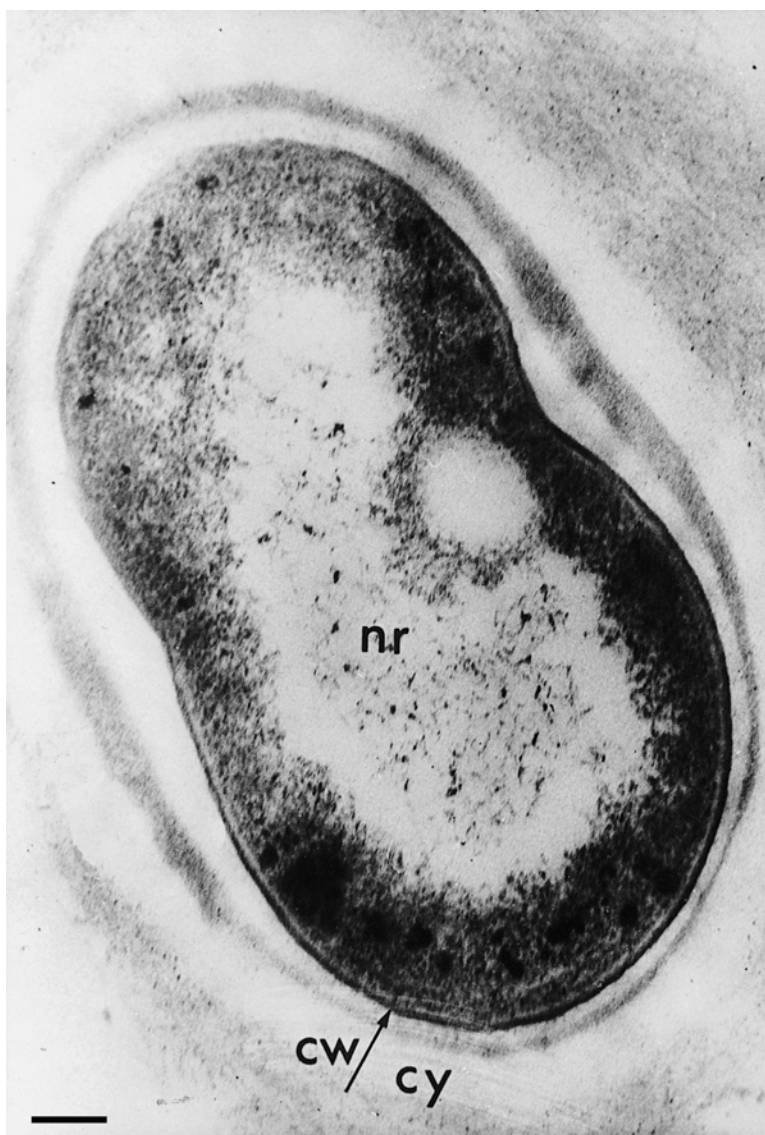


FIGURE BXII.γ.127. Electron micrograph of *Azomonas macrocytogenes*. The nuclear region (*nr*) and cell wall (*cw*) are readily visible. The structure *cy* probably consists of polysaccharide, but can be mistaken for the cyst coat of *Azotobacter*. Bar = 0.2 μm

rocytogenes strains. *A. agilis* and *A. insignis* stain uniformly Gram negative. *A. macrocytogenes* tends to be Gram variable.

All members of the genus are motile, with the flagellar arrangement varying with the species (Table BXII.γ.121).

All species are obligate aerobes, but nitrogen fixation can be enhanced by a decreased oxygen tension (0.04 atm) in some strains. On nitrogen-free agar medium with a sugar as the carbon source, *A. agilis* and *A. insignis* grow at a relatively rapid rate, forming colonies 2–6 mm in diameter after incubation for a few to several days at 30°C. The colonies are opaque, glistening, smooth and convex, and variants rarely appear. Colonies of *A. insignis* are usually smaller than those of *A. agilis*. *A. macrocytogenes* forms colonies ranging from 1–6 mm in diameter after 1 week of incubation; initially, they may be coarsely wrinkled but later they become smooth. On sucrose agar *A. macrocytogenes* produces voluminous and moist colonies due to the formation of colony-retained homopolysaccharides. No species of *Azomonas* produces diffusible homopolysaccharides, i.e., there is no formation of a diffuse halo around the colonies.

Azomonas macrocytogenes and *A. agilis* produce a yellow-green visible, blue-white fluorescent pigment when grown in iron-deficient medium. This siderophore, isolated from *A. macrocytogenes* ATCC 12334 and named azoverdin, is an isopyoverdin-like chromopeptide (Michalke et al., 1996). Azoverdin is produced in Fe-limited cultures grown at 28°C but not at 34°C (Collinson et al., 1990). This strain synthesizes two outer membrane proteins of 74 kDa and 70 kDa in response to Fe starvation. These proteins may act as azoverdin binding and/or transport sites (Collinson and Page, 1989). *A. macrocytogenes* also produces 3,4-dihydroxybenzoic acid, which can solubilize iron from minerals and may represent the red-violet diffusible pigment reported to be produced by all species of *Azomonas* (Tchan and New, 1984; Collinson et al., 1987). *A. insignis* differs from other azomonads by its production of a brown-black pigment in the presence of benzoate. *A. insignis* is also the only species that fails to form a fluorescent pigment.

Nitrogen fixation can occur by alternative nitrogenase enzymes in species of *Azomonas* (see the genus *Azotobacter*). The evidence for such alternative nitrogenases is physiological and genetic. Growth and nitrogen fixation can occur in molybdate-

starved cultures of *A. macrocytogenes*, although at lower rates than in molybdenum-sufficient medium. DNA hybridization studies show that *A. agilis* (ATCC strain 7494) carries genes for all three nitrogenase enzymes encoded by *nif*, *vnf*, and *anf* genes (Fallik et al., 1991). A PCR amplification product specific for the Fe-nitrogenase *anfG* gene but not for the V-nitrogenase *vnfG* gene was detected using *A. macrocytogenes* (ATCC strain 12334) DNA (see *Azotobacter*, Fig. BXII.γ.138). This finding suggests that this species might have one alternative nitrogenase—the Fe enzyme—in addition to the Mo nitrogenase (Loveless and Bishop, 1999). If so, this would be the only member of the *Azotobacter*–*Azomonas* group with this combination of two nitrogenases. Other nitrogen sources for *Azomonas* species are ammonium salts. Only *A. insignis* can generally utilize nitrate. All species are negative for urease activity and can presumably not utilize urea as a nitrogen source—in contrast to *Azotobacter* species. No species can utilize glutamate as an N source.

The optimum temperature for growth is approximately 30°C, although some strains of *A. agilis* prefer 37°C. The minimum temperature for growth is generally close to 14°C; however, some strains, particularly of *A. insignis*, are able to grow at 9–10°C.

Compounds used as sole carbon and energy sources by all species of the genus are shown in Table BXII.γ.122 and are similar to those utilized by all species of *Azotobacter*. Only *A. macrocytogenes* can use D-galactose, melizitose, mannitol, and sorbitol. Maltose is used by *A. macrocytogenes* and most strains of *A. agilis*. Caproate, malonate, and mucate are used by *A. agilis* and *A. insignis* but not by *A. macrocytogenes*. *A. agilis* utilizes benzoate, catechol, 4-hydroxybenzoate, naphthalene, protocatechuate, and 4-toluate; the other species were not tested (Chen et al., 1993). A soil isolate identified as *A. macrocytogenes* degraded and used polyacrylamide as both a nitrogen and a carbon source (Nakamiya and Kinoshita, 1995). None of 17 amino acids tested can serve as a carbon source for *Azomonas* species, possibly due to a lack of ability to transport most amino acids—as in species of *Azotobacter*.

Catalase is present in all species. A positive oxidase reaction is given by the majority of *A. agilis* and *A. insignis* but rarely by *A. macrocytogenes*. Peroxidase is produced by the other species but not by *A. macrocytogenes*.

TABLE BXII.γ.121. Characteristics differentiating the species of the genus *Azomonas*^a

Characteristic	1. <i>A. agilis</i>	2. <i>A. insignis</i>	3. <i>A. macrocytogenes</i>
Presence of enlarged cells in media with ethanol	—	—	+
Flagellar arrangement:			
Peritrichous	+	—	—
Lophotrichous	—	+	—
Monotrichous	—	—	+ ^b
Diffusible pigments:			
Brown-black, on benzoate medium	—	d	—
Blue-white fluorescent, on iron-deficient medium	+	—	d
Utilization as carbon sources:			
Sucrose	+	—	+
Mannitol	—	—	+
Maltose	d	—	+
Malonate	+	+	—
Raffinose	—	—	+
Utilization of nitrate as a nitrogen source	—	+	—

^aSymbols: see standard definitions.

^bRarely, two flagella may occur at one pole.

TABLE BXII.γ.122. Other characteristics of species of the genus *Azomonas*^a

Characteristic	1. <i>A. agilis</i>	2. <i>A. insignis</i>	3. <i>A. macrocytogenes</i>
<i>Nitrogen fixation occurs at pH:</i>			
5.5	—	—	+
6	d	—	+
6.5–10.0	+	+	+
<i>Nitrogen fixation genes:</i> ^b			
<i>nif</i>	+		+
<i>vnf</i>	+		—
<i>anf</i>	+		+
<i>Growth at a temperature of:</i>			
9°C	d	d	d
14–28°C	+	+	+
32°C	+	+	d
37°C	+	—	—
<i>Pigmentation:</i>			
Diffusible red-violet	d	+	+
Diffusible red-purple	—	d	—
Diffusible yellow-green on iron-deficient media ^b	+	d	d
Diffusible blue-white fluorescent on iron-deficient media ^b	+	—	d
<i>Utilization as sole carbon source:</i> ^c			
Fructose, glucose, acetate, pyruvate, fumarate, malate, succinate, α-oxoglutarate, lactate, DL-gluconate, acetyl methyl carbinol	+	+	+
D-Galactose, sorbitol	—	—	+
Trehalose	—	—	d
Glutarate, benzoate	—	—	—
<i>Habitat:</i>			
Soil	—	—	+
Water	+	+	—
<i>Susceptibility to antimicrobial agents:</i>			
Streptomycin, 0.2 µg/ml	S	S	S
Neomycin, kanamycin, 2 µg/ml	d	S	d
Erythromycin, 2 µg/ml	d	R	S
Chloramphenicol, 25 µg/ml	R	R	R
Sulfanilamide, 25 µg/ml	d	d	d
Penicillin G, 5 U/ml	d	S	d
Phenol, 0.95%	R	d	R
Sodium benzoate, 0.5%	R	d	R
Sodium benzoate, 1.0%	d	S	R
Sodium fluoroide, 0.01 M	d	R	d
Mercuric chloride, 10 µg/ml	R	d	d
Iodoacetate, 1mM	d	S	S

^aFor symbols see standard definitions; S, susceptible; R, resistant; blank space, not reported.^bSee Procedures for Testing Special Characters.^cSee also Table BXII.γ.121 for additional compounds.

The genome of *Azomonas macrocytogenes* (ATCC 12334) is much smaller than *Azotobacter* species genomes (2.4 versus 3.9–5.3 Mb) (see *Azotobacter* chapter, Table BXII.γ.125). There are no reports of indigenous plasmids within or transfer of plasmids from other bacteria into *Azomonas* species.

No bacteriophages from *Azomonas* species have been reported. Hegazi and Jensen (1973) reported that bacteriophages isolated with *A. chroococcum* and *A. vinelandii* as hosts were unable to infect or form plaques on *A. agilis*, *A. insignis*, and *A. macrocytogenes*.

Tchan and New (1984) indicated that antisera produced against *A. agilis* and *A. insignis* do not cross-agglutinate with members of *Azotobacter*. Norris and Kingham (1968), however, reported that antisera against *A. agilis* do cross-react with *A. vinelandii*. This discrepancy may be due to strain differences or to variability in immune response between animals. Within the genus *Azomonas*, antiserum against *A. insignis* cross-agglutinates with *A. agilis* (Tchan and New, 1984). Immunodiffusion produces numerous cross-precipitation lines. Rocket-line immunoelectrophoresis indicated that *Azomonas* and *Azotobacter* can be differentiated by the unshared precipitation peaks using an appropriate antiserum (Tchan et al., 1983), although at least one thermoresistant antigen is shared by *Azomonas* and *Azotobacter* (Tchan et al., 1980).

All species of *Azomonas* are sensitive to streptomycin (0.2 µg/ml), as are *Azotobacter* species. Sensitivities to chlortetracycline, oxytetracycline, polymyxin, sulfanilamide, penicillin, and neomycin vary depending on the strain. Only *A. insignis* is entirely sensitive to penicillin (Thompson and Skerman, 1979). As for *Azotobacter* species, those of *Azomonas* are somewhat resistant to chloramphenicol.

Azomonas species are resistant to phenol, benzoate, and mercuric chloride to varying degrees. Most strains tolerate sodium fluoride up to 0.01 M. Cadmium tolerance and the ability to remove this metal from the culture medium was reported for *A. agilis* PYO (You and Park, 1998). The resistance of *A. agilis* to iodoacetate can be used for selective isolation of this species (Thompson and Skerman, 1979).

A. agilis and *A. insignis* are aquatic bacteria. *A. macrocytogenes* is found in soil, although no information is available as to whether it also has an aquatic habitat (for further details, see Becking, 1992). Thus, it appears that the genus is ecologically heterogeneous. Only *A. macrocytogenes* is resistant to desiccation: five of seven strains survived desiccation for at least 1 month (Thompson and Skerman, 1979). All species tolerate a salt concentration of up to 1%, suggesting that the aquatic members are

capable of living in contaminated water where concentrations of organic matter and mineral salts can be relatively high.

ENRICHMENT AND ISOLATION PROCEDURES

The enrichment and isolation techniques depend on the ability of *Azomonas* to grow by fixing nitrogen under aerobic conditions, using organic substrate as energy source. Nonselective enrichment methods are similar to those described in the chapter on *Azotobacter*.

Selective enrichment methods (Thompson and Skerman, 1979) Media containing either 1% sodium benzoate (Derox, 1951; Jensen, 1955a) or 1 mM iodoacetate are selective for *A. agilis*. The growth of *A. macrocytogenes* is favored by incorporating 1% benzoate into nitrogen-free sucrose medium (pH of 6.0) and incubating the cultures at a decreased incubation temperature (10–12°C). No selective enrichment method is available for *A. insignis*.

Isolation methods These are similar to those described previously for *Azotobacter* species. The enrichment cultures are inspected daily. When macroscopic growth becomes apparent, the culture is examined by phase-contrast microscopy to detect the presence of *Azomonas* cells (ovoid cells $2.5 \times 5.0 \mu\text{m}$, or sometimes filamentous forms). Growth will usually occur within 2–5 days. Positive cultures are streaked onto nitrogen-free agar medium with glucose as the carbon source. For selective isolation, nitrogen-free agar media with appropriate selective substances should also be used. Sometimes nitrogen-free iron-deficient agar can be used to encourage pigment production by some species.

MAINTENANCE PROCEDURES

For routine maintenance, *Azomonas* strains should be subcultured at monthly or bimonthly intervals on Winogradsky's agar medium with glucose. Other maintenance methods are similar to those described in the chapter on *Azotobacter*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *AZOMONAS*

The differential characteristics of the three species of *Azomonas* are presented in Table BXII.γ.121.

List of species of the genus Azomonas

1. ***Azomonas agilis*** (Beijerinck 1901) Winogradsky 1938, 400^{AL} (*Azotobacter agilis* Beijerinck 1901, 577.)
a'gi.lis. L. adj. *agilis* quick, agile.

The characteristics are as described for the genus and as listed in Tables BXII.γ.121 and BXII.γ.122. The morphology is as shown in Fig. BXII.γ.126. The cells are typically $2-4 \times 1.6-3.0 \mu\text{m}$. Cells are motile by means of numerous peritrichous flagella—a distinguishing characteristic of this species. The flagella have a wavelength of 2.0–3.0 μm and an amplitude of 0.40–0.59 μm .

Colonies are opaque, low convex, butyrous, and usually mucoid, glistening, and smooth.

Physiological and nutritional characteristics are presented in Tables BXII.γ.121 and BXII.γ.122. Starch is not hydrolyzed. Ammonium salts can be used as a sole source of nitrogen for growth. Neither nitrate nor glutamate can be used as a nitrogen source.

The type strain was isolated by Kluyver and van den Bout (1936).

The mol% G + C of the DNA is: 52–53.2 (T_m).

Type strain: ATCC 7494, DSM 375, WR 83.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Methods for determining presence of genes for alternative nitrogenases, for testing ability of strains to grow diazotrophically under Mo-deprivation conditions, cell morphology, and pigment production are the same as those described in the chapter on the genus *Azotobacter*.

DIFFERENTIATION OF THE GENUS *AZOMONAS* FROM OTHER GENERA

The single most distinguishing feature between *Azomonas* and *Azotobacter* is the failure of the former to form cysts in aging cultures or after induction with butanol. For other features that distinguish *Azomonas* from other diazotrophs and from pseudomonads, to which they are most related, see the chapter on *Azotobacter*.

TAXONOMIC COMMENTS

Analysis of 16S rRNA reveals the close but separate relationship of *Azomonas macrocytogenes* ATCC 12334 to the four *Azotobacter* species characterized (MacDonald and Melton, unpublished results) (see *Azotobacter* chapter, Figs. BXII.γ.139 and BXII.γ.140). Sequences are not available for strains of *A. agilis* and *A. insignis*. Nevertheless, the justification of a single genus with three species remains sound, based on DNA base composition, hybridization analysis, and rRNA cistron similarities (De Smedt et al., 1980). General physiological characteristics, polyamine patterning, distribution of alginate biosynthetic genes, and aromatic biosynthetic pathway enzyme distribution support the common lineage of *Azotobacter* and *Azomonas* species with group I pseudomonads (Byng et al., 1986; Fialho et al., 1990; Goris et al., 1998). While the genus is more heterogeneous than *Azotobacter*, further molecular characterization of *Azomonas* species and strains would be required to justify separation of the group into more than one genus.

2. ***Azomonas insignis*** (Derox 1951) Jensen 1955b, 156^{AL} (*Azotobacter insignis* Derox 1951, 344.)
in.sig'nis. L. adj. *insignis* distinguished by a mark.

The characteristics are as described for the genus and as listed in Tables BXII.γ.121 and BXII.γ.122. Motile by lophotrichous flagella, whose full wavelength may exceed 4 μm . The numerous flagella tend to bind together to form thick, whip-like structures.

Colonies are translucent, low convex, butyrous, and smooth. They are usually small, probably due to low production of extracellular polysaccharides.

Physiological and nutritional characteristics are presented in Tables BXII.γ.121 and BXII.γ.122. Starch is not hydrolyzed. Ammonium or nitrate, but not glutamate, can be utilized as a sole nitrogen source for growth. The ability to use nitrate as a nitrogen source and the inability to use sucrose as a carbon source are characteristics distinctive for this species.

The mol% G + C of the DNA is: 55.1–58.3 (T_m).

Type strain: UQM 1966.

3. *Azomonas macrocytogenes* (ex Baillie, Hodgkiss and Norris 1962, 118) New and Tchan 1982, 381^{VP} (*Azotobacter macrocytogenes* Jensen 1955a, 280; *Azotobacter macrocytogenes* (Jensen 1955a) Thompson and Skerman 1981, 215; *Azotobacter agilis* subsp. *jakutiae* Krasil'nikov 1949, 508.)
mac.ro.cy.to'ge.nes. Gr. adj. *macrus* large; Gr. n. *kytos* cell; Gr. v. *gennaio* produce; M.L. part. adj. *macrocytogenes* large cell producing.

The characteristics are as described for the genus and as listed in Tables BXII.γ.121 and BXII.γ.122. The morphology is shown in Fig. BXII.γ.127. Cells are typical of species of *Azomonas* except that motility occurs by polar rather than peritrichous flagella. Usually one or, less frequently, two flagella per cell are observed. The flagellar wavelength is 1.3–1.7 μm and the amplitude is 0.25–0.52 μm. The cells are often surrounded by large capsules.

Colonies on sucrose media are mucoid and large (15 mm in diameter in 7 d) due to the production of colony-

retained homopolysaccharides. Starch is not hydrolyzed. Ammonium, but not nitrate or glutamate, can serve as a sole nitrogen source for growth.

The gene *anfG*—but not *vnfG*—is present in strain ATCC 123334, suggesting that this species contains the alternative nitrogenase-3 in addition to nitrogenase-1. The possession of *anfG* is possibly a unique characteristic of this species. The genome size of *A. macrocytogenes* ATCC strain 12334 is 2.4 Mb—smaller than the genomes of other *Azomonas* and *Azotobacter* species (see Table BXII.γ.123 in the chapter on *Azotobacter*). The significance of this small size with respect to the diversity of this organism is not known.

The mol% G + C of the DNA is: 58.2–58.6 (T_m).

Type strain: Jensen O, ATCC 12335, DSM 721, NCIB 8700, WR-111.

Additional Remarks: Other strains include strain M (ATCC 12336), a mutant isolated by Jensen, and strain WR-140, cotype of "*A. agilis* subsp. *jakutiae*" Krasil'nikov 1949.

Genus III. *Azotobacter* Beijerinck 1901, 567^{AL}

CHRISTINA KENNEDY, PAUL RUDNICK, MELANIE L. MACDONALD AND THOYD MELTON

A.zo.to.bac'ter. Fr. n. *azote* nitrogen; M.L. masc. n. *bacter* the equivalent of Gr. neut. n. *bactrum* a rod or staff; M.L. masc. n. *Azotobacter* a nitrogen rod.

Cells range from straight rods with rounded ends to more ellipsoidal or coccoid, depending on the culture medium and age. **Cells are up to 2 μm or more in diameter and 4 μm in length.** *A. paspali* cells are usually longer, 5–10 μm in length, and can be filamentous, up to 60 μm long. Cells are usually single but may occur in pairs, irregular clumps (especially with *A. paspali*), or, more rarely, in chains of varying length. **Encystment** occurs during late stationary phase at low frequency or at high frequency after culturing on butanol. **Motile with peritrichous flagella or nonmotile.** **Aerobic, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor.** **Nitrogen is fixed** under microaerobic conditions (2% oxygen), under full aerobiosis, or after adaptation in hyperbaric oxygen. N_2 fixation uses Mo-, V-, or Fe-containing nitrogenase enzymes, depending on the environmental metal supply. **Water-soluble and water-insoluble pigments are produced by some strains of all species.** Growth is heterotrophic; sugars, alcohols, and salts of organic acids are used as carbon sources. Ammonium salts, nitrate, and urea are used as sources of fixed nitrogen. Very few amino acids are used, probably due to a general deficiency in amino acid transport. The minimum pH for growth in the presence of fixed nitrogen sources ranges from 4.8 to 6.0 with maximum pH 8.5. The optimum pH for diazotrophic growth is 7.0–7.5. **Most isolates are from soil, but a few are from water.** One species (*A. paspali*) has been isolated only from roots of the tropical grass *Paspalum notatum*.

The mol% G + C of the DNA is: 63.2–67.5.

Type species: *Azotobacter chroococcum* Beijerinck 1901, 567.

FURTHER DESCRIPTIVE INFORMATION

Preliminary note The immense amount of data accumulated by Thompson and Skerman (1979) remains the foundation for the basic morphological and physiological characterization of both *Azotobacter* and *Azomonas*. The data resulted from an analysis of some 90 strains representing the several species of *Azotobacter* and *Azomonas*—including established strains found in interna-

tional culture collections and their own isolates—for 230 different morphological and physiological attributes. The data were summarized by Tchan and New (1984) in the previous edition of the *Manual*. More current knowledge of *Azotobacter* species is very uneven, particularly with respect to genetic, physiological, and biochemical studies, with *A. vinelandii* being the most studied followed by *A. chroococcum*. Relatively little information has been reported for the other five species, especially for *A. nigricans* and *A. armeniacus*, in each of which only four isolates were previously studied (Thompson and Skerman, 1979). The present authors have drawn on those contributions and on updates from the literature since 1984 to provide a current overview of the characteristics of the genus *Azotobacter*.

Cell morphology In nitrogen-free Winogradsky medium¹ with glucose as the carbon source, the young cells of different species are remarkably similar in appearance (Figs. BXII.γ.128, BXII.γ.129, and BXII.γ.130): rods with rounded ends, 1.6–2.7 × 3.0–7.0 μm. In older cultures, the cells of some species may tend to be ellipsoidal to coccoid. In other species, chains and filamentous forms become more common, and dark-colored granules are observed. *A. paspali* produces long filamentous forms even in young cultures (Fig. BXII.γ.131), and this characteristic—together with the inability of this species to use several carbon sources—differentiates it from the others. In peptone–yeast ex-

1. Winogradsky's nitrogen-free mineral medium is prepared from a 200× stock solution which has the following composition (g/l): KH_2PO_4 , 50.0; $MgSO_4 \cdot 7H_2O$, 25.0; NaCl, 25.0; $FeSO_4 \cdot 7H_2O$, 1.0; $Na_2MoO_4 \cdot 2H_2O$, 1.0; and $MnSO_4 \cdot 4H_2O$, 1.0; the pH is adjusted to 7.2 with NaOH. The medium is prepared by using 5.0 ml of the stock solution and 0.1 g/l of $CaCO_3$ and then sterilized at 120°C for 20 min. To make nitrogen-free organic medium, a suitable quantity of organic substrate is added to the mineral medium. (Certain sugars, including glucose, must be sterilized separately before addition to the sterilized mineral medium.) For a solid medium, 15.0 g/l of agar is added.

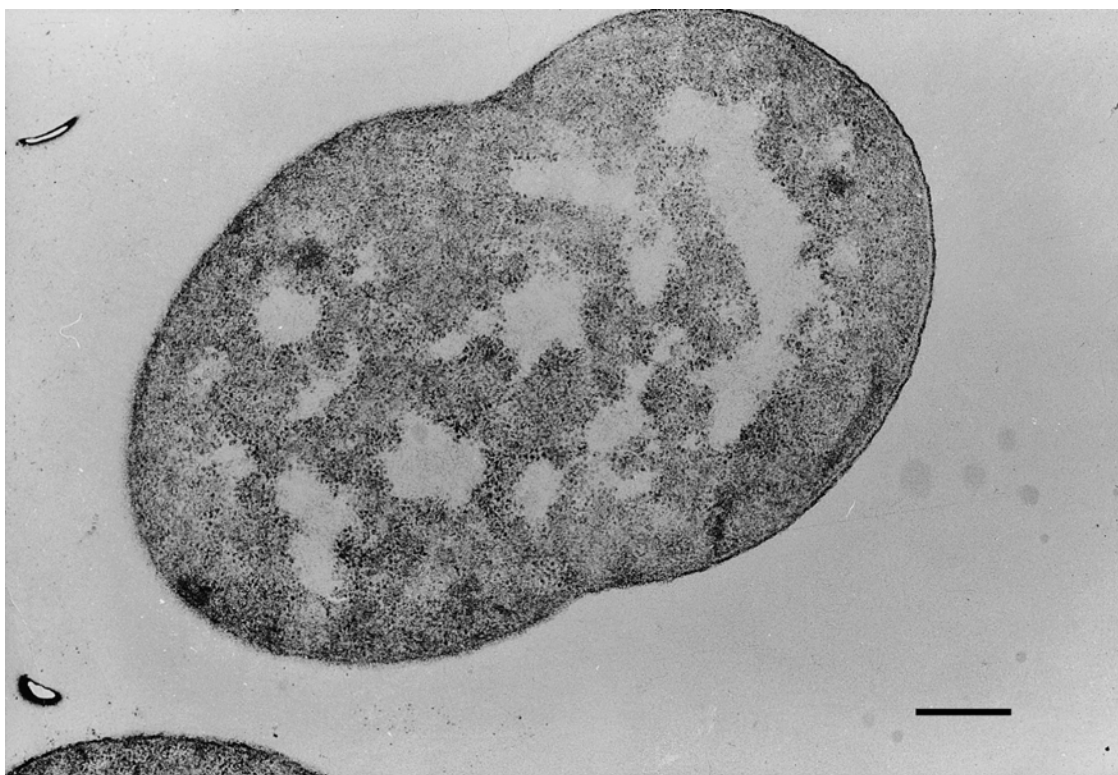


FIGURE BXII.γ.128. Electron micrograph of a vegetative cell of *Azotobacter chroococcum*. Bar = 0.2 μm.

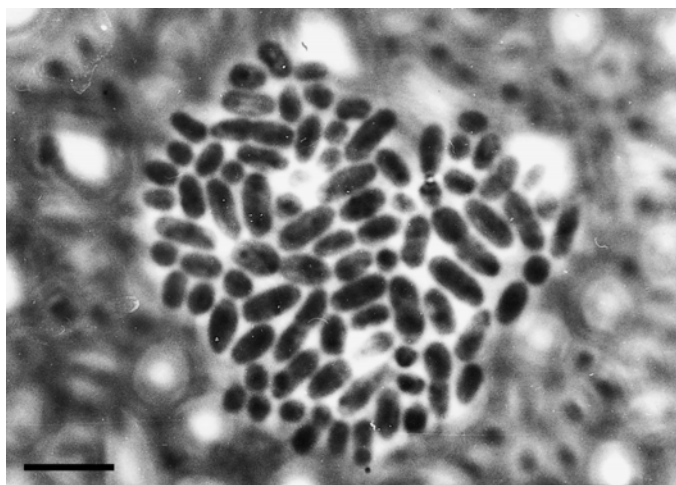


FIGURE BXII.γ.129. Vegetative cells of *Azotobacter chroococcum* (phase-contrast microscopy). Bar = 10 μm.

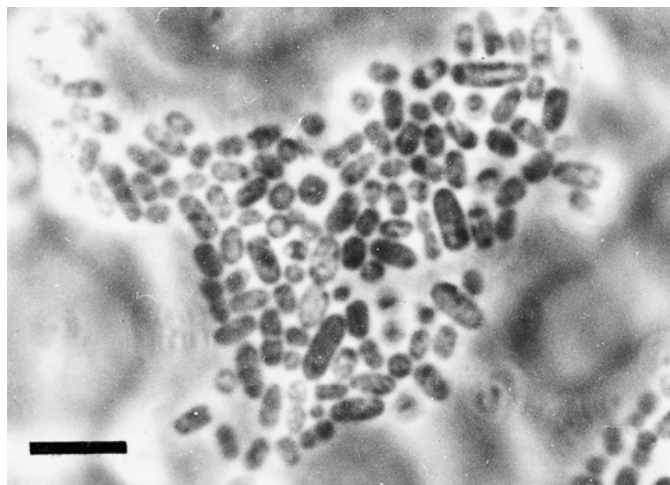


FIGURE BXII.γ.130. Vegetative cells of *Azotobacter nigricans* (phase-contrast microscopy). Bar = 10 μm.

tract agar², all members of the genus produce distorted cells (Tchan and New, 1984). The widely-used, nongummy strain AvOP of *A. vinelandii* (see below) cannot grow at all on Luria-Bertani medium (C. Kennedy, unpublished results), a medium with peptone and yeast extract (Sambrook et al., 1989). Cells grown under nitrogen-fixing conditions are smaller than ammonia-grown cells. *A. paspali* cells aggregate in late-log or sta-

tionary phase, especially with an increasing C:N ratio (Abbass and Okon, 1993a). The large size of *A. vinelandii* cells correlates with their high DNA content (see later), and their volume was estimated to be 16 times greater than that of *Escherichia coli* cells (Efuet et al., 1996) (Fig. BXII.γ.132).

Cell wall composition *Azotobacter* species form a proteinaceous S-layer as the outermost cell surface component. The layer is structurally similar to that of *Aeromonas salmonicida* and two other *Pseudomonas* species (Bingle et al., 1987). Unlike the S-layer in these organisms, however, the S-layer of *A. vinelandii* can be extracted from whole cells or outer membrane fractions with

2. Peptone-yeast extract agar (g/l): Bacto peptone (Difco), 1.0; yeast extract (Difco), 0.5; NaCl, 0.5; Bacto agar (Difco), 15. The pH is adjusted to 7.2 with NaOH prior to autoclaving, giving a final pH of 7.0.

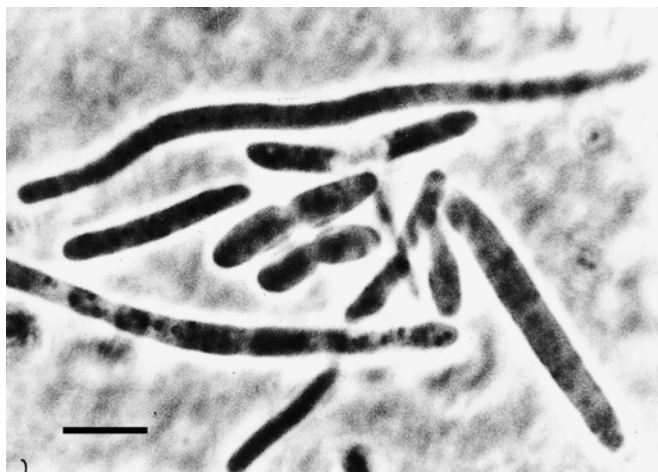


FIGURE BXII.γ.131. Vegetative cells of *A. paspali* (phase-contrast microscopy). Bar = 10 μm.

water. The S-layer of *A. vinelandii* is composed of a ~60 kDa protein, which assembles into tetramers every 12–13 nm. N_2 -fixing cells form an elaborate inner membrane network (Openheim and Marcus, 1970). Such networks are formed concomitantly with the synthesis of nitrogenase and an increase in the rate of respiration, suggesting an involvement in protecting nitrogenase from oxygen damage. Ammonia-grown cells of *A. vinelandii* develop large deposits of poly-β-hydroxybutyrate (PHB), and the cytoplasm appears less dense. PHB deposits are not seen with cells grown with nitrate or N_2 . PHB biosynthesis seems not to be influenced by the fixed nitrogen or carbon supply in *A. chroococcum* (Martinez-Toledo et al., 1995). In *A. beijerinckii*, PHB synthesis is variable and dependent on media and specific culture conditions (Bormann et al., 1998).

The fatty acid composition of vegetative cells differs from that of cysts. In vegetative cells analyzed during mid-exponential

phase, 75% of the total fatty acids are C_{16} derivatives, 4% are C_{14} , and 21% are C_{18} (Su et al., 1979). Seventy percent of the total lipids are replaced with 5-*n*-alkylresorcinols and 6-*n*-alkylpyrones during encystment in *A. vinelandii* (Su and Sadoff, 1981; Reusch and Sadoff, 1983a) and in *A. chroococcum* (Kozubek et al., 1996). These molecules—thought previously to be strictly components of plant membranes—are unique to the cysts and probably contribute to desiccation resistance in both *A. vinelandii* (Reusch and Sadoff, 1983b) and in several strains of *A. chroococcum* (Kozubek et al., 1996).

Motility *A. beijerinckii* and *A. nigricans* are nonmotile. Other species are motile by peritrichous flagella, which are often present in large numbers (Fig. BXII.γ.133). *A. vinelandii* OP is a fast-moving bacterium, with a measured average velocity of 74 μm/sec (Haneline et al., 1991). Chemotaxis occurs and requires oxygen; the speed of the cells increases as a function of attractant concentration. Attraction to fructose, glucose, xylitol, and mannitol occurs, especially after pregrowth in these compounds, indicating that chemotaxis is often inducible. Chemotaxis to other compounds, including hexoses, hexitols, pentitols, pentoses, disaccharides, and amino sugars, occurs.

Cultural characteristics When grown at 30°C with sugar—usually sucrose—as the carbon source at 1–2% (w/v) on either Winogradsky's nitrogen-free agar medium or on Burk's medium³, colonies appear within 48 h at 30°C and reach a diameter of 2–6 mm in a week, depending on the numbers (crowding) per plate. Burk's medium is now more commonly used than Winogradsky's medium for growth of laboratory strains of *A.*

3. Burk's medium has the following composition (g/l): K_2HPO_4 , 0.64; KH_2PO_4 , 0.20; $MgSO_4 \cdot 7H_2O$, 0.2; NaCl, 0.2; $CaSO_4 \cdot 2H_2O$, 0.05. Sucrose or other carbon sources should be added separately after autoclaving from sterile 20% solutions to give a final concentration of 20 g/l. Na_2MoO_4 and $FeSO_4$ should be added from sterile stock solutions to give a final concentration of 0.001 g/l and 0.003 g/l Fe. Agar is added to a final concentration of 1.2–1.5% (w/v).

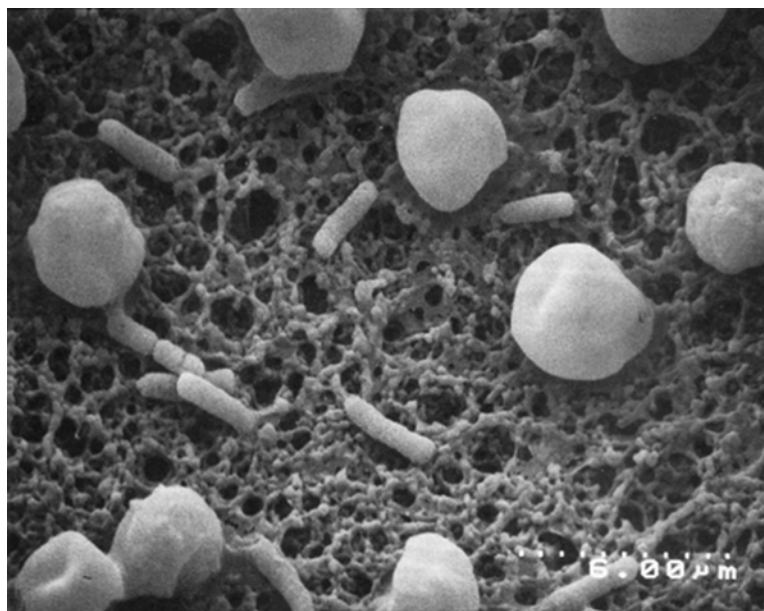


FIGURE BXII.γ.132. Scanning electron micrograph of *E. coli* and *A. vinelandii* vegetative cells, × 1000 magnification. (Reproduced with permission from E.T. Efu et al., *Journal of Basic Microbiology* 36: 229–234, 1996, ©Wiley-VCH.)

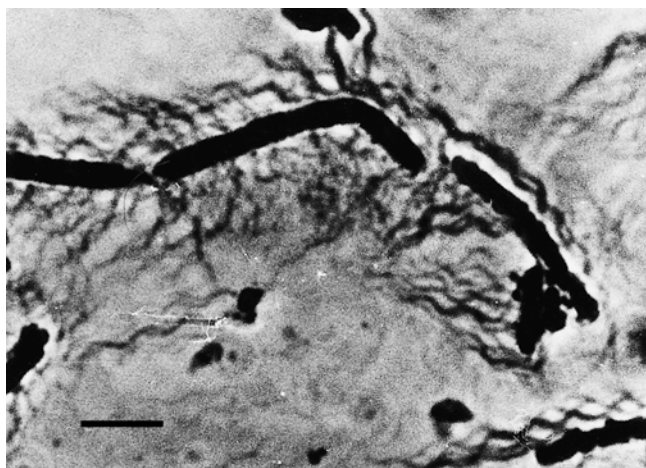


FIGURE BXII.γ.133. Flagella stain of *A. paspali*. Bar = 10 μ m.

vinelandii and *A. chroococcum*. Brown's medium⁴ with glucose results in higher plate counts of *A. chroococcum* colonies (Thompson, 1989). Colonies of most species are generally smooth, glistening, opaque, low convex, and mucoid; however, variations may occur. For *A. vinelandii*, smaller variant colonies may appear due to decreased production of extracellular polysaccharide. *A. armeniacus* sometimes produces translucent colonies and *A. paspali* forms undulate edged and unevenly convex colonies with a dull or rough surface. On sucrose or raffinose agar, the production of diffusible homopolysaccharides—which results in a diffuse halo around the colony—is species dependent. *A. chroococcum* and some strains of *A. beijerinckii* and *A. nigricans* produce diffusible homopolysaccharides from both sugars, whereas *A. armeniacus* and some other strains of *A. beijerinckii* and *A. nigricans* produce them only from sucrose (Thompson and Skerman, 1979).

Cysts Cysts are formed in fewer than 1% of late stationary cells of *Azotobacter* species grown on glucose or mannitol. Encystment can reach >90% 5 d after the addition of butan-1-ol or β -hydroxybutyrate to growing cultures (Lin and Sadoff, 1968) (Fig. BXII.γ.134). Isopropanol appears to be a better inducer for strains of *A. chroococcum* (Marengo, 1983). During encystment, there is a cessation of most metabolic activities. Cysts are significantly more resistant than vegetative cells to desiccation and damage due to physical and chemical agents (Socolofsky and Wyss, 1962). In ultrastructure, they are composed of a central body containing β -hydroxybutyrate surrounded by capsule-like exine and intine layers, of which alginate is a major component (Fig. BXII.γ.135). Deposition of capsular materials and membrane components in the absence of cell division produces the layers of the exine coat (Hitchins and Sadoff, 1970). Intine vesicles migrate towards the exine where they form a layer internal to the exine. These vesicles have also been visualized during vegetative growth (Vela et al., 1970). This observation provides ev-

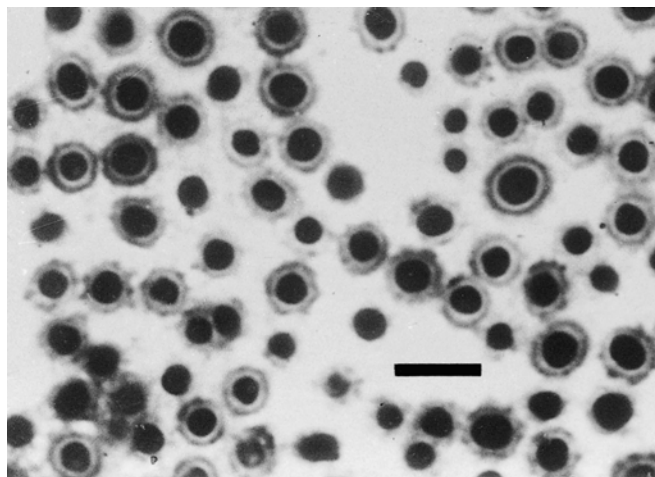


FIGURE BXII.γ.134. Violamine-stained cysts. The central body is surrounded by the cyst coat. Bar = 5 μ m.

idence that cysts are simply resting vegetative cells and are not analogous to bacterial spores, which are more differentiated. The composition of the exine coat in *A. vinelandii* is 32% carbohydrate, 28% protein, and 30% lipid (Lin and Sadoff, 1969).

Alginate synthesis The alginate biosynthetic pathway in *A. vinelandii* has been characterized by Pindar and Bucke (1975). Fructose-6-phosphate is converted to GDP-mannuronic acid, which is polymerized and secreted. The polymer is modified by an O-acetylase and an extracellular C-5 epimerase to form alginate (Rehm et al., 1996). Alginate biosynthesis is apparently essential for encystment in *A. vinelandii* because biosynthetic mutants do not form cysts (Campos et al., 1996). Alginate production is remarkably similar to that in the pathogen *Pseudomonas aeruginosa* (reviewed by May and Chakrabarty, 1994), and genes from each organism can rescue mutations in the other (Moreno et al., 1998). There is industrial interest in the exopolysaccharides synthesized by *A. vinelandii* and *A. chroococcum*, and these polysaccharides can be produced in high concentrations depending on cultural conditions (De la Vega et al., 1991; Pena et al., 2000). The best medium for production of alginate by *A. vinelandii* is that described by Sabry et al. (1996).⁵ In *A. vinelandii*, alginate is made to a concentration of 4.5 g/l when cells are incubated at 300 rpm on a shaker under 5% O₂. Alginate production increases with agitation speeds upwards to 700 rpm; however, increasing the speed leads to a decrease in the polymer size from 680,000 Da at 300 rpm to 352,000 Da at 700 rpm. Reduction in size of the polymer is correlated with alginase activity, detected in the more vigorously aerated cultures. The alginate produced by *A. vinelandii* differs from other bacterial sources in that guluronic acid residues can be introduced after polymerization by conversion of a mannuronic residue, and there are apparently eight enzymes which code for this particular C-5 epimerase activity (Ertesvag et al., 1994, 1995; Rehm et al., 1996; Svanem et al., 1999). The resultant alginate formed by the activity of these enzymes gives it industrially favorable gel-forming properties.

4. Brown's medium has the following composition (g/l): glucose or mannitol, 5.0; K₂HPO₄, 0.8; MgSO₄·7H₂O, 0.2; FeSO₄·7H₂O, 0.04; Na₂MoO₄, 0.005; CaCl₂ (anhydrous), 0.15; agar, 15. The phosphate is prepared as a separate 8% (w/v) solution and added to the agar after sterilization by autoclaving (standard conditions) and cooling to 42°C just before pouring plates.

5. Medium for alginate production (per liter): MgSO₄·7H₂O, 0.3 g; NaCl, 0.4 g; CaCl₂·2H₂O, 42 mg; KH₂PO₄, 4 mg; K₂HPO₄, 16 mg; FeSO₄·7H₂O, 2.5 mg; H₃BO₃, 2.9 mg; ZnSO₄·7H₂O, 2 mg; Na₂MoO₄·2H₂O, 2 mg; CuSO₄·5H₂O, 0.3 mg; and MnCl₂·4H₂O, 0.2 mg.

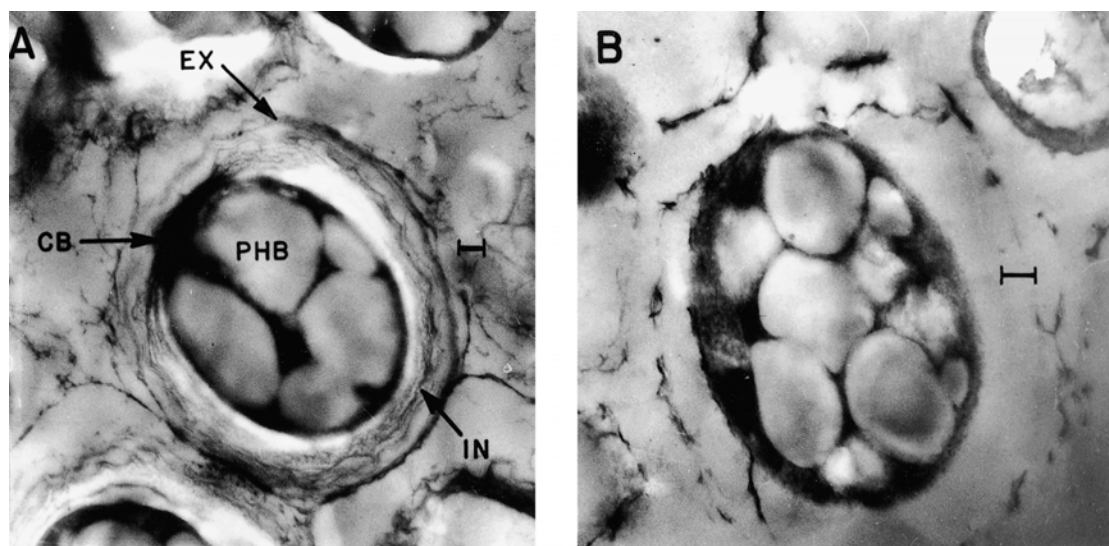


FIGURE BXII-γ.135. Electron micrographs of the cysts formed by strains ATCC 9046 (A) and ATR8 (*algR*⁻) (B). Abbreviations: EX, exine; IN, intine; CB, central body; PHB, poly-hydroxybutyrate. Bars = 0.4 μm. (Reprinted with permission from C. Nuñez et al., *Journal of Bacteriology*, 181: 141–148, 1999, ©American Society for Microbiology.)

The particular exopolysaccharides produced by *A. chroococcum* vary among strains. Cote and Krull (1988) reported the existence of two distinct exopolysaccharides from strain NRRL B-14341 (ATCC 7491), one of which has been identified as an alginate. *A. chroococcum* ATCC 4412 apparently produces only one type of exopolysaccharide, which is alginate-like. Production of this molecule varies with carbon source and air-flow rate, and the best yields occur at 8% sucrose (De la Vega et al., 1991). Since alginate production varies among strains, it may not be possible to differentiate between species based on the production of alginate. *A. beijerinckii* B-1615 produces two different exopolysaccharides in a 9:1 ratio (Likhoshesterov et al., 1991). The major polysaccharide consists of D-galactose, L-rhamnose, and pyruvic acid in a 2:1:1 ratio. The minor polysaccharide contains mannuronic and guluronic acid in a 2:3:1 ratio and is an alginate.

Genes encoding products involved in alginate synthesis are found in all members of *Azotobacter* and *Azomonas* and throughout most of *Pseudomonas* group I. (Fialho et al., 1990). The alginate biosynthetic genes of *A. vinelandii* are clustered as operons (Martinez-Salazar et al., 1996; Nuñez et al., 1999), and a similar arrangement is predicted for *A. chroococcum* (Peciña et al., 1999). Although not all of the *alg* genes have been identified in any member of *Azotobacter* or *Azomonas*, their arrangement is probably similar to that in *P. aeruginosa*, in which three large clusters at distinct locations on the chromosome comprise all functions of alginate biosynthesis and regulation (for review, see Gacesa, 1998). It should be noted that not all *A. vinelandii* strains produce alginate. The commonly used nonmucoid laboratory strain, UW136 (a rifampicin resistant derivative of AvOP/UW), may have arisen from insertion of small IS element in *algU* (Martinez-Salazar et al., 1996).

Siderophores As noted earlier, some *Azotobacter* species produce the yellow-green fluorescent peptide siderophore, azobactin, in response to iron limitation. *A. vinelandii* also produces the three nonfluorescent catecholate siderophores, azotochelin, protochelin, and aminochelin, under similar growth conditions (Cornish and Page, 1998). Iron uptake by each of the sidero-

phores is biphasic and energy-dependent. Fe⁺³ uptake mediated by each of the siderophores is sensitive to metal salts: the addition of 40 mM Na⁺, K⁺, Li⁺, or Mg⁺² inhibits 60% of total uptake. NH₄⁺ also inhibits uptake but to a lesser extent (Knosp et al., 1984). The production of both types of siderophores is repressed by iron compounds in the medium. However, repression of azotobactin synthesis is sensitive to lower concentrations of Fe than is production of the catecholate siderophores. The promoter of the gene encoding the first enzyme in catecholate siderophore synthesis has been characterized (Tindale et al., 2000), and within this promoter is a Fur-box as well as sequences that match the Sox-boxes of *E. coli*. The appearance of these regulatory features, as well as an increase in expression in response to superoxide, suggests that azotochelin may provide protection against oxidative stress under nitrogen-fixing conditions. *A. chroococcum* produces Fe-binding hydroxamates in response to Fe limitation as well as an 85 kDa outer member protein (Fekete et al., 1989). *A. paspali* produces 3,4-dihydroxybenzoic acid (protocatechuate) in both Fe-limited and Fe-sufficient media; this compound solubilizes Fe from minerals (Collinson et al., 1987).

A. salinestrus produces a catechol melanin pigment described as an "iron trap" and may function to protect cells from oxidative damage mediated by H₂O₂ and the Fenton reaction (Page and Shivprasad, 1995). The siderophore-like acquisition of iron mediated by this pigment is proposed to occur in a two-step process. In the first step, iron is concentrated in the periplasm by an energy-dependent process and is promoted by a hydroxamate compound produced in conjunction with the pigment. In the second step, the iron is brought into the cytoplasm; this process is sensitive to inhibitors of Na⁺-dependent activities of the organism.

Nutrition The ability of species of *Azotobacter* to utilize various carbon and nitrogen sources is important for experimental and applied work, particularly considering how carbon sources can influence the efficiency of nitrogen fixation and how various nitrogen sources can be used in the analysis of Nif⁻ and other mutants.

Utilization of carbon sources by *Azotobacter* species is shown in Tables BXII.γ.123 and BXII.γ.124. Some carbon sources are utilized by all species, whereas others differentiate the species. Several other less commonly used compounds can be utilized by some species (Thompson and Skerman, 1979). Few reports that further differentiate the species with respect to carbon source utilization have appeared since the work of Thompson and Skerman (1979) and the last edition of the *Manual*. De La Vega et al. (1991) reported that sucrose is degraded by an extracellular invertase in *A. chroococcum* ATCC 4412, whereas it is degraded by an intracellular invertase in *A. vinelandii* UW (ATCC 13705). Chen et al. (1993) found that *A. chroococcum* (ATCC 9043) and *A. vinelandii* UW utilize not only the aromatic compounds benzoate and 4-OH-benzoate but also catechol, naphthalene, protocatechuate, and 4-toluate, which confirmed and extended the findings of Thompson and Skerman (1979). *A. chroococcum* expressed only *ortho* cleavage dioxygenases during growth on naphthalene and 4-toluate, and only *meta* cleavages of other aromatics. *A. vinelandii* expressed both *ortho* and *meta* cleavage enzymes (Chen et al., 1993). Interestingly, growth rates and nitrogenase activities were nearly as high or higher on certain aromatic compounds—benzoate and naphthalene, in particular—for both *A. chroococcum* and *A. vinelandii*. Among isolates enriched from British soils near Leicester for growth on monochloroacetate, one strain—RC26—had many cultural and physiological characteristics that placed it in the genus *Azotobacter* (Diez et al., 1995). This strain was able to dehalogenate a number of different compounds used as pesticides at rates similar to those determined for species of *Pseudomonas*. While this strain has not been characterized with respect to 16S rDNA sequence, it may represent a soil-inhabiting species of *Azotobacter* with capability for bioremediation of agrochemicals.

Nitrogen fixation Nitrogen fixation under aerobic conditions is a principle characteristic of the genus. The major and

minor clusters of genes encoding the enzyme synthesized under Mo-sufficient conditions in *A. vinelandii*, sequenced and extensively characterized, are shown in Fig. BXII.γ.136. *A. chroococcum* MCD-1 (derived from ATCC 4412) has a similar major cluster, but has been less well characterized in terms of mutational analysis. Some specific functions of the *nif* gene products are mentioned below; a more extensive review of them can be found in Rangaraj et al. (2000). The *nifHDK* genes encode the basic structural proteins of the nitrogenase enzyme. The other gene products encoded by this *nif* cluster are required for (1) synthesis of the Fe-Mo-cofactor, an essential component of the enzyme for substrate (N_2) binding and reduction (*nifV*, *nifEN*, *nifH*); (2) formation of Fe-S clusters required at other sites in the enzyme which are involved in electron transfer to the cofactor (*nifU*, *nifS*); and (3) participation in other aspects of enzyme maturation (*nifM*, *nifW*). In addition to this is another cluster, located elsewhere in the *A. vinelandii* genome, which includes *nifL* and *nifA* in an operon encoding the regulatory proteins NifL and NifA, and other proteins involved in nitrogenase cofactor synthesis and Mo processing (products of the *nifB* *fdxN* *nifO* *nifQ* operon). NifL prevents expression of the other *nif* genes if high levels of ammonium or oxygen are present, conditions that would either make nitrogenase unnecessary for growth or become inactive, respectively. The NifL protein, which carries a flavin cofactor, inhibits the activity of NifA, which is a transcriptional activator of the other *nif* genes and operons under these conditions (Dixon, 1998). The presence of a *nifL* gene is characteristic of diazotrophs in the *Gamma*proteobacteria, which includes another important diazotroph of this taxon, *Klebsiella pneumoniae*, as well as *Azotobacter vinelandii*—the two species most studied. The *nifL* gene has not been identified in the *Alpha*proteobacteria, *Beta*proteobacteria, or *Delta*proteobacteria groups of diazotrophs, with the possible exception of *Azoarcus* species in the class *Beta*proteobacteria (B. Reinhold-Hurek, personal communication). Muta-

TABLE BXII.γ.123. Differential characteristics of the species of the genus *Azotobacter*^a

Characteristic	1. <i>A. chroococcum</i>	2. <i>A. armeniacus</i>	3. <i>A. beijerinckii</i>	4. <i>A. nigricans</i>	5. <i>A. paspali</i>	6. <i>A. salinestris</i> ^b	7. <i>A. vinelandii</i>
Motility	+	+	—	—	+	+	+
Water soluble pigments:							
Yellow-green fluorescent ^c	—	—	—	—	+	—	+
Green	—	—	—	—	—	—	d
Brown-black	—	—	—	d	—	+	—
Brown-black to red-violet	—	+	—	+	—	—	—
Red-violet	—	+	—	d	+	—	d
Nitrate reduced to nitrite	+	—	+	+	—	+	+
Carbon sources: ^d							
D-Glucuronate	—	—	+	—	—	+	—
D-Galacturonate	—	+	+	—	—	nd	—
Glutarate	—	d	—	—	—	nd	+
Glycolate	—	—	—	—	—	nd	+
Phenol	—	—	d	—	—	nd	+
Rhamnose	—	—	—	—	—	—	+
Caproate	+	—	—	—	—	nd	+
Caprylate	—	+	—	—	—	nd	+
meso-Inositol	—	+	+	—	—	nd	+
Malonate	+	—	—	d	—	nd	+
Propan-1-ol	—	—	d	d	—	nd	+
Trehalose	+	+	d	+	—	nd	—
Starch hydrolyzed	+	+	d	d	—	nd	—

^aFor symbols see standard definitions; nd, not determined.

^bAlthough there are limited data for *A. salinestris*, many of the carbon sources used by *A. chroococcum* also support growth of this species (Page and Shivprasad, 1991a).

^cOn iron-deficient medium. See Procedures for Testing Special Characters.

^dThe following carbon sources are not used by any species: lactose, mannose, xylose, arabinose, ribose, and fucose.

TABLE BXII.γ.124. Other characteristics of the species of the genus *Azotobacter*^a

Characteristic	1. <i>A. chroococcum</i>	2. <i>A. armeniacus</i>	3. <i>A. beijerinckii</i>	4. <i>A. nigricans</i>	5. <i>A. paspali</i>	6. <i>A. salinestris</i>	7. <i>A. vinelandii</i>
Peritrichous flagella	+	+	—	—	+	+	+
Nitrogen fixation occurs at pH: ^b							
5.0–5.5	—	—	d	d	—	—	—
6	—	—	+	d	d	d	+
6.5–9.5	+	+	+	+	+	+	+
10	+	d	+	+	+	—	+
Nitrogen fixation genes: ^c							
<i>nif</i>	+	d	+	+	+	d	+
<i>vnf</i>	+	nd	+	+	+	d	+
<i>anf</i>	—	nd	—	—	+	nd	+
Growth at a temperature of:							
9°C	—	—	d	d	—	—	—
14°C	d	—	+	d	+	—	+
18°C	+	D	+	+	+	—	+
32°C	+	+	+	+	+	d	+
37°C	d	d	—	—	+	+	+
Diffusible exopolysaccharides produced	+	d	d	+	—	d	d
Peroxidase	d	d	d	d	—	+	+
Urease	+	+	+	+	+	+	+
Oxidase	+	d	+	+	+	nd	+
Production of H ₂ S from:							
Thiosulfate	d	—	d	d	+	nd	+
Cysteine	—	—	—	—	d	nd	d
Utilization as sole carbon source: ^c							
Fructose, glucose, acetate, pyruvate, fumarate, malate, succinate, α-oxoglutarate, lactate, DL-gluconate, or acetyl methylcarbinol	+	+	+	+	+	+	+
Sucrose	+	d	+	+	+	+	+
Propionate	+	d	+	—	—	nd	+
β-Phenylpropionate	d	—	—	—	—	nd	+
n-Butyrate	+	+	+	d	—	nd	+
Glutarate	—	d	—	—	—	nd	+
DL-β-Hydroxybutyrate	d	+	+	d	—	nd	d
Benzoate	d	—	+	—	—	+	+
Trehalose	d	+	d	+	—	nd	—
Melibiose	+	+	+	d	—	+	+
Maltose	+	+	d	d	—	nd	+
Raffinose	+	d	d	d	—	nd	d
Propan-1-ol	d	—	d	d	d	nd	+
Butan-1-ol	+	d	+	d	—	nd	+
Glycerol	d	—	d	—	—	—	+
Mannitol	+	+	d	d	—	nd	+
Sorbitol	+	+	d	d	—	nd	+
Susceptibility to antimicrobial agents:							
Streptomycin, 0.2 µg/ml	S	S	S	S	S	S	S
Tetracycline, 2 µg/ml	d	R	d	d	R	nd	d
Chloramphenicol, 25 µg/ml	d	d	d	R	d	R	R
Sulfanilamide, 25 µg/ml	d	R	d	d	S	nd	d
Penicillin G, 5 U/ml	d	S	d	R	R	nd	d
Phenol, 0.05%	R	S	d	d	d	nd	R
Sodium benzoate, 0.5%	d	R	d	R	d	nd	R
Sodium fluoride, 0.01 M	d	R	R	d	R	nd	R
Mercuric chloride, 10 µg/ml	d	R	d	d	R	nd	d
Iodoacetate, 1mM	S	S	S	S	S	nd	S
Erythromycin, 2 µg/ml	S	R	d	R	d	nd	R
Neomycin or kanamycin, 1 µg/ml	S	S	S	S	S	S	S

^aFor symbols see standard definitions; S, susceptible; R, resistant; nd, not determined.^bResults of Thompson and Skerman (1979) on solid media. Machado and Döbereiner (1969) found no growth of *A. paspali* above pH 8.0 in their liquid medium.^cSee Procedures for Testing Special Characters.^dAlthough there are no reports of PCR or hybridization results to confirm the presence of these genes, they can be inferred to be present on the basis of tests for diazotrophic growth and/or nitrogenase activity.^eSee Table BXII.γ.123 for additional compounds; also see Thompson and Skerman (1979).

tions in *nifL* can result in constitutive expression of *nif* genes leading to unregulated nitrogenase enzyme overexpression and excretion of ammonium in *A. vinelandii* (Bali et al., 1992). Unlike in most other diazotrophs in the *Proteobacteria*, ammonium inhibition of nitrogen fixation in *A. vinelandii* occurs at only one

target, namely, inactivation of NifA by NifL. Posttranslational control of nitrogenase activity by ammonium might occur in *A. chroococcum* (Munoz-Centeno et al., 1996).

In addition to using the classic Mo-containing enzyme for nitrogen fixation, all *Azotobacter* species examined have the po-

tential to synthesize one or more alternative nitrogenases that do not contain Mo. Evidence for such alternative nitrogenases first came from studies with *A. vinelandii* mutants UW10 and UW9—derivatives of AvOP (UW) (ATCC 13705) carrying mutations in the *nifH* and *nifD* genes which encode protein subunits of nitrogenase. Bishop et al. (1980) found that these mutants gave rise to Nif⁺ colonies at low frequency and that these strains were resistant to tungstate, an antagonist of Mo transport. In addition, both UW10 and UW91 were able to grow and fix nitrogen if Mo was eliminated from the growth medium, and it was proposed that *A. vinelandii* could synthesize a Mo-independent form of nitrogenase. Subsequently, by genetics, biochemistry, and gene sequencing, *A. vinelandii* was found to encode three different nitrogenase enzymes with different structural subunits: (1) the traditional Mo nitrogenase (nitrogenase-1, the structural subunits of the enzyme encoded by the *nifHDK* genes), (2) a V-containing enzyme (nitrogenase-2, encoded by the *vnfH vnfD* genes), and (3) a Fe-containing nitrogenase (nitrogenase-3, encoded by the *anfHDK* gene cluster) (see Fig. BXII.γ.136 for *nif*, *vnf*, and *anf* gene cluster arrangements). These alternate nitrogenase systems have been reviewed by Bishop and Joerger (1990) and Rangaraj et al. (2000). Parallel studies with *A. chroococcum* strain MCD1 have indicated the presence of the Nif and Vnf enzymes only. They are encoded by *nif* and *vnf* genes, which have sequences and arrangements similar to those in *A. vinelandii* (see references in legend to Fig. BXII.γ.136). In *A. vinelandii*, all three enzymes require the activity of five other *nif*

gene products found within the major and minor clusters of *nif* genes, including *nifB*, *nifM*, *nifS*, *nifU*, and *nifV* (Joerger and Bishop, 1988; Kennedy and Dean, 1992). Some of these *nifB* and *nifV* genes are required for Fe-V cofactor biosynthesis (Ruttimann-Johnson et al., 1999) and presumably also for Fe-Fe cofactor formation. The molybdate or vanadate concentration in the growth medium (usually Burk's sucrose medium treated to extract metals; see Procedures for testing Special Characters) with 30 μM Fe determines which nitrogenase is available (Bishop and Joerger, 1990). Thus, Mo (whether or not V is present) at a concentration of about 50 nM represses *vnf* and *anf* gene expression. In cultures grown at 30°C, molybdate deficiency (less than approximately 50 nM) prevents expression of the nitrogenase structural genes, *nifHDK* (Jacobson et al., 1986), and when vanadate is present (approximately 50 nM), the *vnfHDK* genes but not the *anfHDK* genes are expressed. Repression by Mo of *vnf* and *anf* gene expression occurs far less severely in cultures grown at lower temperatures (14°C or 21°C) (Walmsley and Kennedy, 1991), probably because Mo is less well transported at lower temperatures. This characteristic may have ecological significance, possibly correlating with the better efficiency of the V-nitrogenase for reduction of acetylene at lower temperatures (Dilworth et al., 1988).

Three strains of *A. chroococcum* [MCD1 (ATCC 4412), ATCC 480, and ATCC 9043] have been analyzed for alternative nitrogenase genes by Southern hybridization. All were found to have *nif* and *vnf* genes, but not *anf* genes (Fallik et al., 1991). Con-

A. vinelandii

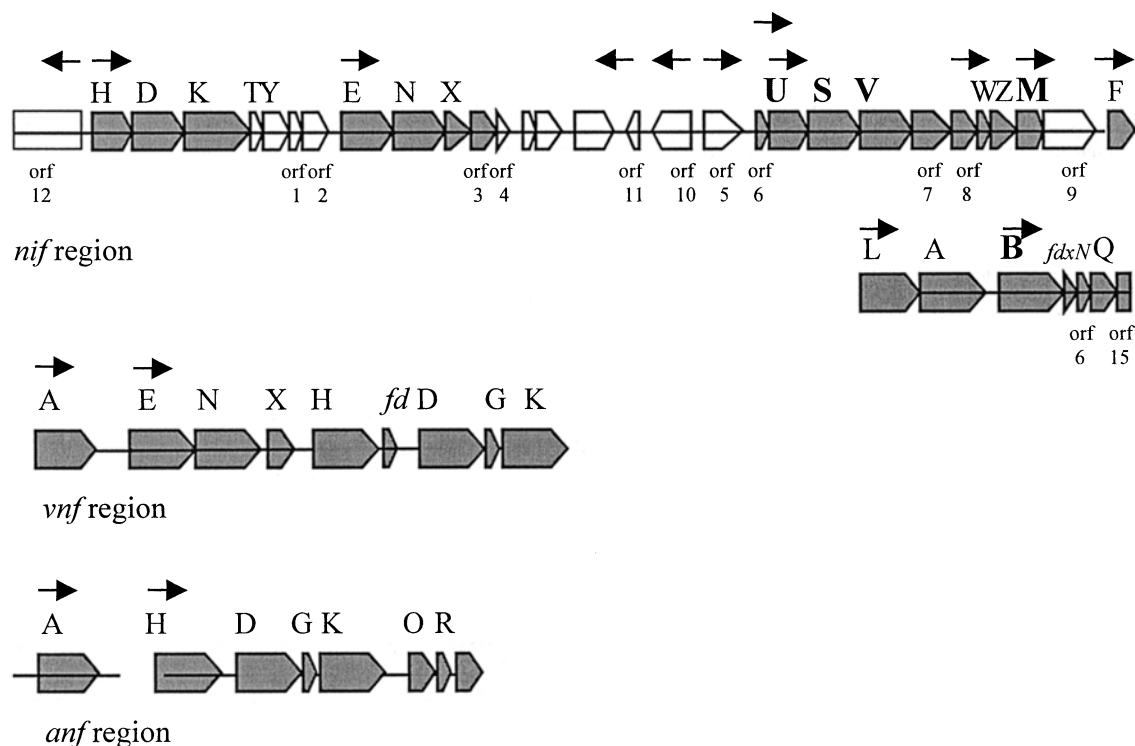


FIGURE BXII.γ.136. Physical map of the *A. vinelandii* nitrogen-fixation genes. *nif* genes encode products involved in the synthesis of the primary Mo-containing enzyme (Jacobson et al., 1989; Joerger and Bishop, 1988; Joerger et al., 1990) while the *vnf* (Joerger et al., 1990) and *anf* clusters encode alternative nitrogenases (see text) (Joerger et al., 1989; Mylona et al., 1996). Bold letters in larger font indicate genes which are essential for all three nitrogenase systems. Open arrows depict genes that are not essential for nitrogen fixation (Jacobson et al., 1989). Small arrows indicate locations of promoters and direction of transcription. *A. chroococcum* *nif* and *vnf* genes are arranged similarly (Evans et al., 1988; Fallik and Robson, 1990).

sequently, this property may characterize the species (Nif^+ , Vnf^+ , Anf^-). *A. beijerinckii* (ATCC 19360), *A. nigrificans* (ATCC 35009), and *A. salinestris* (ATCC 49674) also apparently contain *nif* and *vnf* genes only (Fallik et al., 1991; Loveless and Bishop, 1999). Only *A. vinelandii* and *A. paspali* (strains ATCC 23367 and ATCC 23833) among the *Azotobacter* species (and also *Azomonas agilis*) carry *nif*, *vnf*, and *anf* genes and can synthesize any of the three nitrogenases, depending on environmental conditions of Mo or V supply. Consequently, their phenotype is Nif^+ , Vnf^+ , Anf^- .

Oxygen control of nitrogenase activity in Azotobacters Nitrogen fixation is a process that is oxygen-sensitive due to the O_2 -lability of the nitrogenase component proteins. However, in *Azotobacter* species, N_2 fixation can occur under fully aerobic conditions in all species examined except *A. salinestris*. Even the latter species can adapt to aerobic diazotrophy under conditions of Fe limitation, which increases production of a melanin catechol that may trap Fe and be involved in oxygen tolerance (Page and Shivprasad, 1995). In *A. chroococcum*, a shift of cultures to hyperbaric oxygen conditions ($> 20\%$) is followed by a period of cessation of nitrogenase activity which then returns after further incubation, probably because of conformational protection by the FeSII protein, discussed below and in Robson and Postgate (1980). Aerotolerance in *A. vinelandii* is due mainly to a low O_2 affinity, high K_m cytochrome *d* terminal oxidase encoded by the *cydAB* genes. Mutation in either of these genes results in mutant strains unable to grow aerobically on N_2 , which represents a loss of respiratory protection. Under low (2%) O_2 , the mutants can fix nitrogen and grow (Kelly et al., 1990). The CydR protein represses transcription of *cydAB* under conditions of low O_2 (Wu et al., 1997a) during which the cytochrome *c* pathway provides respiration (Rey and Maier, 1997). A supply of ATP and reducing equivalents are additional factors that may be important to prevent oxygen damage to nitrogenase (Oelze, 2000). Another factor of significance is the FeSII or Shethna protein (Dervartanian et al., 1969). This protein binds to and protects Mo nitrogenase from oxygen damage—but apparently does not bind to or protect either of the alternative Vnf or Anf nitrogenases—during periods of O_2 stress (Dervartanian et al., 1969; Moshiri et al., 1994). *A. vinelandii* mutant strains lacking the FeSII Shethna protein show reduced viability under carbon deprivation if O_2 is present (Maier and Moshiri, 2000). This protein is also found in *A. chroococcum* where it provides “conformational protection” from O_2 damage to nitrogenase (Robson, 1979). An FeSII DNA probe from *A. vinelandii* hybridized only to *A. chroococcum* genomic DNA among several diazotrophs tested (Moshiri et al., 1994). The FeSII protein may therefore be found only in species of *Azotobacter* and thus may be a distinguishing characteristic. Production of alginates and an alginate capsule may form a barrier to oxygen transfer (Sabra et al., 2000), adding to the battery of mechanisms elaborated by azotobacters for protecting nitrogenase from damage by oxygen in an obligate aerobe.

Other nitrogen sources Most *Azotobacter* species can utilize nitrate as a nitrogen source in an assimilatory fashion by reduction to nitrite and then ammonium. Genes for nitrate and nitrite reductases, *nasB* and *nasA*, have been isolated from *A. vinelandii* OP. The *nasAB* operon, regulated from a σ^{54} , NtrC-dependent promoter, is repressed by ammonium and requires nitrate for full expression (Ramos et al., 1993). Soluble nitrate reductases have been purified from *A. vinelandii* and *A. chroococcum* (Guerero et al., 1973; Gangeswaran et al., 1993). Nitrate transport and reduction have been studied in *A. chroococcum*. Although the

genes have not been characterized in this organism, the physiology of nitrate assimilation is similar to that in *A. vinelandii*. Ammonium prevents full synthesis/activity of the enzyme, and nitrate is required for full activity.

All *Azotobacter* species are urease positive in physiological tests—a feature that distinguishes *Azotobacter* from *Azomonas* species (Thompson and Skerman, 1979). This feature is consistent with the fact that urea is a good nitrogen source in both *A. vinelandii* and *A. chroococcum*. It is also a useful trait, because urea added to media at 5–10 mM does not prevent nitrogenase synthesis, in contrast to the effects of ammonium. Consequently, urea allows studies of expression of these genes in liquid cultures (e.g., Walmsley and Kennedy, 1991).

Other aspects of metabolism H_2 uptake systems are similar in both *A. vinelandii* and *A. chroococcum*, having been best characterized physiologically in the latter species. H_2 recycling occurs in N_2 -fixing cultures: H_2 , a byproduct of the reduction of N_2 to NH_4^+ , is re-oxidized to H^+ by a Ni-dependent hydrogenase encoded by the 16-gene *hup* cluster in *A. chroococcum* (Du and Tibelius, 1994; Du et al., 1994) and by a similar 16-gene *hox/hyp* cluster in *A. vinelandii* (Menon et al., 1992; Garg et al., 1994). Hydrogenase activity is apparently beneficial in *A. chroococcum*, as shown by the ability of the Hup^+ strain MCD-1 to outcompete a Hup^- mutant for survival under controlled carbon-limited chemostat conditions (Yates and Campbell, 1989). In *A. vinelandii*, both Hox^+ and Hox^- strains yielded very similar amounts of protein in chemostat cultures (Linkerhagner and Oelze, 1995). In addition, both had similar respiratory activities, and, in the wild-type, only a small fraction of total respiratory activity was due to H_2 -dependent O_2 consumption. This finding led the authors to conclude that hydrogenase does not benefit *A. vinelandii* during carbon-limited growth under N_2 -fixing conditions.

In regard to ferredoxins, it is remarkable that *A. vinelandii* can synthesize at least 12 different ferredoxin-like proteins (Jung et al., 1999), some of which—Fe-SI, FixX, FdxN, FixFd—have been implicated in electron transfer to nitrogenase. However, no mutation in a single ferredoxin-encoding gene prevents electron transfer to nitrogenase; thus, there is an apparent redundancy of function in this respect. The VnfFd ferredoxin is required for activity of the V-containing nitrogenase (Fig. BXII.γ.136) (Raina et al., 1993). Another ferredoxin is the Fe-SII Shethna protein, described above, which protects nitrogenase from inactivation by O_2 . Known ferredoxins in *A. chroococcum* include those encoded by a gene adjacent to *vnfH*, and FdI and FdII, corresponding to ferredoxins characterized in *A. vinelandii* (George et al., 1984; Robson et al., 1986; Thomson, 1991).

Genetic Characteristics *A. vinelandii* can be readily studied genetically because the widely-used, non-gummy strain AvOP (ATCC 13705), derived from AvO (ATCC 12518), is easy to handle. AvOP was also named strain UW in studies at the University of Wisconsin (Bishop and Brill, 1977) and CA in studies at North Carolina State University (P. Bishop laboratory). This strain harbors no indigenous plasmids, is a high-frequency recipient during conjugation, and can be transformed with either linear or plasmid DNA. It becomes competent for DNA uptake after growth on Fe- and Mo-free medium (Page, 1985). Another widely-used derivative of UW is UW136, wild-type for nitrogen fixation but carrying a mutation for rifampicin resistance (Bishop and Brill, 1977). The features of UW and UW136 allow easy genetic studies and manipulation by introduction of wide-host-range plasmids

and isolation of mutant strains, either by transposition or gene replacement techniques (see later). In addition, nitrogenase and other important enzymes have been isolated and analyzed from strain AvOP (UW) because this strain is relatively easy to grow on either a small or large scale for preparation of extracts for protein purification. The strain has more recently become a focus of study for alginate biosynthesis. To date, sequences are available for more than 200 of its genes.

Azotobacter chroococcum has also been well-studied genetically with respect to a number of traits, and although there are no reports of successful transformation in strains of this species, the cells can receive plasmids by conjugation with *Escherichia coli* donors. Strain MCD1, a non-gummy derivative of ATCC 4412 (NCIB 8033) cured of three of five indigenous plasmids, has been used for most genetic studies (Robson et al., 1984). The sequences of approximately 50 genes in *A. chroococcum* have been reported to date.

The genome sizes of several *Azotobacter* species have been determined by pulsed-field gel electrophoresis of large chromosomal DNA fragments generated by rare-cutting restriction endonucleases. The sizes range from 3.10 Mb for *A. chroococcum* NCIB 9043 to 4.57 Mb for *A. vinelandii* UW (MacDonald and Melton, manuscript submitted) (Table BXII.γ.125). In another study, Manna and Das (1994) found the M4 strain of *A. chroococcum* to have a genome size of 5.3 Mb. This size disparity supports previous findings of genomic variation within *A. chroococcum* (Becking, 1992). The presence of a single circular chromosome in *A. vinelandii* is indicated from genetic analysis (Blanco et al., 1990). The size of the *A. paspali* genome is about the same as that of *A. vinelandii*, ~4.5 Mb, even though it might be expected that the *A. paspali* genome would be smaller because the organism uses fewer compounds as carbon sources and has a more restricted rhizosphere habitat.

The origin of replication of *A. vinelandii*, including a 200 bp segment capable of replicating plasmid pBR322 in *E. coli* polA, has been identified and sequenced. A larger fragment of 1652 bp, which includes the 200 bp ori sequence, contains 14 putative DnaA protein binding sites, ten putative binding sites for the IHF protein, and 19 GATC boxes (potential sites for Dam protein methylation), all of which are involved in initiation of DNA replication in *E. coli* and other organisms (Singh et al., 2000). Genes involved in general recombination in *A. vinelandii* include *recA*

and *recF* (Venkatesh and Das, 1992; Badran et al., 1999). The former is more closely related to the *recA* gene of *P. aeruginosa* than to that of *E. coli*, which supports other phylogenetic data discussed below.

Multiple chromosomes are common in the two species examined, *A. vinelandii* and *A. chroococcum*. In *A. vinelandii*, the chromosome number is variable and can reach about 80 copies per cell in late-exponential phase cells; in old stationary-phase cultures the amount of DNA is drastically reduced, possibly in preparation for cyst formation (Maldonado et al., 1994). In exponential cultures of *A. chroococcum* MCD-1, cells contained 20–25 genome equivalents (Robson et al., 1984).

No indigenous plasmids in *A. vinelandii* AvOP (UW) wild-type and derived mutant strains have been detected in the author's laboratory or by several other investigators who study this organism. One strain of *A. paspali* did not harbor any plasmids (Do Nascimento and Tavares, 1987). Robson et al. (1984) reported finding between two and six plasmids in each of eight strains of *A. chroococcum* examined; these plasmids ranged in size from 10.5 kb to 300 kb. The *A. chroococcum* strains included NCIB 8003 (ATCC 4412); strain MCC-1, a Nal^r Sm^r and non-gummy derivative of NCIB 8003; and six new isolates from soils in Sussex, UK. One plasmid in NCIB 8003 appeared to be of the IncP class. Plasmids were cured by growing strains for several generations using the agents ethidium bromide, acriflavin, or mitomycin. The cured strains showed no altered phenotypes that are usually associated with plasmids, except for possibly one trait, the production of agar-diffusible exopolysaccharides. Evidence for a plasmid encoding a gene or genes for degradation of 2,4-dichlorophenoxyacetic acid (2,4-D) in strain MSB1 of *A. chroococcum* was reported (Balajee and Mahadevan, 1989). The presence of single or multiple plasmids might be a distinguishing feature of *A. chroococcum*. Reports of plasmids in other species of *Azotobacter* are not available.

Wide-host-range plasmids of the IncQ and IncP classes can be transferred from *E. coli* by conjugation to *A. chroococcum* strain MCD1 and its derivatives (Kennedy and Robson, 1983; Ramos and Robson, 1987) and by either conjugation or transformation to *A. vinelandii* strain AvOP and mutant derivatives (Kennedy and Robson, 1983; Doran et al., 1987; Kennedy and Toukdarian, 1987; Glick et al., 1989). This genetic manipulation allows for the isolation of numerous genes from IncP cosmid libraries by mutant complementation, and analysis of gene function by specific genes cloned into either IncP or IncQ plasmids. Introducing wide-host-range plasmids of the IncQ compatibility group can result in reduced growth rates in *A. vinelandii* (Glick et al., 1986).

The state of competence for transformation of strains of *A. vinelandii* is associated with siderophore production and can be induced by growth on Fe-deficient or Fe- and Mo-deficient media (Page, 1985; Page and Grant, 1987). Competent cells can be transformed with either plasmid or linear DNA (Doran et al., 1987). Interestingly, the transformation frequency was enhanced 10- to 50-fold when pKT210 (IncQ) carried an insert fragment of *A. vinelandii* *nif* DNA, suggesting that *A. vinelandii* possesses a homology-facilitated transformation system. Mutated genes can be introduced directly on plasmids unable to replicate or on linear chromosomal DNA prepared from mutant strains; recombination with the chromosome at sites flanking the mutation leads to replacement of the wild-type genes (e.g., as described by Jacobson et al., 1989, and in many other reports).

The rRNA genes of several aerotolerant diazotrophs were an-

TABLE BXII.γ.125. Comparison of genome size and rRNA (*rrn*) copy number among isolates of the family *Azotobacter* and *Azomonas*

Strain	Genome size ^a (Mb)	Copies of <i>rrn</i> genes
<i>Azotobacter chroococcum</i> (ATCC 7493)	3.9 ± 0.02	8
<i>Azotobacter chroococcum</i> (ATCC 9043)	3.1 ± 0.32	7
<i>Azotobacter chroococcum</i> ^b M4	5.3	NR ^c
<i>Azotobacter paspali</i> (ATCC 23822)	4.3 ± 0.24	6
<i>Azotobacter paspali</i> API	4.5 ± 0.05	6
<i>Azotobacter paspali</i> (ATCC 11562)	4.6 ± 0.01	6
<i>Azotobacter paspali</i> (ATCC 12095)	4.6 ± 0.01	6
<i>Azotobacter vinelandii</i> UW (ATCC 1218)	4.6 ± 0.35 ^d	6
<i>Azomonas macrocytogenes</i>	2.4 ± 0.01	7

^aGenome size determined by summing restriction fragments generated by rare-cutting enzyme digestion. Values are derived from at least three data sets of two separate enzyme digestions (MacDonald and Melton, manuscript submitted).

^bDetermined by Manna and Das (1994).

^cNR, not reported.

^dManna and Das (1993) reported a genome size of 4.5 Mb for this strain.

alyzed in hybridization experiments by De Smedt et al. (1980) to determine their phylogenetic relationship. The results indicated the validity of the grouping of all the *Azotobacter* species identified in this description and the species described below (with the exception of *A. salinestrus*, which was not available at the time of their analysis). In a further and more recent analysis, the *Azotobacter* species *A. vinelandii* (ATCC 13705 derivative, CA), *A. chroococcum* (ATCC 9043), and *A. salinestrus* (ATCC 49674) all carry an arrangement in the rRNA operon that consists of the 16S rRNA gene (*rrs*), a spacer region containing the tandem tRNA gene sequences for tRNA^{Ile} and tRNA^{Ala}, the 23S rRNA gene (*rrl*), a second spacer region, and the 5S rRNA gene (MacDonald and Melton, manuscript submitted). *A. paspali* (ATCC 23833) also possesses the 16S–23S–5S gene arrangement; however, the tRNAs within the 16S–23S spacer region were not identified. All the *Azotobacter* species examined here contain six, seven, or eight duplicated copies of the rRNA gene cluster (Table BXII.γ.125).

Not surprisingly, the codon usage of *A. vinelandii* is similar to that of *A. chroococcum* (Rudnick and Kennedy, unpublished analysis). The mol% G + C content of the DNA of the former species was earlier estimated from T_m experiments to be 64.9–66.5 and the latter to be 65.8–67.5% (De Ley, 1968). For genes that encode proteins, the sequences of individual genes available for each species numbering about 200 and 50, respectively, indicated a mol% G + C content of 62.5 for *A. vinelandii* and 66.7% for *A. chroococcum*. Interestingly, the *anf* genes of *A. vinelandii* have a mol% G + C value (58.3) lower than that of the *nif* and *vnf* genes (64.4% each). The *anf* cluster of *A. paspali* is also lower in mol% G + C (54.3%) than that of the genome (63.2–64.6%, from T_m experiments). In addition, the clusters of genes encoding uptake hydrogenases, *hox/hyp* in *A. vinelandii* and *hup* in *A. chroococcum*, had a higher mol% G + C value than the total DNA in these species—70.7 and 70.2, respectively. This may reflect the acquisition of these genes—both *anf* and *hup/hox*—by lateral transfer from other organisms during the course of evolution of the genus *Azotobacter*.

Isolation of mutants Transposon mutagenesis is easily achieved in *A. vinelandii* and *A. chroococcum* using Tn5 carried on suicide vectors unable to replicate in these host organisms (for examples, see Kennedy et al., 1986; Joerger et al., 1989; Contreras et al., 1991; Luque et al., 1993; Tibelius et al., 1993; Wu et al., 1997a). In addition, Tn10 was successfully used to isolate mutants of *A. vinelandii*, including methionine auxotrophs (Contreras and Casadesus, 1987). When Tn5 was introduced into *A. beijerinckii* (NCIB 11292), the cells of most of the isolates had abnormal morphologies and rapidly lost viability when subcultured. In contrast, Tn76 was successfully used to make mutants in this strain (Owen and Ward, 1985). Contreras et al. (1991) developed a method to “freeze” Tn5 mutations in *A. vinelandii* by replacing the original transposon with a defective Tn5, resulting in highly stable mutants.

Isolation of auxotrophic mutants of *A. vinelandii*, attempted in many laboratories during the 1970s and 1980s, was largely unsuccessful. It was believed that high DNA content (originally reported by Sadoff et al., 1979), and hence high chromosome copy numbers, leads to difficulty in segregation of mutant phenotypes. However, the successful isolation of many other types of mutations over the past 15 years by transposon mutagenesis or antibiotic resistant cassette insertion in cloned genes, followed by transfer into strains and subsequent gene replacement, has

shown that isolation of mutant strains of *Azotobacter* is not inherently difficult. The variable DNA content of this organism, described above, does indicate that there are growth phases and conditions when the chromosome number is low, thereby allowing for “normal” segregation of introduced, selectable mutations in nonessential genes.

Although isolation of auxotrophs remains difficult, success has been achieved in the construction of Ade[−], Met[−], and Leu[−] mutants (Mishra and Wyss, 1968; Contreras and Casadesus, 1987; Manna and Das, 1997). *A. vinelandii* is unable to transport many amino acids (Toukdarian et al., 1990; D. Dean, personal communication). Although Mishra et al. (1991) reported that threonine was transported at a rate similar to that found for *E. coli* and methionine at a rate more than twice that in *E. coli*, five others tested were transported at a fraction of the rate in *E. coli*. In *A. beijerinckii*, RP4::Tn76 was successfully used to isolate Nif[−] mutants and auxotrophs requiring adenine or leucine (Owen and Ward, 1985). D. Dean and coworkers (personal communication) found that tryptophan auxotrophs could be isolated in *A. vinelandii* only if the strain harbored a wide-host-range plasmid carrying the tryptophan transport genes from *Escherichia coli*. Also consistent with the idea that amino acids are not easily transported into species of *Azotobacter* and *Azomonas* are the data of Thompson and Skerman (1979), who found that none of the 17 amino acids tested could serve as carbon sources in any of the species of these genera. Thus, limited capability of amino acid transport is probably a general characteristic of *Azotobacter* and *Azomonas*.

Mutations have now been introduced into many genes of *A. vinelandii* and *A. chroococcum*, resulting in gene replacement and the construction of mutant strains. A Nif[−] mutant of *A. paspali* was constructed by the conjugation and recombination of a kan^r cassette with the *nifHDK* operon, leading to gene replacement (Fallik et al., 1993). Some mutations are “lethal” in *A. vinelandii*, due to the essential nature of the genes under study. These genes include *glnA*, which encodes glutamine synthetase—the sole pathway for assimilation of ammonium (Kennedy and Toukdarian, 1987; Toukdarian et al., 1990); the *isc* and *fdxD* genes, which encode enzymes and a ferredoxin for Fe–S cluster formation (Zheng et al., 1998; Jung et al., 1999); and the *glnK* and *glnD* genes, which encode proteins that sense and signal the status of the nitrogen supply to glutamine synthetase to activate or inactivate the enzyme (Meletzus et al., 1998; Colnaghi et al., 2001). The ability of *A. vinelandii* to harbor high copy numbers of the chromosome is an advantage in that this feature allows identification of essential genes. Both wild-type and mutated copies marked by antibiotic resistance gene insertion can be maintained and identified by hybridization or PCR analysis; when selective pressure is removed by growth in medium without antibiotics, the chromosomes carrying the mutated essential genes are rapidly lost along with antibiotic resistance.

Bacteriophages Bacteriophages isolated from soil using *A. vinelandii* and *A. chroococcum* as indicator strains are capable of lysing cultures of *A. chroococcum*, *A. vinelandii*, and, to a lesser degree, *A. beijerinckii*; however, they do not affect *Azomonas* (Hegazi and Jensen, 1973). Four serologically distinguishable phages have been identified that can apparently lysogenize *A. vinelandii* strain O (ATCC 12518), resulting in a pseudolysogenic state in which the bacteria are converted to an unencapsulated form (Thompson et al., 1980). Bishop et al. (1977) described a general technique for isolating phage able to infect *A. vinelandii* from

soil samples.⁶ No information on phages is available for *A. nigricans*, *A. armeniacus*, and *A. paspali*.

Antigenic structure The most recent and informative study is that of Tchan et al. (1983), which, by a method of immunoelectrophoresis, related cross-reacting antigenic determinants between and within the species of *Azotobacter* and of *Azomonas*. In these studies, species of *A. vinelandii* and *A. paspali* were found to be immunologically more homogeneous than strains of *A. chroococcum*—probably because of the wider geographical distribution of isolates of the latter. In addition, *A. vinelandii* and *A. paspali* were more similar to each other than to other *Azotobacter* species, but all were related. *Azomonas* species were related at about the same level to each other as to *Azotobacter* species.

Antibiotic sensitivity Table BXII.γ.124 summarizes the data from Thompson and Skerman (1979), from other researchers, and from one of the authors' laboratory. Of special note is that all isolates of *Azotobacter* are very sensitive to streptomycin and kanamycin/neomycin: the minimal inhibitory concentration (MIC) for streptomycin is <0.2 µg/ml, and for kanamycin/neomycin it is <1.0 µg/ml. This was also observed by Page and Shivprasad (1991a) for a more recently distinguished *Azotobacter* species, *A. salinestris*. These concentrations are important for constructing mutants that carry cassettes encoding genes for resistance to these antibiotics. Most strains are also relatively insensitive to chloramphenicol (MIC being 50 µg/ml or more), useful in counterselection of *E. coli* donor strains in conjugation experiments.

Ecology and plant growth stimulation *Azotobacter* species are generally found in soils of slightly acid to alkaline pH (reviewed in Becking, 1992). While the numbers are generally fairly low (<10⁴ cells per g of soil), they are found throughout the world, typically in 30–80% of soils sampled, but rarely in polar regions. Because of their relatively high requirement for phosphorus, they are more commonly found in fertile soils than in sand. The numbers are generally not higher in rhizospheres than in open locations (Jensen, 1965). There are several reports of isolation of *Azotobacter* species from plant leaves and other surfaces. Most isolates are of *A. chroococcum*, which has also been isolated from freshwater canals and watercourses. *A. vinelandii* has been isolated far less frequently and may be more abundant in tropical regions; it has occasionally been isolated from freshwater (Thompson and Skerman, 1979). *A. beijerinckii* is found more often in acid soils than the other *Azotobacter* species, which reflects its laboratory tolerance to lower pH. *A. salinestris*, a Na-dependent species, has been isolated from slightly saline soils in western Canada, in Egypt, and in Australia (Page and Shivprasad, 1991a; Wang et al., 1993). The other species are rarely isolated, and may be restricted regionally. Little information is available for *A.*

nigricans and *A. armeniacus*, and the distribution of these two species in their habitats remains unknown. Only *A. paspali* is known to have a constant association with the root system of a plant, *Paspalum notatum* cv Batatais (Döbereiner, 1966), where availability of organic substances and a more suitable pH is conditioned by the plant in its rhizosphere. The restriction of *A. paspali* to the rhizosphere of *Paspalum* may be related to its reduced ability to utilize many organic substances that can be utilized by other species of *Azotobacter* that are more widely distributed in soils. In addition, *A. paspali* is the only species antagonistic to Gram-positive bacteria, a property that may be advantageous for life in the rhizosphere (Thompson and Skerman, 1979).

The ability of *Azotobacter* species, mainly *A. chroococcum*, to enhance the growth of various crop plants was widely studied in Egypt and Russia in 1950–1970. Although beneficial effects were often observed, on the order of 10–20% better growth, the practice of inoculation of crops has been largely abandoned due to inconsistency of results. The benefits sometimes observed are likely due to the production of plant growth-stimulating substances by azotobacters, rather than to transfer of bacterially fixed nitrogen to the plants. *A. paspali* synthesizes and excretes indole acetic acid, three gibberellins, and two cytokinins (Barea and Brown, 1974). Stimulation of growth of inoculated plants (both tomato and *Paspalum notatum*) was observed by these authors, even though no acetylene reduction activity could be measured. *A. paspali* also enhanced growth of several crop seedlings, particularly in the root surface area (Abbass and Okon, 1993b) (Fig. BXII.γ.137). Although *A. paspali* initially colonized the roots in both sets of experiments, cell survival was low over time (Barea and Brown, 1974; Abbass and Okon, 1993b). Two strains of *A. chroococcum* that were able to stimulate growth of wheat seedlings produced indoleacetic acid and gibberellins (Pati et al., 1995). In another report, growth of *A. chroococcum* was enhanced in the presence of maize root exudates, which also stimulated the production of auxins, gibberellins, and cytokinins by this organism (Martinez-Toledo et al., 1988). These results should prompt a reexamination of whether numbers of *Azotobacter* might be higher in rhizospheres than in nonrhizosphere soils.

ENRICHMENT AND ISOLATION PROCEDURES

The enrichment techniques are based on the ability of *Azotobacter* to grow by fixing nitrogen aerobically or microaerobically and to use organic substrates as an energy source. Soil, macerated roots or leaves, or other samples are suspended in water; then aliquots of diluted suspension are incubated in liquid or semi-soft N-free agar or on the surface of N-free agar plates using Winogradsky's or Burk's medium with 1% glucose or sucrose. All species except *A. salinestris* will grow diazotrophically on agar plates incubated in air, a feature useful for their enrichment and isolation because of the particularly high tolerance to oxygen during diazotrophic growth.

A specific enrichment method for *Azotobacter* species is that of Döbereiner (1995). Plates containing LG growth medium (Burk's medium with 2 ml/l bromothymol blue [0.5% solution in ethanol]) are inoculated with about 100 mg of finely sieved soil suspended in 1 ml of sterile water, or with appropriate dilutions of soil. Colonies of *A. chroococcum* appear after 24 h of incubation at 30°C as moist white colonies that become dark brown after 3–5 d. *A. vinelandii* colonies are similar but do not turn dark brown. *A. paspali* colonies appear only after 48 h; they become yellowish in the center due to assimilation of bromothymol blue and acidification of the medium.

6. Sites selected for enrichment were high in decomposed organic material. Inoculum was prepared by growing a 15-liter culture of *A. vinelandii* to about 5×10^8 cells/ml in Burk's medium. After inoculation, each site was irrigated with 1 liter 2% sucrose every other day. Soil sample extracts for phage assay were prepared by suspending 5.0 g soil in 15 ml phosphate buffer. Soil particles were removed by centrifugation. Extracts were filter-sterilized (45 µm pore size), then 1 ml of the sterile extract was added to 19 ml of a culture of *A. vinelandii* ($\sim 2.5 \times 10^7$ cells/ml). After incubation overnight with vigorous shaking, filter-sterilized samples (0.1 ml) of cultures were tested for the presence of phage by a soft agar overlay method. Four ml of molten 0.5% agar were mixed with 0.1 ml freshly grown *A. vinelandii* cells ($\sim 10^8$) and with aliquots of extracts and dilutions. The mixture was poured onto a fresh plate of Burk's agar medium and incubated for 24–48 h at 30°C and observed for the presence of phage plaques.

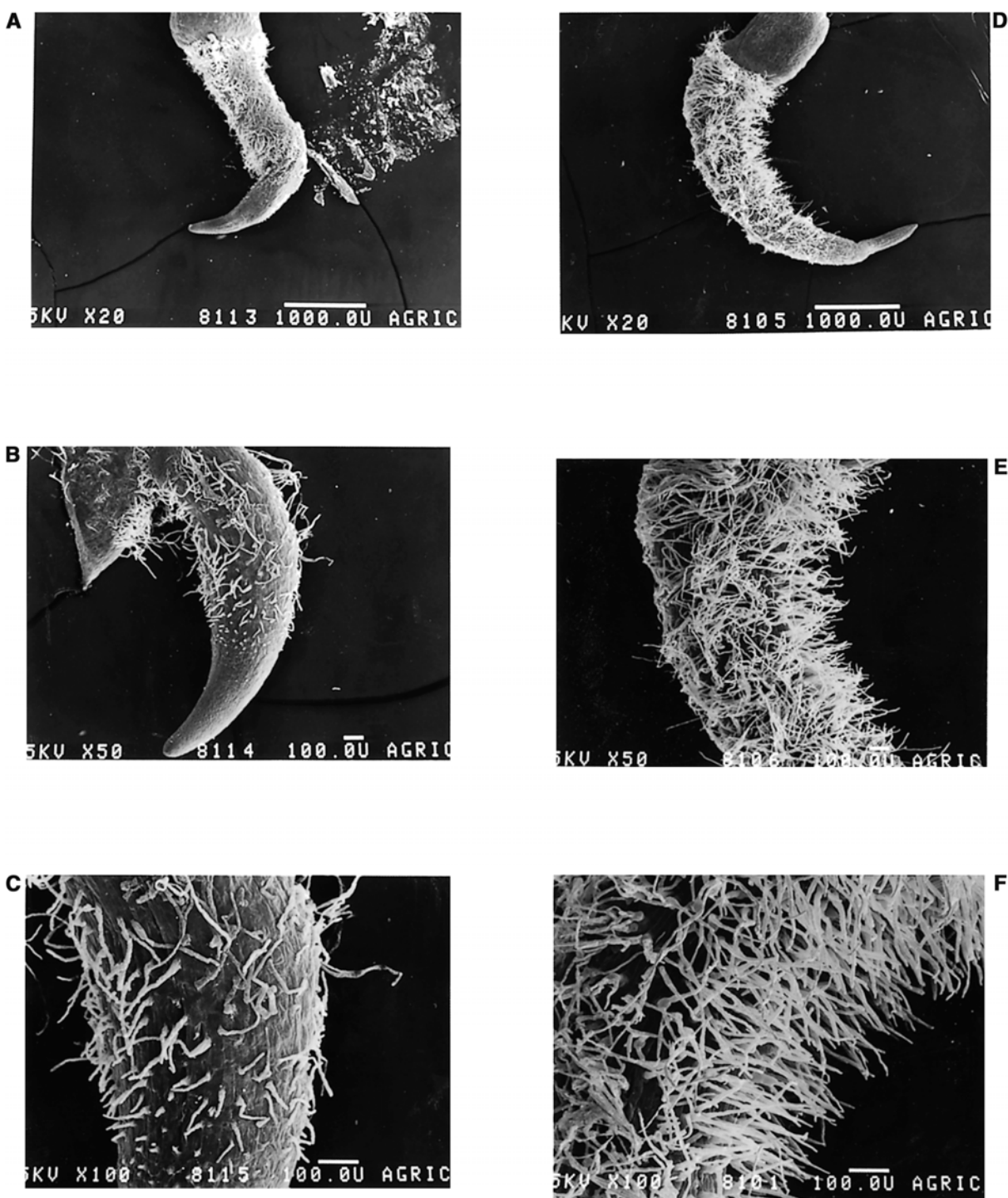


FIGURE BXII.γ.137. Scanning electron microphotographs of the inoculation effects with *A. paspali* (5×10^8 CFU/ml) 48 h after inoculation on the morphology of root hair development of canola seedlings. *Right column*, inoculated seedling; *left column*, uninoculated seedlings.

Thompson and Skerman (1979) described selective enrichment for various *Azotobacter* species based upon the addition to Winogradsky's medium of substances selectively stimulatory or inhibitory to species of *Azotobacter*. *A. vinelandii* can be selectively enriched by using L-rhamnose, ethylene glycol, erythritol, or D-arabitol as the carbon source. Alternatively, 1.0% sodium benzoate or 0.1% phenol can be used to inhibit the growth of other *Azotobacter* species. Incubation at 37°C favors the development of

A. vinelandii. *A. beijerinckii* can be selectively enriched using L-tartrate, o-hydroxybenzoate, D-glucuronate, or D-galacturonate. A pH of 6 or slightly less favors the growth of this species. *A. chroococcum* is the most common species and needs no special enrichment for its isolation. *A. armeniacus* can be enriched with caprylate but *A. vinelandii* will also grow in this medium; however, further isolation can differentiate these two species. *A. paspali* can be isolated from the rhizosphere of *Paspalum notatum*. There

is no selective medium for this species but an incubation temperature of 35–37°C will favor its growth. There is no specific method to isolate *A. nigrificans*.

For isolation of *Azotobacter* strains, liquid enrichment cultures are inspected daily. When macroscopic growth becomes apparent, the culture is examined with phase-contrast microscopy to detect the presence of *Azotobacter* cells, which are recognizable by their general morphology. Growth is usually obtained within 2–5 d. Positive cultures are streaked onto nitrogen-free agar medium with glucose or sucrose as the carbon source. For selective isolation, nitrogen-free agar media with appropriate selective substances should also be used. Sometimes nitrogen-free iron-deficient agar media can be used to encourage pigment production by some species. For soils containing sufficiently high numbers of *Azotobacter*, the different species can be isolated directly on agar plates: 0.1–0.5 g dry soil or an appropriate dilution is spread on the surface of nitrogen-free agar medium as described above.

MAINTENANCE PROCEDURES

For routine maintenance, *Azotobacter* should be subcultured at monthly or bimonthly intervals on Winogradsky's or Burk's agar medium with sucrose, except that glucose should be substituted for sucrose in the case of *A. armeniacus* and certain other strains which prefer glucose.

Thompson (1989) preserved many strains by suspending the cells from agar-grown cultures (10^9 – 10^{10} cells/ml) in 0.5 ml of 10% (v/v) glycerol. The suspensions were placed in glass ampules, rapidly frozen, and stored in liquid nitrogen. On revival, ampules were rapidly warmed by agitation in a water bath at 33°C until thawing occurred. All cultures remained viable for up to 7.5 years, grew with short lag phases, and suffered no apparent phenotypic changes. The following method of preservation has been used in the Kennedy lab: high density cell suspensions of *A. vinelandii* in glycerol (10–50%) have been stored in cryogenic microfuge tubes at –20°C or –80°C, with successful revival after four or more years.

Dimethyl sulfoxide (DMSO) has been used in the laboratory of Dr. Dennis Dean, Virginia Polytechnic Institute and State University, with great success for preservation of *A. vinelandii* wild-type and mutant derivatives of strain AvOP (personal communication). One-ml portions of freshly prepared 7% DMSO in 0.1 M sterile phosphate buffer (pH 7) are transferred to a sterile cryotube and a few loopfuls of fresh cells grown on Burk's medium are suspended in the DMSO solution. The suspension is vortexed to ensure a homogenous suspension without aggregates, then frozen at –80°C. The application of this method to other *Azotobacter* species has not been tested.

A paraffin oil preservation method has been used by Tchan and New (1984). Ten milliliters of a suitable nitrogen-free agar medium is poured into a 28 ml capacity McCartney bottle and allowed to solidify to form a slope approximately 3 cm in length with a base 2 cm in height. The culture is inoculated in the middle of the surface of the slope with a single stroke. When the growth reaches about 1 mm in thickness, the culture is checked for purity. Then 5–7 ml of sterile paraffin (mineral) oil is poured into the bottle to completely cover the slope. (It is important that no agar medium should be in direct contact with air, otherwise the culture can dry out and some cultures may die.) Usually *Azotobacter* survives for many months (even years) by this method. This method is particularly advantageous for frequently used stock cultures.

For preservation by lyophilization, dense culture suspensions in 10% sucrose solution (Bascombe and Jackson, 1965) can be freeze-dried and stored at 5°C. This method may not be suitable for all species.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Identification of alternative nitrogenase genes Although the characterization of species with respect to their ability to grow on Mo-deficient medium, with or without the addition of V, is incomplete in terms of the spectrum of isolates examined, the results of Fallik et al. (1991) and Loveless and Bishop (1999) suggest that (1) *A. chroococcum*, *A. beijerinckii*, *A. nigrificans*, and *A. salinestris* carry the *nif* and *vnf* genes; (2) *A. vinelandii*, *A. paspali*, and *Azomonas agilis* have genes for all three nitrogenases, *nif*, *vnf*, and *anf*; and (3) *Azomonas macrocytogenes* has the *nif* and *anf* genes only. In addition, Loveless et al. (1999) described seven new isolates from aquatic samples or wood-chip mulch in North Carolina by their ability to grow on Mo-deficient medium. These organisms, by 16S rDNA analysis, are taxonomically within the *Gammaproteobacteria*, and most related to the fluorescent pseudomonads. All had genes encoding the Mo nitrogenase (nitrogenase-1) and the V nitrogenase (nitrogenase-2). Four of the isolates also encoded the Fe-only nitrogenase (nitrogenase-3). These isolates may represent a new taxon of diazotrophs. The procedures for identification of these genes by hybridization and/or PCR analysis are therefore important in characterizing isolates in terms of the ability to synthesize the alternative nitrogenases that contain V (nitrogenase-2) or Fe only (nitrogenase-3), in addition to the classical Mo nitrogenase (nitrogenase-1) found in all other diazotrophs characterized. The methods for determining the presence of alternative nitrogenases are as follows:

1. Method of Loveless et al. (1999): blots of genomic DNA isolated from test strains restricted with enzymes such as *EcoRI* are hybridized to specific gene probes isolated from *A. vinelandii* and radiolabeled. The probes are *nifD* (from pTMR18; Bishop et al., 1985), *vnfD* (from pVDSJ1; Jacobitz and Bishop, 1992), and *anfD* (from pPJD3A2; Premakumar et al., 1992).
2. Method of Fallik et al. (1991): hybridization of Southern blots of restriction enzyme-digested DNA from test strains to radiolabeled gene probes. The probes consist of *A. chroococcum* *nifDK* (from pER4; Jones et al., 1984), *vnfDGK* (from pSEQ10; Robson et al., 1989), and *A. vinelandii* *anfDGK* (from pJWD3; Joerger et al., 1989).
3. Method of Loveless and Bishop (1999): PCR analysis for the presence of the genes *vnfG* or *anfG*, which indicate the presence of nitrogenase-2 and nitrogenase-3, respectively. *vnfG* is located between the *vnfD* and *vnfK* genes, and *anfG* between the *anfD* and *anfK* genes (see Fig. BXII.γ.138). The PCR primers flanking the *vnfG* and *anfG* genes—which anneal to conserved sequences within *vnfD* and *vnfK* or within *anfD* and *anfK*—will amplify the *vnfG* or *anfG* genes if they are present. The PCR products of these reactions can be analyzed by determination of their size on agarose gels and subsequent sequence determination. Sequences of specific primers used for identification of *anfG* and *vnfG* in the *Proteobacteria* and the expected product sizes obtained after PCR amplification are shown in Table BXII.γ.126.

Diazotrophic growth in the absence of molybdenum This procedure is carried out to help reveal the presence of alternative nitrogenase-2 (with the addition of V) or of alternative nitro-

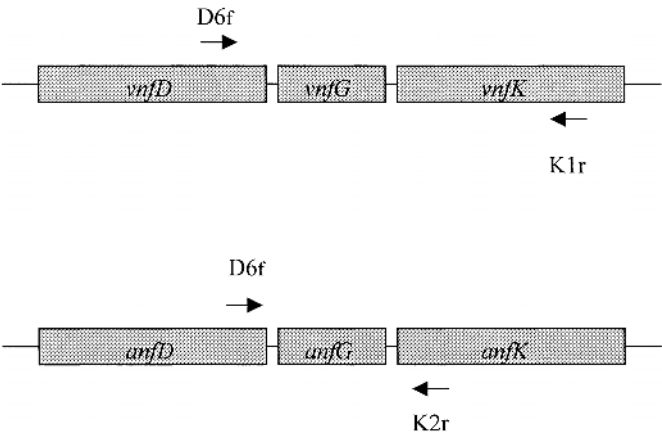


FIGURE BXII.γ.138. PCR primer annealing sites for the amplification of *vnfG* or *anfG* which can be used for the identification of alternative nitrogenases. (Reproduced with permission from T.M. Loveless and P.E. Bishop, Canadian Journal of Microbiology 45: 312–317, 1999, ©National Research Council of Canada.)

genase-3, traits characteristic of species of *Azotobacter* and *Azomonas*. The preparation of media to test the growth of diazotrophs in the absence of Mo is critical because Mo can contaminate many sources of chemicals used to prepare media and the water supply. Alternative nitrogenase synthesis can be inhibited by as little as 20 nM Mo. Two methods are described here that have been successfully used to extract Mo from laboratory-prepared media. The first, modified from Eady and Robson (1984), uses 8-hydroxyquinoline (a chelator of Mo) and dichloromethane to remove 8-hydroxyquinoline from the medium. Chemicals for Burk’s medium are dissolved in double-glass-distilled water, phosphates and other chemicals prepared separately as two different 20× stock solutions (20 fold more concentrated than in the final medium). The 20× solution of chemicals (but not the 20× phosphate buffer) is adjusted to pH 3.5 with 5M HCl. 8-Hydroxyquinoline (0.2 g/l) and dichloromethane (25 ml/l) are added to both 20× stock solutions. The stock solutions are stirred using a Teflon-coated magnetic bar for 2 h. The dichloromethane is allowed to settle and then siphoned off. This extraction procedure is repeated six times. The pH is then adjusted to 7.1 with pellets of NaOH, and the excess 8-hydroxyquinoline removed by repeated extractions with dichloromethane until the medium is colorless. An alternative is to use 1,2-dichloroethane instead of dichloromethane for extraction (Luque et al., 1993).

The second method employs activated charcoal to extract molybdenum from concentrated stock solutions (Schneider et al., 1991). In this method, 50 g of pulverized activated charcoal

(Merck) is suspended in 1 liter of doubly distilled water and stirred for 2 d at room temperature. This suspension is degassed by boiling for 10 min then cooled to room temperature, and 50-ml aliquots are placed on a 5 cm round filter paper disc (e.g., Whatman #1). The water is suctioned through the filter, resulting in a 5-mm layer of activated carbon. After washing the filter with 100 ml doubly distilled water, the stock solutions (50 ml) are suctioned through the filter. The method is more efficient if concentrated single-nutrient-component solutions, adjusted to pH 2–4, are filtered through the activated charcoal then mixed together by dilution into doubly distilled water, followed by pH adjustment.

Regardless of the method used to extract Mo from media, it is essential to use either commercially supplied plasticware, which is usually free of contaminating Mo, or glassware that has been soaked for 24 h or more in a solution of 5% Decon (or similar commercial product) and 0.5% EDTA in pure water (as purified through a MonoQ column, Millipore) for 24 h, followed by rinsing in pure water. Additions of Mo or V compounds are made as necessary, usually Na₂MoO₄·2H₂O or V₂O₅ at 1 μM.

Cell morphology Cultures grown for 24–48 h on both agar and liquid media should be used for the study of general morphology by phase contrast microscopy. Motility may be observed in wet mounts, but it is advisable to incorporate some air bubbles in the mount to insure an adequate supply of oxygen. Motile cells are usually found near the edge of an air bubble.

Cysts These resting forms are best studied with cultures at least 2 weeks old that have been grown on nitrogen-free medium containing 0.2% butanol. If the culture will not grow on butanol (e.g., *A. paspali*), nitrogen-free medium with 0.2% glucose can be used. The cysts may be stained with violamine⁷ (Winogradsky, 1938), acridine orange⁸ (Tchan, unpublished data), or with a mixture of neutral red and light green SF yellowish (Vela and Wyss, 1964). Examination by phase-contrast microscopy is sometimes helpful when violamine stain is used.

Pigment production The production of diffusible pigments in the presence or absence of benzoate is not constant on iron-

7. Violamine stain consists of violamine “2 R”, 1.0 g; phenol, 5.0 g; and water, 100 ml. An air-dried, fixed smear is treated with violamine stain for 45 sec, washed, and counterstained with dilute crystal violet (0.05% or less) for 1 min. After rinsing with water, the preparation is mounted in water for microscopic examination.

8. Acridine orange stain: Cells are suspended in a 0.01% aqueous solution of acridine orange. One drop is deposited on a slide, covered with 2 drops of mineral oil. Drops are then spread and a cover glass applied. A monolayer of cell suspension can be found in the preparation where the depth of liquid is such that motile cells are immobilized to facilitate microscopic examination. When examined by fluorescence microscopy, the vegetative cells are green and the cysts have a green central body and a red coat.

TABLE BXII.γ.126. PCR primer sequences for identification of *vnfG* and *anfG*, which indicate the presence of genes for nitrogenase-2 or nitrogenase-3^a

Primer	Anneals to	Sequence ^b	Expected size of PCR product ^c
D6f	<i>vnfD</i>	5'-CGGGATCCGAAGACTTYGARAAGGTCAT-3'	1700 bp (<i>vnfG</i>)
K1r	<i>vnfK</i>	5'-GGGGGTACCACTTCYAWGCAGAACT-3'	
D7f	<i>anfD</i>	5'-GCTCTAGACGCSATCTAYTCGCCGA-3'	750 bp (<i>anfG</i>)
K2R	<i>anfK</i>	5'-CGGAATTCGGATGCAATCCTTGAT-3'	

^aFrom Loveless and Bishop (1999).

^bSingle letter code: I, inosine; Y, C or T; R, A or G; S, C or G; W, A or T.

^cUse of primers D6f and K1r with test strain DNA should amplify a fragment carrying *vnfG*; primers D7f and K2R should amplify a fragment carrying *anfG* (see Fig. BXII.γ.139). For PCR amplification conditions, see Loveless and Bishop (1999).

containing media, and is unreliable for strain differentiation. Diffusible pigments are readily produced on the following iron-deficient media (Thompson and Skerman, 1979). Any one of the three basal agar media⁹ is modified by omission of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ and reduction of $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$ to 1 $\mu\text{g}/\text{ml}$. Glucose is added to give a concentration of 10 g/l. The medium is inoculated by depositing a small drop of a suspension of culture on the surface of the agar. The plates are examined in daylight for diffusible pigment and under ultraviolet light (wavelength 364 nm) for fluorescent pigments. The medium of Stainer and Scholte (1970) can also be used for the production of nondiffusible pigments or, for those strains that will not grow on this medium, the basal medium of Thompson and Skerman can be used when enriched with sodium gluconate (2.0 g/l).

DIFFERENTIATION OF THE GENUS *AZOTOBACTER* FROM OTHER GENERA

Phenotypic characteristics can differentiate the genus *Azotobacter* from others. Genera most closely related taxonomically, i.e., found in the *Gammaproteobacteria* class, order *Pseudomonadales*, family *Pseudomonadaceae*, are *Azomonas* and several others for which no nitrogen-fixing species/strain has been identified, except for certain isolates of *Pseudomonas stutzeri*. Like *Azotobacter*, *Azomonas* species can fix nitrogen under aerobic conditions while *P. stutzeri* cannot. *Azotobacter* species can utilize urea as nitrogen source and form cysts in older cultures or after growth in butanol (or propanol) while *Azomonas* species cannot, although general cultural and morphological characteristics are similar. In addition, among genera able to grow aerobically on dinitrogen, which includes not only *Azotobacter* and *Azomonas*, but also *Beijerinckia* and *Derrxia*, members of the *Alphaproteobacteria* class, the latter two genera are morphologically and physiologically different from *Azotobacter*. Distinguishing characteristics are summarized in Table BXII.γ.127.

TAXONOMIC COMMENTS

One new species, *A. salinestris*, has been added to the genus since the last edition of the *Manual*. Several isolates of this species were previously classified as strains of *A. chroococcum*, and then distinguished on the basis of Na-dependent growth under most conditions and decreased aerotolerance during diazotrophic growth (Page and Shivprasad, 1991a).

In this edition of the *Manual*, *Azotobacter* and *Azomonas* are classified in the phylum *Proteobacteria*, class *Gammaproteobacteria*,

order *Pseudomonadales*, and family *Pseudomonadaceae*. In the previous edition of the *Manual*, these two genera comprised the family *Azotobacteraceae* (Tchan, 1984). 16S rDNA sequencing of species of the two genera indicated that the family was distinct from, but closely related to, the fluorescent pseudomonads in *Pseudomonas* Superfamily I within the gamma group of *Proteobacteria* (MacDonald and Melton, unpublished studies). By the least squares distance matrix method, *Azotobacter* strains formed an assemblage into one group with a close association to *Azomonas macrocytogenes*, and clusters of pseudomonads in another group; both groups were stemmed to *E. coli* with an evolutionary distance of 0.01 or less (Fig. BXII.γ.139). The maximum parsimony tree exhibited a limited outcropping of organisms (Fig. BXII.γ.140). This can be interpreted as a hierarchy of phylogeny, which begins with the stem organism *E. coli*, progressively branches off the pseudomonads and *Azomonas macrocytogenes*, then clusters the *Azotobacter* species. The parsimony tree had a higher average bootstrap confidence level than the least squares distance tree, 89.5% vs. 75.8%, respectively (MacDonald and Melton, manuscript submitted).

Phylogenetic analysis using other genes or features have placed *A. vinelandii* and *Azomonas macrocytogenes* in the *Pseudomonas* group I lineage with *P. aeruginosa*. These studies included a comparison of four *P. aeruginosa* alginate (*alg*) genes to those from species of the *Pseudomonas* group I, group V, and enteric lineages (Fialho et al., 1990). The comparison indicated the greatest homology with the species of *Pseudomonas* group I, which included *A. vinelandii* but not *P. stutzeri*—which did not have an *algD* gene—or *Azomonas macrocytogenes*—which also did not show homology at this locus. A comparison of aromatic amino acid biosynthetic pathways also indicated the same evolutionary relationship between the group I pseudomonads and the species of *Azotobacteraceae* (Byng et al., 1986); this was consistent with groupings previously defined by rRNA–DNA hybridization studies. As with the rDNA analysis (De Smedt et al., 1980), immunoelectrophoresis—performed with antisera prepared to each of several strains and tested against one another (Tchan et al., 1983)—indicated that the six *Azotobacter* species were more similar to each other than to any *Azomonas* species and that the three *Azomonas* species were as distant from each other as each is from *Azotobacter* (Tchan et al., 1983).

A proposed reclassification of *Azotobacter paspali* to a new genus, *Azorhizophilus* (Thompson and Skerman, 1979), is not justified because this species clearly belongs to the *Azotobacter* genus based on 16S rDNA relatedness (Figs. BXII.γ.139 and BXII.γ.140). Moreover, as in *A. vinelandii*, *A. paspali* carries genes for the synthesis of all three types of nitrogenase. Another genus, *Azomonotrichon*, was suggested for the single species *A. macrocytogenes*, which was originally classified as an *Azomonas* species based on numerical analysis of common features (Thompson and Skerman, 1979). Reclassification as "*Azomonotrichon macrocytogenes*" is not supported by 16S rRNA gene sequence analysis, im-

9. Basal agar media (as cited in Thompson and Skerman, 1979): The medium of Norris and Jensen (1958) consists of (g/l): K_2HPO_4 , 1.0; $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, 0.1; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.2; $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, 0.05; $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$, 0.005; and Ionagar (Oxoid), 10.0; pH 7.3. An acidic modification of this medium consists of (g/l): K_2HPO_4 , 0.3; KH_2PO_4 , 0.7; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.2; $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, 0.05; $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$, 0.005; and Ionagar (Oxoid), 10.0; pH 6.2. An enriched modification of the medium consists of Norris and Jensen's medium containing 0.5% yeast extract (Difco).

TABLE BXII.γ.127. Differential characteristics of the genus *Azotobacter* and other aerobic diazotrophic genera^a

Characteristic	<i>Azotobacter</i>	<i>Azomonas</i>	<i>Beijerinckia</i>	<i>Derrxia</i>
Dumbbell-shaped cells with polar granules	—	—	+	—
Monotrichous flagellum	—	D	—	+
Cyst formation	+	—	—	—
Autotrophic growth on N_2	—	—	—	+
Mol% G + C of DNA	63.2–67.5	52–58.6	55–61	69–73

^aFor symbols see standard definitions.

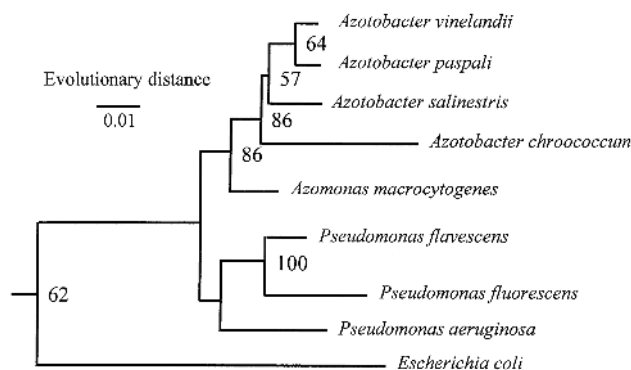


FIGURE BXII.γ.139. Least squares distance matrix tree showing phylogenetic relationships of *Azotobacter* and *Azomonas*. The tree is representative of the best fit of 100 data sets by the least squares distance algorithm and 100 bootstrapped pseudosamples. The evolutionary distance is 0.01. Bootstrap confidence values of 50% or greater are represented at each node of the tree (Courtesy of M.L. MacDonald and T.T. Melton, unpublished studies).

munological analysis (Tchan et al., 1983), and comparison of protein profiles (Jana et al., 1992). Previous inclusion of the nitrogen-fixing genera *Beijerinckia* and *Derxia* in the family *Azotobacteraceae* is now fully discounted by 16S rRNA gene sequence analysis. In this edition of the *Manual*, *Beijerinckia* is included in the class *Alphaproteobacteria*, and *Derxia* is placed in the class *Betaproteobacteria*, but distant from *Azotobacter* or *Azomonas* species.

Most species of *Azotobacter* and *Azomonas* were not extensively studied after 1979 with the exception of *A. vinelandii** and *A. chroococcum*, as reviewed in this chapter. A detailed history of the species, together with cultural, morphological, and physiological

*Editorial Note: During 2002 and 2003, a rough draft of the sequence of the genome of *A. vinelandii* became available through the U.S. Department of Energy Joint Genome Institute. Automated annotation of this genome has been carried out at Oak Ridge National Laboratories by Frank Larrimer and associates and at The Institute for Genome Research. A fully annotated finished genome may be available in 2004.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *AZOTOBACTER*

The differential characteristics of the species of *Azotobacter* are presented in Table BXII.γ.123. Other characteristics of the species are indicated in Table BXII.γ.124.

List of species of the genus *Azotobacter*

1. ***Azotobacter chroococcum*** Beijerinck 1901, 567^{AL}.
chro.o.coc' cum. Gr. n. *chroa* color; Gr. n. *coccus* a grain; M.L. neut. n. *chroococcum* colored coccus.

The morphology is shown in Figs. BXII.γ.128 and BXII.γ.129. Cells appear as blunt rods to ellipsoid forms, 1.6–2.5 μm in diameter and 3–5 μm in length. Chains of cells are occasionally observed in some strains. Cysts (sometimes referred to as “microcysts”) are formed in old cultures or after addition of isopropanol to exponential cultures. Cells have randomly placed granules, described as meta-chromatic or sudanophilic (Thompson and Skerman, 1979). Motile cells from 1- to 2-d-old cultures have peritrichous flagella with a wavelength of 2.5–3.3 μm and an amplitude of 0.37–0.76 μm. Colonies are opaque and convex, and, for most strains, are mucoid, glistening, and

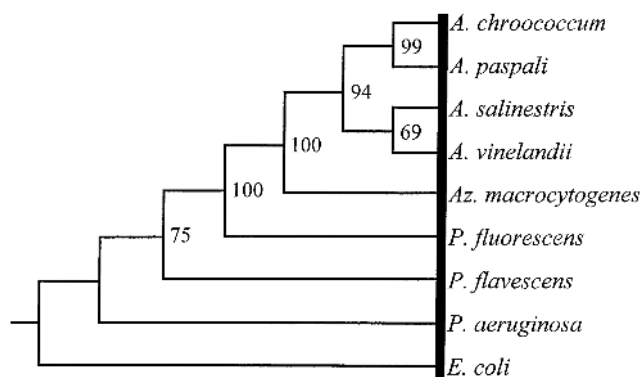


FIGURE BXII.γ.140. Maximum parsimony tree showing phylogenetic relationships of *Azotobacter* and *Azomonas*. The tree is representative of the best fit of 100 data sets by the maximum parsimony algorithm and 100 bootstrapped pseudosamples. Tree branch length is not significant for this algorithm. Bootstrap confidence values of 50% or greater are represented at each node of the tree (Courtesy of M.L. MacDonald and T.T. Melton, unpublished studies).

characteristics of approximately 90 isolates, can be found in Thompson and Skerman (1979) and references therein. See also Becking (1992) for further descriptive information.

ACKNOWLEDGMENTS

We thank Adam Field and Angela Smedke for assistance in compiling DNA sequences for analyzing codon usage and mol% G + C content. We also thank Mali Gunatilaka for help at many stages of preparation of the manuscript. Finally, we would like to dedicate this chapter to Thoyd Melton who passed away before its publication. His contributions to the studies of the genetics and physiology of *Azotobacter vinelandii* were significant. He was a devoted scientist who was ardent about research, teaching, and his fellow colleagues.

FURTHER READING

Becking, J.H. 1992. The family *Azotobacteraceae*. In Balows, Trüper, Dworin, Harder and Schleifer (Editors), *The Prokaryotes*, 2nd ed., Vol. 3, Springer-Verlag, New York. 3144–3170.

smooth. A brown-black nondiffusible pigment is produced in aging colonies. Many organic compounds can be used as carbon sources. Distinguishing physiological and nutritional characteristics of the species are presented in Tables BXII.γ.123 and BXII.γ.124. N₂, ammonium, nitrate, and urea are used as nitrogen sources. N₂ is fixed under aerobic or microaerobic conditions. Ammonium prevents N₂ fixation by inhibition of expression of *nif* genes and possibly by inactivation of Mo nitrogenase (nitrogenase-1). In three strains tested, one “alternative” nitrogenase is synthesized when Mo supply is very low (nitrogenase-2, carrying the Fe-V cofactor). This correlates with the presence of *vnf* but not *anf* genes in these strains. For other characteristics, less distinguishing, see Thompson and Skerman (1979). The genome size varies between 3.1 and 5.3 Mb. There are 6 or 7 copies of the *rrn* operon.

The mol% G + C of the DNA is: 65.8–67.5 (T_m).

Type strain: R. L. Starkey's strain 43, ATCC 9043, DSM 2286, WR-88.

2. **Azotobacter armeniacus** Thompson and Skerman 1981, 215^{AL} (Effective publication: Thompson and Skerman 1979, 272 (*Azotobacter agilis* subsp. *armeniae* Kirakosyan and Melkonyan 1964, 41; *Azotobacter vitreus* subsp. *armeniae* Kirakosyan and Melkonyan 1964, 41.)

ar.me.ni.a'cus. M.L. neut. n. *armeniacus* named after Armenia, U.S.S.R., where the species was isolated.

The morphology is as described for the genus and most other species. Cells are motile with peritrichous flagella having a wavelength of 2.0–2.9 μm and an amplitude of 0.40–0.59 μm . Colonies are opaque, mucoid, and more highly convex than those of other species. Brown-black to red-violet water-soluble pigments are produced. No strain has been tested for the presence of alternative nitrogenases. Distinctive features include inability to utilize nitrate or ammonium for growth, but strains are urease positive. Physiological and nutritional characteristics of the species are presented in Tables BXII. γ .123 and BXII. γ .124.

The mol% G + C of the DNA is: 63.5–65.0 (T_m).

Type strain: N28, DSM 2284, WR-136.

3. **Azotobacter beijerinckii** Lipman 1904, 248^{AL}
bei.jer.inck'i.i. M.L. gen. n. *beijerinckii* of Beijerinck; named after M.W. Beijerinck of Delft, the Dutch microbiologist.

The morphology of cells is similar to that of *A. chroococcum* and can vary depending on culture conditions. The cells are nonmotile and lack flagella. Colonies are smooth, but variants may arise due to the quantity of extracellular polysaccharide produced. Growth at 37°C is poor or does not occur. A yellowish or cinnamon-colored pigment is produced in aging cultures. Many isolates are acid tolerant. Physiological and nutritional characteristics of the species are presented in Tables BXII. γ .123 and BXII. γ .124. Ammonium and nitrate are used as nitrogen sources. Type strain ATCC 19360 carries both *nif* and *vnf* genes but not *anf*, indicating that *A. beijerinckii* can synthesize nitrogenase-1 and nitrogenase-2.

The mol% G + C of the DNA is: 66.0–66.2 (T_m).

Type strain: ATCC 19360, DSM 378, NCIB 8948, WR-135.

Two subgroups of *A. beijerinckii* have been delineated. The acid-tolerant subgroup corresponds to "*A. beijerinckii* subsp. *acidotolerans*" Tchan. The other subgroup is distinguished by its sensitivity to 0.05% phenol or 40 $\mu\text{g}/\text{ml}$ of diamond fuchsin, its inability to use sorbitol or aconitate, and its failure to produce diffusible homopolysaccharides.

4. **Azotobacter nigricans** Krasil'nikov 1949, 506^{AL}
ni'gri.cans. L. adj. *nigricans* black.

The cells (Fig. BXII. γ .130) are round-ended rods, as are typical of other species of *Azotobacter*. The cells are nonmotile and lack flagella. Colonies are smooth, but variants occur due to the quantity of extracellular polysaccharide produced. Most colonies are opaque but can be translucent. Pigments are produced and are usually red-violet. Growth at 37°C is poor or does not occur. Strains have been isolated from soils in Eastern Europe. Physiological and nutritional characteristics are presented in Tables BXII. γ .123 and BXII. γ .124. Ammonium and nitrate are used as nitrogen source in preference to nitrogen. The type strain carries

nif and *vnf* but not *anf* genes, indicating an ability to synthesize nitrogenase-1 and nitrogenase-2.

The mol% G + C of the DNA is: 64.5 (T_m).

Type strain: ATCC 35009, DSM 2288, UQM 1967.

- a. **Azotobacter nigricans** subsp. **nigricans** Krasil'nikov 1949, 506^{AL}

In aging colonies, a brown-black water-soluble pigment is usually produced.

The mol% G + C of the DNA is: 64.5 (T_m).

Type strain: ATCC 35009, DSM 2288, UQM 1967.

- b. **Azotobacter nigricans** subsp. **achromogenes** Thompson and Skerman 1981, 215^{VP} (Effective publication: Thompson and Skerman 1979, 12.)

a.chro.mo'ge.nes. Gr. pref. *a* not; Gr. n. *chroma* color; Gr. suff. *-genes* born of; N.L. neut. adj. *chromogenes* not producing color.

Aging colonies produces only a yellow nondiffusible pigment within the colony.

The mol% G + C of the DNA is: 64.5 (T_m).

Type strain: A-2 of Jensen (WR-41).

5. **Azotobacter paspali** Döbereiner 1966, 364^{AL} (*Azorhizophilus paspali* (Döbereiner 1966) Thompson and Skerman 1981, 215.)

pas.pal'i. M.L. gen. n. *paspali* of Paspalum; named after *Paspalum*, the generic name of a grass.

The general morphology is similar to that of other *Azotobacter* species, but the cells are longer than usual, 5–10 μm in the exponential phase. Long filamentous cells up to 60 μm can occur (Fig. BXII. γ .131). Cells are motile by numerous peritrichous flagella having a wavelength of 2.3–2.8 μm and an amplitude of 0.40–0.59 μm (Fig. BXII. γ .133). Some strains have "curly" flagella with a wavelength of 1.1–1.4 μm and an amplitude of 0.25–0.34 μm . Colonies are opaque with an entire or undulate edge, raised, usually not evenly convex, and butyrous with a dull or rough surface. Physiological and nutritional characteristics are presented in Tables BXII. γ .123 and BXII. γ .124. Ammonium but not glutamate can be used as the sole nitrogen source for growth. Thompson and Skerman (1979) reported the ability of *A. paspali* isolates to utilize nitrate as nitrogen source but failed to detect nitrite production from nitrate or nitrite disappearance in standard tests. This conclusion should be reexamined because nitrate assimilation is not known to occur other than by nitrate reduction to nitrite and nitrite reduction to ammonium. Urea is hydrolyzed. Genes for both alternative nitrogenases (nitrogenase-2 *vnf* and nitrogenase-3 *anf*) were detected in two strains (the type strain ATCC 23833 and also ATCC 23367). Distinctive features include (1) production of a yellow-green fluorescent pigment (as for *A. vinelandii*); (2) utilization of fewer compounds as carbon sources than other *Azotobacter* species. The carbon sources used include maltose, trehalose, melibiose, raffinose, mannitol, and others (Thompson and Skerman, 1979). The organism has never been isolated from environments other than the rhizosphere of the grass *Paspalum notatum* cv. Batatais. Genome size is uniform, 4.3–4.6 Mb in four strains tested, as is number of copies (six) of the *rrn* operon.

The mol% G + C of the DNA is: 63.2–64.6 (T_m).

Type strain: Ax-8, ATCC 23833, DSM 2283, WR-129.

6. *Azotobacter salinestris* Page and Shivprasad 1991b, 374^{VP} *sal.in.es' tris*. N.L. adj. *salinus* saline; L. suff. *-estris* belonging to, living in; N.L. adj. *salinestris* living in saline environments.

The morphology is described as coccobacilli (oval with pointed ends); cells are approximately $2 \times 3\text{--}4\text{ }\mu\text{m}$ in young or exponential cultures. Older cells are round with a diameter of $3\text{--}5\text{ }\mu\text{m}$, pleomorphic, and are often found in pairs and chains of six-to-eight cells. Cysts are formed. Cells are motile by means of peritrichous flagella $9\text{--}10\text{ }\mu\text{m}$ long, with a wavelength of $2.7\text{--}2.9\text{ }\mu\text{m}$ and an amplitude of $0.36\text{--}0.55\text{ }\mu\text{m}$. Cells from old cultures are nonmotile. Physiological characteristics include growth at temperatures between 30° and 36°C . The optimal temperature for N_2 fixation is 35°C . Cells and colonies grow at pH 6.2–8.0. The optimal pH for growth on N_2 is 7.2; on nitrate it is pH 7.3, and on ammonium it is pH 7.5. N_2 fixation occurs best in microaerophilic cultures, but cells can be adapted to aerobic diazotrophy. Most strains, except the type strain, produce a capsule and synthesize poly- β -hydroxybutyrate. The type strain, ATCC 49674, and strain 253 contain *nif* and *vnf* genes and can fix nitrogen in the absence of Mo with V. Growth under most conditions requires the presence of 1 mM Na^+ . Strains are urease positive, amylase positive, and catalase positive (with Na^+). No fluorescent compounds are synthesized, even when cultures are Fe limited. Carbon sources include fructose, galactose, glucose, sucrose, mannitol, melibiose, 0.25% sodium benzoate, and starch. Nitrogen sources are N_2 , ammonium, and nitrate. A brown-black to tan-brown pigment, allomelanin (a catechol melanin), is produced by most strains when grown at high aeration under N-fixing conditions. Growth is inhibited by K^+ and Rb^+ , with Rb^+ having the greatest effect. For other characteristics, see Tables BXII.γ.123 and BXII.γ.124.¹⁰

The mol% G + C of the DNA is: 65.19–65.98 (T_m).

10. Not all traits tested by Thompson and Skerman (1979) for the other *Azotobacter* species have been tested in *A. salinestris*.

Type strain: 184, ATCC 49674, DSM 11553.

GenBank accession number (16S rRNA): AF035213.

7. *Azotobacter vinelandii* Lipman 1903, 238^{AL} (*Azotobacter miscellum* Pshenin 1964, 617.)

vine.lan' di.i. M.L. gen. n. *vinelandii* of Vineland; named after Vineland, New Jersey, where the species was first isolated.

The morphology is similar to that of *A. chroococcum* and can vary from rods to more coccoid forms. The cells typically are $1.6\text{--}2.5 \times 3\text{--}5\text{ }\mu\text{m}$. Ammonium-grown cells are more coccoid in stationary phase (Fig. BXII.γ.133). Cysts are formed in old cultures or after addition of butanol to exponential cultures. Motile cells from 1- to 2-d-old cultures have peritrichous flagella with a wavelength of $2.0\text{--}3.0\text{ }\mu\text{m}$ and an amplitude of $0.40\text{--}0.59\text{ }\mu\text{m}$. The velocity of motile cells can be $74\text{ }\mu\text{m/sec}$. A few strains are nonmotile. Colonies are opaque and convex, and, for most strains, are mucoid, glistening, and smooth. Distinguishing physiological and nutritional characteristics of the species are presented in Tables BXII.γ.123 and BXII.γ.124; utilization of rhamnose and phenol as carbon sources is distinctive. A yellow-green fluorescent pigment is produced, especially under conditions of Fe limitation. Other diffusible pigments can also be produced. N_2 , ammonium, nitrate, and urea, but not glutamate, are used as nitrogen sources. N_2 is fixed under aerobic or microaerobic conditions. Ammonium prevents N_2 fixation by inhibition of expression of *nif* genes and possibly by inactivation of Mo nitrogenase (nitrogenase-1, carrying the Fe–Mo cofactor). In strain AvOP (ATCC 13705), two “alternative” nitrogenases can be synthesized when Mo supply is very low (nitrogenase-2, carrying the Fe–V cofactor and nitrogenase-3, with a Fe–Fe cofactor, synthesized in the absence of Mo and V). This correlates with the presence of *vnf* and *anf* genes. For other characteristics, less distinguishing, see Thompson and Skerman (1979). The genome size is 4.6 Mb. Six copies of the *rrn* operon are present.

The mol% G + C of the DNA is: 64.9–66.5 (T_m).

Type strain: ATCC 478, DSM 2289, WR-100.

Genus IV. *Cellvibrio* Blackall, Hayward and Sly 1986, 354^{VP} (Effective publication: Blackall, Hayward and Sly 1985, 95)

THE EDITORIAL BOARD

Cell.vib' ri.o. M.L. n. *cell* abbreviation of M.L. n. *cellulosum* cellulose, which is degraded by the organism; M.L. masc. n. *Vibrio* a generic name; M.L. masc. n. *Cellvibrio* cellulose (degrading) vibrio.

Aerobic Gram-negative **slender curved rods** ($0.2\text{--}0.5 \times 1.0\text{--}1.3\text{ }\mu\text{m}$). Motile with **mixed flagellation**. **Oxidize glucose and hydrolyze cellulose. Produce curdian polysaccharide from glucose.** Catalase and oxidase positive. Do not utilize organic acids. No growth factors required.

The mol% G + C of the DNA is: 46.1–54.1.

Type species: *Cellvibrio mixtus* Blackall, Hayward and Sly 1986, 354 (Effective publication: Blackall, Hayward and Sly 1985, 95.)

FURTHER DESCRIPTIVE INFORMATION

Cells possess mixed flagellation, with up to 11 lateral flagella; the lateral flagella, which are not always present, have a smaller diameter and a shorter wavelength than the single polar flagellum.

The enzymes involved in the degradation of polysaccharides and in the metabolism of the degradation products have been studied in *Cellvibrio mixtus*, “*Cellvibrio vulgaris*”, “*Cellvibrio fulvus*”, and “*Cellvibrio gilvus*”, but the taxonomic position of the latter three strains has not been determined (see Taxonomic Comments).

ENRICHMENT AND ISOLATION PROCEDURES

Enrichment and isolation can be carried out by incubation of rhizosphere soil samples in liquid mineral salts medium containing cellulose or dextran as a carbon source followed by streaking onto agar medium containing the same carbon source (Blackall et al., 1985).

MAINTENANCE PROCEDURES

Strains can be stored on slants of sucrose peptone or peptone yeast extract agar at 7°C (Blackall et al., 1985).

TAXONOMIC COMMENTS

The genus *Cellvibrio* was first described by Winogradsky (1929), but did not appear on the Approved Lists (Skerman et al., 1980). Blackall et al. (1985) revived the genus based on the description of a number of newly isolated cellulose- and dextran-hydrolyzing strains which they compared to selected *Pseudomonas*, *Aeromonas*,

Vibrio, *Xanthomonas*, *Cytophaga*, and *Cellulomonas* strains as well as to strains of "*Cellvibrio vulgaris*" and "*Cellvibrio fulvus*". In a numerical taxonomic analysis of phenotypic characters the new isolates clustered with each other and with "*Cellvibrio vulgaris*" and "*Cellvibrio fulvus*" but separately from all the other organisms. However, the new isolates differed from the "*Cellvibrio vulgaris*" and "*Cellvibrio fulvus*" strains and from the organisms described by Winogradsky in failing to produce pigment on cellulose agar; they also differed from "*Cellvibrio vulgaris*" and "*Cellvibrio fulvus*" in the mol% G + C of the DNA (50.1 ± 4.0 for the new isolates and 44.9 and 44.6 for the latter two strains).

List of species of the genus *Cellvibrio*

1. ***Cellvibrio mixtus*** Blackall, Hayward and Sly 1986, 354^{VP} (Effective publication: Blackall, Hayward and Sly 1985, 95.) *mix' tus*. M.L. masc. adj. *mixtus* mixed referring to the type of flagellation.

On semi-opaque mineral salts–cellulose medium all strains form circular white colonies with entire edges surrounded by a zone of clearing. Produce acid from L-arabinose, dextrin, galactose, glucose, lactose, maltose, mannose, melibiose, raffinose, sucrose, trehalose, and xylose. Hydrolyze chitin, esculin, pectate (at neutral and alkaline pH only), and starch, but not casein, DNA, gelatin, or Tween 80. Do not produce gas from nitrate, H₂S from cysteine or indole from tryptophan. Voges–Proskauer and methyl red tests negative. Produce α -glucosidase, β -glucosidase, and β -xylosidase.

The mol% G + C of the DNA is: 53–55 (T_m).

Type strain: UQM 2601.

- a. ***Cellvibrio mixtus* subsp. *mixtus*** Blackall, Hayward and Sly 1986, 354^{VP} (Effective publication: Blackall, Hayward and Sly 1985, 95.)

Properties are as for the species with the following additions: produce acid from cellobiose and melezitose. Produce β -glucuronidase and β -galactosidase. Do not hydrolyze dextran. Gram-negative; cells soluble in 3% KOH.

The mol% G + C of the DNA is: 52.6 (T_m).

Type strain: UQM 2601.

GenBank accession number (16S rRNA): AF448515.

- b. ***Cellvibrio mixtus* subsp. *dextranolyticus*** Blackall, Hayward and Sly 1986, 354^{VP} (Effective publication: Blackall, Hayward and Sly 1985, 95.) *dex.trano.lyticus*. *dextran* dextran, a chemical; Gr. adj. *lyticus* dissolving; M.L. masc. adj. *dextranolyticus* dextran dissolving.

Properties are as for the species with the following additions: hydrolyze dextran and produce acid from fructose. Gram-negative, but cells not soluble in 3% KOH.

The mol% G + C of the DNA is: 55.2 (T_m).

Type strain: UQM 1666.

Genus V. *Mesophilobacter* Nishimura, Kinpara and Iizuka 1989, 380^{VP}

THE EDITORIAL BOARD

Me.so.phi.lo.bac'ter. Gr. n. *mesos* middle; Gr. adj. *philus* loving; M.L. *bacter* rod, staff; M.L. masc. n. *Mesophilobacter* mesophilic rod.

Gram-negative nonmotile aerobic pleomorphic rods. (0.5–0.6 $\times$ 1.0–2.0 μ m). Catalase positive; oxidase negative. Nonsporeforming, non-acid-fast, nonfermentative, nonencapsulated. Resistant to penicillin. **Optimum growth at 33–37°C.**

The mol% G + C of the DNA is: 44.0–46.9.

Type species: Mesophilobacter marinus Nishimura, Kinpara and Iizuka 1989, 380.

FURTHER DESCRIPTIVE INFORMATION

The strains were isolated from seawater (Nishimura et al., 1986b, 1989).

Cells of young cultures may be stout (0.8–1.0 $\times$ 1.5–3.0 μ m) or may be elongated up to 15 μ m in length. Cells in stationary

cultures are coccoid with diameters 0.6–0.9 μ m (Nishimura et al., 1989).

ENRICHMENT AND ISOLATION PROCEDURES

Strains of this species were isolated from seawater on seawater agar or modified nutrient agar as described by Nishimura et al. (1986b).

DIFFERENTIATION OF THE GENUS *MESOPHILOBACTER* FROM OTHER GENERA

Nishimura et al. (1989) provide a table of characteristics that differentiate *Mesophilobacter* from the genera *Moraxella*, *Acinetobacter*, and *Psychrobacter*.

List of species of the genus *Mesophilobacter*

1. ***Mesophilobacter marinus*** Nishimura, Kinpara and Iizuka 1989, 380^{VP} *ma.ri'nus*. L. adj. *marinus* of the sea, marine.

Description as for the genus with the following additional characteristics. No change in litmus milk. No H₂S produced. Voges–Proskauer test negative. Methyl red test

positive. Reduces nitrate to nitrite. Assimilates acetate, citrate, D-fructose, fumarate, gluconate, D-glucose, malate, maltose, D-ribose, succinate, sucrose, and trehalose. Produces acid from cellobiose, D-fructose, D-glucose, D-man-

nitol, and D-ribose. Grows in presence of 7% NaCl. Major fatty acids C_{16:0}, C_{16:1}, and C_{18:1}.

The mol% G + C of the DNA is: 44.0–46.9 (HPLC).

Type strain: 14S-4, DSM 9142, IAM 13185, NCIMB 13183.

Genus VI. Rhizobacter Goto and Kuwata 1988, 238^{VP}

MASAO GOTO

Rhi.zo.bac'ter. Gr. n. *rhiza* a root; M.L. masc. n. *bacter* masc. equivalent of Gr. neut. n. *bacterion* rod, staff; M.L. masc. n. *Rhizobacter* root rod.

Straight to slightly curved, nonsporeforming, encapsulated rods 0.9–1.3 × 2.1–2.5 μm. **Poly-β-hydroxybutyrate granules are formed.** Cells stain Gram negative. **Motile by polar flagella or lateral flagella or both** but motile cells are rare in the populations. **Aerobic** with respiratory metabolism of glucose. Good growth is obtained at 28–30°C; no growth at 35°C. Growth occurs at pH between 5.0 and 9.0. Utilizes D-glucose as a sole source of carbon and energy. **White or yellowish white, plicated, tough, or viscid colonies on agar plates;** if yellow, the pigment differs from xanthomonadines. **Floccular growth consisting of globular units in liquid media; fingerlike projections never occur.** **Oxidase and catalase positive.** The following are positive: H₂S from cysteine and auxin, reduction of nitrate to nitrite, *o*-nitrophenyl-β-D-ga-

lactopyranoside (ONPG) test, and growth inhibition by KCN. Negative for denitrification, methyl red, and production of arginine dihydrolase, nitrogenase, fluorescent pigment, indole, and acetoin formation. Susceptible to 10 μg of vibriostatic agent O/129 phosphate. **Hydrolyzes starch, dextrin, and glycogen.** Cannot utilize benzene derivatives as carbon sources. Does not require NaCl or growth factors. Contains ubiquinone Q8. The type species occurs in soil and is a plant pathogen, causing galls on carrot roots in nature.

The mol% G + C of the DNA is: 67–71.

Type species: ***Rhizobacter dauci* corrig.** Goto and Kuwata 1988, 238 (*Rhizobacter daucus* (sic) Goto and Kuwata 1988, 238.)

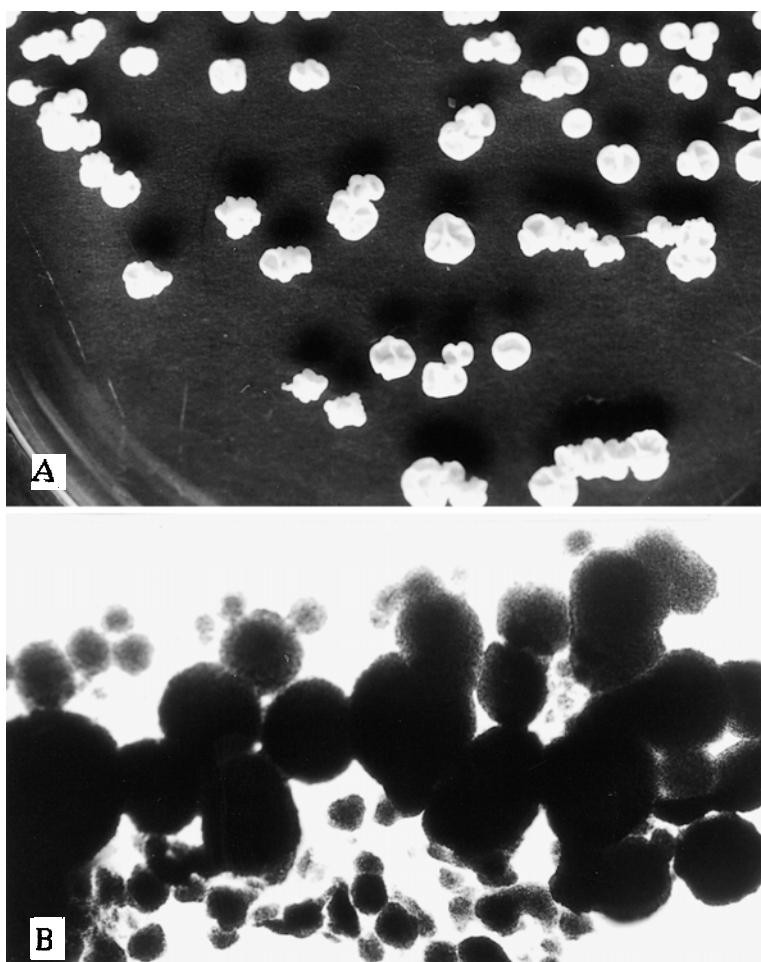


FIGURE BXII.γ.141. Colony morphology of *R. dauci* H6^T. (A) Plicated colonies of wild type on DPPG agar. (B) Globular aggregates of bacterial cells in DPPG liquid medium. (× 100)

FURTHER DESCRIPTIVE INFORMATION

The growth on agar plates depends on the concentrations of ingredients: growth is best on a 10-fold dilution of potato-peptone-glucose (DPPG) agar, a 10-fold dilution of yeast extract-peptone-glucose (DYPG) agar, or Casitone-yeast extract agar¹. This oligotrophic trait seems to be due to the presence of peptone, which significantly retards the growth of the bacterium.

Although this bacterium is aerobic, it can grow slowly under anaerobic conditions. After several days, the colonies on DPPG agar plates are 1.5–2.0 mm in diameter, white, opaque, tough, and pulvinate or capitate with entire margins. Later the colonies develop highly plicated surfaces and undulating margins (Fig. BXII.γ.141A).

Plication is less distinct or absent when glucose is omitted from the medium. The colonies on DYPG agar plates are yellowish white but otherwise identical to those on DPPG agar plates. The colonies are so tough and adhere so firmly to the agar surfaces that they are difficult to remove with a platinum loop. The agar surface is slightly depressed for about 1–2 mm around a colony.

Growth in liquid media is floccular, and the medium never clouds uniformly. The flocs are composed of globular aggregates of bacterial cells 60–200 μm in diameter (average, 120 μm) (Fig. BXII.γ.141B). The flocs adhere to the test tube wall in the early stages of growth and then develop into a film or pellicle, which eventually fall to the tube bottom.

The results of the oxidation-fermentation test are not clear-cut. Because of the slow growth, as well as the considerable amount of growth and acid production that take place in the tubes sealed with mineral oil, the oxidation-fermentation test may be misread in the early stages of growth. After several days of incubation, however, growth and acid production in the open tubes become conspicuously greater than growth and acid production in the sealed tubes, making determination of oxidative metabolism easier.

This bacterium uses diverse kinds of compounds as a sole source of carbon: L-arabinose, D-arabinose, xylose, fructose, galactose, glucose, lactose, maltose, melibiose, sucrose, raffinose, carboxymethylcellulose, dextrin, glycogen, adonitol, inositol, mannitol, sorbitol, salicin, gluconate, mucate, quinate, and tartrate. The following are used weakly: isoleucine, leucine, methanol, and propanol.

Other physiological and biochemical characteristics include positive reactions for gelatinase, lecithinase, urease, pit formation on pectate gel, hydrolysis of Tween 80 and esculin, production of H₂S, and indoleacetic acid and/or its derivatives, and negative reactions for casein hydrolysis, tyrosinase, deoxyribonuclease, and phenylalanine deaminase. No detectable change occurs in litmus milk cultures.

Slimy colony mutants are often detected during the storage of agar slant cultures at room temperature. These mutants grow faster than the parents and form abundant viscous slime on agar media.

The bacterium causes bacterial carrot gall that is a typical soil-borne disease. In natural infection, galls develop along the entire

length of the storage roots from the crown to the root tip. Small galls may also be found on the fibrous roots. Galls vary in size from a few millimeters to approximately 1 cm in diameter (average, several millimeters). Large galls may be formed by the union of several smaller galls. Galls are often distributed along lenticles, in severe cases forming a ring around the root (Fig. BXII.γ.142A). The galls are light brown and have very rough surfaces. Another characteristic of the disease is the development of numerous adventitious fibrous roots.

On artificially inoculated carrot seedlings, hypertrophied areas become apparent on the wounded roots 7–14 d after inoculation, and outgrowths like those on naturally infected plants develop about 1 month after inoculation. The disease is particularly severe on the roots, which are injured by scratching the epidermis with a sterilized sharp needle. The bacterial cells occupy the narrow fissures formed in gall tissues, and deposit heav-

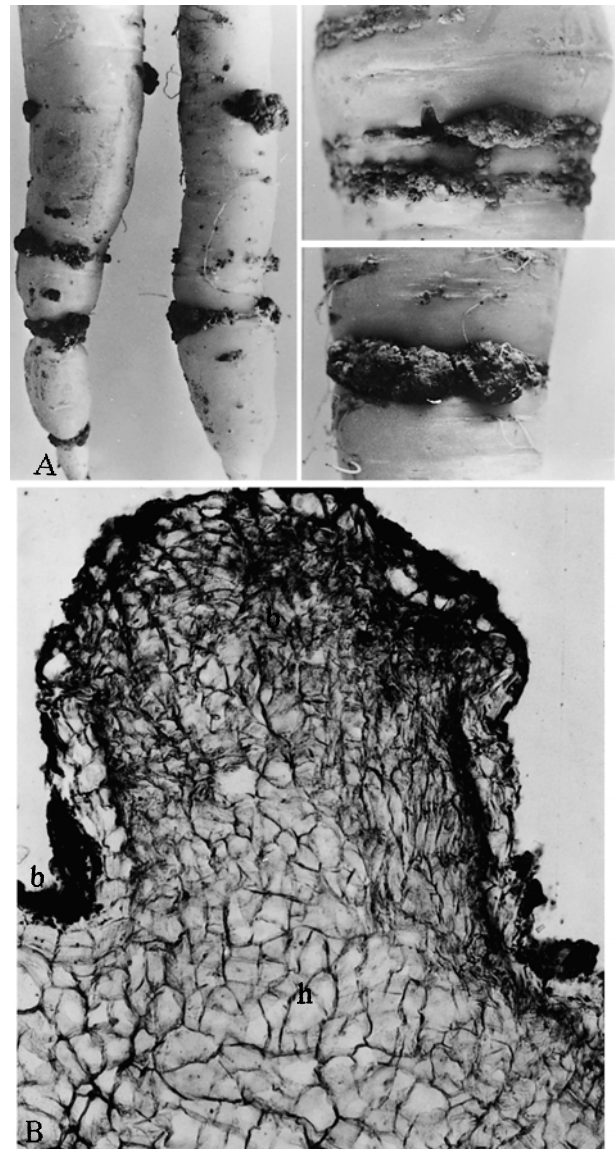


FIGURE BXII.γ.142. Characteristics of carrot bacterial gall. (A) Symptoms of galls formed on carrot roots. (B) Transverse section of an outgrowth showing hyperplastic tissues (*h*) and bacterial cells found in bacterial fissures and on gall surface (*b*).

1. DPPG agar consists of potato extract (20 g of potato in 1000 ml of water), 1000 ml; peptone, 1 g; glucose, 10 g; and agar, 15 g; pH 6.8. DYPG agar consists of peptone, 1 g; yeast extract, 0.5 g; glucose, 10 g; agar, 15 g; and water, 1000 ml; pH 6.8. Casitone-yeast extract agar consists of Casitone (Difco), 5 g; yeast extract, 2.5 g; agar, 15 g; and water, 1000 ml; pH 6.8.

ily on the surface of galls (Fig. BXII.γ.142B). When tomato seeds are immersed for several minutes in bacterial suspension and planted in sterilized soil, the conspicuous outgrowths develop along the main root, giving a thick club-shaped appearance and causing a remarkable growth inhibition of whole plant parts.

ENRICHMENT AND ISOLATION PROCEDURES

Isolation of *Rhizobacter dauci* is difficult because of its preference for oligotrophic media and its slow growth on agar plates, which can be overgrown by soil bacteria that contaminate the habitat. In practice, a small piece (approximately a 2-mm cube) of outgrowth including the surface is cut out of a fresh gall, washed briefly, macerated in 10 ml of sterilized distilled water, and subjected to 10-fold serial dilution with sterilized distilled water. The suspensions are plated onto a DPPG agar. The plates are incubated aerobically at 28°C. The bacterium is easily recognizable after 1 week or so because of its slow-growing, white, plicated, tough colonies. It may be further purified by single-colony isolation procedures.

MAINTENANCE PROCEDURES

For routine work, the bacterium may be stored either as the suspensions in sterilized distilled water at room temperature or as the stab cultures on DPPG agar medium at 4°C. Lyophilization is preferable for long-term preservation. For reviving the lyophilized cultures, freeze-dried cells are suspended in a few milliliters of sterilized distilled water and plated immediately on DPPG agar, and again after storing the suspension overnight at room temperature. The typical wild-type colonies with plicated surface are often formed on the latter plates.

DIFFERENTIATION OF THE GENUS *RHIZOBACTER* FROM OTHER GENERA

See Table BXII.γ.128.

TAXONOMIC COMMENTS

This bacterium is characterized by the lateral flagellation and the susceptibility to vibriostatic agent 0/129. These features have been known in the members of the family *Vibrionaceae*, facultative anaerobes (Baumann and Schubert, 1984), but are not common in the aerobes (Goto, 1988). Until the phylogenetic data became available, the bacterium was most appropriately placed in the family *Pseudomonadaceae* based on various other phenotypic characteristics.

Rhizobacter dauci resembles *Rhizomonas suberifaciens* in its preference for oligotrophic media, wrinkled colony formation, and polar or lateral flagellation. However, the former is clearly distinguished from the latter by the higher mol% G + C of the DNA, a different major ubiquinone, the large number of carbon sources utilized, and significantly larger dimensions of the cells (van Bruggen et al. 1990).

Rhizobacter dauci resembles *Zoogloea* spp. in formation of zoogloae but is differentiated by its plant pathogenicity, shape of the zoogloae, susceptibility to the vibriostatic agent 0/129 phosphate, lower tolerance to NaCl, lack of growth at 35°C, positive lecithinase reaction, lack of a litmus milk reaction, and a wider spectrum of carbon sources used (Crabtree and McCoy, 1967; Friedman and Dugan, 1968).

Colonies of *Rhizobacter dauci* resembles those of *Beijerinckia* spp. grown on nitrogen-free glucose mineral agar (Becking, 1984). However, *Beijerinckia* spp. are distinct from *R. dauci* in plant pathogenicity, positive nitrogenase activity, and the significantly lower mol% G + C.

List of species of the genus *Rhizobacter*

- 1. ***Rhizobacter dauci*** *corrig.* Goto and Kuwata 1988, 238^{VP} (*Rhizobacter daucus* (sic) Goto and Kuwata 1988, 238.)
dau'ci. M.L. n. *Daucus* generic name of carrot, host plant;
M.L. masc. gen. n. *dauci* of the carrot.

The characteristics are as described for the genus and as listed in Table BXII.γ.128. The morphology is as depicted as Fig. BXII.γ.141.
The mol% G + C of the DNA is: 67–71 (*T_m*).
Type strain: H6, ATCC 43778, ICMF 9400, LMG 9036.

TABLE BXII.γ.128. Differential characteristics of the genus *Rhizobacter* and other genera of the family *Pseudomonadaceae*

Characteristics	<i>Rhizobacter</i> ^a	<i>Rhizomonas</i> ^b	<i>Zoogloea</i> ^c	<i>Beijerinckia</i> ^d
Cell diameter, >1 μm	+	—	+	+
Lateral flagellation	+	+	—	+
Susceptibility to 0/157	+	nd	—	nd
Zoogloea	+ ^e	—	+ ^f	—
Acid production from glucose	+	+	—	+
Denitrification to N ₂	—	—	+	nd
Inhibition by peptone	+	+	+	+
Lecithinase activity	+	nd	— ^b	nd
Nitrogenase activity	—	—	nd	+
Carbon sources utilized	Wide	Narrow	Narrow	Wide
Isoprenoid quinone	Q8	Q10	nd	nd
Mol% G + C content of DNA	66.9–70.6	59	65.3	55–61
Plant Pathogenicity	+	+	—	nd

^aData from Goto and Kuwata (1988).

^bData from Van Bruggen et al. (1990).

^cData from Unz (1984).

^dData from Becking (1984).

^eGlobular zoogloea.

^fFingerlike zoogloea.

Genus VII. Rugamonas Austin and Moss 1987, 179^{AL} (Effective publication: Austin and Moss 1986, 1907)

MAURICE O. MOSS AND DAWN A. AUSTIN

Ru.gamo' nas. L. fem. n. *ruga* wrinkle; Gr. fem. n. *monas* unit, monad; M.L. n. *Rugamonas* wrinkled unit.

Rods 0.8–0.9 × 2.4–4.0 µm, in young cultures. **Intracellular granules of prodigiosenes develop in cells more than 7 d old.** Cells in cultures less than 7 d are motile by one or more polar or subpolar flagella. **Colonies on Bennett's agar¹ after 4 d at 20°C are pink to deep red, wrinkled, and rubbery in consistency after 5 d,** with a diameter of approximately 3 mm. Oxidase and catalase positive, strictly aerobic and oxidative metabolism, unable to metabolize carbohydrates fermentatively. Nitrates are reduced to N₂ anaerobically. Growth occurs in the presence of 0.0–0.5% (w/v) NaCl. Arginine decarboxylase negative. Lysine and ornithine decarboxylases are produced by some strains. Esculin, chitin, DNA, RNA, gelatin, lecithin, Tween 20, 40, 60, and 80, tyrosine, and urea are degraded but not allantoin, cellulose, elastin, hypoxanthine, starch, and xanthine. H₂S is not produced on triple sugar iron agar. The gluconate and Koser's citrate tests are positive; Voges-Proskauer, methyl red, and melonate tests are negative. β-Galactosidase (ONPG method) positive. Has been isolated from river water.

The mol% G + C of the DNA is: 66–67.

Type species: Rugamonas rubra Austin and Moss 1987, 179 (Effective publication: Austin and Moss 1986, 1908.)

FURTHER DESCRIPTIVE INFORMATION

As cultures age, cells show reduced motility and become increasingly pleomorphic. Cultures grown in Bennett's broth form pink/red flocs after 7 d. Colonies on Bennett's agar at 20°C are white, shiny, and circular after 2–3 d, pink after 4 d, and deep red, wrinkled, and rubbery in consistency after 5 d, with a diameter of approximately 3 mm. The tough rubbery consistency is characteristic of all isolates of *R. rubra* obtained so far, as is the production of red pigments of the prodigiosene family. The type strain was shown to produce 6-methoxy-2-methyl-3-pentylprodigeosene (prodigiosin) but the predominant pigment was the 3-heptyl analogue (Fig. BXII.γ.143).

On rich media such as Bennett's agar, the initially pillar-box red colonies become a deep maroon color and colonies often have a green metallic sheen. These color changes reflect both the reduction in pH and overproduction of the prodigiosenes, which can be seen as dark granular inclusions in wet mounts prepared for motility studies. At this stage viability drops rapidly and no viable organisms can usually be recovered from colonies after 14 d incubation on Bennett's agar.

Temperature range for growth after 168 h is 1.2–30.6°C; optimum, 20°C. No growth at 37°C. The cardinal temperatures are sensitive to the period of incubation and after 48 h the following have been recorded: T_{min}, 5.6°C; T_{opt}, 26.4°C; T_{max}, 30.6°C (Moss, 1983).

R. rubra has only been reported from a lowland, freshwater river site where it occurred in low but persistent numbers during a 5-month period (January to May 1984), especially after rainfall 24–48 h before collection of samples (Austin and Moss, 1986).

Although it is not possible to assign this organism to a particular ecological niche, the utilization of a wide range of sugars, amino acids, and organic acids as sole carbon sources suggests that it may have the ability to exploit a range of environments.

ENRICHMENT AND ISOLATION PROCEDURES

Maximum recovery of isolates from river water was achieved by making fivefold serial dilutions up to 1/125 using ¼ strength Ringer's buffer and plating 0.5-ml samples onto large (12-cm) plates of modified Bennett's agar supplemented with neomycin hydrochloride, cycloheximide, and nystatin, all at 50 µg/ml. Plates are incubated at 20°C for up to 14 d, and tough membranous colonies are streaked onto plates of Bennett's agar to ensure purity (Austin and Moss, 1986).

MAINTENANCE PROCEDURES

For medium-term storage, cultures are inoculated into dilute peptone water, incubated to allow turbidity to develop, and then stored at ~4°C. For long-term storage, freeze drying is not successful but storage in liquid nitrogen has given adequate survival. The type strain has been successfully recovered after 15 years' storage in liquid nitrogen using the following double-strength cryoprotectant (composition per liter of distilled water): K₂HPO₄, 12.6 g; sodium citrate, 0.9 g; MgSO₄·7H₂O, 0.18 g; (NH₄)₂SO₄, 1.8 g; KH₂PO₄, 3.6 g; glycerol, 44 ml; sterilized by filtration. A 1-ml sample of a 3-d-old culture of the organism in Bennett's broth is thoroughly mixed with 1 ml of cryoprotectant in a freeze vial (Nunc) and stored in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *RUGAMONAS* FROM OTHER GENERA

One of the most striking features of *Rugamonas* is the production of flocs in liquid culture and tough rubbery colonies on solid media. Table BXII.γ.129 shows some of the characteristics that distinguish three well-characterized Gram-negative, motile, prodigiosin-producing species of *Rugamonas*, *Alteromonas*, and *Serratia*.

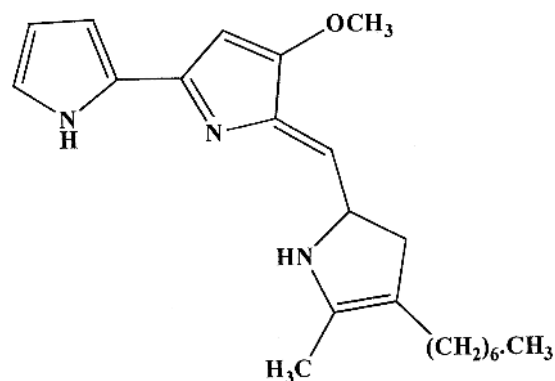


FIGURE BXII.γ.143. The structure of 6-methoxy-2-methyl-3-heptylprodigeosene, which is produced by *Rugamonas* strains.

1. Bennett's broth (Keeble and Cross, 1977) contains (g/l distilled water): yeast extract (Difco), 1.0; beef extract (Lab Lemco), 1.0; Bacto casitone (Difco), 2.0; and glucose, 10.0; pH 7.3. For a solid medium, 10.8 g/l of agar (LabM no. 2 or Oxoid no. 1) are added.

TABLE BXII.γ.129. Characteristics differentiating three Gram-negative, motile, prodigiosin-producing species

Characteristic	<i>Rugamonas rubra</i>	<i>Alteromonas rubra</i>	<i>Serratia rubidaea</i>
Oxidase	+	+	—
Fermentative metabolism	—	—	+
Growth at 35°C	—	+	+
Anaerobic growth	—	—	+
Utilizes D-sorbitol	+	—	—
Flagella	1–2 Polar or subpolar	1 Polar	Peritrichous
G + C DNA	66.7	46–48	53–59

TAXONOMIC COMMENTS

The general characteristics of *Rugamonas* isolates suggest an affinity with the *Pseudomonadaceae* but they cannot be comfortably

placed in any existing genera. They are distinguished from *Pseudomonas* and *Alteromonas* by their floc-forming ability in liquid cultures and from *Zoogloea* by the mol% G + C of the DNA.

List of species of the genus *Rugamonas*

1. ***Rugamonas rubra*** Austin and Moss 1987, 179^{AL} (Effective publication: Austin and Moss 1986, 1908.)
rub'ra. L. fem. adj. *rubra* red.

The characteristics are as given for the genus. In addition, the following enzymes are produced: alkaline phosphatase, acid phosphatase, esterase-lipase, leucine arylamidase, and phosphoamidase. Esterase is produced by some strains. Phenylalanine deaminase is not produced. Sole carbon sources include D-alanine, L-arabinose, DL-arginine, meso-inositol, sodium DL-lactate, cellobiose, D-fructose, D-galactose, L-histidine, maltose, D-mannose, D-raffinose, D-ribose, DL-serine, sodium acetate, trisodium citrate, D-sorbitol, sucrose, and trehalose. The following are not used: adonitol, *i*-erythritol, sodium malonate, D-melibiose, D-mel-

izitose, D-rhamnose, DL-valine, and D-xylose. Lactose is used by some strains.
By disk assay, strains are sensitive to chloramphenicol, chlortetracycline, erythromycin, gentamicin, oxytetracycline, streptomycin, and tetracycline (each at 10 µg/disk); furazolidone (50 µg), kanamycin (30 µg), novobiocin (5 µg), sulfafurazole (500 µg), and cotrimoxazole (25 µg). Resistant to ampicillin (2 µg), cloxacillin (5 µg), colistin sulfate (10 µg), neomycin (10 µg), nitrofurantoin (200 µg), and penicillin G (1.5 U).
Isolated from river water.
The mol% G + C of the DNA is: 66.7 ± 0.1 (*T_m*).
Type strain: Strain MOM 28/2/79, ATCC 43154, CCM 3730, NCIMB 12552.

Genus VIII. *Serpens* Hespell 1977, 380^{AL*}

ROBERT B. HESPELL

Ser'pens. L. fem. n. *serpens* snake, serpent.

Rod-shaped cells, 0.3–0.4 µm wide by 8–12 µm long. Occur singly in pairs. Cysts or coccoid bodies not formed, but cells in the stationary phase of growth are longer (16–25 µm) and often possess blebs or spherical protuberances. Gram negative. **Extremely flexible and capable of serpentine-like motility in agar gels. Possess bipolar tufts of 4–10 flagella and also a few lateral flagella.** Poly-β-hydroxybutyrate or other internal granules not formed. Have a strictly respiratory type of metabolism with oxygen as the sole electron acceptor. **Grow aerobically but prefer oxygen concentrations less than that of an air atmosphere. Catalase and oxidase positive.** Chemoorganotrophic. **Lactate is the only effective carbon and energy source**, although very slight growth occurs with acetate or α-ketoglutarate. Carbohydrates, fatty acids, and sugar alcohols are not catabolized. Casein hydrolysate, peptone, yeast extract, and, for most strains, ammonium chloride can serve as nitrogen sources; nitrates and nitrites are not used. Vitamins are stimulatory but not required. Optimum temperature, 28–30°C. On media containing 1.8–2.0% agar, colonies are cream colored, round, 3–6 mm in diameter, and have a filamentous edge. **On media with less than 1.5% agar,**

only subsurface spreading colonies occur. Found in the sediments of eutrophic freshwater ponds.
The mol% G + C of the DNA is: 66.
Type species: ***Serpens flexibilis*** Hespell 1977, 381.

FURTHER DESCRIPTIVE INFORMATION

The most striking morphological feature is the complex, serpentine-like movement of the cells when observed in slides of agar pieces removed from the edge of a spreading colony (Fig. BXII.γ.144). The organisms move very rapidly and display a furious lashing and bending of the cells; coiling into a knot-like configuration is also common. In liquids, the movement is less violent and the cells move rapidly with reversals along a straight line axis with a gentle flexing of the body, and the distal or trailing tip of the cell vibrates intensely (Fig. BXII.γ.144E–H). No tumbling type of movement is evident. Motility occurs in buffer solutions of polyvinylpyrrolidone having a viscosity greater than 1000 centipoise (Greenberg and Canale-Parola, 1977). In comparison, viscosities of 60 and 1000 centipoise prevent the motility of *Escherichia coli* and *Spirochaeta halophila*, respectively. The structural basis permitting such cell flexibility is unknown, but the organisms possess a thin peptidoglycan layer and a morphologically typical Gram-negative outer membrane.

*Editorial Note: The chapter on *Serpens* is reprinted from the 1st edition of *Bergey's Manual of Systematic Bacteriology*.

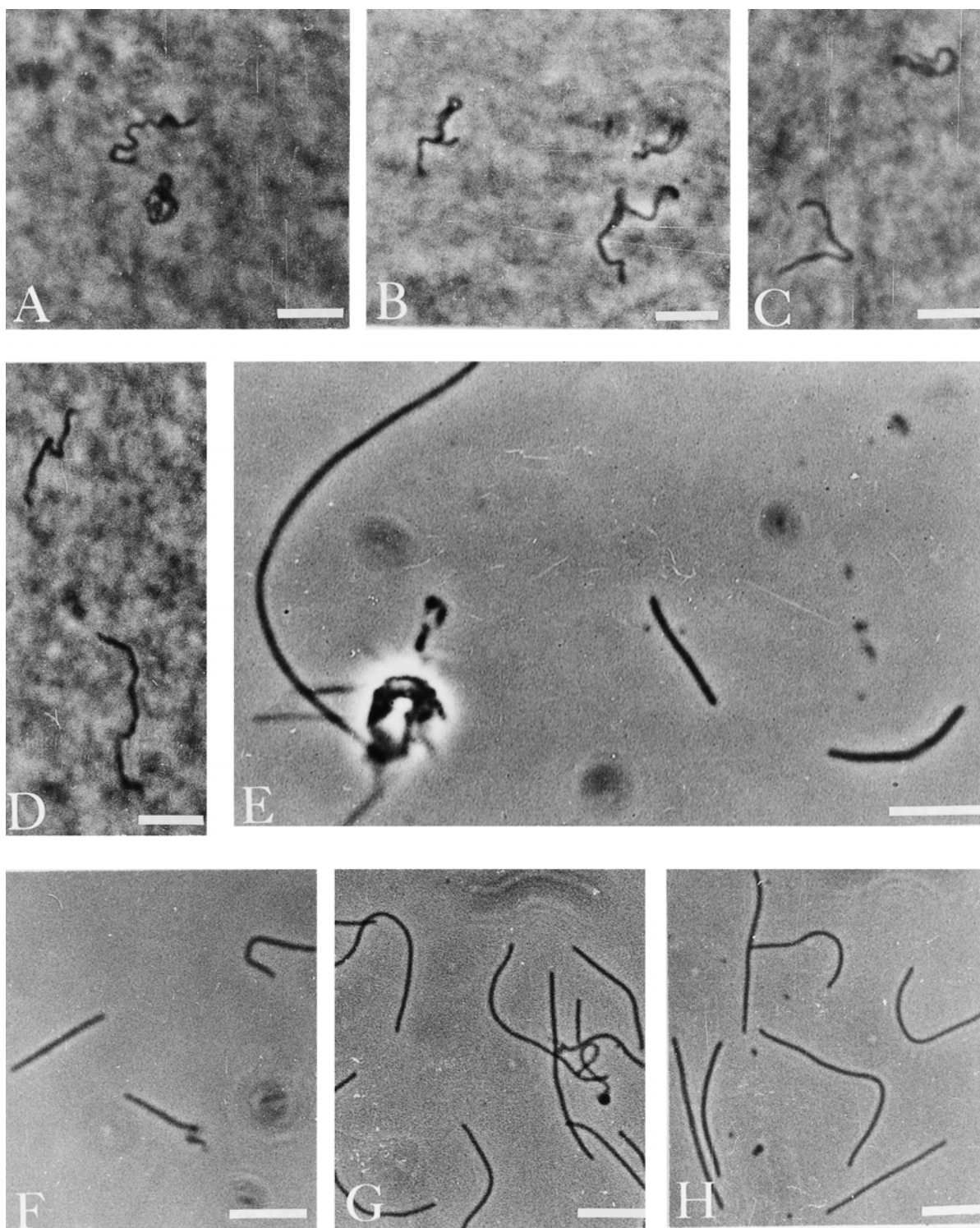


FIGURE BXII.γ.144. Living cells of *Serpens flexibilis* PFR-1. Wet-mount preparations, phase contrast. Cells in agar pieces excised from subsurface, spreading growth (A–D) displayed rapid motion with bending, lashing, and serpentine-like movements. Cells rapidly reversed themselves between coiled and uncoiled forms (A). In liquid menstrua (E–H) logarithmic-phase cells (F and H) showed straight-line movements with flexing of the cell body. Stationary cells often clumped (E) and had protuberances (G). Bars = 3.0 μm. (Reproduced with permission from R.B. Hespell, International Journal of Systematic Bacteriology 27: 371–381, 1977, ©International Union of Microbiological Societies.)

Although growth occurs aerobically, *S. flexibilis* prefers oxygen tensions less than that of an air atmosphere, growing subsurface (1–3 mm deep) in stationary liquid media or in media with agar concentrations of less than 1.5%. The type strain and all other

strains examined can use only lactate as an effective oxidizable carbon and energy source. Cell extracts contain only trace levels of the key enzymes of the Embden–Meyerhof–Parnas or hexose monophosphate pathways, but the tricarboxylic acid cycle en-

zymes are in high levels. Vitamins are not required for growth, but yeast extract is often stimulatory. Most strains can use ammonium chloride as a sole nitrogen source, but increased growth rates and cell yields occur with media containing peptone or casein hydrolysate.

ENRICHMENT AND ISOLATION PROCEDURES

S. flexibilis can be readily isolated from the upper few centimeters of the sediments of eutrophic freshwater ponds. Traditional enrichment culture techniques employing lactate as the carbon source are not useful for isolation as the organisms often become



FIGURE BXII.γ.145. Transmission electron micrograph of *S. flexibilis* PFR-1 cells. The insets (A and B) are higher magnifications of the ends of the same cell stained with phosphotungstic acid. Although the overall cell shape has been preserved, some flagella have detached, and an aggregate of four flagella remains at one end of the cell. Several laterally inserted flagella are present, and their insertion points (arrows) are more clearly evident in the insets. Bars = 0.25 μm. (Reproduced with permission from R.B. Hespell, *International Journal of Systematic Bacteriology* 27: 371–381, 1977, ©International Union of Microbiological Societies.)

overgrown by other heterotrophic bacteria. However, selective isolation can be accomplished by the use of membrane filters overlayed on a Petri dish of appropriate media, a technique that has also been employed for the isolation of thin spirilla and spirochetes (Canale-Parola et al., 1966). A small inoculum of pond water/mud slurry is deposited in the center of a sterile cellulose filter disk (0.3–0.45 μ m pore diameter) that has been placed on the surface of a plate of isolation medium.¹ After incubation for 6–12 h at 30°C, the disk is removed and the plate is incubated for 2–4 d. The organisms grow as a subsurface, whitish veil that diffuses from the center of the plate. By picking from the edge of the veil and streaking onto a second plate, cloned cultures may be obtained. Although aerobic spirilla and spirochetes can also be obtained in this manner, these organisms can usually be distinguished from *S. flexibilis* by light microscopic

observations, but definitive confirmation should be done by electron microscopy.

MAINTENANCE PROCEDURES

S. flexibilis is grown routinely in LYPP broth² and can be maintained by biweekly transfers on slants of LYPP agar. The organism can be preserved indefinitely by lyophilization or by storage in liquid nitrogen.

TAXONOMIC COMMENTS

The genus presently consists of a single species, *S. flexibilis*.*

2. LYPP broth (per 100 ml water): 60% sodium lactate syrup, 1.0 ml; yeast extract, 0.3 g; peptone, 0.2 g; and K₂HPO₄, 0.35 g. The pH of the medium is adjusted to 7.2 prior to autoclaving. LYPP agar is prepared by adding 1.5 g of agar.

*Editorial Note: Woese et al. (1982) reported that *S. flexibilis* is closely related to *Pseudomonas pseudoalcaligenes* (SAB = 0.9) and might be considered as a variant pseudomonad that had initially developed a defect in its system for synthesizing septa. Ahmad and Jensen (1987) found that *S. flexibilis* lacked the "overflow" pathway in the biosynthetic pathway for aromatic amino acids and shared this trait with most members of the *Pseudomonas aeruginosa* group.

1. Isolation medium (per 90 ml water): yeast extract, 0.2 g; peptone, 0.1 g; hay extract, 10 ml; and agar 1.0 g. The hay extract is prepared by boiling 10 g of hay in 100 ml of water for 15 min and clarifying the mixture by centrifugation. The pH of the isolation medium is adjusted to 7.0 with KOH before sterilization.

List of species of the genus *Serpens*

1. *Serpens flexibilis* Hespell 1977, 381^{AL}
flex.i.bi'lis. L. adj. *flexibilis* flexible, pliant.

The description of the species is the same as that for

the genus. The morphological features of the type strain are depicted in Figs. BXII.γ.144 and BXII.γ.145.

The mol% G + C of the DNA is: 66 (Bd).

Type strain: PFR-1, ATCC 29606.

Family II. *Moraxellaceae* Rossau, Van Landschoot, Gillis and De Ley 1991, 317^{VP}

ELLIOT JUNI AND KJELL BØVRE

Mo.ra.xel.la'ce.ae. M.L. fem. pl. n. *Moraxella* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Moraxellaceae* the *Moraxella* family.

Organisms are **rod shaped, coccoid**, or coccil or may exhibit a characteristic multicellular micromorphology. Cells usually single but may also occur in pairs or short chains. Endospores are not formed. Gram negative, but there may be a tendency to resist decolorization. The cells are **nonmotile** in liquid media but pilus-associated surface-bound motility ("twitching motility") may be observed. Psychrotrophic or mesophilic. Strains of all recognized species usually have an optimum growth temperature between 14 and 36°C. Some species have capsules. Pili (fimbriae), pilus-associated colony morphology, and **competence for natural transformation** are frequently detected. Colonies are not pigmented. Chemoorganotrophic and aerobic. Several species have complex growth factor requirements while some species grow readily in simple defined media containing a single organic carbon and energy source. Except for acinetobacters, all strains are oxidase positive. **Usually catalase positive. Indole is not produced.** Except for some strains of *Acinetobacter* and *Psychrobacter*, **no acid is produced from carbohydrates**. True waxes may be present. Fatty acid profiles show the presence of mainly unbranched, saturated and mono- or di-unsaturated fatty acids composed of 16 and 18 carbon atoms. Most strains reside indigenously on mucosal membranes of humans and animals. Except for *Moraxella catarrhalis*, *Moraxella nonliquefaciens*, *Moraxella bovis*, and *Moraxella osloensis*, most strains are rarely involved in disease of humans or other

animals. Strains of *Acinetobacter* and *Psychrobacter* are found in soil and water or in association with food products and have been isolated from the skin or mucous membranes of humans and other animals. The *Moraxellaceae* belong to the *Gammaproteobacteria* and to rRNA superfamily II.

The mol% G + C of the DNA is: 38–50.

Type genus: *Moraxella* Lwoff 1939, 173 emend. Henriksen and Bøvre 1968, 391.

Further comments With the exception of *M. caviae*, *M. caprae*, *M. phenylpyruvica*, and *M. lincolnii*, strains competent for natural transformation have been found for the other genera of the *Moraxellaceae*, thus making it possible to establish genetic relationships among all members of the family. Genetic studies involving the highly conserved streptomycin-resistance locus (resulting from mutation of S12, a protein component of the small ribosomal subunit) have made it possible to detect intergeneric, as well as interspecies, relationships. Because of the definitive, although frequently weak, genetic interactions involving the streptomycin-resistance marker, it is possible to recognize newly discovered and previously uncharacterized members of the *Moraxellaceae*. Using genus-specific transformation of nutritionally deficient mutants of certain conserved genes, it has been possible to assign strains of all species of a genus to that particular genus in a few of the genera of the *Moraxellaceae*.

TAXONOMIC COMMENTS

The first description of a rod-shaped bacterium involved in eye infection in humans was made independently by Morax (1896) and by Axenfeld (1897). This so-called Morax–Axenfeld bacillus was Gram negative, nonmotile, occurred frequently as diploid pairs, and required serum or other complex media for growth. In subsequent years, organisms with similar properties were isolated from patients with conjunctivitis. In 1939, Lwoff proposed that these organisms, as well as some others, be placed in a new genus, *Moraxella*. Seeking to find a family that included the genus *Moraxella*, Henriksen (1952) suggested, and it was later formally proposed (Henriksen and Bøvre, 1968), that based on similarities in morphology, Gram stain and oxidase reaction, as well as other properties, the family *Neisseriaceae* was a likely candidate. However, from results of studies involving genetic transformation (Bøvre and Hagen, 1981), DNA–DNA hybridization (Tønnum et al., 1989), DNA–rRNA hybridization (Rossau et al., 1986, 1989), and 16S rDNA sequencing (Enright et al., 1994), it can be concluded that there is presently no basis for including the moraxellae in the family *Neisseriaceae*.

In 1962, Bøvre and Henriksen demonstrated transformation of streptomycin resistance among the oxidase-positive moraxellas. In independent studies by Bøvre (1963) and Catlin (1964), it was shown that interspecies transformation of the high-level streptomycin resistance marker could take place between *Neisseria catarrhalis* and strains of moraxellae. It was proposed that organisms previously called *Neisseria catarrhalis* be renamed as *Moraxella catarrhalis* (Henriksen and Bøvre, 1968). In 1970, Catlin proposed that organisms named *Neisseria catarrhalis* be transferred to the genus *Branhamella*. For pedagogic reasons Bøvre (1979) suggested that the genus *Moraxella* be divided into the subgenus *Moraxella* for the six rod-shaped species (*Moraxella (Moraxella) lacunata*, *M. (M.) bovis*, *M. (M.) nonliquefaciens*, *M. (M.) osloensis*, *M. (M.) phenylpyruvica*, and *M. (M.) atlantae* and the subgenus *Branhamella* for the coccal species (*Moraxella (Branhamella) catarrhalis*, *M. (B.) caviae*, *M. (B.) ovis*, and *M. (B.) cuniculi*: the latter three coccal groups frequently being referred to as the “false neisseriae”). In the light of recent findings, use of the subgeneric terms *Moraxella* and *Branhamella* should be discontinued. Transformation studies (Bøvre and Hagen, 1981) have shown that all the coccal species are members of the genus *Moraxella*. Other groups considered to have been species of the genus *Moraxella* are, in fact, separate genera of the family *Moraxellaceae*.

Early studies of the nutritional and physiological properties of the moraxellae (Baumann et al., 1968a) revealed that one group (*M. osloensis*) was nutritionally unexacting and three groups (*M. lacunata*, *M. nonliquefaciens*, and *M. bovis*) displayed various levels of nutritional requirements.

For many years, bacteria now known to be strains of *Acinetobacter* were associated with the genus *Achromobacter*. In 1923, the Bergey's Manual Committee proposed a new genus, *Achromobacter*, to include all nonpigmented, Gram-negative, and aerobic saprophytes (see Ingram and Shewan, 1960). In 1954, Brisou and Prévot proposed that nonmotile achromobacters be considered species of the newly defined genus *Acinetobacter*. With the introduction of the oxidase test in diagnostic bacteriology, it was noted that organisms previously classified as strains of *Acinetobacter* included both oxidase-positive and oxidase-negative species (Henriksen, 1952; Piéchaud, 1961). In 1967, Thornley published the results of an extensive investigation of both oxidase-positive and

oxidase-negative bacteria, all considered strains of *Acinetobacter*. *Acinetobacters* have been difficult to classify in the past because they do not possess a sufficient number of unique phenotypic properties that enable them to be differentiated with certainty from other bacteria of similar appearance. In addition to being referred to as strains of *Achromobacter*, organisms subsequently demonstrated to be acinetobacters had been assigned at least 15 different names; those used most frequently include “*Bacterium anitratum*”, “*Herellea vaginicola*”, and “*Mima polymorpha*” (Baumann et al., 1968b). In a resolution based on classical taxonomic and genetic studies, the Subcommittee on the Taxonomy of *Moraxella* and Allied Bacteria suggested that the genus *Acinetobacter* should include only the oxidase-negative strains (Lessel, 1971). It was also proposed that the generic name *Achromobacter* be rejected (Hendrie et al., 1974). The strongest evidence that acinetobacters and moraxellas belong to the same family is that DNAs from streptomycin-resistant acinetobacters showed a weak, but consistently positive, transformation of *Moraxella catarrhalis* and *Moraxella osloensis* (Bøvre, 1967), and that relatively strong DNA–rRNA hybridization between the oxidase-positive and oxidase-negative organisms (Johnson et al., 1970) has been demonstrated. These results were supported by 16S rDNA sequence analysis (Pettersson et al., 1998).

The genus *Psychrobacter* contains most of the oxidase-positive strains of the original *Achromobacter* group. *Moraxella*-like psychrotrophic bacteria have been isolated from fish, poultry, and meat products (Gennari et al., 1992). Unlike the moraxellae and similar to some acinetobacters, some psychrobacters are able to form acid from glucose and other sugars. The demonstration of weak transformation of *M. osloensis* and other *Moraxella* species with DNA from streptomycin-resistant psychrotrophic strains (Bøvre and Hagen, 1981) established the relationship of these bacteria to the moraxellae. These organisms were subsequently classified as members of a new genus, *Psychrobacter* (Juni and Heym, 1986b).

Except for strains of *Acinetobacter* and *Psychrobacter*, all other members of the *Moraxellaceae* have been isolated as parasites of humans and other mammals. Several such organisms from sheep, cattle, and pigs have been studied (Bøvre and Hagen, 1981) and two of these, *M. caprae* (Kodjo et al., 1995) and *M. boeurei* (Kodjo et al., 1997), both isolated from goats, have been named. It seems most likely that there are many more species of the *Moraxellaceae* remaining to be discovered, particularly in less well-studied animals.

From studies of DNA–rRNA hybridization a new family, the *Moraxellaceae*, was proposed to accommodate the genera *Moraxella*, *Acinetobacter*, and *Psychrobacter* (Rossau et al., 1991). The family name *Moraxellaceae* is a most appropriate designation since it includes all organisms that have been shown to have genetic interactions with each other using the high-level-streptomycin resistance marker. The family name *Branhamaceae* was proposed to accommodate the spherical and rod-shaped species of *Moraxella* (Catlin, 1991). Unlike the family name *Moraxellaceae*, the proposed family name *Branhamaceae* fails to include *Acinetobacter* and *Psychrobacter*, genera that are clearly members of the same family. A summary of properties of individual members of some of the *Moraxellaceae* is given in Table BXII.γ.130.

Criteria for establishing membership in the *Moraxellaceae*
Strains are considered to belong to the family *Moraxellaceae*, provided that preferably all, but at least the first, of the criteria listed below are fulfilled:

TABLE BXII.γ.130. Phenotypic characteristics of the moraxellae^{a,b,c}

Characteristics	<i>M. lacunata</i>	<i>M. nonliquefaciens</i>	<i>M. bovis</i>	<i>M. caprae</i>	<i>M. ovis</i>	<i>M. caviae</i>	<i>M. catarrhalis</i>	<i>M. canis</i>	<i>M. cuniculi</i>	<i>M. osloensis</i>	<i>M. phenylpyruvica</i>	<i>M. atlantae</i>	<i>M. boerrii</i>	<i>M. lincolnii</i>
Morphology	R	R	R	R	C	C	C	C	C	R	R	R	SR	C, R
Motility	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Catalase activity	+	+	(+)	+	+	+	+	+	+	+	+	+	+	+
Oxidase activity	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Growth on MacConkey agar	—	—	—	—	—	—	—	—	—	v	nd	+	—	nd
Acids produced from glucose	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Growth on minimal medium ^d	—	—	—	—	—	—	—	(+)	—	+	—	—	—	nd
Hemolysis	—	—	+	+	(+)	w	—	+	—	—	—	—	+	—
Nitrate reduction	+	+	(—)	+	+	+	(+)	+	—	(—)	(+)	—	+	—
Liquefaction of gelatin	+	—	+	—	—	—	—	—	—	—	—	—	+	—
DNase activity	—	—	—	—	(—)	(—)	+	(+)	—	—	—	—	—	—
Proteolysis on Löffler slants	+	—	+	—	—	—	—	—	—	—	—	—	+	—
Indole	—	—	—	—	—	—	—	—	—	—	—	—	—	nd
Phenylalanine deaminase activity	—	—	—	—	—	—	—	—	—	—	+	nd	—	—
Hydrolysis of Tween 80	+	—	+	+	—	—	—	—	—	—	nd	—	+	—
Alkaline phosphatase activity	+	—	—	—	+	+	+	+	+	+	nd	+	—	—
Esterase activity	+	+	+	—	+	+	+	+	+	+	nd	+	+	+
Acid phosphatase activity	w	—	w	—	—	—	—	—	w	+	nd	+	—	—
Mol% G + C content	40–44.5	40–44	41–44.5	40–41.5	44.5–46.5	44.5–47.5	40–43	45.5–49.6	44.5	43–46	42.5–43.5	46.5–47.5	41–41.5	44

^aReproduced with permission from A. Kodjo et al., International Journal of Systematic Bacteriology 47: 115–121, 1997, ©International Union of Microbiological Societies.^bC, coccus; R, rod; SR, short rod.^c—, negative reaction; +, positive reaction; w, weak reaction; (+), most strains are positive; (—), most strains are negative; nd, not tested.^dMinimal medium containing ammonium and acetate.

1. All strains of the family must show either a weak or a strong genetic interaction of the conserved high-level streptomycin resistance marker with at least one established *Moraxellaceae* strain. Since competence for natural transformation is widespread in the *Moraxellaceae*, there should be no difficulty in satisfying this criterion.
2. 16S rRNA gene sequence similarity of all strains within the family should be significantly greater than is found between a given strain and any other bacterium that does not satisfy criterion 1.
3. In DNA–rRNA hybridizations, using DNA from any member of the *Moraxellaceae* and rRNA from a strain being tested, $T_{m(e)}$ values should be significantly greater for rRNAs derived from all members of the family than the corresponding value obtained using rRNA from any other bacterial strain.

At the present time, available data support criteria 2 and 3. However, because of the heterogeneity of *Acinetobacter* and *Psychrobacter*, as demonstrated by wide variation in DNA base composition, large numbers of species (as determined by DNA–DNA homology), and fairly widespread 16S rRNA gene sequence differences within the *Moraxellaceae*, it is possible that strains belonging to the family that do not fulfill one, or both, of criteria 2 and 3 may be found.

Criteria for distinguishing genera in the *Moraxellaceae* A genus of the *Moraxellaceae* can be distinguished from other genera

in the family if it can be determined that one or more of the properties listed below apply to all members of the genus:

1. Distinct phenotypic properties. (Example: only members of the genus *Acinetobacter* are oxidase negative.)
2. Very similar rRNAs, as determined by sequence similarities or by DNA–rRNA relatedness. (Examples: high 16S rRNA gene sequence similarity for strains of the species *M. lacunata*, *M. bovis*, and *M. nonliquefaciens* (Pettersson et al., 1998), all of which are members of the genus *Moraxella*, and unique $T_{m(e)}$ range for strains of the genus *Acinetobacter* and also for strains of the genus *M. osloensis* [Rossau et al., 1991]).
3. In quantitative transformation of the high-level streptomycin resistance marker, demonstration of a ratio of interstrain to intrastrain transformation equal to or greater than 10^{-4} , for a strain being tested and compared with at least one member strain of the genus. (Examples: strains of all species of *Moraxella*, all strains of *M. osloensis*, and all strains of *M. atlantae* [Bøvre and Hagen, 1981; Kodjo et al., 1997]).
4. Ability of DNAs from all strains of a genus to transform a conserved nutritional marker of a competent member of the genus. (Examples: transformation of a *trpE*[−] auxotroph of *Acinetobacter*, strain BD413 [Juni, 1972], transformation of a *trpE*[−] auxotroph of a competent strain of *M. osloensis* [Juni, 1974], and transformation of a hypoxanthine and thiamin auxotroph of a competent strain of *Psychrobacter* [Juni and Heym, 1980]).

Criteria for establishing named species in the *Moraxellaceae*

1. Strains of the same species should show 70% or greater DNA–DNA relatedness with 5°C or less ΔT_m (Wayne et al., 1987).
2. Strains of the same species may have one or more phenotypic properties that are unique for members of the species, as defined by DNA–DNA hybridization. (Example: selected phenotypic properties of the named species of *Acinetobacter*.)

Genera and species of *Moraxellaceae***I. Genus *Moraxella***

- A. *M. lacunata*
 1. “*M. lacunata* subsp. *lacunata*”
 2. “*M. lacunata* subsp. *liquefaciens*”
- B. *M. bovis*
- C. *M. canis*
- D. *M. caprae*
- E. *M. catarrhalis*
- F. *M. caviae*
- G. *M. cuniculi*
- H. *M. equi*
- I. *M. nonliquefaciens*
- J. *M. ovis*

Species Incertae Sedis

- A. *M. atlantae*
- B. *M. boevei*
- C. *M. lincolnii*
- D. *M. osloensis*
- E. *M. phenylpyruvica*

II. Genus *Acinetobacter*

- A. *A. baumannii*
- B. *A. calcoaceticus*
- C. *A. haemolyticus*
- D. *A. johnsonii*
- E. *A. junii*
- F. *A. lwoffii*
- G. *A. radioresistens*
- H. *A. venetianus*
- I. Other species: 14 as yet unnamed species (DNA groups 3, 6, 9, 10, 11, TU13, TU14 (= BJ13), TU15, CTTU13, 1-3, BJ14, BJ15, BJ16, and BJ17)

III. Genus *Psychrobacter*

- A. *P. faecalis*
- B. *P. frigidicola*
- C. *P. glacincola*
- D. *P. immobilis*
- E. *P. marincola*
- F. *P. pacificensis*
- G. *P. proteolyticus*
- H. *P. submarinus*
- I. *P. urativorans*

Note concerning the significance of natural transformation in bacterial taxonomy Natural transformation, when it occurs, provides a unique mechanism for recombination of bacterial genes. Most early transformation studies made use of antibiotic resistance markers. Streptomycin resistance, a frequently employed genetic marker, can be expected to give somewhat different results compared with those obtained using other chromosomal markers. Single step mutation to high levels of streptomycin resistance (or dependence) results from specific base changes in

DNA coding for the S12 (formerly P10) protein of the small ribosomal subunit (Ozaki et al., 1969). Since genes coding for ribosomal components (proteins and RNAs) have been shown to be highly conserved (Dubnau et al., 1965), transformation of a competent strain with a ribosomal marker DNA from any member of the same bacterial family is possible. Intergeneric transformation of such a marker is generally considerably weaker than intrageneric (interspecies) transformation. It has also been shown, using the streptomycin-resistance marker, that interspecies transformation is usually weaker than intraspecies transformation (Bøvre and Hagen, 1981). Extremely weak intergeneric transformation of the high-level streptomycin-resistance marker has, in the past, led to the conclusion that the interacting strains are members of the same genus. This view is no longer tenable in the light of more recent findings for the existence of several independent genera in the *Moraxellaceae*.

In contrast to ribosomal markers, nutritional markers appear to be limited to intrageneric (interspecies) transformation of competent strains. Studies of a variety of nutritional markers in competent strains have revealed that all of them can undergo intrastrain transformation. Only a relatively small number of them, however, are conserved in the sense that they can be transformed by DNAs from strains of all species of a particular genus. The reason for the conserved behavior of this latter class of nutritional markers is currently not understood. It has been proposed that the ability of DNA from one strain to transform another competent strain signifies that both strains are members of the same genus (Ravin, 1963). Using conserved nutritional markers this hypothesis has been verified for several genera of the *Moraxellaceae*: *Acinetobacter* (*trpE*[−] marker) (Juni, 1972), *Psychrobacter* (a hypoxanthine and thiamine marker) (Juni and Heym, 1980), and *M. osloensis* (*trpE*[−] marker) (Juni, 1974). The ability of DNAs from a large number of strains of *Oligella* (formally *Moraxella urethralis*) to transform a *trp*[−] marker has also been demonstrated (Juni, 1977). In these examples, transformation of the appropriate nutritional mutant by DNA from any strain of the genus has never failed. Furthermore, a false positive reaction with DNA from an organism that is not a member of the same genus has never been observed in these transformation assays. It would thus appear that the ability to transform conserved nutritional markers (but not conserved ribosomal markers) in competent strains of a bacterial genus might, in fact, be the best way to establish membership in that genus.

Using nonconserved nutritional markers, it has been shown that interstrain transformation takes place with high efficiency for strains of the *Moraxella* species *M. catarrhalis* (Juni, 1990), *M. nonliquefaciens* (Juni et al., 1987), and *M. bovis* (Juni et al., 1988). DNAs from other species of *Moraxella* failed to transform either of two *M. catarrhalis* auxotrophs (Juni, 1990). Auxotrophic mutants of *M. nonliquefaciens* and *M. bovis* were also transformed with lower efficiencies by DNAs from some, but not all, species of *Moraxella* (Juni et al., 1987, 1988).

It is possible to transform competent strains of any genus in the family for the ability to grow in the presence of streptomycin, using DNA from a streptomycin-resistant (or dependent) mutant of any strain of the family. Otherwise, the high-level streptomycin-resistance marker resembles conserved nutritional markers in its ability to transform any competent member of the genus from which the streptomycin-resistance marker was obtained. As with the case of conserved nutritional markers, interstrain transformation of the streptomycin-resistance marker can be less efficient than intrastrain transformation (Bøvre and Hagen, 1981). In-

tergeneric transformation of the high-level streptomycin-resistance marker probably occurs because of the conserved nature of all ribosomal genes, which are even more similar in sequence than conserved nutritional markers.

For most genetic markers, interfamilial transformation has not been observed. In a rare exception, however, it has been shown that 23S rDNA from a variety of unrelated bacteria can stably transform any of the seven rDNA operons in competent *Acinetobacter* because of the highly conserved nature of rRNA genes (Strätz et al., 1996). It was also shown that the further the phylogenetic distance (as determined from 16S rRNA gene sequence data), the lower the frequency of such interfamilial transformation (Strätz et al., 1996).

Phylogenetic relationships of the Moraxellaceae

DNA-rRNA hybridization analysis DNA-rRNA hybridization and 16S rRNA (or rDNA) nucleotide sequencing of 1000 or more bases are two of the main procedures that have been used to obtain data for establishing phylogenetic relationships (Murray et al., 1990b). The (family) relationships of all genera of the *Moraxellaceae* were originally established by genetic transformation experiments (Bøvre and Hagen, 1981).

DNA-rRNA hybridization studies of the family *Neisseriaceae*, as this family was previously constituted, led to the conclusion that organisms presently considered to be members of the *Moraxellaceae* are not related to the *Neisseriaceae* (Rossau et al., 1986, 1989). Hybridization of rRNA from a strain of *Moraxella lacunata* and a strain of *Psychrobacter immobilis*, to DNAs from strains of all

members of the *Moraxellaceae* resulted in a distinct cluster of $T_{m(e)}$ values ($T_{m(e)}$ is the temperature at which one-half of a DNA-rRNA duplex is denatured) ranging from 69.9 to 80.0°C. Significantly lower $T_{m(e)}$ values were obtained for DNAs from strains that were members of other families of rRNA superfamily I, of rRNA superfamily II, and of rRNA superfamilies V and VI (Rossau et al., 1991). $T_{m(e)}$ values for acinetobacters formed a distinct group. The other strains were separated into subgroups of a so-called *Moraxella* group (Rossau et al., 1991). The *Moraxella lacunata* subgroup included strains of the classical "*Moraxella lacunata* group" as well as the coccal species of *Moraxella*, which could not be distinguished from each other by this procedure.

Only three strains of *M. osloensis* were examined. They formed a distinct subgroup when their DNAs were hybridized to "*M. lacunata* subsp. *liquefaciens*" rRNA but were clustered together with the *M. lacunata* subgroup when these DNAs were hybridized to *Psychrobacter immobilis* rRNA (Rossau et al., 1991). Using *M. lacunata* rRNA and DNA from other strains, *M. phenylpyruvica* clustered with strains of *Psychrobacter*. When *Psychrobacter* rRNA was used for hybridization, however, strains of *M. phenylpyruvica* appeared to reside in a unique subcluster. The one strain of *M. atlantae* examined could be assigned to either the *Psychrobacter* or the *Moraxella lacunata* subgroup, depending upon which rRNA was used for hybridization (Rossau et al., 1991).

Although DNA-rRNA hybridization is a useful procedure for establishing membership in the *Moraxellaceae*, results from such studies do not necessarily provide the means for distinguishing genera, or species of a particular genus, within the family.

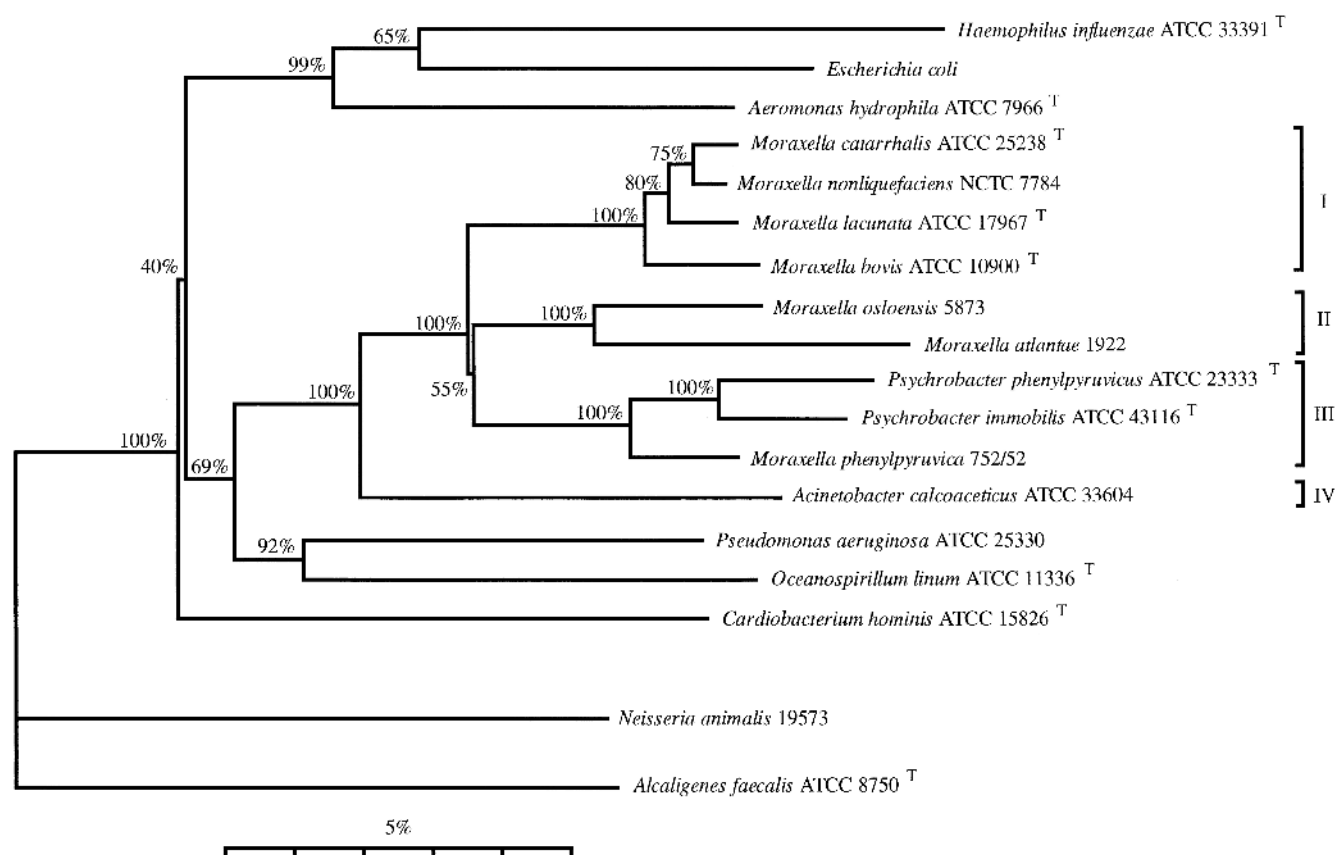


FIGURE BXII.γ.146. Phylogenetic tree based on distance matrix analysis of 1355 positions in the 16S rRNA genes for selected members of *Moraxellaceae*. (Reproduced with permission from B. Pettersson et al., *International Journal of Systematic Bacteriology* 48: 75–89, 1998, ©International Union of Microbiological Societies.)

16S rRNA gene sequence analysis 16S rRNA gene sequence analysis is similar to DNA-rRNA hybridization in that it enables isolated strains to be recognized as members of the *Moraxellaceae*. With both procedures, it was possible to demonstrate that the genera *Acinetobacter*, *Moraxella*, and *Psychrobacter* are not members of the *Neisseriaceae* (Rossau et al., 1991; Enright et al., 1994).

A phylogenetic study of 33 strains, including 11 (previously designated) species of the family *Moraxellaceae*, by 16S rDNA sequence analysis (Pettersson et al., 1998) showed that the family *Moraxellaceae* formed a distinct clade consisting of four phylogenetic groups, as determined from branch lengths, bootstrap values, and signature nucleotides (Fig. BXII.γ.146). Group I contained the classical and coccal *Moraxella* species. Group II consisted of *Moraxella osloensis* and *Moraxella atlantae*. Group III contained strains of *Psychrobacter* and a single strain of *Moraxella phenylpyruvica*. Group IV contained two strains of *Acinetobacter*. The phylogenetic tree showed that the members of the *Moraxellaceae* constituted a monophyletic taxon forming a distinct line of descent within the *Gammaproteobacteria*. A phylogenetic tree for organisms belonging to group I is shown in Fig. BXII.γ.147.

16S rRNA nucleotides U and A in positions 514 and 537, respectively, flank a highly conserved oligonucleotide that should be useful for the development of diagnostic probes for detection of members of the *Moraxellaceae* (Pettersson et al., 1998). Phylogenetic conclusions regarding individual genera of the *Moraxellaceae* will be discussed below, along with the other features of each genus. Studies of other bacterial families have shown that information obtained through phylogenetic analysis of 16S rRNA

does not necessarily parallel results obtained by DNA-DNA hybridization at the species level (Stackebrandt and Goebel, 1994).

As this volume was going to press, it was noticed that the genus *Enhydrobacter* (Staley et al., 1987) resembles a genus of the *Moraxellaceae*, based on 16S rRNA sequence similarity (sequence similarity was determined using the RDP Sequence Match program and comparing *Enhydrobacter aerosaccus* strain LMG 21877 = ATCC 27094^T (AJ550856) to the NCBI database). The nearest neighbors of *Enhydrobacter aerosaccus* were *Moraxella osloensis* strains Ben 58 (X95304), 5873 (AF005190), PIV-10-1 (AJ505859), and AU1220 (AY043376) with S_{AB} values ranging from 0.956–0.973 and *M. osloensis* strain 3 (Y15855), S_{AB} = 0.893.

It should be noted, however, that *Enhydrobacter* DNA has a mol% G + C of 66, whereas the mol% G + C values of other genera in the family *Moraxellaceae* range from 38–50. In addition, *Enhydrobacter*, unlike all members of the *Moraxellaceae*, has several other unique properties, such as an extremely slow growth rate, growth at an optimum temperature of 37–39°C, ability to carry out sugar fermentations, and being gas-vacuolated. Although the relationship of *Enhydrobacter* 16S rRNA to that of *M. osloensis* suggests that *Enhydrobacter* may be a member of the *Moraxellaceae*, the several startlingly unusual properties of *Enhydrobacter* cast doubt on this conclusion.

It might be considered that 16S rDNA in *Enhydrobacter* may have been derived from *M. osloensis* by horizontal gene transfer or by mutation of 16S rDNA in an unrelated strain over eons of time resulting in the rRNA sequence in *Enhydrobacter* that is presently observed. It is also possible that a strain of *M. osloensis* may

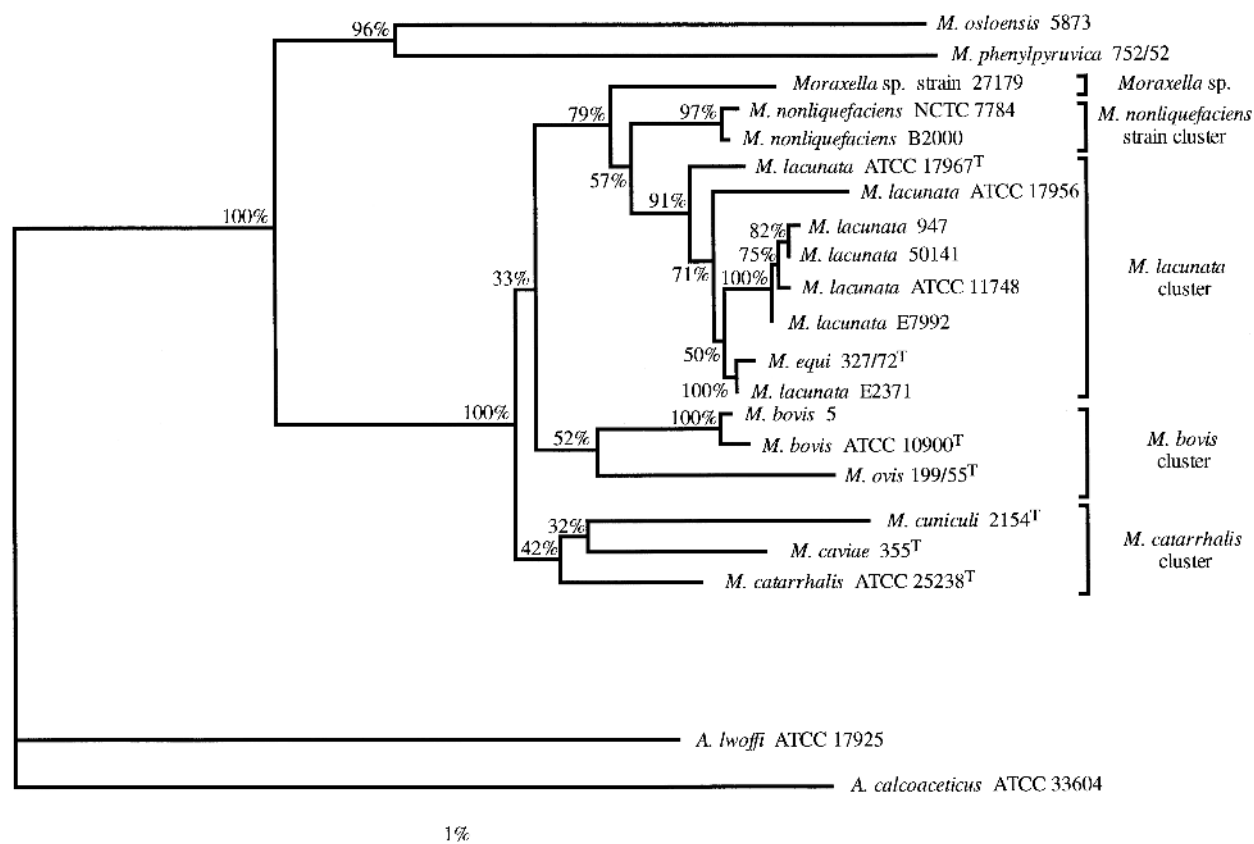


FIGURE BXII.γ.147. Phylogenetic tree based on distance matrix analysis of 1440 positions in the 16S rRNA genes of the moraxellae. (Reproduced with permission from B. Pettersson et al., *International Journal of Systematic Bacteriology* 48: 75–89, 1998, ©International Union of Microbiological Societies.)

have undergone very extensive genetic modification of the many genes responsible for the unique phenotypes described above resulting in the organism now referred to as *Enhydrobacter*. In any case, this dilemma can be resolved readily by checking for ability of DNA from a high level streptomycin-resistant mutant of *Enhydrobacter* to transform competent strains of *M. osloensis*, or any of several other competent strains of several genera of the *Moraxellaceae* where even a weak, but reproducible, transformation would prove that *Enhydrobacter* is indeed a member of the *Moraxellaceae*. Also, *Enhydrobacter* DNA should be tested for ability to transform the genus-specific *trpE*⁻ mutant (Juni, 1974) where, once again, a positive result would prove that *Enhydrobacter* is a member of the *Moraxellaceae*. Until such tests are performed, it is premature to consider that *Enhydrobacter* is a member of the *Moraxellaceae* based solely on 16S rRNA data.

ACKNOWLEDGMENTS

This manuscript is dedicated to the memory of Kjell Bøvre, friend and colleague. It is largely because of his critical studies that we now have a much better understanding of the moraxellae.

FURTHER READING

- Bøvre, K. 1984. Genus II. *Moraxella* Lwoff 1939. In Krieg and Holt (Editors), *Bergey's Manual of Systematic Bacteriology*, 1st Ed., Vol. 1, The Williams & Wilkins Co., Baltimore. 296–303.
- Bøvre, K. and N. Hagen. 1981. The family *Neisseriaceae*: rod-shaped species of the genera *Moraxella*, *Acinetobacter*, *Kingella*, and *Neisseria*, and the *Branhamella* group of cocci. In Starr, Stolp, Trüper, Balows and Schlegel (Editors), *The Prokaryotes. A Handbook on Habitats, Isolation and Identification of Bacteria*, 1st Ed., Vol. 2, Springer-Verlag, New York. pp. 1506–1529.
- Henriksen, S.D. 1973. *Moraxella*, *Acinetobacter*, and the *Mimeae*. *Bacteriol. Rev.* 37: 522–561.
- Henriksen, S.D. 1976. *Moraxella*, *Neisseria*, *Branhamella*, and *Acinetobacter*. *Annu. Rev. Microbiol.* 30: 63–83.
- Juni, E. 1978. Genetics and physiology of *Acinetobacter*. *Annu. Rev. Microbiol.* 32: 349–371.
- Juni, E. 1984. Genus III. *Acinetobacter* Brisou and Prévot 1954. In Krieg and Holt (Editors), *Bergey's Manual of Systematic Bacteriology*, 1st Ed., Vol. 1, The Williams & Wilkins Co., Baltimore. 303–307.
- Juni, E. 1992. The genus *Psychrobacter*. In Balows, Trüper, Dworkin, Harder and Schleifer (Editors), *The Prokaryotes*, 2nd Ed., Vol. IV, Springer-Verlag, New York. 3241–3246.

Genus I. *Moraxella* Lwoff 1939, 173, emend. Henriksen and Bøvre 1968, 391^{AL}

ELLIOT JUNI AND KJELL BØVRE

Mo.rax.el'la. M.L. dim. *-ella* ending; M.L. fem. n. *Moraxella* named after V. Morax, a Swiss ophthalmologist who pioneered the recognition of the type species.

Rods or cocci. The rods are often very short and plump, frequently approaching a coccus shape (1.0–1.5 × 1.5–2.5 µm); they usually occur in pairs and short chains (one plane of division). Variation in cell size, shape, and filament or chain formation is often seen in cultures, the pleomorphism being enhanced by lack of oxygen and by incubation temperatures above the optimum. **The cocci are usually smaller** (0.6–1.0 µm in diameter) and occur as single cells or in pairs with the adjacent sides flattened (differing planes of division); division in two planes at right angles to each other sometimes results in the formation of tetrads. Gram negative, but often with a tendency to resist Gram decolorization. Flagella are absent. Both rod-shaped and coccoid species may be fimbriated. Swimming motility is absent, but surface-bound “twitching motility” has been observed in some rod-shaped species. **Aerobic, but some strains may grow weakly under anaerobic conditions.** May be encapsulated. Chemoorganotrophic. Most species are nutritionally fastidious and all grow on complex media; some are stimulated

significantly by fatty acids (bile salts, Tween 80). Optimum temperature for growth, 33–35°C. **Colonies are not pigmented. Oxidase positive** (with either tetra- or dimethyl-*p*-phenylenediamine reagent). **Usually catalase positive. No acid is produced from carbohydrates.** Usually highly sensitive to penicillin. Parasitic on the mucous membranes of humans and other warm-blooded animals. Strains of the several species of this genus have been shown to be genetically closely related by transformation of the high-level streptomycin resistance marker (Bøvre and Hagen, 1981) as well as by transformation of nutritional markers (Juni et al., 1987, 1988; Juni, 1990). DNA–DNA hybridization studies have shown high relative interspecies binding ratios for DNAs from strains of the genus *Moraxella* (Tønjum et al., 1989). Species of *Moraxella* share 95% similarity in their 16S rRNA gene sequences.

The mol % G + C of the DNA is: 40.0–47.5 (*T_m*, Bd, HPLC).

Type species: Moraxella lacunata (Eyre 1900) Lwoff 1939, 173 (*Bacillus lacunatus* Eyre 1900, 5; (*Moraxella* (*Moraxella*) *lacunata* (Eyre 1900) Bøvre 1979, 404.)

List of species of the genus *Moraxella*

1. *Moraxella lacunata* (Eyre 1900) Lwoff 1939, 173^{AL} (*Bacillus lacunatus* Eyre 1900, 5; (*Moraxella* (*Moraxella*) *lacunata* (Eyre 1900) Bøvre 1979, 404.)*
la.cu.na'ta. L. n. *lacuna* a shallow depression; M.L. fem. adj. *lacunata* pitted.

Two subspecies of *Moraxella lacunata* have been recognized (Henriksen, 1969).

The mol % G + C of the DNA is: 40.0–44.5.

Type strain: ATCC 17967, NCTC 11011.

GenBank accession number (16S rRNA): AF005160, AF005170, D64049.

- a. “*Moraxella lacunata* subsp. *lacunata*”

Medium thick to plump rods, 0.8–1.2 µm in diameter, coccoid to distinctly bacillary, occurring predominantly in pairs and short chains. Frequently pleomorphic. May form narrow capsules. Colonies on blood agar are small (0.1–0.3 mm in diameter). Colonies are translucent to semiopaque. Pitting of the agar may be observed. No hemolysis occurs, but on heated blood (chocolate) agar, large dark zones around the colonies usually occur.

Strains may show improved growth on chocolate agar medium and may not grow on certain rich media without serum or oleic acid. They are unable to grow on medium MB, a defined medium that supports growth of strains of *M. bovis* (Juni et al., 1988). Other characteristics of

*Editorial Note: Neither “*Moraxella lacunata* subspecies *lacunata*” nor “*Moraxella lacunata* subspecies *liquefaciens*” are validly published names.

the species are given in Table BXII.γ.130 of the description of the family *Moraxellaceae*. Human strains have been derived mainly from inflamed, as well as healthy, conjunctiva and from sites in the upper respiratory tract (Graham et al., 1990). They have occasionally been isolated from the blood. Attachment to epithelial tissues is mediated by type 4 pili (Ruehl et al., 1988), which are regulated by the *piv* gene. This gene encodes a site-specific DNA invertase (Marrs et al., 1990), the transcription of which is initiated at *pivp*, a σ^{70} -dependent promoter (Heinrich and Glasgow, 1997). The organism appears to be a significant causative agent of human conjunctivitis (and keratitis) and was more frequently isolated in the past. It is only rarely isolated at the present time.

The mol% G + C of the DNA is: 40.0–44.5 (Bd).

Type strain: none.

GenBank accession number (16S rRNA): AF005170.

Additional Remarks: There is presently no type strain of this subspecies; the previously listed type strain, ATCC 17967 (NCTC 11011), is actually a strain of "*M. lacunata* subsp. *liquefaciens*". Strain ATCC 11748 (NCTC 10748) should be proposed as the neotype strain since it has typical properties of the species, as well as being competent for natural transformation.

b. "*Moraxella lacunata* subsp. *liquefaciens*"

li.que.fa'ciens. M.L. part. adj. *liquefaciens* dissolving.

This subspecies is very similar to "*M. lacunata* subsp. *lacunata*" in morphology and in its response to most conventional tests, but differs in being able to grow on media without serum and to liquefy the usual gelatin media. "*M. lacunata* subsp. *liquefaciens*" was first described as the cause of keratitis. It has also been isolated from cases of endocarditis and sinusitis. Less fastidious than "*M. lacunata* subsp. *lacunata*" (Bøvre, 1965c); can grow in a defined medium (medium MB) containing 11 growth factors (Juni et al., 1988). Colonies on blood agar medium have a greater diameter (~3 mm) than colonies of "*M. lacunata* subsp. *lacunata*". Properties enabling further distinction between strains of "*M. lacunata* subsp. *liquefaciens*" and "*M. lacunata* subsp. *lacunata*" are discussed below. A number of isolates from the conjunctiva of healthy guinea pigs were shown to have the same morphological, cultural and biochemical reactions as "*M. lacunata* subsp. *liquefaciens*" (Ryan, 1964). Furthermore, quantitative transformation studies, using high-level streptomycin-resistance markers, revealed that two of these guinea pig isolates were very closely related to authentic strains of "*M. lacunata* subsp. *liquefaciens*" (Bøvre, 1965c). Because there are antigenic differences between guinea pig isolates and an authentic strain of "*M. lacunata* subsp. *liquefaciens*" (Ryan, 1964), it is possible that the animal strains represent another distinct subspecies of *M. lacunata*.

The mol% G + C of the DNA is: 41.5–43 (Bd).

Type strain: ATCC 17952, NCTC 7911 should be proposed as the neotype strain.

GenBank accession number (16S rRNA): AF005169.

Further comments The occurrence of genetically competent strains permits identification by quantitative genetic transformation (Bøvre and Hagen, 1981; Tøn-

jum et al., 1992). Although "*M. lacunata* subsp. *lacunata*" and "*M. lacunata* subsp. *liquefaciens*" have many phenotypic similarities and cannot be differentiated readily based on 16S rRNA gene sequence analysis (Pettersson et al., 1998), it is possible to accomplish this by comparison of certain phenotypic properties. It has long been known that strains of "*M. lacunata* subsp. *lacunata*" are more nutritionally fastidious and generally have smaller colonies on blood agar medium than strains of "*M. lacunata* subsp. *liquefaciens*" (Bøvre, 1965c; Bøvre and Hagen, 1981). Studies of a collection of bacteria, received as strains of *M. lacunata*, showed that not all of these strains could grow on a defined medium (medium MB) suitable for growth of strains of *M. bovis* (Juni et al., 1986; 1988); one group of these strains (group 1) being able to grow on medium MB whereas strains of a second group (group 2) being unable to grow on this medium. It was subsequently shown that group 1 consisted of strains of "*M. lacunata* subsp. *liquefaciens*" whereas strains in group 2 consisted of strains of "*M. lacunata* subsp. *lacunata*" (Tønjum et al., 1992).

Furthermore, other studies of a collection of strains identified as *M. lacunata* based on results of classical phenotypic tests disclosed that these organisms fell into two classes based on their respective cellular fatty acid profiles (Moss et al., 1988). Profiles I and II are characteristic for strains of "*M. lacunata* subsp. *liquefaciens*" and "*M. lacunata* subsp. *lacunata*", respectively. Many of the same strains were used in both the nutritional (Juni et al., 1988) and fatty acid studies (Moss et al., 1988).

Although there was some overlap, the clustering of strains of "*M. lacunata* subsp. *lacunata*" and "*M. lacunata* subsp. *liquefaciens*" into distinct groups was also apparent from total genomic DNA–DNA hybridization, (Tønjum et al., 1992). These two subspecies were also distinguished based on genetic variation of 12 enzyme loci (Tønjum et al., 1992). In quantitative transformation studies using the high-level streptomycin-resistant marker, it was shown that "*M. lacunata* subsp. *lacunata*" and "*M. lacunata* subsp. *liquefaciens*" were closely related but could, nevertheless, be distinguished from each other (Tønjum et al., 1992).

The different nutritional requirements and different cellular fatty acid patterns, as well as variation of enzyme loci of the subspecies, suggest that these two organisms normally occupy different ecological niches. "*M. lacunata* subsp. *lacunata*" may normally reside at a site so well supplied with nutrients that it has, over eons of time, evolved and lost considerable biosynthetic ability. By contrast, "*M. lacunata* subsp. *liquefaciens*" may reside at a site deficient in certain nutrients and has evolved to synthesize those growth factors not supplied at its normal host site, while losing the ability to synthesize growth factors not required at this site. This accounts for its relatively increased biosynthetic ability. Evolution in membrane structure, as indicated by different fatty acid compositions of the two subspecies, may result in membranes that, in some manner, contribute to the ability to compete successfully with other bacteria for access to a particular site. If these properties do, in fact, contribute to the ability to preferentially occupy certain sites, it would seem prudent to compare phenotypic properties, such

as those described above, when attempting to allocate bacteria to species since these may be the very factors that determine speciation. The subspecies nomenclature is retained pending a more precise definition of the terms species/subspecies as well as further possible findings regarding other different characteristics of the entities.

Phylogeny of the *Moraxella lacunata* cluster A phylogenetic study of the *Moraxellaceae* by 16S rDNA sequence analysis revealed four distinct groups (Pettersson et al., 1998) (Fig. BXII.γ.146). Strains of *Moraxella* were shown to reside in three clusters of group 1 (Fig. BXII.γ.147). A so-called *M. lacunata* cluster included 13 strains of "*M. lacunata* subsp. *lacunata*", "*M. lacunata* subsp. *liquefaciens*", and *M. equi*. Eight strains of this cluster had unique 16S rRNA gene sequences, and interstrain similarity values within the cluster ranged from 98.4 to 99.9% (Pettersson et al., 1998). Five strains of "*M. lacunata* subsp. *lacunata*", obtained from different geographic regions, clustered tightly together, and shared 99.8% 16S rRNA gene nucleotide similarity (Pettersson et al., 1998).

2. ***Moraxella bovis*** (Hauduroy, Ehringer, Urbain, Guillot and Magrou 1937) Murray 1948, 591^{AL} (*Haemophilus bovis* Hauduroy, Ehringer, Urbain, Guillot and Magrou 1937, 247; *Moraxella (Moraxella) bovis* (Hauduroy, Ehringer, Urbain, Guillot and Magrou 1937) Bøvre 1979, 404.)
bo'vis. L. gen. n. *bovis* of cattle.

Rods, showing variation in shape, size and arrangement. The cells are often fimbriated and show "twitching motility." A distinct variation in colony type occurs. Freshly isolated strains form hemispherical to flat colonies (~1 mm in diameter in 48 h) on blood agar; they corrode the agar and often show surface spreading. These characteristics are associated with fimbriation of the cells and "twitching motility" (Bøvre and Frøholm, 1972b; Henrichsen et al., 1972; Pedersen et al., 1972). The texture of such colonies is often friable. Nonfimbriated variants occur spontaneously in cultures and form noncorroding, nonspreading colonies which are convex and usually larger than their corroding counterparts (often 3.0 mm after 48 h), and have a butyrous consistency. Fimbriated cells of collection strains may also form large colonies, but these may be less corroding than the colonies of fresh isolates. With few exceptions, bovine strains produce distinct hemolytic zones around colonies. Dark zones are produced on chocolate agar. Nutritionally fastidious. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Most frequently isolated from bovine eyes in cases of infectious keratoconjunctivitis, but has also been isolated from unaffected eyes and the nasal cavity of cattle, the eyes of sheep and mice, but not the eyes of rats, rabbits, or guinea pigs. Considered as potentially pathogenic for animals, dependent on accessory factors to elicit disease. Conjunctival colonization and disease development in UV-irradiated bovine eyes have been found associated with fimbriation of the bacteria (Pedersen et al., 1972).

Further comments Growth of most strains of *M. bovis* occurs in a defined medium containing 11 growth factors (Juni and Heym, 1986a). The cells are frequently transformable, genetic competence being associated with fim-

briation (piliation) and corrosion (Bøvre and Frøholm, 1972a). A *M. bovis* pilin gene has been cloned and sequenced (Marrs et al., 1985; Rozsa and Marrs, 1991). Strains may be identified by quantitative genetic transformation of the high-level streptomycin resistance marker (Bøvre, 1965a, c), or qualitatively by transformation of an auxotrophic mutant of a competent strain (Juni et al., 1988). The species is slightly genetically heterogeneous, with interstrain transformation affinities as low as 10% of intra-strain transformation affinities (Bøvre, unpublished results). A so-called *M. bovis* cluster consisting of strains of *M. bovis* and *M. ovis* has been observed in phylogenetic studies involving 16S rRNA gene sequence analysis (Fig. BXII.γ.147); the 16S rRNA genes of *M. bovis* and *M. ovis* were 97.1–97.4% similar (Pettersson et al., 1998).

Two sequence-specific endonucleases (*MboI* and *MboII*) have been demonstrated in *M. bovis*, ATCC 10900 (Gelinas et al., 1977). The genes encoding the *MboI* restriction-modification enzymes are adjacent to each other with this gene cluster encoding two distinct methyl transferases (Ueno et al., 1993) and have been cloned and expressed in *E. coli*. An enzyme from *M. bovis* that methylates the *MboII* recognition sequence has been purified (McClelland et al., 1985). The two genes encoding this class II restriction-modification system are located adjacent to each other and have each been cloned separately in *E. coli* (Bocklage et al., 1991). Details of the interaction of the *MboII* restriction endonuclease with its DNA target site have been investigated (Sektas et al., 1995).

The mol% G + C of the DNA is: 42.5–43 (Bd).

Type strain: ATCC 10900.

GenBank accession number (16S rRNA): AF005182.

3. ***Moraxella canis*** Jannes, Vaneechoutte, Lannoo, Gillis, Vancanneyt, Vandamme, Verschraegen, Van Heuverswyn and Rossau 1993, 448^{VP}
ca'nis. L. gen. n. *canis* of a dog.

Cells are cocci (diameter ~0.8–1.3 µm) and usually occur in pairs or short chains. Gram negative, but there may be a tendency to resist decolorization. A brown pigment may be observed on Mueller–Hinton agar. Cells are non-motile in liquid medium. Chemoorganotrophic and aerobic. Mesophilic. Cell growth is usually observed on a medium containing mineral salts and ammonium acetate, on lactose–bromothymol blue agar, on eosine–methylene blue agar, on Mueller–Hinton medium, and on MacConkey agar at 37°C. Growth occurs on Mueller–Hinton agar at room temperature. No growth occurs in the presence of 6% NaCl or at 5°C. Usually indifferent to bile salts. Oxidase and catalase positive. Extracellular DNase, butyrase, esterase, and γ-glutamyl-aminopeptidase are produced. Phenylalanine deaminase, indole, urease, and carbonic anhydrase are not produced. No acid is produced from glucose. Tween 80 and starch are not hydrolyzed. Autoagglutination in physiological solution is usually not observed. Gelatin and serum are not liquefied. No hemolysis of human and sheep blood occurs. The major fatty acids are oleic acid (C_{18:1} ω9c) and palmitoleic acid (C_{16:1} ω7c). Isoheptadecanoic acid (C_{17:0} iso) is present in moderate amounts, and myristic acid (C_{14:0}) is absent. Typical characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Isolated from the saliva of dogs or cats and occasionally

from humans (dog bite wounds or blood). Pathogenicity is not known.

Further comments Strains are frequently competent for genetic transformation. Transformation studies of strain NCTC 4103 (ATCC 23246), originally considered a strain of *M. catarrhalis*, showed that this strain interacted weakly with authentic strains of *M. catarrhalis* (Catlin and Cunningham, 1964). In an investigation involving transformation of nutritional mutants of *M. catarrhalis* and NCTC 4103, it was shown that strain NCTC 4103 is more closely related to *M. canis* than to *M. catarrhalis* (Juni, 1990). Significant levels of DNA–DNA relatedness (52–61%) were found between strain NCTC 4103 and strains of *M. canis*, and it has been suggested that strain NCTC 4103 occupies a position somewhere between *M. catarrhalis* and *M. canis* (Jannes et al., 1993). From the results of a polyphasic taxonomic investigation, it was concluded that strains of *M. canis* constitute a distinct cluster within the genus *Moraxella* (Jannes et al., 1993).

A review of the properties of *M. catarrhalis* appeared in 1992 (Doern, 1992).

The mol% G + C of the DNA is: 45.9–47.5 (T_m).

Type strain: N7, CCUG 8415A, LMG 11194.

4. ***Moraxella caprae*** Kodjo, Tønnum, Richard and Bøvre 1995, 471^{VP}
ca'prae. L. gen. n. *caprae* of a goat.

Gram negative. Medium to large straight rods with a tendency to grow as diplobacilli or in short chains. Non-motile, aerobic, and grow well at temperatures between 30 and 37°C. Colonies are grayish white, smooth, and convex, vary in size from 1 to 1.5 mm, and are surrounded by a wide area of hemolysis after 36–48 h of aerobic incubation on 5% blood agar plates. Grow poorly on nutrient agar. Oxidase and catalase positive. All strains hydrolyze Tween 80 and tributyrin and reduce nitrate. No acid is produced from carbohydrates. All strains are nonproteolytic (Loeffler slant and gelatin negative). Indole and urease reactions are also negative. The main features that distinguish *M. caprae* from *M. bovis* are reduction of nitrate and absence of proteolysis. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*. The clinical significance of this organism is not known. All strains were isolated from the nasal cavity of healthy goats.

Further comments Strains of *M. caprae* have a high degree of genetic affinity to *M. bovis*, as shown by quantitative and qualitative genetic transformation assays, and exhibit high DNA–DNA relative binding ratios to each other, but lower levels of DNA relatedness to all other species of *Moraxella* (Kodjo et al., 1995). None of the eight strains of *M. caprae* tested was competent for genetic transformation.

The mol% G + C of the DNA is: 40–41.5 (HPLC).

Type strain: CCUG 33296, NCTC 12877.

5. ***Moraxella catarrhalis*** (Frosch and Kolle 1896) Henriksen and Bøvre 1968, 391^{AL} (*Mikrokokkus catarrhalis* Frosch and Kolle in Flüge 1896, 15; *Branhamella catarrhalis* (Frosch and Kolle 1896) Catlin 1970, 157; *Moraxella* (*Branhamella*) *catarrhalis* Bøvre 1979, 404.)
ca.tarrh.a'lis. Gr. adj. *catarrhus* downflowing, *catarrh*; M.L.adj. *catarrhalis* of *catarrh*.

Cocci; division in two planes at right angles to each other

and tetrad formation may be observed. May be fimbriated. Colonies are approximately 2.0 mm in diameter in 48 h, opaque, hemispherical, becoming considerably larger and convex, almost flat, on prolonged incubation. They usually have a friable texture and can be shifted around and picked from the agar without losing their semi-convex shape and without adherence to the agar. They are nonhemolytic. Extracellular DNase is produced. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Previously, as a rule, highly sensitive to penicillin, but strains resistant because of β -lactamase production are now generally encountered. Frequently isolated from the human nasopharynx, which appears to be its main natural habitat (Murphy, 1996). Has also been isolated from inflammatory secretions of the middle ear and maxillary sinus, from bronchial aspirate in bronchitis and pneumonia, and occasionally from systemic infections. The organism is most often considered to be a well-adapted parasite, but it is also a significant cause of disease in both the upper and lower respiratory tracts. It is responsible for otitis media and sinusitis in children and infants (Murphy, 1996; Enright and McKenzie, 1997; Catlin, 1990).

Further comments *M. catarrhalis* has been shown to possess several outer membrane proteins; the outer membrane protein profiles of this organism are highly conserved (Murphy, 1996). It produces a rough-type lipopolysaccharide attached to a lipid A moiety, and lacks high-molecular-mass O-polysaccharide chains (Johnson et al., 1976). There are at least three serotypes of *M. catarrhalis* lipopolysaccharide (Vanechoutte et al., 1990).

The cells are frequently competent for genetic transformation and are easily identified by transformation procedures. In a qualitative transformation assay, in which nutritional mutants of a competent strain of *M. catarrhalis* were transformed to prototrophy with homologous wild-type DNA, DNAs from 20 independently isolated strains of *M. catarrhalis* all transformed these mutants with the same efficiency (Juni, 1990). In quantitative transformations, comparing ratios of interstrain to intrastrain transformation of the high-level streptomycin-resistance marker, a competent strain of *M. catarrhalis* showed extremely poor transformation (ratios $<5 \times 10^{-5}$) with DNAs from representative strains of most species of *Moraxella*. The exception was *M. cuniculi*, where a ratio of 1×10^{-3} was obtained (Bøvre and Hagen, 1981). In another study of transformation of the high-level streptomycin-resistance marker, an affinity ratio $>10^{-4}$ for *M. catarrhalis* and *M. bovis* was observed (Kodjo et al., 1997).

The amino acids glycine and arginine are the principal growth factors in a defined medium for all strains of *M. catarrhalis* (Juni et al., 1986).

The species appears genetically homogeneous, but one deviating strain, NCTC 4103 (ATCC 23246), has been detected (Catlin and Cunningham, 1964; Bøvre, 1965b, 1980; Bøvre and Hagen, 1981). Strain NCTC 4103 was originally reported to show a significant level ($\leq 59\%$) of DNA–DNA hybridization with *M. catarrhalis* (Rossau et al., 1991), but these results could not be reproduced in a subsequent study (Jannes et al., 1993).

Phylogenetic investigations based on 16S rRNA gene sequence analysis place *M. catarrhalis*, *M. cuniculi*, and *M. caviae* together in a somewhat diverse cluster (Pettersson et

al., 1998). A review of the properties of *M. catarrhalis* appeared in 1992 (Doern, 1992).

The mol% G + C of the DNA is: 40–43 (Bd).

Type strain: Ne 11, ATCC 25238, NCTC 11020.

GenBank accession number (16S rRNA): AF005185.

6. **Moraxella caviae** (Pelczar 1953) Henriksen and Bøvre 1968, 391^{AL} (*Neisseria caviae* Pelczar 1953, 744; *Moraxella (Branhamella) caviae* (Pelczar 1953) Bøvre 1979, 404.)
ca'vi.ae. M.L. fem. n. *Cavia* generic name of the guinea pig; M.L. gen. n. *caviae* of *Cavia*.

Cocci commonly arranged in pairs with adjacent sides flattened; division in two planes at right angles to each other. Tetrad formation may be observed. The colonies are low, conical, and have a diameter of approximately 2.5 mm in 48 h. They are semiopaque and have a butyrous consistency. Weak hemolysis may be observed, and in this respect, the colonies may resemble those of *M. ovis*.

Sensitive to penicillin, but in some studies the organism has been shown to resist 0.1 U/ml penicillin. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*. Isolated from the pharynx and mouth of healthy guinea pigs.

Further comments All strains examined are noncompetent for genetic transformation. In quantitative transformation of the high-level streptomycin-resistance marker to competent strains of *M. catarrhalis* and *M. cuniculi*, the ratios of interstrain to intrastrain transformation were extremely low (2×10^{-5} and 5×10^{-5} , respectively), but a higher ratio of 5×10^{-4} for transformation of *M. ovis* (Bøvre and Hagen, 1981) was found. Similar transformation studies involving DNA from streptomycin-resistant *M. caviae* resulted in extremely low interactions with competent strains of *M. bovis* (Henriksen, 1973). Phylogenetic studies of 16S rRNA gene sequences place one strain of *M. caviae* in the same, somewhat unstable, cluster, which includes *M. catarrhalis* and *M. cuniculi* (Pettersson et al., 1998). It would appear that *M. caviae* might be the most distantly related species of *Moraxella*.

The mol% G + C of the DNA is: 44.5–47.5 (Bd).

Type strain: ATCC 14659, CCUG 355, NCTC 10293.

7. **Moraxella cuniculi** (Berger 1962) Bøvre and Hagen 1984, 355^{VP} (Effective publication: Bøvre and Hagen 1981, 1522) (*Neisseria cuniculi* Berger 1962, 455.)
cun.i'cu.li. L. gen. n. *cuniculi* of the rabbit.

Cocci commonly arranged in pairs with adjacent sides flattened; division in two planes at right angles to each other. Tetrad formation may be observed. The colony size in 48 h is similar to, or slightly smaller than, that of *M. catarrhalis*. The colonies are conical, opaque, and often friable. Nonhemolytic. Isolated from the mouths of healthy rabbits. No evidence of pathogenicity. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Further comments Using the high-level streptomycin-resistance marker, ratios of interstrain to intrastrain transformation of 2×10^{-3} , 2×10^{-4} , and 1×10^{-4} for interaction with *M. catarrhalis*, *M. ovis*, and *M. lacunata*, respectively, were obtained (Bøvre and Hagen, 1981); these results verify that *M. cuniculi* is a species of the genus *Moraxella*. Only low level DNA–DNA hybridization between *M. cuniculi* and either *M. ovis* or *M. caviae* could be demonstrated (Véron et al., 1993). The type strain is competent for genetic

transformation, making genetic identification of new isolates possible.

The mol% G + C of the DNA is: 44.5 (Bd).

Type strain: ATCC 14688, NCTC 10297.

8. **Moraxella equi** Hughes and Pugh 1970, 457^{AL}
e'qui. L. gen. n. *equi* of the horse.

Nonmotile diplococcus. Colonies on blood agar are 0.5–1.0 mm in diameter, resembling *M. bovis*, but are nonhemolytic and butyrous. Other properties resemble those of *M. bovis*. Oxidase and catalase positive. Produces an alkaline reaction in several carbohydrate fermentation media, grows well on Herellea agar, unlike *M. bovis* which reportedly fails to grow on Herellea agar (Hughes and Pugh, 1970). Liquefies gelatin at 37°C, does not produce ornithine or lysine decarboxylases, does not reduce nitrate to nitrite, does not produce H₂S or indole, and does not grow on Simmons citrate medium, KCN medium, or urea broth. Precipitin lines indistinguishable from those with control homologous antiserum were produced when *M. equi* cells were tested against three *M. bovis* antisera (Hughes and Pugh, 1970).

Isolated from the eyes of two groups of horses and ponies that had signs of conjunctivitis and erosion of the epithelium of their eyelid margins. Becomes established when introduced into equine eyes but not into bovine eyes, in contrast to *M. bovis*, which does not survive in equine eyes (Hughes and Pugh, 1970).

Further comments *M. equi* has shown overlapping affinities in streptomycin-resistance transformation with some bovine isolates of *M. bovis* (Bøvre, unpublished results); an interstrain to intrastrain transformation ratio of 8×10^{-2} was found for interaction with the type strain of *M. bovis* (Kodjo et al., 1995). *M. equi* was indistinguishable from strains of “*M. lacunata* subsp. *liquefaciens*” in ability to grow on a defined medium containing 11 growth factors, and in transformation of nutritional mutants of *M. bovis* (Juni et al., 1988), such transformations being significantly less efficient than those observed with DNAs from strains of *M. bovis* (Juni et al., 1988). Furthermore, the 16S rRNA gene of *M. equi* has a sequence similarity of 99.9% to that of a strain of “*M. lacunata* subsp. *liquefaciens*” (Pettersson et al., 1998). Nevertheless, in DNA–DNA hybridization and multi-locus enzyme electrophoresis studies *M. equi* and *M. bovis* are virtually indistinguishable (Tønjum et al., 1992). *M. equi* and *M. bovis* do not show a close relationship to “*M. lacunata* subsp. *liquefaciens*” in DNA–DNA hybridization (Tønjum et al., 1992). These findings confirm that *M. equi* is a unique *Moraxella* species. It might be considered that *M. equi* first appeared when DNA from either *M. bovis* or “*M. lacunata* subsp. *liquefaciens*” was transferred from one to the other and integrated into the chromosome of the recipient strain, resulting in a hybrid strain. Such a strain may have survived because it is better suited to occupy certain specific sites in the horse. Currently only a single strain of *M. equi* (ATCC 25576) is available. Further studies, involving isolation of moraxellae from horses in different geographical regions, are required to determine whether *M. equi* is universally distributed.

The mol% G + C of the DNA is: not reported.

Type strain: ATCC 25576.

9. **Moraxella nonliquefaciens** (Scarlett 1916) Lwoff 1939, 175^{AL} (*Bacillus duplex non liquefaciens* Scarlett 1916, 107; *Moraxella (Moraxella) nonliquefaciens* Bøvre and Henriksen 1967a, 134.)

non.li.que.fa'ci.ens. L. pref. *non* not; L. part. adj. *liquefaciens* dissolving; L. part. adj. *nonliquefaciens* not dissolving.

Rods, showing variation in shape, size, and arrangement. The colony size and colony type variation in relation to fimbriation and "twitching motility" are similar to that described for *M. bovis*, but the corroding colonies of freshly isolated strains may often be larger and more spreading (Bøvre et al., 1970; Bøvre and Frøholm, 1972b; Henriksen et al., 1972). Colonies are translucent to semiopaque. Some strains are very mucoid. Cell walls of nonmucoid strains often contain polysaccharide that is chemically and immunologically identical to capsular polysaccharide of serogroup B meningococci (Bøvre et al., 1983). No hemolysis occurs on blood agar. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

M. nonliquefaciens is nutritionally fastidious; 22 independently isolated strains were able to grow in a defined medium containing nine amino acids and three vitamins (Juni et al., 1984). It fails to grow on Hugh and Leifson's OF medium. Previously, as a rule, highly sensitive to penicillin, but strains resistant because of β-lactamase production are now often encountered. Most frequently isolated from the nasal cavity, which is probably the main natural habitat. Also found in other samples from the respiratory tract, including bronchial aspirates in chronic bronchitis and upper respiratory disease (Graham et al., 1990), the latter in small children. It is considered a well-established parasite of humans with good adaptation to the host and, although rarely reported, may play a role, similar to *M. catarrhalis*, as an agent of disease. It has been found in individuals with endophthalmitis (Ebright et al., 1982; Lobue et al., 1985) and occasionally in blood in systemic disease (Brorson et al., 1983; Tønnum et al., 1992).

Further comments The cells are frequently transformable, with competence for genetic transformation being associated with fimbriation of cells and with corroding, spreading colonies (Bøvre and Frøholm, 1972a). Strains of *M. nonliquefaciens* have type 4 pilin genes (*tfa*) (Tønnum et al., 1991). *M. nonliquefaciens* strains can be identified by quantitative (Bøvre, 1964, 1965a, c) and qualitative (Juni et al., 1987) genetic transformation. The species is genetically homogeneous (Bøvre and Henriksen, 1967a). Strains of *M. nonliquefaciens* have high interstrain 16S rRNA gene sequence similarities (99.8%) and can be distinguished from other strains of the *M. lacunata* cluster by such analysis (Pettersson et al., 1998).

The mol% G + C of the DNA is: 40–44 (Bd).

Type strain: 4663/62, ATCC 19975, NCTC 10464.

10. ***Moraxella ovis*** (Lindqvist 1960) Henriksen and Bøvre 1968, 391^{AL} (*Neisseria ovis* Lindqvist 1960, 165; *Moraxella (Branhamella) ovis* (Lindqvist 1960) Bøvre 1979, 404.)
o'vis. L. gen. n. *ovis* of the sheep.

Cocci commonly arranged in pairs with adjacent sides flattened; division in two planes at right angles to each other. Tetrad formation may be observed. Colonies on blood agar are ~2.5 mm in diameter in 48 h, low convex, becoming almost flat after longer incubation. They are grayish white and slightly smaller and more opaque than the largest colony forms of *M. bovis*. The colonies are friable or soft, and are usually surrounded by a narrow zone of

clear hemolysis. Nonhemolytic variants may arise spontaneously in cultures, and primarily nonhemolytic strains (genetically confirmed as belonging to the species) have been isolated directly from sheep and cattle (and a horse). Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Isolated from the conjunctiva of sheep and cattle and from upper respiratory sites in sheep (and a horse). Frequently found in cultures from eyes in infectious keratoconjunctivitis of sheep, and often found together with *M. bovis* in cases of such disease in cattle. Considered to be of low pathogenicity, most frequently occurring as a harmless parasite.

Further comments The strains are frequently competent for genetic transformation and can be easily identified by quantitative transformation (Bøvre and Hagen, 1981). Demonstrated to be a species of *Moraxella* by virtue of transformation of the high-level streptomycin-resistance marker (Bøvre and Hagen, 1981) as well as by transformation of an auxotrophic mutant of *M. bovis* (Juni et al., 1988). Other closely related coccal and rod-shaped moraxellae, not yet described, have been isolated from sheep and cattle and these must be taken into consideration when identifying *M. ovis*, as well as *M. bovis*, in these animals (Bøvre, 1980; Bøvre and Hagen, 1981). Strains of *M. ovis* grow on a defined medium (medium MB) suitable for growth of strains of *M. bovis* (Juni et al., 1988). There is a significant amount of DNA–DNA hybridization between *M. ovis* and *M. bovis* (Tønnum et al., 1989). The 16S rRNA genes of *M. ovis* and *M. bovis* have been shown to be 97.1–97.4% similar (Pettersson et al., 1998).

The mol% G + C of the DNA is: 44.5–46.5 (Bd).

Type strain: 199/55, ATCC 33078, NCTC 11227.

GenBank accession number (16S rRNA): AF005186.

*Species Incertae Sedis*¹

1. ***Moraxella atlantae*** Bøvre, Fuglesang, Hagen, Jantzen and Frøholm 1976, 520^{AL} (*Moraxella (Moraxella) atlantae* Bøvre 1979, 404.)

at.lan'tae. M.L. gen. n. *atlantae* of Atlanta, the American city where strains of the species were first recognized as a distinct group by Elizabeth O. King.

This group is comprised of strains originally designated *Moraxella atlantae*. Rods, showing variation in shape and size. There is little tendency to form chains. Cells are often fimbriated and may show "twitching motility." Colonies are small (0.2–0.5 mm in diameter, occasionally up to 1 mm) in 48 h on either blood agar or chocolate agar. Agar corrosion and spreading of colonies may be seen. Permanent dissociation of such colonies formed by fimbriated cells into nonfimbriated, noncorroding variants has not been observed. Colonies are semiopaque and may appear slightly pink, probably due to blood pigment accumulation. Non-hemolytic. Bile salts (or Tween 80) stimulation of growth occurs with levels up to 1%. Penicillin sensitive, but usually resists 0.05 U/ml penicillin. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

1. Although originally considered to be species of *Moraxella*, each of the five species listed in this section appears to represent an independent genus of the *Moraxellaceae*.

Only rarely isolated. Found in human blood, cerebrospinal fluid, and spleen. The natural habitats and pathogenicity are not yet defined.

Further comments The strains are often competent for genetic transformation and can be identified by transformation methods (Bøvre et al., 1976). The cellular lipid composition closely resembles that of *M. phenylpyruvica*; however, *M. atlantae* differs by containing true waxes (Bøvre et al., 1976; Bryn et al., 1977). Although previously considered a species of *Moraxella*, this organism constitutes a unique genus in the family *Moraxellaceae*, based on the following evidence: 1. Ratios of interstrain to intrastrain transformation of the high-level streptomycin-resistance marker for all other strains in the family tested are $\leq 2 \times 10^{-5}$ (Bøvre and Hagen, 1981). 2. There is only weak DNA–DNA hybridization with *M. phenylpyruvica* and no significant level with other members of the family (Tønnum et al., 1989). 3. Can be differentiated from other *Moraxellaceae* species by the presence of mannose in its whole-cell monosaccharide pattern (Jantzen et al., 1974). 4. 16S rRNA gene sequence similarity with other members of the *Moraxellaceae* is highest for *M. osloensis* (93%) and is lower for other members of the family (Pettersson et al., 1998).

M. atlantae should be given a new genus designation.

The mol% G + C of the DNA is: 46.5–47.5 (Bd).

Type strain: ATCC 29525, CDC 5118, NCTC 11091.

2. ***Moraxella boevrei*** Kodjo, Richard, and Tønnum 1997, 120^{VP} *boev' re.i.* L. gen. n. *boevrei* of Bøvre, in honor of Kjell Bøvre.

This group is comprised of strains originally designated *Moraxella boevrei*. Cells are thin rods to coccus-like structures and have a tendency to occur as diplobacilli or in short chains. Colonies on blood agar are gray-white, rough, and small (diameter after 48 h, 0.5 mm) and are surrounded by a wide area of hemolysis. Isolates are nonmotile, aerobic, and mesophilic. Growth occurs in an aerobic atmosphere (preferably containing 5% CO₂) on 5% sheep or human blood agar plates or chocolate agar plates. Growth does not occur or occurs weakly on nutrient agar. The oxidase and catalase reactions are strongly positive. All strains hydrolyze Tween 80 and reduce nitrate. No acid is produced from carbohydrates. All strains are proteolytic (Loeffler slants and gelatin positive). Indole and urease reactions are negative. The only conventional characteristic that distinguishes *M. boevrei* from *M. bovis* is the reduction of nitrate. Otherwise, all other conventional and hydrolytic enzymes of *M. boevrei* and *M. bovis* are identical. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Isolated from the nasal flora of goats. Pathogenicity has not been demonstrated.

Further comments The strains are competent for genetic transformation and can be identified by quantitative transformation of the high-level streptomycin-resistance marker. The basis for considering *M. boevrei* to be a new genus includes the following: 1. Using *M. boevrei* DNA with *M. bovis*, *M. lacunata*, *M. atlantae*, and *M. catarrhalis* as recipients, ratios of interstrain to intrastrain transformation to streptomycin resistance were less than 5×10^{-5} to 6.5×10^{-5} . 2. There is little or no DNA–DNA hybridization of *M. boevrei* with most other species of *Moraxellaceae* and other Gram-negative bacteria tested. 3. The results of rDNA restriction

pattern analysis showed that the strains of *M. boevrei* were similar to each other and different from 10 other member strains of the *Moraxellaceae* (Kodjo et al., 1997).

It should be noted, however, that DNA–DNA hybridization resulted in relative binding ratios of 12–23 between *M. boevrei* and strains of *M. bovis*, *M. nonliquefaciens*, and *M. ovis* as compared with ratios of 58 to >100 among *M. boevrei* isolates (Kodjo et al., 1997). In view of these findings, *M. boevrei* may actually be a distantly related species of the genus *Moraxella* (*sensu stricto*). Pending further studies to establish the precise taxonomic position of *M. boevrei*, this entity is presently considered a possibly unique genus of the *Moraxellaceae*, based on the criteria used to classify members of this family.

The mol% G + C of the DNA is: 41–41.5 (HPLC).

Type strain: 88365, ATTC 700022, CIP 104716, CCUG 35435, NCTC 12925.

3. ***Moraxella lincolnii*** Vandamme, Gillis, Vancanneyt, Hoste, Kersters, and Falsen 1993, 479^{VP} *lin' col.ni.i.* N.L. gen. n. *lincolnii* of K. Lincoln, a Swedish microbiologist.

This group is comprised of strains originally designated *Moraxella lincolnii*. Cells are nonmotile, coccus-like to plump rods 1–1.5 × 1.5–2.5 μm. They often occur in pairs and may form short chains. After 2 d incubation, colonies are whitish, smooth, convex, and circular and have a diameter of 1–3 mm. The colonies of some strains may have a flattened edge. No hemolysis and no production of pigment or odor. Aerobic. Optimal growth occurs at 28–33°C. Growth also occurs at 36–37°C but not at 42°C. Growth occurs in the absence of NaCl; no growth occurs in the presence of 1.5% NaCl. Oxidase and catalase activities are present. All strains are susceptible to penicillin (10 μg discs). Most strains reduce nitrite but fail to reduce nitrate. Acid is not produced from D-glucose, D-fructose, maltose, or sucrose. Lactate cannot be used as a carbon source. No denitrification, liquefaction of gelatin, proteolysis on Loeffler slants, hydrolysis of Tween 80, or indole production. No urease, DNase, or β-galactosidase activity. The major fatty acids are C_{10:0}, C_{12:0} 3OH, C_{16:1} ω7c, C_{16:0}, C_{18:1} ω9c, and C_{18:0}. Other characteristics are listed in Table BXII.γ.130 of the description of the family *Moraxellaceae*.

Has been isolated mainly from the respiratory tract of humans. Its clinical significance, if any, is unknown.

Further comments Assignment of these organisms to *Moraxellaceae* is based on the results of DNA–rRNA hybridization studies in which DNAs from strains of *M. lincolnii* were hybridized with radioactively labeled rRNA from *M. lacunata* (Vandamme et al., 1993). *T_{m(e)}* values of 74.4–74.8°C obtained are well within the range for all members of the *Moraxellaceae* (*T_{m(e)}* range: ~70–80.0°C [Rossau et al., 1991]). Further studies are required to establish possible relationships to other members of the family.

M. lincolnii will require a new genus designation should further investigations verify that it is, indeed, a distinct genus of the *Moraxellaceae*.

The mol% G + C of the DNA is: 44 (*T_m*).

Type strain: CCUG 9405, LMG 5127.

4. ***Moraxella osloensis*** Bøvre and Henriksen 1967a, 131^{AL} (*Moraxella* (*Moraxella*) *osloensis* Bøvre 1979, 404.)

os.lo.en'sis. M.L. adj. *osloensis* pertaining to Oslo, Norway, where the species was first recognized.

This group is comprised of strains originally designated *Moraxella osloensis*. Rods, showing variation in size, shape, and arrangement similar to that of the genus *Moraxella* (*sensu stricto*). The cells may be indistinguishable from cocci in some specimens or when freshly isolated. A distinct rod shape occurs with pronounced formation of fusiform cells when cultured in sublethal concentrations of penicillin; diplocells may sometimes appear like "waterbuffalo horns" (Bøvre, unpublished). Intracellular inclusions of poly- β -hydroxybutyrate are usually formed, especially when the nitrogen source is limiting.

Colonies are 2.0–2.5 mm in diameter in 48 h and are semiopaque. They do not exhibit colony type variation.

Strains usually grow in mineral media with ammonium ions and acetate, but one genetically verified exception has been found. Highly or, most often, moderately sensitive to penicillin, but resistant strains which possess β -lactamase have been found. Other characteristics are listed in Table BXII. γ .130 of the description of the family *Moraxellaceae*. Isolated from the upper respiratory tract, genitourethral specimens, blood, cerebrospinal fluid, and pyogenic manifestations in joints, bursae, and other sites from humans. Not yet isolated with certainty from nonhuman sources. Usually considered a harmless parasite, but significant pathogenicity is possible.

Further comments Although originally designated as members of *M. nonliquefaciens*, two strains (19116/51 and 752/52) of the 22 studied were considered to represent two new taxonomic groups, based on quantitative transformation studies using the high-level streptomycin-resistance marker (Bøvre, 1964). Further studies of strains similar to strain 19116/51 led to the designation of *M. osloensis* (Bøvre, 1965d; Bøvre and Henriksen, 1967a). Strains are frequently competent for genetic transformation and procedures for their identification by transformation are readily performed (Bøvre, 1965d; Bøvre et al., 1977; Juni, 1974). Although previously considered to be a species of *Moraxella*, this organism constitutes a unique genus in the family *Moraxellaceae*, based on the following evidence: 1. Ratios of interstrain to intrastrain transformation of the high-level streptomycin-resistance marker for all other strains in the family tested (Bøvre and Hagen, 1981) are $\leq 4 \times 10^{-5}$. 2. Except for a weak reaction with DNA from a strain of *M. phenylpyruvica* (752/52), there is no detectable DNA–DNA hybridization with DNA from most members of the family (Tønjum et al., 1991). 3. DNAs from strains of *M. osloensis* have a unique average intergroup $T_{m(e)}$ value (75.6°C) when hybridized with rRNA from "*M. lacunata* subsp. *liquefaciens*", ATCC 17952 (Rossau et al., 1991). 4. A *trpE*[−] mutant of *M. osloensis* was transformed to prototrophy only by DNAs from strains of *M. osloensis* (Juni, 1974). 5. Unique cellular fatty acid composition (Jantzen et al., 1974; Bøvre et al., 1977; Moss et al., 1988). 6. Ability to grow on a defined medium containing a single organic carbon and energy source with an ammonium salt as the nitrogen source (Bøvre and Henriksen, 1967a; Juni, 1974). A specific endonuclease, *MosI*, has been purified from *M. osloensis*, ATCC 19976, and shown to recognize the same sequence as *MboI* (Gelinas et al., 1977).

Phylogenetic analysis of 16S rRNA gene sequences has

shown that *M. osloensis* and *M. atlantae* form a common and distinct clade of the *Moraxellaceae* (Pettersson et al., 1998) (Fig. BXII. γ .147). Strains of *M. osloensis* and *M. atlantae* were shown to be distantly related to each other, their 16S rRNA genes displaying 93.4% similarity and $\leq 93\%$ similarity, respectively, to other members of the *Moraxellaceae* (Pettersson et al., 1998).

M. osloensis should be given a new genus designation.

The mol% G + C of the DNA is: 43–46 (Bd).

Type strain: ATCC 19976, CDC A1920, NCTC 10465.

Additional Remarks: A frequently used (transformable) reference strain is CDC 5873.

5. ***Moraxella phenylpyruvica*** Bøvre and Henriksen 1967b, 344^{AL}. Epit. spec. cons. Opin. 42, Jud. Comm. 1971, 107 (*Moraxella* (*Moraxella*) *phenylpyruvica* Bøvre 1979, 404.) *phe.nyl.py.ru'vi.ca*. M.L. n. *acidum phenylpyruvicum* phenylpyruvic acid; M.L. fem. adj. *phenylpyruvica* pertaining to phenylpyruvic acid, the product of deamination of phenylalanine by this organism.

This group is comprised of strains originally designated *Moraxella phenylpyruvica*. Rods, showing variation in shape, size, and arrangement. The colonies are relatively small (0.9–1.0 mm in diameter) in 48 h on blood agar. They often appear semiopaque with a very slight pink hue, probably due to blood pigment accumulation. Nonhemolytic. Phenylalanine deaminase and urease positive, but strains negative for one or, very rarely, both activities may occur. Grows distinctly, but slowly, at 4–10°C. Usually grows at high salt or bile concentrations; the minimum inhibitory concentrations are usually 7.5–9.0% NaCl and 5% bile salts (Oxoid). Bile salt (Tween 80) stimulation of growth occurs, with levels up to 4% (Snell et al., 1972; Bøvre et al., 1976). Although usually highly sensitive to penicillin, several strains of human and animal origin have been found to be penicillin resistant due to β -lactamase production. Other characteristics are listed in Table BXII. γ .130 of the description of the family *Moraxellaceae*.

Isolated from human blood and cerebrospinal fluid, the genitourethral tract, and from other sites and specimens of humans; also isolated from the genital tract and brain of sheep and cattle, the intestine of a goat, and the genital tract of pigs. Pathogenicity for humans, and perhaps animals, has not been assessed adequately, although judging from its sources of isolation this species may be a significant potential pathogen.

Further comments Strains competent for genetic transformation have not yet been found, but genetic relationships to other taxa of the family *Moraxellaceae* have been detected by the use of strains as DNA donors in transformation. The cellular lipid composition is unique among recognized *Moraxellaceae* species in that true waxes are absent; in other respects, the lipid composition resembles that of *M. atlantae* (Jantzen et al., 1974; Bøvre et al., 1976; Bryn et al., 1977; Moss et al., 1988). Although previously considered a species of *Moraxella*, this organism constitutes a unique genus in the family *Moraxellaceae*, based on the following evidence: 1. Ratios of interstrain to intrastrain transformation of the high-level streptomycin-resistance marker for all other strains in the family tested are equal to, or less than, 2×10^{-5} (Bøvre and Hagen, 1981). 2. There is only slight DNA–DNA hybridization with *M. osloensis* and *M. at-*

lantae, and nondetectable levels with other members of the family (Tønjum et al., 1989). 3. A streptomycin-dependent marker of *M. phenylpyruvica*, which had been transformed into the competent strain of *Acinetobacter* (a rare occurrence), was transformed to streptomycin independence by DNAs from 35 strains of *M. phenylpyruvica*; DNAs from other members of the *Moraxellaceae* were inactive (Juni, 1990).

M. phenylpyruvica may be heterogeneous since strains 752/52 and ATCC 23333^T have a 16S rRNA gene sequence similarity of only 95.9% (Pettersson et al., 1998). Both of these strains react in the same positive manner in trans-

formation of an *M. phenylpyruvica* streptomycin-dependent marker located in the competent strain of *Acinetobacter*, indicating a strong relationship (Juni, 1990).

M. phenylpyruvica should be given a new genus designation.*

The mol% G + C of the DNA is: 42.5–43.5 (Bd).

Type strain: ATCC 23333, CDC 2863, NCTC 10526.

*Editorial Note: Bowman et al. (1996) have proposed the transfer of *Moraxella phenylpyruvica* to the genus *Psychrobacter* as *Psychrobacter phenylpyruvicus*.

Genus II. *Acinetobacter* Brisou and Prévot 1954, 727^{AL}

ELLIOT JUNI

A.ci.ne' to.bac.ter. Gr. adj. *akinetos* unable to move; M.L. n. *bacter* the masculine form of the Gr. neut. n. *bactrum* a rod; M.L. masc. n. *Acinetobacter* nonmotile rod.

Rods 0.9–1.6 × 1.5–2.5 μm, becoming spherical in the stationary phase of growth. Colonies are generally nonpigmented and are mucoid when the cells are encapsulated. Cells commonly occur in pairs and in chains of variable length. Do not form spores. Gram negative but occasionally difficult to destain. Swimming motility does not occur but the cells display "twitching motility", presumably because of the presence of fimbriae. Aerobic, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. Most strains do not reduce nitrate to nitrite. Most strains grow between 20 and 37°C, having temperature optima of 33–35°C. Some strains cannot grow at 37°C. **Oxidase negative. Catalase positive.** Grow well on most complex media. Most strains grow in defined media containing a single carbon and energy source, such as acetate or lactate, using ammonium or nitrate salts, or one of several common amino acids, as a supply of nitrogen. Frequently amino acids such as glutamic acid or aspartic acid can serve as a single source of carbon, energy, and nitrogen in a defined mineral medium. With rare exceptions, they display no growth factor requirements. Most frequently saprophytic, occurring naturally in soil, water, sewage, and foods such as raw vegetables. Can also reside, possibly indigenously, on the human skin and in the human respiratory tract. Can cause nosocomial infections such as bacteremia, secondary meningitis, pneumonia, and urinary tract infections in humans.

The mol% G + C of the DNA is: 38–47.

Type species: *Acinetobacter calcoaceticus* (Beijerinck 1911) Baumann, Doudoroff and Stanier 1968b, 1538 emend. Bouvet and Grimont 1986, 238 (*Micrococcus calcoaceticus* Beijerinck 1911,1067.)

FURTHER DESCRIPTIVE INFORMATION

Habitat *Acinetobacter*s occur naturally in soil and water (Baumann, 1968) and are present in sewage (Warskow and Juni, 1972). It has been estimated that at least 0.001% of the total culturable, heterotrophic, aerobic population in soil and water are *acinetobacter*s (Baumann, 1968). *Acinetobacter*s have been demonstrated to associate with soybean root nodules in relatively large numbers (Wong et al., 1986). They have been shown to occur in raw, washed, and frozen vegetables (Gennari and Stegagno, 1986), in fresh, frozen, and stored fish products (Gennari and Stegagno, 1985), and in spoiled meat, milk, and cheese (Gennari et al., 1992). Body sites, such as the skin and respiratory

tract, also appear to harbor certain *acinetobacter*s (Bergogne-Bérézin and Towner, 1996).

General Physiology

CELL STRUCTURES Rapidly growing cells tend to be plump rods, whereas cells in the stationary phase of growth are spherical and have a somewhat smaller diameter than the rods (Baumann et al., 1968b). An occasional rod-shaped cell can frequently be observed in a microscopic field of stationary phase cells. Cells occur typically in pairs, and occasionally in chains. Many strains are encapsulated and capsules may be seen readily in India ink wet mounts. Colonies are generally nonpigmented and are mucoid when the cells are encapsulated.

Electron microscopy of thin sections of cells has revealed a cell wall ultrastructure that is typical of Gram-negative bacteria (Breuil et al., 1975; Scott et al., 1976). The peptidoglycan contains muramic acid, glucosamine, alanine, D-glutamic acid, and meso-diaminopimelic acid (Martin et al., 1973; Horisberger, 1977). Unlike the cell walls of several Gram-negative bacteria, the murein of one strain of *Acinetobacter* had an unusually high proportion of cross-linked peptide side chains (Quintela et al., 1995). It is known that antibodies against *Chlamydia* cross-react with *Acinetobacter* (Brade and Brunner, 1979). Nurminen et al. (1984) present evidence that this cross reaction involves a lipopolysaccharide component; other findings suggest that *Chlamydia* antiserum reacts with the cell wall of *Acinetobacter* (Espinosa et al., 1996).

When originally investigated, one strain of *Acinetobacter* was shown to possess a lipopolysaccharide containing D-glucose, glucosamine, galactosamine, lipid A, ethanolamine, fatty acids, phosphate, and protein (Adams et al., 1970). Lipopolysaccharide composition is not the same in all strains studied (Borneleit and Kleber, 1991). Numerous investigations have elucidated the structures of O-antigen lipopolysaccharides from a variety of *Acinetobacter* strains (Vinogradov et al., 1997; Haseley and Wilkinson, 1997; Haseley et al., 1997, 1998; and other references to be found in these papers). It has been suggested that because of the presence of deoxy sugars, amino acids, and branched polymers, *Acinetobacter* O-antigens display a marked hydrophobicity, which is consistent with the ability of *acinetobacter*s to grow on hydrophobic substrates (Haseley et al., 1997).

Evolution of *acinetobacter*s has resulted in a large variety of surface antigens, many of them capsular, on different strains. A

serologic system for identification of strains by immunofluorescent staining revealed 28 serovars among strains capable of forming acid aerobically from glucose, as well as a series of other serovars among acinetobacters that do not form acid from glucose (Marcus et al., 1969). The capsular polysaccharide of one strain of *Acinetobacter* has been shown to interact with antisera prepared against group B and group G streptococci, as well as with antipneumococcal type XX serum (Heidelberger et al., 1969).

Outer membranes of *A. calcoaceticus* and *A. radioresistens* have been analyzed and found to contain several unique proteins ranging in molecular weight from 12,000 to 50,000 daltons (Fischer et al., 1984; Nishimura et al., 1986a). The major outer membrane protein of *A. calcoaceticus* has a molecular weight of 38,000 daltons, in contrast to that of *A. radioresistens*, which has two major outer membrane proteins with molecular weights of 39,000 and 42,000 daltons, respectively (Nishimura et al., 1986a). Two other studies, which examined larger numbers of strains, showed a good correlation between outer membrane cell envelope protein patterns and DNA-DNA hybridization groups (Ino and Nishimura, 1989; Dijkshoorn et al., 1990).

METABOLISM Nitrite and nitrate can serve as nitrogen sources for acinetobacters, and these organisms possess an assimilatory nitrate reductase (Jyssum and Joner, 1965). Although most acinetobacters are not able to reduce nitrate to nitrite in the conventional nitrate reduction assay, a few strains can do so (Riley and Weaver, 1974); these strains are unable to grow anaerobically with nitrate as the terminal electron acceptor (Juni, 1978). Since all acinetobacters are oxidase negative, they lack cytochrome *c* (Baumann et al., 1968a). However, they do contain cytochromes *b*, *o*, and occasionally *d* and P-450, as well as flavoproteins, and appear to have a normal electron transport system (Asperger and Kleber, 1991).

Enzymes of the tricarboxylic acid cycle have been shown to be present in acinetobacter cell-free extracts. The presence of NADP-isocitrate dehydrogenase has been demonstrated (Weitzman, 1991). The absence of an NAD-isocitrate dehydrogenase, which is typical for most bacteria, and the relatively slow growth rate with acetate as the sole carbon source, indicate that for acinetobacters, as well as for the vast majority of bacteria, the tricarboxylic acid cycle is more important in the synthesis of amino acids (specifically aspartate, glutamate, and proline) and porphyrin than it is in general cellular respiration. Enzymes of the glyoxylate cycle have been demonstrated in *Acinetobacter* and all evidence verifies the obligate requirement of this cycle for growth with two-carbon compounds such as acetate as sole carbon and energy source (Juni, 1978). The enzymes of the Embden-Meyerhof pathway, for conversion of phosphoenolpyruvate to hexose phosphates, are present in *Acinetobacter* and are used for gluconeogenesis from C₄-dicarboxylic acids, but not for the degradation of sugars (Taylor and Juni, 1961).

Relatively few strains can use D-glucose as a carbon source for growth (Baumann et al., 1968b; Juni, 1972). One strain that can use glucose as a carbon source has been shown to catabolize it via the Entner-Doudoroff pathway, involving the key enzymes 6-phosphogluconate dehydratase and 2-keto-3-deoxy-6-phosphogluconate aldolase (Taylor and Juni, 1961). Many acinetobacters that are unable to grow on glucose contain an aldose dehydrogenase and are able to acidify glucose media rapidly (gluconic acid being formed), as well as media containing other sugars such as D-xylose, L-arabinose, D-galactose, D-mannose, L-rham-

nose, maltose, lactose, and cellobiose (Hauge, 1960); this enzyme is responsible for the positive diagnostic test for the production of acid from glucose. Strains of *Acinetobacter* able to form acid from glucose have been demonstrated to produce two glucose dehydrogenases, a soluble and a membrane-bound enzyme (Matsushita et al., 1989; Duine et al., 1982). The two glucose dehydrogenases are distinct enzymes; the membrane-bound form appears to be the only enzyme used by intact cells for the oxidation of glucose (Duine, 1991). Both enzymes use pyrroloquinoline quinone (PQQ, methoxatin) as a coenzyme (Duine, 1991). Genes involved in synthesis of PQQ and coding for membrane-integrated quinoprotein glucose dehydrogenase in *Acinetobacter* have been cloned in *E. coli* (Goosen et al., 1987, 1989a, b; Cleton-Jansen et al., 1989; Dewanti and Duine, 1998).

GROWTH AND NUTRITION Most strains of *Acinetobacter* can grow in a simple mineral medium containing a single carbon and energy source such as ethanol, acetate, lactate, pyruvate, malate, or α -ketoglutarate. Ammonium and nitrate salts serve as nitrogen sources. Some organic compounds, such as amino acids, can serve as single sources of carbon, energy, and nitrogen. Rare strains have growth factor requirements; such strains frequently revert spontaneously to prototrophy when cultured in the laboratory (Warskow and Juni, 1972). Acinetobacters resemble saprophytic pseudomonads in being able to use a wide variety of organic compounds as carbon, energy, and/or nitrogen source for growth in mineral media, and generally oxidize such compounds to completion. Utilizable organic nutrients include certain sugars, fatty acids, aliphatic alcohols, unbranched hydrocarbons, dicarboxylic acids, amino acids, and many aromatic and alicyclic compounds (Baumann et al., 1968b); the spectrum of compounds utilized is frequently species specific (Bouvet and Grimont, 1986). Acinetobacters, normally associated in relatively large numbers with soybean root nodules, have also been shown to obtain energy for growth by oxidizing H₂, which is released by the nodules (Wong et al., 1986). Some strains can utilize a single hexose, D-glucose, and the pentoses D-ribose, D-xylose, and L-arabinose. Other carbohydrates, disaccharides, and sugar alcohols (including glycerol) are not used. There is no evidence that acinetobacters can obtain energy for growth by fermentation. Since acinetobacters are nonmotile, growth in liquid media requires vigorous aeration. Growth can take place from pH 5.0 to 8.0, with a pH optimum for growth of approximately 6.5 (Baumann, 1968; Juni, 1982).

Biochemistry Because of their ability to metabolize many organic compounds, acinetobacters normally synthesize a wide variety of degradative enzymes. Most metabolic and genetic studies in *Acinetobacter* have been directed towards elucidating the pathways involved, as well as regulation of expression of the relevant enzymes. Since aromatic compounds are available through normal breakdown of dead plant and animal tissues, it is not surprising to find that acinetobacters can degrade organic compounds such as benzoate, *p*-hydroxybenzoate, mandelate, quinate, and tryptophan (Fewson, 1991).

The normal soil and water habitats of acinetobacters, coupled with their ability to degrade a wide variety of organic compounds, make it likely that acinetobacters can be used for bioremediation. Although toxic to most microorganisms, aromatic compounds such as salicylate, halogenated aromatics, and phenol (Schirmer et al., 1997) can be degraded by acinetobacters, occasionally by strains harboring specific degradative plasmids (Fewson, 1991).

Because of their ability to utilize hydrocarbons, acinetobacters

are capable of playing a role in the degradation of crude oil. It has been demonstrated that certain acinetobacters elaborate and excrete polymers that emulsify hydrocarbons, such as oils, thus making these substances available for degradation in an aqueous environment (Gutnick et al., 1991; Navon-Venezia et al., 1995). Acinetobacters that degrade alkanes are frequently isolated from areas contaminated with petroleum. Various aspects of alkane metabolism have been investigated (Asperger and Kleber, 1991; Maeng et al., 1996; Parche et al., 1997; Ratajczak et al., 1998). Many acinetobacters possess the ability to degrade alkanes by pathways that, in some cases, involve rubredoxin, rubredoxin reductase, and cytochrome P-450 (Asperger and Kleber, 1991; Geissdörfer et al., 1995).

Some acinetobacters elaborate an extracellular lipase, as determined by hydrolysis of Tween 80 (Baumann et al., 1968b) and other lipids (Breuil and Kushner, 1975; Fischer, 1986; Kok et al., 1995a,b, 1996). Strains that show hemolysis on blood agar plates excrete phospholipase (Lehmann, 1971, 1973).

Early studies of pathways for degradation of aromatic compounds in *Acinetobacter* demonstrated that these compounds are generally converted to β -ketoadipate, and that succinate and acetyl CoA are the products of degradation of β -ketoadipate (Stanier and Ornston, 1973). Evolution of genes for the β -ketoadipate pathway has been reviewed (Ornston and Neidle, 1991). There is evidence for supraoperonic clustering on the chromosome of certain genes involved in aromatic catabolism, where the product of one gene is further metabolized by a second gene in an independent and closely located transcriptional operon (Ornston and Neidle, 1991; Averhoff et al., 1992). An apparent DNA slip-page structure has been proposed to explain the extensive evolutionary divergence of *pcaD* and *catD*, genes that encode identical catalytic activities in *Acinetobacter* but are located in different regions of the chromosome (Hartnett and Ornston, 1994). Evidence has been presented for gene conversion in the nonreciprocal genetic exchange involved in repair of defective *pcaJF* by nucleotide sequences of a functional *catJF* gene in *Acinetobacter* (Kowalchuk et al., 1995), a process that requires a functional *recA* gene (Gregg-Jolly and Ornston, 1994). Transcriptional activation of aromatic genes, similar to *LysR* type activation (Collier et al., 1998), as well as other transcriptional activators, has been reported (DiMarco and Ornston, 1994; Gerischer et al., 1998). Using a mutant of *Acinetobacter* sp. strain ADP1 that was unable to grow on benzene as the sole carbon source, since it lacked both transcriptional activators of the *cat* genes, it was demonstrated that spontaneous mutants able to grow with benzoate carried 10–20 amplified chromosomal regions (amplicons), each encompassing the *cat* genes (Reams and Neidle, 2003).

It has been demonstrated that certain features of aromatic amino acid biosynthesis are unique and characteristic for all acinetobacters (Byng et al., 1985).

Acinetobacter chromosomes have been reported to encode genes for synthesis of several restriction endonucleases and, most probably, their corresponding modification enzymes. These include *AccI* and *AccII* (Roberts, 1985), *AccIII* (Kita et al., 1985), *AlwNI* (Morgan et al., 1987), *AcII* (Degtyarev et al., 1992), *AjoI* (Nowak et al., 1994), *AspM* (Zelinskaia et al., 1996), and *AccBSI* (Abdurashitov et al., 1997). The genes of the *AccI* restriction-modification system have been sequenced following cloning and expression in *E. coli* (Kawakami et al., 1991).

Polyphosphate has been shown to be accumulated in wastewater by some acinetobacters (Gutnick et al., 1991; Kortstee et al., 1994). These organisms appear to be responsible for remov-

ing inorganic phosphate from sewage through their association with activated sludge (Fuhs and Chen, 1975; Buchan, 1983); the presence of K^+ seems to be essential for phosphate uptake (Van Groenestijn et al., 1988). A strain from activated sludge was demonstrated to accumulate large amounts of inorganic polyphosphate (3.7–5.7 mg P/100 mg dry weight of cells) and has been identified as *A. johnsonii* (Bonting et al., 1992). It has been suggested that a polyphosphate-synthesizing system is encoded on a 20-kb plasmid (Bayly et al., 1991). A strain of *A. johnsonii* that accumulates inorganic phosphate under aerobic conditions was shown to contain a high-affinity phosphate-uptake system that is induced in the absence of inorganic phosphate (Van Veen et al., 1993; Kortstee et al., 1994). A polyphosphate kinase gene from strain ADP1 (BD413) has been isolated and shown to be induced by phosphate starvation (Geissdörfer et al., 1998). Inorganic polyphosphate can serve as an energy reserve, as can poly- β -hydroxybutyrate and simple wax esters, all of which have been shown to accumulate in many strains of *Acinetobacter* (Fixter and Sherwani, 1991).

Studies of amino acid transport in *Acinetobacter* have revealed that alanine, lysine, and proline are taken up and concentrated by high affinity systems that are coupled to membrane-bound glucose dehydrogenase (Van Veen et al., 1994).

A phosphotyrosine-protein phosphatase from *A. johnsonii*, which specifically dephosphorylates an endogenous protein kinase known to autophosphorylate at multiple tyrosine residues in the inner membranes of this organism, has been studied and shown to be a member of the phosphotyrosine-protein phosphatase family (Grangeasse et al., 1998).

Mutarotase, an enzyme that catalyzes the anomeric interconversion of D-glucose and other aldoses, has been demonstrated in *Acinetobacter*; its gene has been cloned and expressed in *E. coli* (Gatz et al., 1986), and the nucleotide sequence of the gene directing the synthesis of this enzyme has been determined (Gatz and Hillen, 1986).

Genetics It is to be expected that a relatively large part of the *Acinetobacter* chromosome will code for enzymes involved in degradative pathways, because of the large number of organic compounds that can be degraded by these organisms. A measure of the percentage of the chromosome devoted to these activities comes from the studies of Gutnick et al. (1969), where it was shown that for about 600 organic compounds screened as possible carbon and nitrogen sources for *Salmonella typhimurium* LT-2, an organism previously considered to have only a limited number of possible carbon and energy sources, about 100 utilizable compounds were found. It might be expected that the portion of the chromosomes of *Acinetobacter* strains devoted to the genes controlling synthesis of enzymes involved in degrading organic compounds must be considerably greater than the 10% estimated for *S. typhimurium* (Gutnick et al., 1969).

Using plasmids, it has been possible to mobilize the *Acinetobacter* chromosome and conjugally transfer chromosomal genes to a recipient strain (Towner and Vivian, 1976b). In this manner, it has been demonstrated, using a series of genetic markers, that the *Acinetobacter* chromosome is circular (Towner and Vivian, 1976a) and loci of 23 different mutations were mapped on a circular linkage group (Towner, 1978). The orientation and precise location of more than 40 genes have been physically mapped on the circular chromosome of *Acinetobacter* sp. strain EBF/65, which has been shown to consist of 3780 ± 191 kb (Gralton et al., 1997). Analysis of codon usage in *Acinetobacter* structural genes

has revealed that all possible triplet codons are used, some preferentially (White et al., 1991).

Transformation Genetic exchange in *Acinetobacter* can be mediated by transformation, transduction, and conjugation (Vivian, 1991). Although rare, competence for genetic transformation in *Acinetobacter* has been observed in two strains (Juni and Janik, 1969; Ahlquist et al., 1980). Two clinically isolated strains of *Acinetobacter* were reported to be weakly competent (Bergan and Vaksvik, 1983). Nutritional, as well as antibiotic resistance, markers are readily transformed in competent strain BD413, and transformation of a *trpE*⁻ auxotroph of this strain to prototrophy can be accomplished specifically by DNAs from acinetobacters (Juni, 1972). Genes involved in capsule biosynthesis of a competent strain have been studied by transformation (Juni and Janik, 1969). A partial mapping of genes directing synthesis of enzymes for the tryptophan biosynthetic pathway in *Acinetobacter* has been accomplished using transformation (Sawula and Crawford, 1972). A genetic analysis of the genes for proline biosynthesis has also been reported (Ginther, 1978). Natural transformation has been useful in the analysis of catabolic pathways in *Acinetobacter* (Kloos et al., 1995).

Using natural transformation in *Acinetobacter*, it has been possible to introduce randomly mutated genes, resulting from errors produced during PCR replication, into the chromosome of a competent recipient strain without prior cloning of PCR fragments (Kok et al., 1997). It has been demonstrated that plant DNA can be taken up and recombinationally incorporated into suitable plasmids resident in competent *Acinetobacter* (de Vries and Wackernagel, 1998; Gebhard and Smalla, 1998). A general method, where randomly mutated PCR products are transformed into competent *Acinetobacter* and captured by marker-replacement recombination in a plasmid resident in the transformation recipient, has been described (Melnikov and Youngman, 1999).

Physiological characterization of natural transformation in *Acinetobacter* has been reported (Palmen et al., 1993). The DNA uptake process in *Acinetobacter* does not make use of specific sequence recognition for binding and uptake of DNA (Lorenz et al., 1992; Palmen et al., 1993). Investigations of the molecular basis for DNA uptake in competent strains of *Acinetobacter* have implicated elements in the assembly of the type IV pilus (Palmen and Hellingwerf, 1997; Link et al., 1998), findings that are consistent with earlier observations correlating competence and piliation in other bacteria (Bøvre and Frøholm, 1971).

Phages Lytic phages for acinetobacters are isolated readily from sewage (Blouse and Twarog, 1966; Twarog and Blouse, 1968; Herman and Juni, 1974). A generalized transducing phage (P78) specific for one strain of *Acinetobacter* has been isolated and demonstrated to lysogenize the host strain (Herman and Juni, 1974). Weigle reactivation has been demonstrated for UV-irradiated temperate phage P78, suggesting that *Acinetobacter* possesses an inducible DNA repair pathway (Berenstein, 1982). Ultraviolet light was shown to induce temperate prophage P78 (Berenstein, 1986). Several *Acinetobacter* phages have been classified according to structural characteristics of phage heads and tails (Ackermann et al., 1994).

Plasmids More than 80% of all acinetobacters isolated harbor multiple plasmids (Hartstein et al., 1988; Gerner-Smidt, 1989; Seifert et al., 1994b), some of which are cryptic (Hunger et al., 1990; Minas and Gutnick, 1993; Seifert et al., 1994b); others carry antibiotic resistance markers (Murray and Moellering, 1979;

Hinchliffe and Vivian, 1980; Gerner-Smidt, 1989; Garcia et al., 1996). Transmissible antibiotic resistance in *Acinetobacter* has been shown to be associated with plasmids that belong to broad host-range incompatibility groups (Towner, 1991). Plasmids from many incompatibility groups have been transferred to *Acinetobacter*, by conjugation or mobilization, where most of them are not completely stable (Towner, 1991). By making use of the ability of P1 incompatibility plasmid RP4 to mobilize the *Acinetobacter* chromosome, it was first demonstrated that the chromosomal DNA is circular (Towner and Vivian, 1976a; Towner, 1978). Some *Acinetobacter* plasmids encode genes involved in degradation of certain organic compounds (Winstanley et al., 1987, 1991; Fujii et al., 1997). Plasmid-mediated heavy metal resistance in *Acinetobacter* has been reported (Towner, 1991). A shuttle plasmid for *Acinetobacter* and *Escherichia coli* was constructed by ligation of a cryptic *Acinetobacter* plasmid and pBR322; the *trpE* gene of *Acinetobacter* was cloned in this vector (Hunger et al., 1990). The same shuttle vector was used to construct a gene bank of *Acinetobacter* chromosomal DNA (Hunger et al., 1990). Three cryptic plasmids have been observed in competent strain BD413 and two of these, when ligated with pUC19, resulted in shuttle vectors that replicated in both *Acinetobacter* and in *Escherichia coli* (Minas and Gutnick, 1993). Plasmids have been introduced into strain BD413 by natural transformation (Singer et al., 1986; Hunger et al., 1990). The means by which resident plasmids were originally introduced into isolated strains are not known.

Transposable elements Some acinetobacters carry transposons on their chromosomes and these encode antibiotic resistance genes (Towner, 1991). Studies of the gene for chloramphenicol acetyltransferase (CAT) in a strain of *A. baumannii* revealed that this chromosomal gene is part of a transposon similar to Tn2670 from plasmid NR1; the 5' end of the cloned CAT gene is linked to IS 1 and the 3' end is linked to components of Tn 21 (Elisha and Steyn, 1991). A newly discovered insertion sequence, IS1236, which resembles IS3, has been discovered in *Acinetobacter* sp. strain BD413 (ADP1) where it exists in seven copies (Gerischer et al., 1996). Transposon Tn 3171, which may be identical to Tn7, was shown to be incorporated in the chromosome of *Acinetobacter* in a nonrandom fashion (Towner, 1983). When introduced into *Acinetobacter* on a suicide plasmid, Tn5 was shown to insert nonrandomly into the chromosome at a limited number of sites (Singer and Finnerty, 1984). However, Tn 5 has been inserted into a variety of cloned fragments of *Acinetobacter* DNA that can be transformed into strain BD413 (Doten et al., 1987; Goosen et al., 1987; Reddy et al., 1989; Gregg-Jolly and Ornston, 1994). When present in *Acinetobacter*, Tn 5 confers low-level resistance to streptomycin due to the presence of a Tn5-encoded streptomycin phosphotransferase (O'Neill et al., 1984). Low transposition frequencies for Tn7 and Tn10 to the *Acinetobacter* chromosome have been reported (Ely, 1985). Several different kinds of mutations in an *Acinetobacter* strain were produced by introduction of a derivative of Tn 10 (mini-Tn10 PttKm) (Leahy et al., 1993).

Integrans, genetic elements that integrate cassettes encoding antibiotic resistance, have been demonstrated to be present in the majority (48 of 59 strains) of clinical isolates of *A. baumannii* biotype 9 (Gonzalez et al., 1998).

Pathogenesis and antibiotic susceptibility Although considered normally nonpathogenic, acinetobacters are causative agents of nosocomial infections, particularly in debilitated individuals and those residing in intensive care units (Bergogne-

Bérézin and Towner, 1996). From 1971 to 1981, species of *Acinetobacter* accounted for 1.4% of all nosocomial infections in a university hospital in the United States (Larson, 1984). Infection with *Acinetobacter* has been shown to result in septicemia, meningitis, endocarditis, brain abscess, lung abscess, pneumonia, empyema, urinary tract infections, skin and wound infections, as well as other clinical manifestations (Glew et al., 1977; Bergogne-Bérézin and Towner, 1996). There appears to be a clear association between hospital instrumentation, such as the use of mechanical ventilators, endotracheal and gastric tubes, vascular catheterization, lumbar punctures, myelography, ventriculography, indwelling urinary catheterization, and penetrating keratoplasty, as well as the fitting of contact lenses and subsequent infection with *Acinetobacter*. Evidence has been presented indicating that nosocomial infection with *Acinetobacter* has a distinct seasonal variation (Retailliau et al., 1979).

Although acinetobacters are frequently isolated from several areas of the human body, there is some uncertainty as to whether they are present as contaminants rather than as commensals. Bacteria able to survive and multiply in human tissue must have a mechanism for sequestering iron. Several different siderophores are synthesized by acinetobacters (Smith et al., 1990; Echenique et al., 1992; Actis et al., 1993; Okujo et al., 1994; Yamamoto et al., 1994); a high affinity iron transport system has been demonstrated in a strain of *A. baumannii* (Echenique et al., 1992).

Virtually all acinetobacters can grow in simple mineral media containing one of a large number of organic carbon and energy sources (Warskow and Juni, 1972; Baumann, 1968), a characteristic not usually observed for bacteria whose normal habitat is an animal site. It therefore seems likely that, although some of these bacteria can act as opportunistic pathogens, they should probably not be considered as part of the indigenous microflora. The finding that acinetobacters are frequently isolated from the skin of hospital inpatients (Al-Khoja and Darrell, 1979; Seifert et al., 1997) can account for the occurrence of nosocomial infections with strains of this organism. Although isolated readily from the hands of hospital personnel, there is no evidence for persistent colonization with *Acinetobacter* (Buxton et al., 1978). Of the 20 or more recognized *Acinetobacter* genomic species (DNA groups) and named species, *A. baumannii* and the unnamed groups 3 and 13 *sensu* Tjernberg and Ursing (TU) are the ones most frequently isolated from hospitalized patients (Dijkshoorn et al., 1993; Seifert et al., 1993b).

High rates of antibiotic resistance, frequently to multiple drugs, are found in *Acinetobacter* spp. (Bergogne-Bérézin and Towner, 1996). Unlike most of the *Moraxellaceae*, acinetobacters are generally resistant to penicillin. Growth of most strains is not inhibited by 1 U/ml of penicillin G and the majority of strains are resistant to 100 U/ml (Baumann et al., 1968a). Since the 1970s, there has been an increase in the number of antibiotics, particularly the older antibiotics, to which hospital-isolated acinetobacters have become resistant. Most strains of *A. baumannii* now show resistance to aminopenicillins, ureidopenicillins, cephalosporins, cephamycins, many aminoglycosides-aminocyclitols, chloramphenicol, and tetracyclines (Bergogne-Bérézin and Towner, 1996). Currently, imipenem appears to be the most effective drug, although imipenem-resistant strains have been isolated already (Tankovic et al., 1994; Lytikäinen et al., 1995).

Taxonomic relationships of acinetobacters

Assignment of isolated strains to the genus *Acinetobacter* The following phenotypic properties are generally used to assign iso-

lated strains to the genus *Acinetobacter*: Gram-negative coccobacilli, oxidase negative, aerobic (nonfermenting), nonmotile, and, for most strains, negative nitrate reduction.

The ability of DNA from strains of *Acinetobacter* to transform a conserved *trpE*⁻ mutant of the naturally competent BD413 strain to prototrophy (Juni, 1972) has never failed to be demonstrated during the many years since the introduction of this test. Several hundred independently isolated strains have already been identified with this assay. Transformation of *trpE*⁻ has been shown to be genus specific and no exceptions have been reported. The specificity of this assay has been confirmed in a clinical setting (Brooks and Sodeman, 1974).

A genus-specific 16S rRNA-targeted oligonucleotide probe has been used to detect acinetobacters (Wagner et al., 1994b).

DNA-rRNA hybridization The highly conserved nature of 16S rRNA makes it useful for recognition of generic relationships among isolated bacterial strains. In DNA-rRNA hybridization studies, employing rRNA from *Acinetobacter calcoaceticus* ATCC 23055^T and filter-fixed DNAs from 23 strains of *Acinetobacter*, $T_{m(e)}$ (midpoint of thermal elution) values from 75.5 to 79.5°C were obtained (Van Landschoot et al., 1986). Similar studies, where DNAs from five established strains of *Acinetobacter* were hybridized with rRNA from *Moraxella lacunata*, resulted in hybrids having $T_{m(e)}$ values ranging from 71 to 73°C; corresponding values ranging from 73.1 to 80°C were obtained using DNAs from strains of all other genera in the *Moraxellaceae* (Rossau et al., 1991). When DNA-rRNA hybridizations were performed using the same five established strains of *Acinetobacter* and rRNA from *Psychrobacter immobilis*, $T_{m(e)}$ values ranging from 69.9 to 71.7°C were obtained; this range being unique for the five strains of *Acinetobacter* (Rossau et al., 1991). In both of these studies, corresponding DNA-rRNA hybridizations using DNA from bacteria not in *Moraxellaceae* resulted in $T_{m(e)}$ values from 53.5 to 67.5°C.

In studies employing rRNA from *A. calcoaceticus* ATCC 23055^T, it was proposed that two strains previously designated as *Pseudomonas cruciviae* and *Pseudomonas pavonacea* were "probably not members of the genus *Acinetobacter*", since although $T_{m(e)}$ s of their hybrids (75.5°C and 76.0°C, respectively) indicated that they belong to the *Acinetobacter* rRNA branch, these strains differed significantly in phenotypic properties from some typical *Acinetobacter* species (Van Landschoot et al., 1986). The finding that DNA from each of these atypical strains transformed *Acinetobacter* strain BD413 *trpE*⁻ to prototrophy (Juni, unpublished results) proves that these strains are indeed acinetobacters, and serves to confirm the significance of $T_{m(e)}$, as well as transformation, for identification of acinetobacters.

***Acinetobacter* species and DNA groups delineated by DNA-DNA hybridization** It has been suggested, and generally accepted, that bacterial species be defined primarily on the basis of DNA re-association, wherein members of a particular species show approximately 70% or greater DNA-DNA relatedness and with 5°C or less ΔT_m (Wayne et al., 1987). It was also recommended that genomic species that cannot be identified by phenotypic properties should not be given a species name. Early DNA-DNA hybridization studies of 45 strains of *Acinetobacter*, using a nitrocellulose filter method, distinguished groups with members that showed 50%, or greater, relatedness to one of five reference strains, and a sixth group consisted of strains showing less than 50% relatedness to any of the reference strains (Johnson et al., 1970). The results of this investigation have been correlated with hybridizations studies using the S1 nuclease method where,

within species, the levels of DNA relatedness to reference strains ranged from 62 to 95%, with ΔT_m values of less than 3.5°C (Bouvet and Grimont, 1986). To date, extensive DNA–DNA hybridization investigations have revealed the existence of approximately 21 species of *Acinetobacter* (Bouvet and Grimont, 1986; Bouvet and Jeanjean, 1989; Tjernberg and Ursing, 1989; Gerner-Smidt and Tjernberg, 1993). Six species were named (and assigned species numbers) in one study (Bouvet and Grimont, 1986), and a seventh species, *A. radioresistens*, was named by another laboratory (Nishimura et al., 1988). The other genomic (DNA) groups, which could not be differentiated by phenotypic methods, or contained fewer than 10 strains, were assigned a number. It should be noted that genomic species 13–15 of Bouvet and Jeanjean (1989) and Tjernberg and Ursing (1989) were described independently, and in order to avoid confusion, it is current practice to add the suffices BJ or TU to denote the groups of the respective studies. Genomic species 13 BJ and DNA group 14 TU have one reference strain in common, and have been considered one genomic species, while the other groups are apparently distinct.

Numerous reports refer to strains that cannot be allocated to any of the described (genomic) species. Since almost all of the acinetobacters studied were derived from hospital sources and because the most common habitats of these organisms are soil and water, it seems likely that many naturally occurring strains have yet to be isolated and that the current listing is probably not complete.

A distinct correlation between the PAGE-types of the outer membrane proteins and several *Acinetobacter* DNA relatedness groups has been reported (Ino and Nishimura, 1989; Dijkshoorn et al., 1990).

Although the highly competent strain BD413, a soil derivative, was originally reported to be a strain of *A. calcoaceticus*, phenotypic characterization (Bouvet and Grimont, 1986), studies of outer membrane protein patterns (Ino and Nishimura, 1989), and 16S rDNA sequence analysis (Strätz et al., 1996) suggest that this strain may, in fact, belong to a unique species. Furthermore, phylogenetic analysis of nucleotide sequences of *gyrB* genes revealed that strain BD413 was not strongly linked to any currently recognized genomic species (Yamamoto and Harayama, 1996) and may, in fact, represent a new genomic species (Yamamoto et al., 1999).

A group of *n*-alkane-degrading *Acinetobacter* strains have been described as members of a new species, "*A. venetianus*"; these strains harbored plasmids containing genes homologous to *alkBFGH* of *Pseudomonas oleovorans*, which encode the enzymatic steps of *n*-alkane oxidation (Di Cello et al., 1997). It has been shown that the oil-degrading *Acinetobacter* strain RAG-1 is a member of this species and that all strains of "*A. venetianus*" have a unique restriction pattern of amplified 16S rDNA and share similarities in DNA–DNA hybridization and biochemical characteristics (Vanechoutte et al., 1999).

It is most likely that *Acinetobacter* species, particularly those which include many virtually identical independently isolated strains, consist of organisms that have evolved to survive in some particular ecological niche(s). In each instance, the particular genetic constitution of strains of a species may represent an arrangement that is especially advantageous to the survival of those strains in their natural niche; the similarity of such arrangements in isolated strains may account for the observed high levels of DNA–DNA hybridization. It is possible that mutations in 16S

rDNA may not interfere with the ability of strains of a particular species to retain a selective advantage for occupation of their normal niche, thus accounting for frequent lack of correlation between data from DNA–DNA hybridization and 16S rDNA sequence analysis, as described below.

16S rDNA analysis 16S rRNA gene sequences from single strains of 21 reported *Acinetobacter* (genomic) species have been shown to have binary similarities of more than 94% and to constitute a phylogenetically coherent group of organisms within the *Gammaproteobacteria* (Ibrahim et al., 1997; Rainey et al., 1994a) (Figs. BXII.γ.148 and BXII.γ.149). Most strains of *Acinetobacter* were found to belong to one of five clusters (Fig. BXII.γ.149). Cluster I included DNA groups 2 and TU13; cluster II included DNA groups 3, CTTU13, and 1-3; cluster III included DNA groups 6, 8, and 9; cluster IV included DNA groups BJ15 through BJ17; and cluster V included DNA groups 1, BJ14, 10, and 11 (Ibrahim et al., 1997). Several DNA groups (4, 5, 7, 12, TU14, and TU15) are not associated significantly with any of the five clusters. Although reassociation of DNA from group 10 to DNA from group 11 is low (Bouvet and Grimont, 1986), strains from these groups have almost identical 16S rDNA sequences (Ibrahim et al., 1997). Neither study (Ibrahim et al., 1997; Rainey et al., 1994a) had complete correspondence between DNA–DNA hybridization values and clustering of 16S rDNA sequences, similar to observations for other groups of bacteria (Stackebrandt and Goebel, 1994). The finding that two authentic strains of *A. calcoaceticus* have 16S rRNAs with a similarity coefficient (S_{AB}) of only 0.66 (Byng et al., 1985) is yet another demonstration of the lack of correspondence of data obtained by DNA–DNA hybridization and 16S rRNA gene sequences.

Although there appear to be unambiguous sequence differences in the highly variable regions of the 16S rDNAs of each of the 21 DNA groups examined (Ibrahim et al., 1997), it seems premature to use these signatures for absolute species identification until more strains from each DNA group are analyzed in a similar manner. In addition, since the *Acinetobacter* chromosome encodes at least seven rRNA operons (Strätz et al., 1996; Gralton et al., 1997) it is possible that one, or more, of the rDNA sequences might have experienced mutational and recombinational changes during evolution of individual members of a species (Sneath, 1993).

Phylogenetic studies involving *gyrB*, *rpoD*, and *recA* genes It has been suggested that the resolution of 16S rRNA-based phylogenetic analysis is not sufficient to distinguish closely related genomic species, because of the extremely slow rate of base substitution in 16S rDNAs (Yamamoto and Harayama, 1996). Genes that code for proteins evolve more rapidly than the genes coding for rRNA. It has been proposed that the highly conserved nature of type II DNA topoisomerase genes makes them excellent candidates for studying bacterial diversity (Huang, 1996). The DNA topoisomerase *gyrB* of acinetobacters has been subjected to phylogenetic analysis based on nucleotide sequences of these genes and the corresponding amino acid sequences of the gene products (Yamamoto and Harayama, 1996). It was shown that the percentage of nucleotide substitutions in the *gyrB* genes of 15 strains of *Acinetobacter* varied from 0.3 to 30.4%, but that the percentage of substitutions in the corresponding amino acid sequences was between 0.5 and 13.3%, indicating that nucleotide substitution which results in amino acid substitution is less frequent than synonymous substitution (Yamamoto and Harayama,

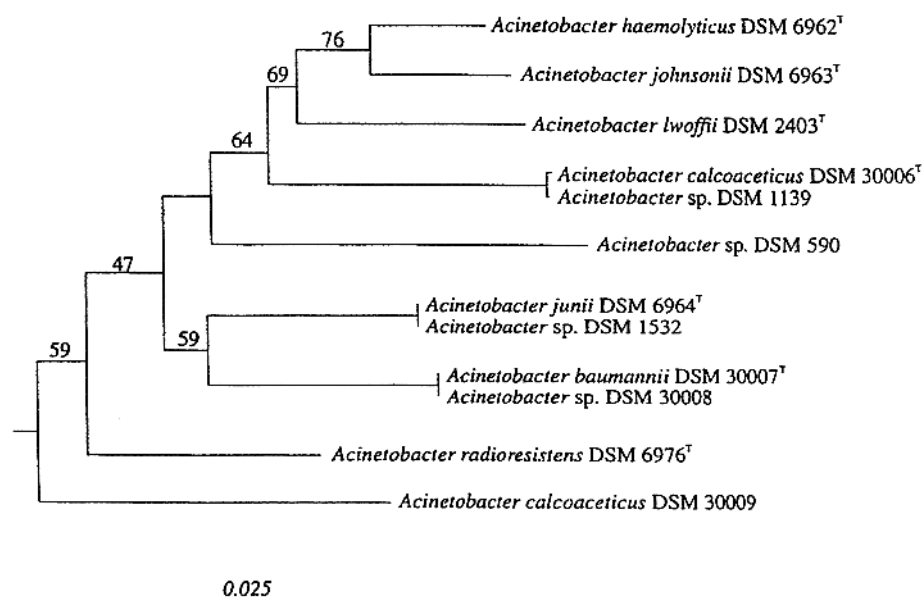


FIGURE BXII.γ.148. Intragenetic relationships of the genus *Acinetobacter* as determined by 16S rDNA sequence analysis. (Reproduced with permission from F.A. Rainey et al., FEMS Microbiology Letters. 124: 349–354, 1994, ©Elsevier Science B.V.)

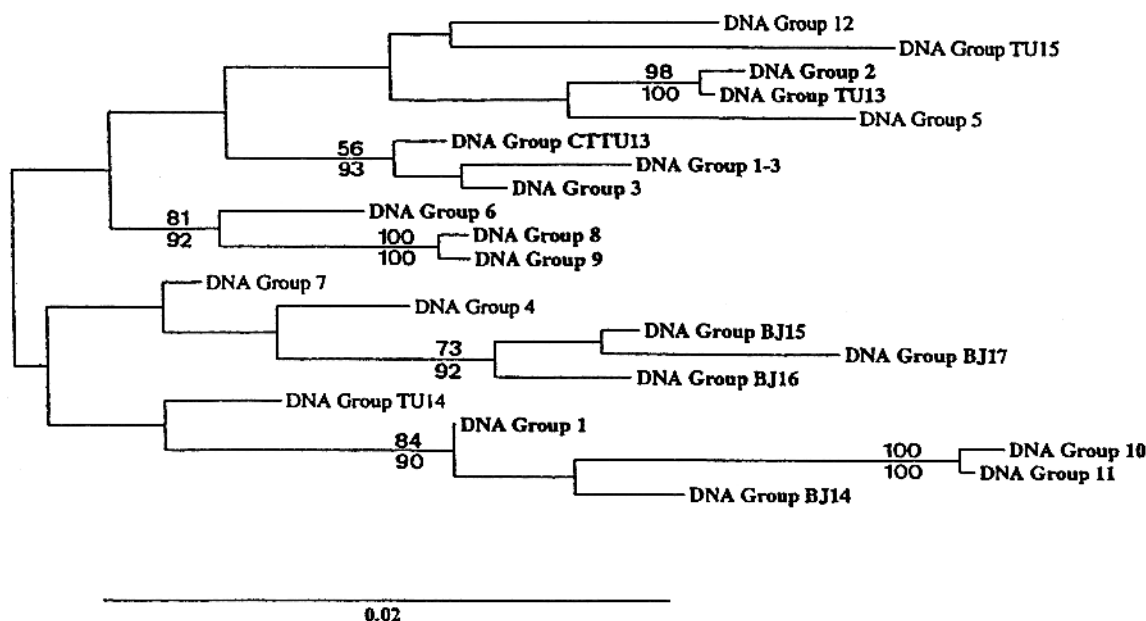


FIGURE BXII.γ.149. Unrooted phylogenetic dendrogram constructed for 21 strains representing all known DNA groups described for the genus *Acinetobacter*. (Reproduced with permission from A. Ibrahim et al., International Journal of Systematic Bacteriology 47: 837–841, 1997, ©International Union of Microbiological Societies.)

1996). The results of phylogenetic analysis using *gyrB* nucleotide sequences was consistent with those using DNA–DNA hybridization and phenotypic comparisons.

In a more extensive investigation using 49 *Acinetobacter* strains, 46 of which had previously been classified into 18 genomic species using the results of DNA–DNA hybridization, it was shown that the phylogenetic grouping of such strains based on *gyrB* genes was almost congruent with that based on DNA–DNA hybridization studies (Yamamoto et al., 1999) (Fig. BXII.γ.150). Three strains that had not been characterized previously by

DNA–DNA hybridization appeared to belong to two new genomic species (Fig. BXII.γ.150). Since minor discrepancies existed in the grouping of strains of genomic species 8, 9, and BJ17, the phylogenetic tree for these strains was reconstructed from the sequence of *rpoD*, the structural gene for RNA polymerase σ^{70} factor (Yamamoto et al., 1999); this tree was completely congruent with the grouping based on DNA–DNA hybridization (Fig. BXII.γ.150).

Using *Acinetobacter recA* obtained by polymerase chain reaction amplification, it was possible to distinguish 17 genomic species

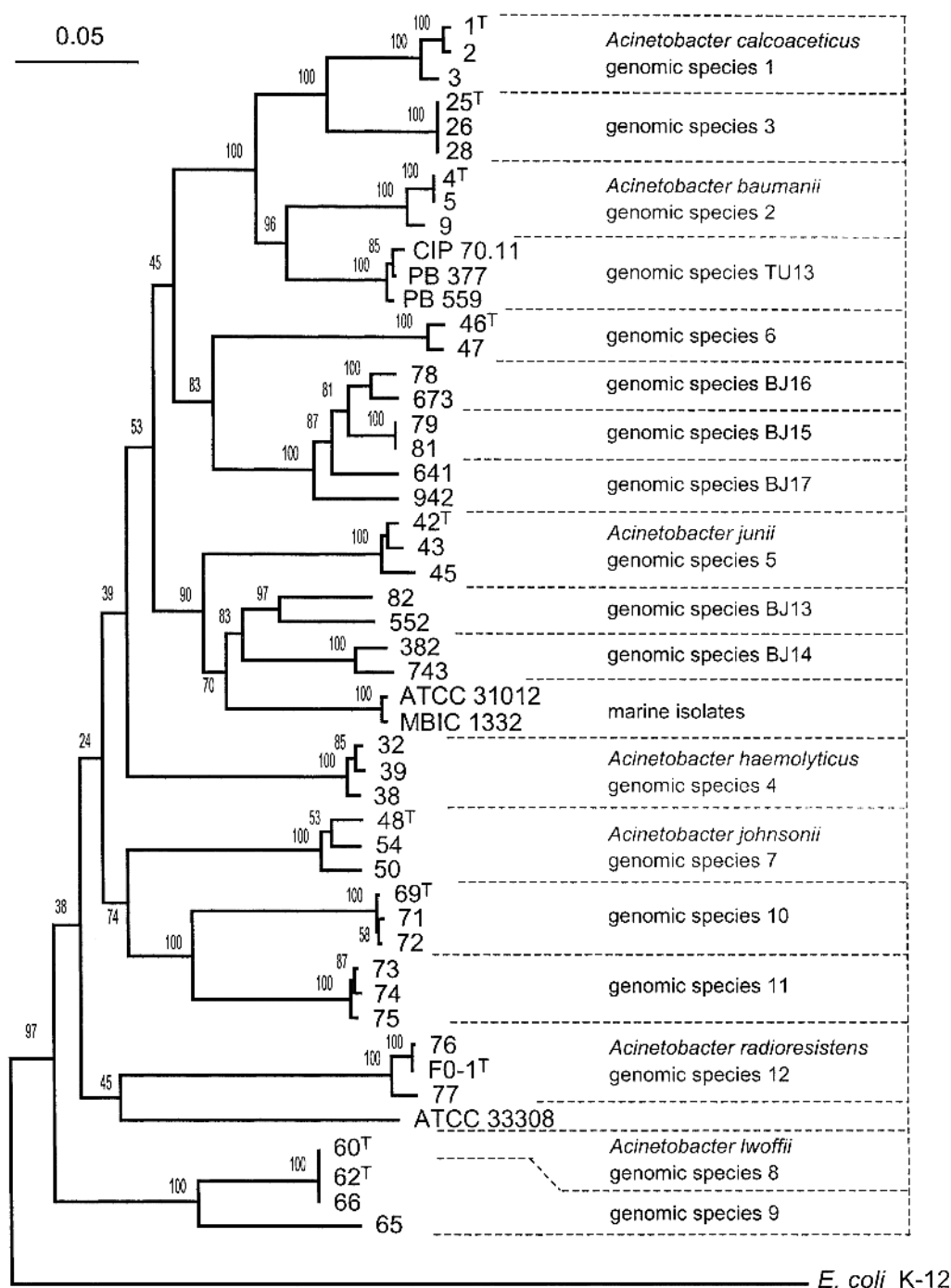


FIGURE BXII.γ.150. Phylogenetic trees of 49 *Acinetobacter* strains from the nucleotide sequences of the *gyrB* genes. Reproduced with permission from S. Yamamoto et al., International Journal of Systematic Bacteriology 49: 87–95, 1999, ©International Union of Microbiological Societies.)

by restriction fragment length polymorphism analysis with two restriction enzymes (Nowak et al., 1995).

Methods used for *Acinetobacter* (genomic) species identification

The recognition of distinct DNA relatedness groups within the genus *Acinetobacter* has raised interest in the ecology and pathology of the different groups. A great many approaches to identification of these (genomic) species by methods easier to perform than DNA–DNA hybridization have been taken. Table BXII.γ.131 presents phenotypic properties that frequently make

it possible to distinguish strains from most of 19 DNA groups. It should be noted though, that phenotypic systems, including nutritional and physiological tests (Bouvet and Grimont, 1986, 1987b; Gerner-Smidt et al., 1991; Kämpfer et al., 1993) or test galleries of commercial systems (Soddell et al., 1993; Bernards et al., 1995, 1996), do not allow for identification of all DNA groups. In particular, using phenotypic properties, it is difficult to discriminate between the closely related (genomic) species *A. calcoaceticus* (DNA group 1), *A. baumannii* (DNA group 2), and

TABLE BXII.γ.131. Differentiation of genomic species of *Acinetobacter* by phenotypic tests^{a,b,c}

Characteristics	Genomic species												
	1,2,3, and 13 TU ^d	4 and 6	5	7–15 TU ^d	8–9	10	11	12	13 BJ–14 TU	14 BJ	15 BJ	16BJ	17BJ
Most discriminatory tests (n = 0):													
<i>Utilization of:</i>													
Azalate	[+]	–	–	[–]	D	D	+	[+]	–	+	–	(+) ^e	–
β-Alanine	[+]	[]	–	[–]	–	+	+	–	D	+	+	–	+
L-Leucine	[+]	[+]	D	[–]	–	–	–	+	[–]	+	(+)	(+)	(+)
L-Phenylalanine	[+]	–	–	–	D	–	–	+	+	+	+	+	+
L-Tryptophan	[+]	–	–	–	–	–	–	[v]	+	+	+	–	+
4-Hydroxybenzoate	[+]	[+]	–	[–]	–	D	[+]	[–]	+	+	+	–	+
Phenylacetate	[+]	–	–	–	[+]	–	D	[+]	+	+	+	+	+
Growth at 37°C	[+]	[+]	+	–	D	+	–	+	[–]	+	+	+	+
Acid from glucose	[+]	D	–	–	D	+	–	[–]	+	+	–	–	–
Gelatinase	–	[+]	–	–	–	–	–	–	+	+	+	+	+
Additional tests (n = 4):													
<i>Hemolysis of:</i>													
Sheep blood	–	[+]	D	–	–	–	–	–	+	+	+	+	+
Human blood	–	[+]	D	–	–	–	–	–	+	+	+	+	+
<i>Utilization of:</i>													
Citrate	[+]	D	D	D	[–]	+	D	–	+	+	–	+	+
Glutarate	[+]	[–]	[–]	D	D	+	+	+	D	+	–	(+)	(+)
L-Arginine	+	[+]	[+]	D	[–]	–	–	[+]	[+]	+	+	+	+
L-Aspartate	[+]	D	D	D	–	+	+	D	D	–	–	(+)	–
L-Histidine	+	–	[+]	–	–	+	+	–	+	+	+	+	+
Phenylpyruvate	[+]	+	–	–	D	–	–	[+]	+	+	+	+	+
Protocatechuate	+	+	[–]	D	–	+	+	–	+	+	+	+	+
Vanillate	[+]	[+]	–	[–]	–	D	D	–	–	+	–	+	–
<i>Hydrolysis of</i> ^f													
pNP-β-D-Xylopranoside ^g	[+]	D	–	–	–	D	–	–	–	–	–	+	+
γ-L-Glutamate-pNA ^h	[+]	D	–	–	–	–	–	–	D	–	–	+	+
<i>Susceptibility to:</i>													
Chloramphenicol	[–]	D	D	D	+	–	D	[–]	–	nd	nd	nd	nd
Penicillin G	[–]	–	D	D	[+]	–	–	D	–	nd	nd	nd	nd

^aTable reproduced with permission from Towner, K.J., 1996. Biology of *Acinetobacter* spp. In *Acinetobacter*: Microbiology, Epidemiology, Infections, Management, edited by Bergogne-Bérézin, Joly-Guillou, and Towner, © CRC Press, Boca Raton, pp. 44–45.

^bResults for genomic species 4–17BJ are based on only one strain and should be verified by testing more authentic strains.

^cAdapted from Kämpfer et al. (1993), except for susceptibility tests which are according to Gerner-Smidt and Frederiksen (1993). Symbols: +, all strains positive; [+], 80% or more positive; D, 20–79% positive; [–], 20% or less positive; –, all strains negative. The following genomic species have been named: 1, *A. calcoaceticus*; 2, *A. baumannii*; 4, *A. haemolyticus*; 5, *A. junii*; 7, *A. johnsonii*; 8, *A. lwoffii*; 12, *A. radioresistens*.

^dGenomic species grouped according to common characteristics.

^e[+] Weak positive.

^fOnly one strain investigated.

^gpNp, *p*-nitrophenyl; pNA, *p*-nitroanilide.

^hWeak reaction.

the unnamed groups 3 and 13TU, sometimes referred to as the *A. calcoaceticus*-*A. baumannii* (Acb) complex. Table BXII.γ.132 lists phenotypic properties that permit some differentiation of the species of this complex.

Other methods used for species identification of acinetobacters (Dijkshoorn, 1996) are based on structural characteristics of strains and include serotyping (Traub and Leonhard, 1994), fatty acid compositional analysis (Kämpfer, 1993), outer membrane protein patterns (Ino and Nishimura, 1989; Dijkshoorn et al., 1990), and enzyme electrophoresis (Bouvet and Jeanjean, 1995). None of these methods has been completely successful for assignment of isolated strains to the currently described (genomic) species.

Due to the difficulties in phenotypic identification, and given the clinical relevance of some genomic species, a variety of molecular methods has been explored for their usefulness in identification of *Acinetobacter* species. These methods include ribotyping (Gerner-Smidt, 1992; Dijkshoorn et al., 1993), amplified rDNA restriction analysis (ARDRA) (Vanechoutte et al., 1995; Dijkshoorn et al., 1998; Jawad et al., 1998) and AFLP[®] (amplified

fragment length polymorphism) (Janssen et al., 1997), 16S-23S rDNA spacer analysis (Dolzani et al., 1995; Nowak et al., 1995), and tRNA spacer fingerprinting (Ehrenstein et al., 1996).

Typing methods Primarily because of an interest in tracing strains of *Acinetobacter* involved in nosocomial infections and epidemic spread, a great effort has been made to distinguish strains. Early phenotypic methods used to allocate strains to definitive types include phage typing (Bouvet et al., 1990) and bacteriocin typing (Andrews, 1986). Other phenotypic methods include biotyping (Bouvet and Grimont, 1987b), multilocus enzyme electrophoretic typing (Picard et al., 1989; Thurm and Ritter, 1993), cell envelope protein typing (Dijkshoorn et al., 1987), antibiotic susceptibility typing (Dijkshoorn et al., 1993; Seifert et al., 1993a; Marcos et al., 1994; Lytikäinen et al., 1995), and serotyping (Traub and Leonhard, 1994; Pantophlet et al., 1998). Antibio-gram typing is widely used in screening for strain identity, although gain or loss of resistance genes may lead to false conclusions. Biotyping has been shown to be a practical method for allocating strains of the Acb complex to as many as 19 biotypes (Bouvet and Grimont, 1987b; Bouvet et al., 1990) and has ena-

TABLE BXII.γ.132. Phenotypic differentiation of genomic species of the *A. calcoaceticus*–*A. baumannii* complex^{a,b}

Characteristic	Genomic species			
	1 ^c	2 ^d	3	13 TU ^e
Growth at 41°C	[–]	+	d	+
Growth at 44°C	–	+	–	d
Growth with:				
L-Arabinose	d	[+]	d	+
D-Glucarate	d	[+]	[+]	[–]
D-Ribose	d	[+]	d	[+]
Sorbinic acid	[–]	d	[–]	[+]
Glycerate	[+]	[–]	d	[–]
Levulinate	d	d	–	[–]
L-Tartrate	[–]	d	[+]	–
Acetyl-L-glutamine	[–]	d	d	d
D-Asparagine	+	–	–	[–]
L-Hydroxyproline	[–]	[+]	[–]	d

^aTable reproduced with permission from Towner, K.J., 1996. Biology of *Acinetobacter* spp. In *Acinetobacter*: Microbiology, Epidemiology, Infections, Management, edited by Bergogne-Bérézin, Joly-Guillou, and Towner, © CRC Press, Boca Raton, pp. 46.

^bAdapted from Kämpfer et al. (1993). For symbols, see Table BXII.γ.131.

^c*A. calcoaceticus*.

^d*A. baumannii*.

^eDNA–DNA hybridization group 13 (Tjernberg and Ursing, 1989).

bled the recognition of specific types from different countries (Marcos et al., 1994; Dominguez et al., 1995; Ratto et al., 1995).

Rapid developments in molecular biology have led to the application of numerous DNA-based typing methods for *Acinetobacter* strain identification, including discrimination of strains of the Acb complex. These include plasmid DNA fingerprinting (Gerner-Smidt, 1989; Seifert et al., 1994b), ribotyping (Gerner-Smidt, 1992), PCR fingerprinting methods with detection of amplification products by agarose gel electrophoresis (Reboli et al., 1994) or automated laser fluorescence (Webster et al., 1996), pulsed field gel electrophoresis (Tankovic et al., 1994), and selective amplification of restriction fragments by AFLP[®] (Janssen and Dijkshoorn, 1996; Janssen et al., 1997). Nosocomial acinetobacters can be very similar; therefore, it is current practice to use a combination of methods for strain identification (Dijkshoorn et al., 1993; Struelens et al., 1993; Seifert et al., 1994a; Lytikäinen et al., 1995). By application of such a polyphasic strategy, including ribotyping, protein electrophoretic typing, AFLP[®], biotyping and antibiogram typing, it was possible to recognize and differentiate two highly similar strains found in different NW European hospitals and countries (Dijkshoorn et al., 1996). It was shown recently that multiply-resistant strains with features similar to these two clones are also common in the Czech Republic (Nemec et al., 1999). It is not yet known whether these strains are also responsible for outbreaks in other parts of the world.

Analysis of 170 acinetobacter strains isolated from various foods revealed that although strains belonging to almost every species were represented, 46% were identified as *A. johnsonii* and 15% as *A. lwoffii* (Gennari and Lombardi, 1993). A majority (78%) of *Acinetobacter* strains isolated from activated sludge, in a pilot-scale treatment of sewage for enhanced removal of phosphate, was identified as *A. johnsonii* (Duncan et al., 1988).

Study of quantitative transformation of the high-level streptomycin-resistance marker Since quantitative transformation of the high-level streptomycin-resistance marker was instrumental in elucidation of the taxonomic relationships within the *Moraxellaceae* (Bøvre and Hagen, 1981), similar studies with 54 clinical isolates

and 17 reference strains of *Acinetobacter*, using competent strain BD4 as the recipient, were initiated (Bergan and Vaksvik, 1983). Ratios of interstrain to intrastrain transformation of 0.06–6.4% were obtained, DNAs from streptomycin-resistant mutants of 13 of the clinical isolates failing to transform the competent streptomycin-sensitive strain. Although these studies were limited, the results obtained suggest that evolutionary change in ribosomal protein S12 (*StrA*) in the genus *Acinetobacter* may be as extensive as that found in the genus *Moraxella* (Bøvre and Hagen, 1981).

ENRICHMENT AND ISOLATION PROCEDURES

Isolation of acinetobacters can be accomplished using ordinary laboratory media such as heart infusion agar, brain heart infusion agar, or trypticase soy agar. Although many acinetobacters grow well at 37°C, the optimum growth temperature for most strains is 33–35°C. Acinetobacters having lower optimum temperatures and unable to grow at 37°C have been reported (Breuil et al., 1975). A general enrichment procedure for isolation of acinetobacters from soil and water involves inoculation of 20 ml of an acetate-mineral medium with 5 ml of a water sample, or of a filtered 10% soil suspension, followed by vigorous aeration during incubation at 30°C, or at room temperature (Baumann, 1968). Aeration is especially favorable for enrichment of nonmotile acinetobacters since, in a nonaerated culture, motile oxygen-consuming pseudomonads tend to move to the surface layer, thereby decreasing the amount of oxygen diffusing into the interior of the liquid volume. Acinetobacters have a slightly acid pH optimum for growth, and aeration at a pH from 5.5 to 6.0 favors enrichment of these organisms from soil and water samples (Baumann, 1968; Juni, 1982). Presumptive identification of colonies of *Acinetobacter* growing on any medium involves a demonstration of nonmotile coccobacilli in a wet mount, an oxidase-negative reaction, and a Gram-negative stain (Warskow and Juni, 1972). When these criteria are satisfied, definitive identification can be achieved through use of the transformation assay (Juni, 1972).

Selective media for isolation of acinetobacters from environmental and clinical sources are available and include Herellea

Agar (Difco) (Mandel et al., 1964) and Leeds *Acinetobacter* Medium (Jawad et al., 1994).

MAINTENANCE PROCEDURES

Cultures of *Acinetobacter* can be maintained on heart infusion agar plates (Difco) stored in plastic bags (to delay evaporation and drying) at room temperature for periods as long as six months before being transferred (Marcus et al., 1969). This procedure is not recommended, however, since it has been observed that the ability to use certain carbon sources is lost, probably because of spontaneous mutation, when cultures are transferred infrequently in this manner (Juni, unpublished). Storage at low temperatures, or by lyophilization, appears to be best for preservation

of properties initially observed for a particular strain. Cultures can be conveniently stored by preparing a heavy suspension of organisms in buffer, or in broth, and adding 0.5 ml of this suspension to 1.0 ml of sterile glycerol in a screw-capped test tube. After mixing, this suspension is stored at approximately -65°C . Since 67% glycerol remains liquid at temperatures above -46°C , a stored culture can be sampled conveniently by withdrawing a loopful, or 0.1 ml, of the glycerol stock and streaking this material directly on a heart infusion, or other suitable medium, plate without significant warming of the glycerol suspension.

ACKNOWLEDGMENTS

I would like to thank Lenie Dijkshoorn for invaluable assistance in preparation of the section dealing with taxonomy of acinetobacters.

List of species of the genus *Acinetobacter*

1. ***Acinetobacter calcoaceticus*** (Beijerinck 1911) Baumann, Doudoroff and Stanier 1968b, 1538^{AL} emend. Bouvet and Grimont 1986, 238 (*Micrococcus calcoaceticus* Beijerinck 1911, 1067.)

cal.co.a.ce'ti.cus. L. n. *calx* chalk; L. n. *acetum* acetic acid; M.L. n. *calcoaceticus* calcium acetate, which was used by Beijerinck in the enrichment medium from which he isolated the organism.

Characteristics as given for the genus. Colonies on tryptocase soy agar are circular, convex, smooth, and slightly opaque with entire margins; colonies are 0.5–1.5 mm in diameter after 24 h and 2.5–3.5 mm in diameter after 48 h at 30°C . Good growth occurs at $15\text{--}37^{\circ}\text{C}$ in one day. No growth occurs at 41°C and above. Nitrates are not reduced to nitrites in complex medium. Grows on citrate (Simmons). Acid is produced from D-glucose. γ -Glutamyltransferase is produced, but β -xylosidase is not produced. Horse or sheep blood is not hemolyzed; gelatin is not hydrolyzed.

The following compounds are utilized as sole sources of carbon and energy: L-phenylalanine, L-histidine, phenylacetate, malonate, azelate, β -alanine, *trans*-aconitate, L-arginine, L-ornithine, DL-lactate, 2,3-butanediol, DL-4-aminobutyrate, glutarate, L-aspartate, and L-tyrosine. D-malate and histamine are not utilized. L-Leucine is utilized by some strains.

Isolated from soil.

Genomic species 1.

The mol% G + C of the DNA is: 40–42 (T_m).

Type strain: ATCC 23055, CIP 81.08, DSM 30006.

GenBank accession number (16S rRNA): X81661, Z93434.

2. ***Acinetobacter baumannii*** Bouvet and Grimont 1986, 239^{VP} *baumannii*. M.L. gen. *baumannii* in honor of Paul and Linda Baumann.

Characteristics as given for the genus. Colonies on tryptocase soy agar are circular, convex, smooth, and slightly opaque with entire margins and sometimes have a butyrous aspect. Colonies are 1.5–2.0 mm in diameter after 24 h and 3.0–4.0 mm in diameter after 48 h at 30°C . Good growth occurs at $15\text{--}44^{\circ}\text{C}$. Occasional strains may require methionine for growth on minimal medium. Nitrates are not reduced to nitrites in complex medium (exceptions occur). Acid is produced from D-glucose by most strains. Strains which produce acid from D-glucose also produce β -xylosidase. γ -Glutamyltransferase is produced by most strains. Horse or sheep blood is not hemolyzed; gelatin is not hy-

drolyzed. Citrate (Simmons) is utilized by prototrophic strains. The following compounds are utilized: DL-lactate, glutarate, L-aspartate, L-tyrosine, ethanol, 2,3-butanediol, and DL-4-aminobutyrate. The following compounds are utilized by most strains: L-phenylalanine and phenylacetate (growth on L-phenylalanine is associated with growth on phenylacetate in this species), L-histidine, azelate, D-malate, L-leucine, β -alanine, *trans*-aconitate, L-arginine, and L-ornithine. Histamine is not utilized.

Isolated from human clinical specimens or the natural environment. Most *Acinetobacter* strains isolated from nosocomial infections belong to this species.

Genomic species 2.

The mol% G + C of the DNA is: 40–43 (T_m).

Type strain: ATCC 19606, CIP 70.34, DSM 30007.

GenBank accession number (16S rRNA): X81660.

3. ***Acinetobacter haemolyticus*** Bouvet and Grimont 1986, 239^{VP} *haemolyticus*. Gr. n. *haema* blood, dissolving.

Characteristics as given for the genus. Colonies on tryptocase soy agar are circular, convex, smooth, and slightly opaque with entire margins and sometimes have a sticky consistency. Colonies are 1.5–2.0 mm in diameter after 48 h at 30°C . Good growth occurs at $15\text{--}37^{\circ}\text{C}$. No growth occurs at 41°C . Nitrates are not reduced to nitrites in complex medium (exceptions occur). On horse or sheep blood agar, colonies are surrounded by a clear zone which is limited by a dark red ring after 24 h at 37°C or after 48 h at 30°C . Gelatin is hydrolyzed (exceptions occur). Some strains produce acid from D-glucose and produce β -xylosidase. Some strains produce γ -glutamyltransferase. Most strains grow on Simmons citrate agar.

DL-4-Aminobutyrate is utilized by all strains. The following compounds are utilized by most strains: L-histidine, D-malate, L-leucine, ethanol, and L-arginine. The following compounds are not utilized: DL-lactate, azelate, histamine, β -alanine, 2,3-butanediol, and L-ornithine.

Isolated from clinical specimens and the environment.

Genomic species 4.

The mol% G + C of the DNA is: 40–43 (T_m).

Type strain: Strain Mannheim 2446/60, B40, ATCC 17906, CIP 64.3, DSM 6962, NCTC 10305.

GenBank accession number (16S rRNA): X81662.

4. ***Acinetobacter johnsonii*** Bouvet and Grimont 1986, 239^{VP} *johnsonii*. M.L. gen. *johnsonii* in honor of John L. Johnson.

Characteristics as given for the genus. Colonies on tryptic soy agar are circular, convex, smooth, and slightly opaque with entire margins; colonies are 1.0–1.5 mm in diameter after 24 h and 2.0–3.0 mm in diameter after 48 h at 30°C. Good growth occurs at 15 to 30°C; no growth occurs at 37°C. Nitrates are not reduced to nitrites in complex medium (exceptions occur). Acid is not produced from D-glucose; β -xylosidase and γ -glutamyltransferase are not produced. Horse or sheep blood is not hemolyzed; gelatin is not hydrolyzed. No growth factor is required (in a few cases methionine is required). Grows on Simmons citrate. DL-Lactate and ethanol are utilized as sole sources of carbon and energy.

The following compounds are utilized by many strains: malonate, D-malate, L-aspartate, L-tyrosine, 2,3-butanediol, L-arginine, L-ornithine, and DL-4-aminobutyrate. The following compounds are not utilized: glutarate, L-phenylalanine, phenylacetate, L-histidine, azelate, L-leucine, histamine, β -alanine, and *trans*-aconitate.

Isolated from clinical specimens, activated sludge, foods, and eviscerated chickens.

Genomic species 7.

The mol% G + C of the DNA is: 44–45 (T_m).

Type strain: ATCC 17909, CIP 64.6, DSM 6963, NCTC 10308.

GenBank accession number (16S rRNA): X81663.

5. **Acinetobacter junii** Bouvet and Grimont 1986, 239^{VP}
ju'ni.i. M.L. gen. *junii* in honor of Elliot Juni.

Characteristics as given for the genus. Colonies on tryptic soy agar are circular, convex, and smooth with entire margins. Colonies are 1.0–1.5 mm in diameter and translucent after 24 h and 2.0–2.5 mm in diameter and slightly opaque after 48 h at 30°C. Good growth occurs at 15–37°C; occasionally growth occurs at 41°C. No growth occurs at 44°C. No growth factor is required. Nitrates are not reduced to nitrites in complex media. Acid is not produced from D-glucose. β -Xylosidase and γ -glutamyltransferase are not produced. Horse or sheep blood is not hemolyzed; gelatin is not hydrolyzed.

Citrate (Simmons) is occasionally utilized. DL-Lactate, L-histidine, D-malate, and ethanol are utilized as sole sources of carbon and energy. The following compounds are utilized by many strains: L-aspartate, L-leucine, L-tyrosine, L-arginine, and DL-4-aminobutyrate. The following compounds are not utilized: glutarate, L-phenylalanine, phenylacetate, malonate, azelate, histamine, β -alanine, 2,3-butanediol, *trans*-aconitate, and L-ornithine.

Isolated from human clinical specimens.

Genomic species 5.

The mol% G + C of the DNA is: 42 (T_m).

Type strain: Mannheim 2723/59, ATCC 17908, CIP 64.5, DSM 6964, NCTC 10307.

GenBank accession number (16S rRNA): X81664.

6. **Acinetobacter lwoffii** (Audureau 1940) Brisou and Prévot 1954, 727^{AL} emend. Bouvet and Grimont 1986, 238.
lwof.fi.i. M.L. gen. *lwoffii* in honor of Andre Lwoff.

Characteristics as given for the genus. Colonies on tryptic soy agar are circular, convex, smooth, and slightly opaque with entire margins; colonies are 1.0–1.5 mm in diameter after 24 h and 3.0–4.0 mm in diameter after 48 h at 30°C. Good growth occurs at 15–37°C in one day. No

growth occurs at 41°C and above. Nitrates are not reduced to nitrites in complex medium. No growth occurs on citrate (Simmons). Acid is not produced from D-glucose; β -xylosidase and γ -glutamyltransferase is not produced. Horse or sheep blood is not hemolyzed; gelatin is not hydrolyzed.

Azelate and DL-lactate are utilized as sole sources of carbon and energy. The following compounds are not utilized: L-phenylalanine, L-histidine, malonate, L-aspartate, L-leucine, histamine, β -alanine, 2,3-butanediol, glutarate, *trans*-aconitate, and L-arginine. Some strains can utilize phenylacetate, L-tyrosine, ethanol, L-ornithine, and DL-4-aminobutyrate.

Habitat unknown.

Genomic species 8.

The mol% G + C of the DNA is: 46 (T_m).

Type strain: ATCC 15309, CIP 64.10, DSM 2403, NCTC 5866.

GenBank accession number (16S rRNA): X81665.

7. **Acinetobacter radioresistens** Nishimura, Ino, and Iizuka 1988, 210^{VP}

ra.di.o.re.sis'tens. L. n. *radius* ray, beam; L. part. adj. *resistens* resisting; M.L. part. adj. *radioresistens* ray resisting, because of high resistance to gamma-ray irradiation.

Cells are rods 0.6–0.7 $\times$ 0.8–1.0 μ m. Pleomorphic. A plump form is 0.8–0.9 $\times$ 1.2–1.8 μ m; elongate cells (15 μ m or more long) occur in young cultures; and coccoid cells (0.6–0.9 μ m in diameter) occur in old cultures. The cells occur singly or in pairs. Nonsporeforming, nonencapsulated, nonmotile, non-acid fast, Gram negative. Colonies on nutrient agar smooth, entire, convex, glistening, opaque, and yellowish white to pale yellow. A pellicle is formed on the surface of nutrient broth.

Metabolism is strictly respiratory. Catalase is produced. Cytochrome oxidase is not produced. Nitrate is reduced to nitrite in succinate-nitrate medium but not in Casitone-nitrate medium. Indole and hydrogen sulfide are not produced. Tween 80 is hydrolyzed. Lysine and ornithine decarboxidase, phenylalanine deaminase, arginine dihydrolase, gelatinase, and urease are not produced. Hemolysis on sheep blood agar is negative. Growth occurs in media containing 5% NaCl. Good growth occurs at 27–31°C. Growth occurs at 42°C. L-Arabinose, D-ribose, D-xylose, D-galactose, D-glucose, D-fructose, D-mannose, L-rhamnose, D-glucosamine, D-cellobiose, lactose, maltose, melibiose, sucrose, and raffinose are not oxidized and are not assimilated. Acetate, fumarate, DL-lactate, L-malate, malonate, succinate, hexadecane, heptadecane, octadecane, eicosane, ethanol, *n*-butanol, L-alanine, L-glutamate, L-leucine, and L-proline are assimilated. Citrate, gluconate, decane, nonadecane, methanol, *n*-propanol, isopropanol, ethanolamine, β -alanine, DL-phenylalanine, L-histidine, L-isoleucine, DL-norleucine, DL-serine, L-threonine, dulcitol, *D*-inositol, D-mannitol, and D-sorbitol are not assimilated.

Highly resistant to gamma-ray irradiation (D_{10} value, 125 to 220 krad under air equilibrium in 0.067 M phosphate buffer). Resistant to benzylpenicillin (100 IU).

The major cellular fatty acids are the even-numbered straight-chain acids C_{18:1}, C_{16:1}, and C_{16:0}. The ubiquinone system is Q-9.

Isolated from cotton and from soils.

Assigned to genomic species 12 (Tjernberg and Ursing, 1989).

The mol% G + C of the DNA is: 44.1–44.8 (T_m).

Type strain: Strain FO-1, DSM 6976, IAM 13186.

GenBank accession number (16S rRNA): X81666.

8. “*Acinetobacter venetianus*” Di Cello, Pepi, Baldi and Fani 1997, 245.

venetianus. venetianus because it was isolated from the Venice lagoon

Characteristics are as given for the genus. Strains tested (2) are β -hemolytic on sheep blood agar. Growth occurs at 37°C but not at 41°C or 44°C. Aerobic acidification of glucose on OF medium is negative. β -Xylosidase is not produced. Strains test positive for gelatinase production; caprate, malate, and citrate hydrolysis; and assimilation of acetate, alanine, caprate, valerate, citrate, histidine, and 3-hydroxybutyrate. There is no growth on rhamnose, N-acetyl-

glucosamine, ribose, inositol, fucose, sorbitol, arabinose, propionate, 5-ketogluconate, glycogen, *m*-hydroxybutyrate, and 2-ketogluconate.

Strains grow on fuel oil and the *n*-alkanes (C_{10} , C_{14} , and C_{20}), and their respective oxidation products as sole carbon sources. Strains harbor plasmids that encode the enzyme activities required for oxidation of alkanes and dehydrogenation of aliphatic alcohols.

Isolated from lagoons and other waters that are contaminated with fuel oil or other products derived from petroleum.

The mol% G + C of the DNA is: 43.6–43.9 (T_m).

Deposited strain:

GenBank accession number (16S rRNA): X80825 (for strain RAG-1).

Additional Remarks: No type strain has been designated. Strains investigated include VE-C3, RAG-1, and T4.

Other Organisms

The following genomic groups have been recognized by virtue of possessing unique DNA–DNA hybridization groups. They are not yet named because none of them has been shown to display a set of phenotypic properties that distinguish them from the other species of *Acinetobacter*. Some of the phenotypic properties

of these genomic species are shown in Tables BXII.γ.131 and BXII.γ.132.

Unnamed genomic species of *Acinetobacter*: 3, 6, 9, 10, 11, TU13, TU14 (= BJ13), TU15, CTTU13, 1-3, BJ14, BJ15, BJ16, and BJ17.

Genus III. *Psychrobacter* Juni and Heym 1986b, 389^{VP}

ELLIOT JUNI

Psy.chro' bac.ter. Gr. adj. *psychros* cold; M.L. masc. n. *bacter* rod; M.L. masc. n. *Psychrobacter* a rod that grows at low temperatures.

Rods or cocci, 0.9–1.3 × 1.5–3.8 μ m. Rods can vary in length from extremely short (coccobacilli) to relatively long. For some strains the rods tend to be somewhat swollen. Also observed to grow as cocci or coccobacilli. Nonpigmented. Colonies on heart infusion agar are usually cream-colored, smooth, and opaque. Gram negative; the stain frequently tends to be retained. Nonmotile, and aerobic. **Most strains are psychrotrophic; they are able to grow at 5°C, have temperature optima near 20°C, and generally are unable to grow at 35–37°C.** Strains able to grow well at 35–37°C usually cannot grow at 5°C. Grow well on most common complex media. Most strains can grow on a mineral-agar medium containing a single carbon and energy source and ammonium salts as the nitrogen source. Some strains form acid aerobically from D-glucose, D-mannose, D-galactose, L-arabinose, D-xylose, and L-rhamnose. They are **halotolerant and grow in the presence of 6.5% or greater NaCl**. Many strains are radiation resistant. **Oxidase positive. Catalase positive.** Many strains are **competent for natural genetic transformation**. Have been isolated from the open sea, the deep sea, sea ice, ornithogenic soils, skins and gills of fish, skins of poultry, various food products, irradiated foods, air contaminants, and from a variety of human sources. Not known to be clinically significant. Cultures of *Psychrobacter* can be maintained in the same manner as for cultures of *Acinetobacter*, as described above.

The mol% G + C of the DNA is: 41–50.7.

Type species: *Psychrobacter immobilis* Juni and Heym 1986b, 388.

FURTHER DESCRIPTIVE INFORMATION

Bacterial strains that were oxidase positive and psychrotrophic (subsequently shown to be psychrobacters) were proposed as

members of the *Moraxellaceae* because of weak but distinct genetic interactions with authentic *Moraxella* species, involving the high-level streptomycin-resistance marker (Bøvre and Hagen, 1981). Based on 16S rRNA gene sequence information, *Psychrobacter immobilis* has been placed in group III (of four groups) of the family *Moraxellaceae* (See Fig. BXII.γ.146 of the description of the family *Moraxellaceae*) (Pettersson et al., 1998). Nine distinct species of *Psychrobacter* have been revealed from analysis of phenotypic properties, DNA–DNA hybridization, and 16S rRNA sequences (see Fig. BXII.γ.151) (Bowman et al., 1996, 1997a, 1997d; Maruyama et al., 2000; Denner et al., 2001a; Kämpfer et al., 2002a; Romanenko et al., 2002a). 16S rRNA sequences having similarities of 93.9–99.9% have been reported for strains of all species of *Psychrobacter* (Bowman et al., 1996, 1997d; Denner et al., 2001a; Kämpfer et al., 2002a; Romanenko et al., 2002a). Based on the extent of 16S rDNA similarity, psychrobacters are more closely related to the moraxellae than they are to the acinetobacters (Bowman et al., 1996; Pettersson et al., 1998) (Fig. BXII.γ.146). The wide range of DNA composition of psychrobacters (mol% G + C 41–50.7) provides evidence for heterogeneity of the genus.

It has been reported (Romanenko et al., 2002a) that strains of *P. submarinus* and *P. marincola* have 99.9% similarity in 16S rRNA sequences but showed DNA–DNA reassociation of only 15% similarity. Furthermore, the mol% G + C of the DNAs of these strains differed by 3.9%. These data indicate that *P. submarinus* and *P. marincola* must have a long history of independent evolution, from a presumed common ancestor, in spite of conservation of the sequences of their respective 16S rRNAs. By contrast, it was demonstrated (Denner et al., 2001a) that 16S rRNA similarities between *P. proteolyticus* and those of *P. glacincola*

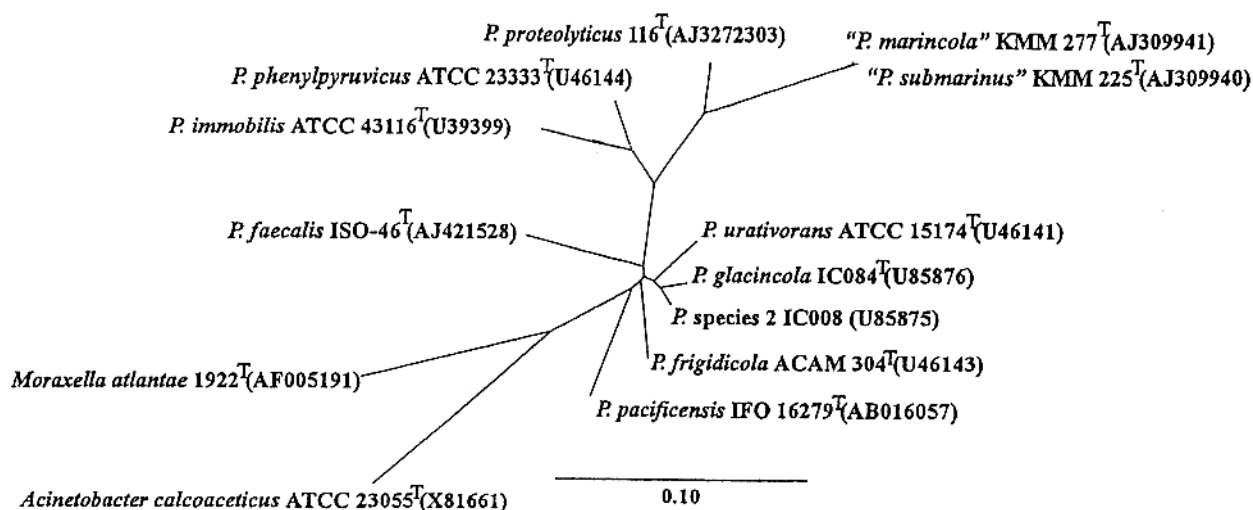


FIGURE BXII.γ.151. Phylogenetic tree for the species of the genus *Psychrobacter*. (Reproduced with permission from P. Kämpfer et al., Systematic and Applied Microbiology 25: 31–36, 2002, ©Urban & Fischer.)

and *P. urativorans* were 97.4 and 96.9%, respectively, whereas the corresponding DNA–DNA reassociation values were 62.2 and 35.9%, respectively. It is clear, as already noted (Stackebrandt and Goebel, 1994), that the relative evolutionary stability of 16S rRNA does not mirror corresponding stability of chromosomal DNA.

The organism originally designated as "*Micrococcus cryophilus*" (McLean et al., 1951) has been shown to be a psychrobacter (Juni and Heym, 1980), *P. urativorans* (Bowman et al., 1996).

The relatively large individual cells of many *Psychrobacter* strains are in contrast to the smaller sizes of most other Gram-negative bacteria. Major fatty acid profiles include $C_{16:1\omega7c}$, 35–47%; $C_{17:1\omega8c}$, 3–7%; and $C_{18:1\omega9c}$, 37–70% (Nishimura et al., 1979; Moss et al., 1988; Bowman et al., 1996). A novel fatty acid, $C_{12:1\omega9c}$, has been shown to be present in *P. immobilis* (Moss and Daneshvar, 1992). Contain wax esters (Gallagher, 1971; Russell and Volkman, 1980; Bowman et al., 1996, 1997d). The major ubiquinone of *Psychrobacter* contains eight isoprene units (Q8) (Moss et al., 1988). Cell membranes resemble those of other Gram-negative bacteria (Thorne, et al., 1973).

Most strains grow well at room temperature on plates of a variety of complex media. Strains of *P. frigidicola* grow best at 14–16°C; some fail to grow at room temperature and may be considered psychrophilic. Growth in liquid media requires aeration. Most psychrobacters can grow on mineral media containing sodium lactate and/or monosodium glutamate and an ammonium salt as the source of nitrogen. Many organic compounds can serve as sole carbon and energy sources for growth of psychrobacters (Shaw and Latty, 1988). The growth lag phase in such mineral media is decreased by addition of amino acids and, frequently, Tween 80 (Juni, 1992). Psychrobacters are aerobic and appear to contain all the enzymes of the tricarboxylic acid cycle (Tai and Jackson, 1969). They have been isolated from foods following gamma-ray irradiation at 300 and 500 krad (Ito et al., 1976), but strains isolated from nonirradiated foods are frequently radiation sensitive (Ito and Iizuka, 1983). They have also been isolated from sea life and the deep sea (Georgala, 1958; Maruyama et al., 2000; Denner et al., 2001a; Romanenko et al., 2002a), the skin of chickens (Lahellec et al., 1975), food products (Shaw and

Latty, 1988; Gennari et al., 1992), Antarctic ornithogenic soils (Bowman et al., 1996), Antarctic sea ice (Bowman et al., 1997d), and a bioaerosol originating from pigeon feces (Kämpfer et al., 2002a).

DNAs from strains of all species of *Psychrobacter* tested are able to transform a hypoxanthine and thiamine auxotroph (ATCC 43117), of a competent *Psychrobacter immobilis* strain (ATCC 43116), for ability to grow on a defined medium (Juni and Heym, 1980). DNAs from strains of all other members of the *Moraxellaceae* fail to transform this auxotroph to prototrophy; the transformation assay appears to be genus specific. Approximately one-third of strains isolated from humans and previously classified as CDC group EO-2 were shown to be psychrobacters, based on their phenotypic properties and the results of the transformation assay (Hudson et al., 1987).

DNA–rRNA hybridization studies have shown that *Moraxella phenylpyruvica* is more closely related to psychrobacters than to the other members of the *Moraxellaceae* (Rossau et al., 1991). The overall levels of 16S rDNA sequence similarity between one strain of *M. phenylpyruvica* and the several species of *Psychrobacter* was shown to be 93–96% (Bowman et al., 1996, 1997d; Denner et al., 2001a; Kämpfer et al., 2002a). *M. phenylpyruvica* also resembles *Psychrobacter* in being able to grow (slowly) at 4°C and tolerate NaCl concentrations up to 9%, and it has been proposed that *M. phenylpyruvica* be transferred to the genus *Psychrobacter* as *Psychrobacter phenylpyruvicus* comb. nov. (Bowman et al., 1996). There are three lines of evidence suggesting that *M. phenylpyruvica* is, in fact, not a species of *Psychrobacter*: 1. DNAs from strains of *M. phenylpyruvica* fail to transform the hypoxanthine and thiamine auxotroph of *Psychrobacter* to prototrophy (Juni and Heym, 1980), this assay being genus specific for all strains of *Psychrobacter* tested. 2. In a transformation assay that makes use of a streptomycin-dependent marker from a strain of *M. phenylpyruvica* that was transformed into the competent *Acinetobacter* strain BD413 (a rare transformant), it was shown that DNAs from 34 independently isolated strains of *M. phenylpyruvica* (including strain ATCC 23333^T, the type strain of *P. phenylpyruvicus*) all transformed the streptomycin-dependent *Acinetobacter* to streptomycin-independence (Juni, 1990). DNAs from strains of *Psychrobacter* were inactive (Juni, unpublished results). 3. There are

characteristic differences in fatty acid composition between strains of *Psychrobacter* and strains of *M. phenylpyruvica* (Moss et al., 1988; Bowman et al., 1996, 1997d) and, unlike *Psychrobacter* and other members of the *Moraxellaceae*, *M. phenylpyruvica* does not possess true waxes (Bowman et al., 1996). In the description of *M. phenylpyruvica* (listed under species of the genus *Moraxella*),

it is suggested that strains bearing this designation are members of a unique genus of the family *Moraxellaceae*.

A *Psp*PI type-II restriction-modification system of a strain of *Psychrobacter* has been investigated; the gene encoding the *Psp*PI methyltransferase has been cloned, expressed in *E. coli*, and its nucleotide sequence has been determined (Rina et al., 1997).

List of species of the genus *Psychrobacter*

1. *Psychrobacter immobilis* Juni and Heym 1986b, 388^{VP}.

im.mo' bi.lis. L. adj. *immobilis* motionless.

Plump rods or coccobacilli frequently showing diploforms, 0.9–1.3 µm in diameter. Nonmotile. Aerobic. Gram positive. Oxidase and catalase positive. Most strains grow on a defined medium with a single carbon and energy source and a mineral or organic source of nitrogen (i.e., an amino acid such as glutamate). Many organic compounds can serve as carbon and energy sources. Most strains form acid aerobically from D-glucose, D-mannose, D-galactose, L-arabinose, D-xylose, and L-rhamnose and use some of these compounds as carbon and energy sources for growth. Phenylalanine is deaminated. Other phenotypic characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 44–47 (*T_m*).

Type strain: ATCC 43116, CCUG 9708, DSM 7229, LMG 7091, LMG 7203.

GenBank accession number (16S rRNA): U39399.

2. *Psychrobacter faecalis* Kämpfer, Albrecht, Buczolits and Busse 2002b, 1438^{VP} (Effective publication: Kämpfer, Albrecht, Buczolits and Busse 2002a, 35.)

fae.ca' lis. L. n. *faex*, *faecis* dregs; M.L. adj. *faecalis* fecal.

Aerobic, cells are straight rods 0.8–1.2 × 1–2 µm and occur singly; they are nonmotile, Gram negative, oxidase- and catalase-positive. Colonies are circular, slightly raised, and beige with entire margins on nutrient agar. Grows well at 4–36°C on different complex media; no growth was found at 45 and 55°C on nutrient agar. No growth occurs on Salmonella-Shigella Agar. H₂S is not produced. Indole production, growth on Simmons citrate, urease, arginine dihydrolase, lysine decarboxylase, and ornithine decarboxylase, were all negative. No acid production from the following sugars could be detected: glucose, lactose, sucrose, D-mannitol, dulcitol, salicin, adonitol, inositol, sorbitol, L-arabinose, raffinose, L-rhamnose, maltose, D-xylose, trehalose, cellobiose, methyl-D-glucoside, erythritol, melibiose, D-arabitol, and D-mannose. The following compounds are not utilized as sole sources of carbon: L-arbutin, cellobiose, D-galactose, gluconate, D-maltose, D-mannose, α-D-melibiose, L-rhamnose, D-ribose, sucrose, salicin, D-trehalose, D-xylose, adonitol, *l*-inositol, D-sorbitol, glycerol, D-fructose, D-mannitol, maltitol, putrescine, *trans*-aconitate, mesaconate, adipate, azelate, itaconate, suberate, β-alanine, L-phenylalanine, L-serine, L-proline, L-histidine, L-tryptophan, 3-hydroxybenzoate, and phenylacetate. Ubiquinone Q-8 is the major quinone type, in addition to small amounts of ubiquinone Q-9 and Q-7; spermidine is the major polyamine. The polar lipid profile is characterized by the presence of phosphatidylethanolamine, phosphatidylglycerol, and diphosphatidylglycerol. The fatty acids C_{18:1} ω₉c, C_{17:1} ω₈c, C_{16:1} ω₇c, C_{15:0} iso 2OH, and C_{12:0} 3OH are produced. Isolated from an aerosol originating from pigeon feces. Other phenotypic characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: not determined.

Type strain: Iso-46, DSM 14664, CIP 107288.

GenBank accession number (16S rRNA): AJ421528.

3. *Psychrobacter frigidicola* Bowman, Cavanagh, Austin and Sanderson 1996, 847^{VP}.

fri' gid.i.col.a. L. adj. *frigidus* cold; L. n. *cola* dweller; M.L. n. *frigidicola* cold dweller.

Cells are coccoid, 1.5–2.0 µm in diameter; occur in pairs, tetrads, or rarely in short chains. Aerobic. Nonmotile. No growth factors are required. Able to grow in the absence of NaCl but growth is stimulated by 0.25–0.30 M NaCl. Can tolerate up to 9% NaCl. Optimal temperature for growth 14–16°C. Maximum temperature for growth 19–22°C. Phenylalanine and tryptophan are deaminated. Urease, nitrate reduction, and production of acid from carbohydrates are negative. Isolated from ornithogenic soils from Antarctica. Other phenotypic characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 41–42 (*T_m*).

Type strain: ACAM 304, CCUG 34377.

GenBank accession number (16S rRNA): U46143.

4. *Psychrobacter glacincola* Bowman, Nichols and McMeekin 1997e, 1274^{VP} (Effective publication: Bowman, Nichols and McMeekin 1997d, 214.)

gla.cin.co' la. L. n. *glacies* ice; L. *incola* inhabitant; M.L. n. *glacincola* the ice inhabitant.

Cells are coccoidal with a diameter of 1.0–1.5 µm. Nonmotile. Colonies are up to 4 mm in diameter, circular, have an entire edge, low convex elevation, butyrous consistency, and an off-white color. No fluorescent pigments formed. Seawater is required for optimal growth but most strains will grow at between 0 and 2.1 M NaCl. Optimal growth occurs at 19–22°C. Growth factors are not required. Fail to utilize or form acid from carbohydrates. Utilize nitrate and ammonium salts as nitrogen sources. Some strains produce urease and reduce nitrate to nitrite. Tween 40 and Tween 80 are hydrolyzed. Phenylalanine is not deaminated. Isolated from Antarctic ice. Other phenotypic characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 43–44 (*T_m*).

Type strain: ACAM 483, DSM 12194.

GenBank accession number (16S rRNA): U46145.

5. *Psychrobacter marincola* Romanenko, Schumann, Rohde, Lysenko, Mikhailov and Stackebrandt 2002a, 1296^{VP}

ma.rin.co' la. L. gen. n. *maris* of the sea; L. n. *incola* inhabitant; N.L. n. *marincola* inhabitant of the sea.

Aerobic, Gram negative, nonmotile nonpigmented, non-sporeforming, coccoidal cells, 0.7–1.0 µm in diameter. Oxidase- and catalase-positive. Colonies on nutrient agar are whitish, circular with an entire edge, convex, 1–2 mm in diameter. No fluorescent pigments are formed. Moderately halophilic. Seawater is required for growth; can grow at 0.5–

TABLE BXII.γ.133. Properties of *Psychrobacter* species^a

Characteristic	<i>P. immobilis</i> ^b	<i>P. faecalis</i> ^c	<i>P. frigidicola</i> ^b	<i>P. glacincola</i> ^b	<i>P. marincola</i> ^d	<i>P. pacificensis</i> ^e	<i>P. proteolyticus</i> ^f	<i>P. submarinus</i> ^d	<i>P. urativorans</i> ^b
Maximum growth temperature, °C	35–39	36	19–22	19–22	35	33–38	<35	35	25–27
Optimum growth temperature, °C	27–31	15–30	14–16	13–15	25–28	25–28	ND	25–28	17–19
Tolerates 2.0 M NaCl	v+	+	–	+	+	–	+	+	–
Tolerates 5% bile salts	+	+	–	+	+	ND	–	+	–
Stimulated by bile salts	–	ND	–	–	ND	ND	ND	ND	–
Acid from carbohydrates	+	–	–	–	–	+	+	+	–
Arginine arylamidase	+	ND	v+	–	ND	ND	ND	ND	+
Phenylalanine arylamidase	v+	ND	+	+	ND	ND	ND	ND	–
Phenylalanine deaminase	+	ND	–	–	–	–	ND	–	+
Nitrate reduction	v–	ND	–	v+	–	–	–	–	v–
Urease	v+	–	–	v–	–	+	+	–	v+
Lecithinase (egg yolk reaction)	v+	ND	–	+	ND	ND	ND	ND	–
Tween 40 hydrolysis	+	ND	–	+	–	ND	ND	+	–
Tween 80 hydrolysis	+	ND	–	+	+	ND	+	ND	–
<i>Carbon sources:</i>									
3-Hydroxybutyrate	+	+	–	v+	ND	v	+	ND	+
Pyruvate	+	+	–	+	ND	ND	+	ND	+
L-Malate	+	+	+	–	–	+	–	–	v+
Oxaloacetate	+	ND	–	+	ND	ND	ND	ND	–
DL-Lactate	+	+	–	v+	–	v	–	–	v+
Propionate	v+	(+)	–	+	–	–	+	–	–
Isovalerate	v–	ND	–	v+	ND	ND	ND	ND	–
Heptanoate	v+	ND	–	+	ND	ND	ND	ND	v–
Pimelate	–	ND	–	v+	ND	ND	ND	ND	–
Azelate	v–	–	+	+	ND	ND	+	ND	–
L-Alanine	+	+	–	v+	–	v	+	–	–
Hydroxy-L-proline	v–	ND	–	+	ND	ND	ND	ND	–
L-Phenylalanine	+	–	+	–	ND	ND	–	ND	v+
Urate	v+	ND	+	+	ND	ND	ND	ND	+
Mol% G + C (<i>T_m</i>) of the DNA	44–47	ND	41–42	43–44	50.7	44–45	43.6	46.7	44–46

^aSymbols: +, positive for 90–100% of strains; v, positive for 11–89% of strains; v+, positive for 11–89% of strains but type strain is positive; v–, positive for 11–89% of strains but type strain is negative; –, positive for 0–10% of strains; ND, not determined.

^bData for *P. immobilis*, *Psychrobacter frigidicola*, *P. glacincola*, and *P. urativorans* from Bowman et al., 1997d.

^cData for *P. faecalis* from Kämpfer et al., 2002a.

^dData for *P. marincola* and *P. submarinus* from Romanenko et al., 2002a.

^eData for *P. pacificensis* from Maruyama et al., 2000.

^fData for *P. proteolyticus* from Denner et al., 2001a.

15.5% (w/v) NaCl. Growth occurs at 7–35°C, with an optimum at 25–28°C. Does not grow at 4, 37, or 40°C. The pH range for growth is 5.5–9.5, with optimum growth at 6.5–8.5. Negative for Simmons citrate test, Voges–Proskauer reaction, ONPG test, H₂S production and denitrification. Susceptible to ampicillin (10 µg), benzylpenicillin (10 U), gentamycin (10 µg), kanamycin (30 µg), carbenicillin (25 µg), lincomycin (15 µg), oleandomycin (15 µg), polymyxin (300 U), streptomycin (30 µg), tetracycline (30 µg) and neomycin (15 µg). The major fatty acid is C_{18:1} ω_{9c}. The phospholipids are phosphatidylethanolamine, phosphatidylglycerol, phosphatidylinositol and phosphatidylserine. Isolated from internal tissues of the ascidian *Polysyncrator* sp., collected in the Indian Ocean. Other characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 50.7 (*T_m*).

Type strain: KMM277, DSM 14160.

GenBank accession number (16S rRNA): AJ309941.

6. ***Psychrobacter pacificensis*** Maruyama, Honda, Yamamoto, Kitamura and Higashihara 2000, 845^{VP}
pa.ci.fi.cen'sis. M.L. adj. *pacificensis* pertaining to the Pacific Ocean.

Strains are aerobic, Gram negative, nonmotile, nonpigmented, nonsporulating, oxidase-positive, catalase-positive coccobacilli 1.0–1.5 µm long and about 1 µm wide. Strains

produce many fimbriae as appendages, but no flagella. Circular, convex, off-white colonies with a complete edge occur on agar plates containing polypeptone and yeast extract. No fluorescent pigments formed. Seawater or NaCl at about 3% is required for optimal growth, with most strains not growing at 0 or 8% NaCl. It takes 1–2 weeks to reach stationary phase at 4°C, but isolates give growth yields at 4°C comparable to those at 20°C. Optimal growth at about 25°C, with the maximum growth temperature being 38°C. Acid is formed aerobically from glucose, xylose and arabinose. Urease is positive, but phenylalanine deaminase and tryptophan deaminase are negative. The species is negative for the following biochemical tests: glucose fermentation, indole production, esculin hydrolysis, gelatin liquefaction, and arginine dihydrolase. Strains use L-histidine and DL-malate as sole carbon and energy sources. Some strains use acetate, L-alanine, 3-hydroxybutyrate, lactate, malonate and substrate, but none use *p*-hydroxybutyrate, citrate, gluconate, propionate, L-serine or *n*-valerate. The major fatty acid is C_{18:1} ω_{9c} and Q-8 is the major quinone. Isolated from 6,000 m deep seawater in the Japan Trench off Hachijo Island, Japan. Other characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 43–44 (HPLC).

Type strain: NIH B P2K6.

GenBank accession number (16S rRNA): AB016057.

7. **Psychrobacter proteolyticus** Denner, Mark, Busse, Turkiewicz and Lubitz 2001b, 1619.^{VP} (Effective publication: Denner, Mark, Busse, Turkiewicz and Lubitz 2001a, 51.) *pro.te.o.ly' ti.cus*. M.L. n. adj. *proteolyticus*, proteolytic.

Cells are Gram negative, nonpigmented, nonmotile, nonsporeforming, rod-shaped to coccoidal, and 0.5–1.25 µm wide by 1.0–2.5 µm long. Colonies on complex standard media are cream-colored, circular, entire, convex, and of butyrous consistency; diameters are as large as 1.0 mm after 2 d incubation at room temperature. The following biochemical tests are positive: oxidase, catalase, urease, alkaline phosphatase, C4 esterase, C8 esterase, lipase, leucine arylamidase, valine arylamidase, acid phosphatase, and naphthol-AS-BI-phosphohydrolase. Gelatin, casein, and tributyrin are hydrolyzed whereas esculin, starch, and DNA are not. The following biochemical tests are negative: arginine dihydrolase, lysine decarboxylase, ornithine decarboxylase, C14 lipase, cystine arylamidase, trypsin, chymotrypsin, α-galactosidase, β-galactosidase, β-glucuronidase, α-glucosidase, β-glucosidase, α-mannosidase, α-fucosidase, H₂S production, nitrate reduction, and indole production. Citrate is used as a single carbon source. Acid is produced from the carbohydrates: glucose, rhamnose, melibiose, amygdalin, and arabinose. No acid is produced from the carbohydrates: mannitol, inositol, sorbitol, and saccharose. Ubiquinone-8 is the main respiratory lipoquinone. Phosphatidylethanolamine, phosphatidylglycerol, and diphosphatidylglycerol are the major polar lipids. The predominant nonpolar fatty acids are: C_{18:1 ω9c} and C_{16:1 c}; C_{17:1 ω8c}, C_{16:0}, C_{18:0}, and C_{12:0} are present in minor amounts. The predominant 2-hydroxy fatty acid is C_{12:0 2OH}. Isolated from a stomach specimen of an Antarctic krill. Other phenotypic characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 43.6 (HPLC).

Type strain: 116, CIP 106830, DSM 13887.

GenBank accession number (16S rRNA): AJ272303.

8. **Psychrobacter submarinus** Romanenko, Schumann, Rohde, Lysenko, Mikhailov and Stackebrandt 2002a, 1296^{VP} *sub.ma.ri' nus*. L. prep. *sub* under; L. adj. *marinus* of the sea; N.L. adj. *submarinus* from under the sea.

Aerobic, Gram negative, nonmotile, nonpigmented, nonsporeforming, ovoid cells, 1.6–1.9 µm long and 0.7–1.0

µm in diameter. Oxidase- and catalase-positive. Colonies on nutrient agar are whitish, circular with an entire margin, convex, 2–3 mm in diameter. No fluorescent pigments are formed. Moderately halophilic. Seawater is required for growth; growth occurs in the presence of 0.5–15.5% (w/v) NaCl. Temperature range for growth is 4–35°C, with an optimum at 25–28°C. Does not grow at 37 or 40°C. pH range for growth is 5.5–9.5 with optimum growth at 6.5–8.5. The strain is negative for Simmons citrate test, ONPG test, acetoin production, H₂S production, and denitrification. Susceptible to ampicillin (10 µg), gentamicin (10 µg), kanamycin (30 µg), carbenicillin (25 µg), polymyxin (300 U), streptomycin (30 µg), and neomycin (15 µg). Resistant to benzylpenicillin (10 U), lincomycin (15 µg), oleandomycin (15 µg), tetracycline (15 µg). The major fatty acid is C_{18:1 ω9c}. The phospholipids are phosphatidylethanolamine, phosphatidylglycerol, diphosphatidylglycerol, phosphatidylinositol, and phosphatidylserine. Other characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 46.7 (*T_m*).

Type strain: KMM 225, DSM 14161.

GenBank accession number (16S rRNA): AJ309940.

9. **Psychrobacter urativorans** Bowman, Cavanagh, Austin and Sanderson 1996, 847^{VP} *u.ra.ti.vor' ans*. M.L. n. *uratum* salt of uric acid; M.L. part. adj. *vorans* devouring; M.L. part. adj. *urativorans* uric acid devouring.

Cells are coccoid, 1.5–2.0 µm in diameter, and occur in pairs and tetrads. Nonmotile. Aerobic. No growth factors are required. Growth is inhibited by 1% bile. Tolerates 9% NaCl. Growth is not stimulated by NaCl. Optimal temperature for growth 18–20°C. Maximum temperature for growth 25–27°C. Does not deaminate phenylalanine or tryptophan. Acid is not produced from carbohydrates. Strains may produce urease and reduce nitrate to nitrite. Other phenotypic characteristics appear in Table BXII.γ.133.

The mol% G + C of the DNA is: 44–46 (*T_m*).

Type strain: ACAM 534, ATCC 15174, CCUG 4982, DSM 20429.

GenBank accession number (16S rRNA): U46141.

Family III. *Incertae Sedis*

Genus I. Enhydrobacter Staley, Irgens and Brenner 1987, 290^{VP}

JAMES T. STALEY AND DON J. BRENNER

En.hy'dro.bac.ter. Gr. adj. *enhydros* aquatic Gr. neut. n. *bacter* masc. form of Gr. neut. n. *bacterion* a rod; N.L. masc. n. *Enhydrobacter* aquatic rod.

Coccobacillary to rod-shaped cells, 0.5–0.7 × 1.0–5.0 µm. **Uncellular; pairs and short chains occur.** No resting stages known. **Gram negative. Facultative aerobic.** Chemoheterotrophic. **Oligotrophic. Grow very slowly.** Sugars are fermented anaerobically, and organic acids are used aerobically. Growth occurs on mineral medium with D-glucose as the sole carbon source and inorganic ammonium compounds as the sole nitrogen sources. No growth

on nutrient agar, Clark–Lubs medium or EMB agar. Vitamins are required by the type strain when grown on defined media. Oxidase and catalase positive. Type strain isolated from the oxygen-depleted zone of a eutrophic lake.

The mol% G + C of the DNA is: 66.

Type species: **Enhydrobacter aerosaccus** Staley, Irgens and Brenner 1987, 290.

FURTHER DESCRIPTIVE INFORMATION

A recent analysis of 16S rRNA sequences indicated that *Enhydrobacter aerosaccus* is closely related to *Moraxella osloensis* (Thompson et al., 2003a). The genus *Enhydrobacter* has therefore been placed in the family *Moraxellaceae* as a *genus incertae sedis*; no further testing has been done to relate it to or to differentiate it from other genera in this family.

Unlike other described members of aeromonads, *Enhydrobacter aerosaccus* is a true oligotrophic bacterium that grows very slowly on typical media used for cultivation of *Proteobacteria*. Biochemical reactions on Hugh–Leifson medium take 30–60 days at room temperature.

This monospecific genus most closely resembles members of the genus *Aeromonas* in that it grows on a defined medium with D-glucose as the sole carbon source and uses ammonium salts as the sole nitrogen sources for growth. Like *Aeromonas* spp., *Enhydrobacter* can use some amino acids such as L-alanine, L-serine, and L-arginine as carbon sources for growth. The type strain of *E. aerosaccus* is nonmotile, a feature that is also found in some species of *Aeromonas*, such as *A. salmonicida*.

However, *E. aerosaccus* differs from *Aeromonas* spp. in several ways. The mol% G + C content of the DNA is 66, considerably higher than the 57–63% found in *Aeromonas* spp. Furthermore, it does not grow on typical media used for cultivation of *Aeromonas* spp., such as methyl red–Voges–Proskauer (Clark–Lubs) medium, EMB medium, nutrient agar, and King A and King B media. Furthermore, *E. aerosaccus* is gas vacuolate, a property not found in any *Aeromonas* spp.

Finally, DNA–DNA reassociation experiments indicate that *Enhydrobacter* is sufficiently distantly related to *Aeromonas* to justify the creation of a separate genus (Staley et al., 1987).

ENRICHMENT AND ISOLATION PROCEDURES

The sole strain of this genus has been isolated from a eutrophic lake, Wintergreen Lake, near Kalamazoo, Michigan (Van Ert and Staley, 1971). A dilute peptone enrichment culture is made using 100 ml of lake water from the thermocline of the oxygen-depleted zone (< 1 mg/l dissolved O₂), supplemented with 10 mg of sterile Bacto Peptone (Difco laboratories, Detroit, MI). After incubation for 18 days at room temperature, 0.1 ml of a 1:100 dilution is spread on plates containing 0.01% peptone with mineral salts and vitamins (Van Ert and Staley, 1971). After incubation for 27 d, material from the plate, containing colonies of planctomycetes, is restreaked on PYG medium (0.025% peptone, 0.025% yeast extract, and 0.025% glucose with vitamins and mineral salts; see genus *Ancalomicrobium* description, this volume). A small, chalky white colony containing gas-vacuolated, rod-shaped cells appears after one month. This is streaked to purity on a medium containing 0.1% glucose, 0.1% Difco Vitamin Free Cas-amino acids, mineral salts, and vitamins.

MAINTENANCE PROCEDURES

The organism has been maintained by lyophilization for 30 years.

DIFFERENTIATION OF THE GENUS *ENHYDROBACTER* FROM OTHER GENERA

Table BXII.γ.134 shows the differences between *Enhydrobacter* and *Aeromonas*. Although the genera resemble one another in morphology, the type strain and species of *Enhydrobacter* is gas vacuolate. *Enhydrobacter aerosaccus* does not grow on many of the typical media used for cultivation of *Aeromonas* spp., and the time for growth in biochemical tests is 30–60 d.

List of species of the genus *Enhydrobacter*

1. ***Enhydrobacter aerosaccus*** Staley, Irgens and Brenner 1987, 290.
aer' o.sac.cus. Gr. n. *aer* air; Gr. n. *sakkos* sack; N.L. *aerosaccus* gas vacuolate.

Nonmotile, gas-vacuolated, nonflagellated rods. Gas vacuoles are not produced when growing on sugars, but are produced when cells are grown on certain organic acids, such as acetate, pyruvate, and succinate. Facultatively aerobic, but best growth aerobically is under microaerophilic conditions. No growth at 7 or 43°C; growth occurs at 20–41°C, with optimum growth at 37–39°C. Grows at pH 5.0–9.5. Nitrate is reduced to nitrite. Folic acid and biotin are required for growth. Lysine, ornithine, and arginine are

decarboxylated after 5 weeks at room temperature. The following carbon sources are utilized after incubation at 30°C for 60 d: L-arabinose, D-xylose, L-fucose, L-rhamnose, D-ribose, D-glucose, D-fructose, D-mannose, L-sorbose, D-galactose, D-xylose, sucrose, maltose, acetate, citrate, formate, fumarate, glycerol, succinate, inulin, ethanol, pyruvate, lactate, *meso*-malate, oxaloacetate, and *meso*-tartrate. Colonies, which adhere to the agar, are colorless with an entire margin and convex elevation. The type strain was isolated from the oxygen-depleted zone of a eutrophic lake.

The mol% G + C of the DNA is: 66 (Bd).
Type strain: ATCC 27094, DSM 8914, LMG 81277.
GenBank accession number (16S rRNA): AJ550856.

TABLE BXII.γ.134. Differentiation of the genera *Enhydrobacter* and *Aeromonas*^a

Characteristic	<i>Enhydrobacter</i>	<i>Aeromonas</i>
<i>Growth on:</i>		
Nutrient agar	–	+
EMB agar	–	+
Clark–Lubs medium	–	+
Average time for biochemical reactions (d)	30–60	1–2
Growth rate	very slow	rapid
Gas vacuole formation	+ ^b	–
Mol% G + C of DNA	66	57–63

^aFor symbols, see standard definitions.
^bOnly one strain has been described, and it is gas vacuolate.

Order X. **Alteromonadales** *ord. nov.*

JOHN P. BOWMAN AND THOMAS A. McMEEKIN

Al.ter.o.mon.a.da'les. M.L. fem. n. *Alteromonas* the type genus of the order; L. suff. *-ales* ending which denotes an order; M.L. fem. n. *Alteromonadales* the *Alteromonas* order.

Gram negative, straight or curved rods. Motile by means of a **single polar flagellum**. Do not form endospores or cysts. **Chemoheterotrophs** which are **either facultative anaerobes or strictly aerobic**. If able to grow anaerobically, growth occurs by respiration and, less commonly, by fermentation. Oxygen is used universally as an electron acceptor. **Most strains are catalase and oxidase positive**. Possess psychrophilic, psychrotolerant, and mesophilic growth regimes. Most species **require Na⁺ ions** for growth and most species grow optimally on seawater-based media. Most species are able to utilize ammonia as a nitrogen source. Some species have a requirement for amino acids. **Primarily marine inhabitants** found in a wide range of oceanic ecosystems ranging from surface waters and sea ice to abyssal sediments. Usually nonpathogenic. The major coenzyme Q type is ubiquinone-8.

The mol% G + C of the DNA is: 36–54 (*T_m*, Bd).

Type genus: **Alteromonas** Baumann, Baumann, Mandel and Allen 1972, 418 emend. Gauthier, Gauthier and Christen 1995a, 760

FURTHER DESCRIPTIVE INFORMATION

The order *Alteromonadales* and family *Alteromonadaceae* are created here to accommodate an assemblage of marine Gram-negative bacteria that form a phylogenetic clade in the *Gammaproteobacteria*, positioned between the families *Pseudomonadaceae* and *Enterobacteriaceae*. Members of this group include the genera *Colwellia*, *Moritella* (*Vibrio marinus*), *Shewanella*, *Ferrimonas*, *Alteromonas*, and *Pseudoalteromonas*. These genera are characteristically rod-like in morphology, motile by a single polar flagellum, oxidase positive, require or stimulated by sodium ions, and possess DNA base compositions ranging from 36 to 54 mol% G + C. Cells generally lack inclusions such as poly-β-hydroxybutyrate, resting stages such as spores and microcysts, and cellular extensions such as prostheca, buds, and spinae. The metabolism of the members of the order *Alteromonadales* is quite varied. Some members are facultative anaerobes able to grow anaerobically by fermentation and respiration (*Colwellia*, *Moritella*, and some *Shewanella* species) or by respiration alone (*Ferrimonas* and most *Shewanella* species).

Still others are strictly aerobic (*Alteromonas* and *Pseudoalteromonas*).

Overall, there are few if any common properties (except those stated) which the genera in the order *Alteromonadales* have in common. Based on the molecular clock theory (Ochman and Wilson, 1987), the divergence in 16S rRNA gene sequences suggests that members of the order *Alteromonadales* have been evolving in the oceans for at least 500–600 million years (assuming a 1% sequence difference equals ~50 million years). During that time, a multiplicity of marine (and some terrestrial) habitats have been colonized, with the development of a wide range of eco-physiological specialization and versatility. For example, the order *Alteromonadales* is noteworthy for containing all known psychrophilic bacteria belonging to the *Gammaproteobacteria*, the majority of barophilic species, and most bacterial species capable of forming omega-3 polyunsaturated fatty acids.

TAXONOMIC COMMENTS

Before phylogenetic analysis became commonplace, members of the order *Alteromonadales* were grouped either within the family *Vibrionaceae* (e.g., *Shewanella*; MacDonell and Colwell, 1985; Farmer, 1992) or were simply ungrouped (e.g., *Alteromonas*, Baumann et al., 1984d; Gauthier and Breittmayer, 1992). The advent of 16S rRNA-based phylogenetic analysis has resulted in considerable reorganization of the species in this group. This includes major reassignment of species of *Alteromonas* to *Shewanella* (MacDonell and Colwell, 1985; Coyne et al., 1989), *Pseudoalteromonas* (Gauthier et al., 1995a), and *Marinomonas* (van Landschoot and De Ley, 1983). *Colwellia* (Deming et al., 1988a) was created to accommodate a psychrophilic *Vibrio* species (D'Aoust and Kushner, 1972) which, on the basis of 5S and 16S rRNA gene sequence analysis, was clearly distinct from other *Vibrio* species. *Vibrio marinus* (Baumann et al., 1984b) also groups phylogenetically within the family *Alteromonadaceae* and has been recently transferred to a new genus, *Moritella*, as *Moritella marina* (Urakawa et al., 1998). *Ferrimonas*, a genus containing iron-reducing species, is similar in many respects to *Shewanella* and is represented by only a single strain (Rosselló-Mora et al., 1995a).

Family I. **Alteromonadaceae** Ivanova and Mikhailov 2001b, 1229^{VP} (Effective publication: Ivanova and Mikhailov 2001a, 15.)

JOHN P. BOWMAN AND THOMAS A. McMEEKIN

Al.ter.o.mo.na.da'ce.ae. M.L. fem. n. *Alteromonas* the type genus of the family; L. suff. *-aceae* which denotes a family; M.L. fem. n. *Alteromonadaceae* the *Alteromonas* family.

The description is the same as for the order *Alteromonadales*. See Table BXII.γ.135 for characteristics differentiating some genera of the family *Alteromonadaceae*.*

Type genus: **Alteromonas** Baumann, Baumann, Mandel and Allen 1972, 418 emend. Gauthier, Gauthier and Christen 1995a, 760.

*Editorial Note: Only the genera *Alteromonas*, *Colwellia*, *Idiomarina*, and *Pseudoalteromonas* were included in the family by Ivanova and Mikhailov (2001a). The genera *Alishewanella*, *Ferrimonas*, *Glaciecola*, *Marinobacter*, *Marinobacterium*, *Microbulbifer*, *Moritella*, *Psychromonas*, *Shewanella*, and *Thalassomonas* have been added to this family based on 16S rDNA sequence analysis.

TABLE BXII.γ.135. Characteristics differentiating some genera belonging to the family *Alteromonadaceae*^a

Characteristics	<i>Alteromonas</i>	<i>Cokwellia</i>	<i>Ferrimonas</i>	<i>Moritella</i> (<i>Vibrio marinus</i>)	<i>Pseudoalteromonas</i>	<i>Shewanella</i>
<i>Growth in presence of:</i>						
0% NaCl	—	—	—	—	—	D
10% NaCl	+	—	—	—	D	D
<i>Growth at:</i>						
4°C	—	+	—	+	D	D
25°C	+	—	+	D	+	D
35°C	+	—	+	—	D	D
Barophilic	—	D	—	D	—	D
Nitrate reduction	—	+	+	+	D	+
Denitrification	—	—	+	—	D	D
<i>Anaerobic growth:</i>						
Fermentative	—	+	—	+	—	D
Respiratory	—	+	+	+	—	+
<i>Hydrolysis of:</i>						
Gelatin	+	+	—	+	+	+
Starch	+	+	—	—	D	D
<i>Utilization of:</i>						
D-glucose, maltose	+	+	nd	+	+	D
Salicin, galacturonate, DL-glycerate	—	—	nd	—	—	—
PUFA	—	DHA	—	DHA	—	EPA
Mol% G + C of DNA	44–47	35–46	54	40–42	36–48	38–54

^aFor symbols, see standard definitions; nd, not determined; PUFA, polyunsaturated fatty acid; EPA, eicosapentaenoic acid (20:5 ω3c); DHA, docosahexaenoic acid (C_{20:6} ω3c).

Key to Some Genera of the Family Alteromonadaceae

- I. Straight to curved rod-shaped cells, motile, that require or are stimulated by NaCl and are oxidase positive. Strictly aerobic; no anaerobic growth with nitrate as the electron acceptor.
 - A. Utilizes D-turanose, salicin, D-glucuronate, and/or DL-glycerate.

Genus *Alteromonas*
 - B. Does not utilize D-turanose, salicin, D-galacturonate, and/or DL-glycerate.

Genus *Pseudoalteromonas*
- II. Straight to curved rod-shaped cells, motile, that require or are stimulated by NaCl and are oxidase positive. Anaerobic growth using nitrate as an electron acceptor.
 - A. Gelatinase positive.
 1. Dissimilatory iron-reduction positive and/or form eicosapentaenoic acid (EPA); most species amylase negative.

Genus *Shewanella*
 2. Dissimilatory iron-reduction negative; form docosahexaenoic acid (DHA).
 - a. Amylase positive.

Genus *Cokwellia*
 - b. Amylase negative.

Moritella (*Vibrio marinus*)
 - B. Gelatinase negative.

Genus *Ferrimonas*

Genus I. *Alteromonas* Baumann, Baumann, Mandel and Allen 1972, 418, emend. Gauthier, Gauthier and Christen 1995a, 760

JOHN P. BOWMAN AND THOMAS A. McMEEKIN

Al.te.ro.mo' nas. L. adj. *alter* another; Gr. n. *monas* a unit, monad; M.L. fem. n. *Alteromonas* another monad.

Straight rods, approximately 1 × 2–3 μm, occurring singly or in pairs. Do not form spores or cysts. Do not accumulate poly-β-hydroxybutyrate. **Gram negative. Motile**, propelled by a single polar unsheathed flagellum. Colonies are nonpigmented. **Chemoheterotrophic. Aerobic, having a strictly respiratory type of metabolism** with oxygen as the terminal electron acceptor. Catalase and oxidase positive. **Nitrate is not reduced, nor can it be used as an alternative electron acceptor.** Do not grow by hydrogen autotrophy. A constitutive arginine dihydrolase system is absent. A range of carbon substrates can be used, including car-

bohydrates, alcohols, organic acids, and amino acids. Ammonia serves as a nitrogen source. Growth factors are not required. **Require Na⁺ ions for growth.** Growth occurs between 10 and 40°C; **no growth at 4°C or 45°C.** Major quinone is ubiquinone-8 (Q-8). Major fatty acids are C_{16:0}, C_{16:1} ω7c, C_{17:1} ω8c, and C_{18:1} ω7c. Isolated from seawater. Member of the order *Alteromonadales*, family *Alteromonadaceae*, class *Gammaproteobacteria*.

The mol% G + C of the DNA is: 44–48 (T_m).

Type species: *Alteromonas macleodii* Baumann, Baumann, Mandel and Allen 1972, 418.

FURTHER DESCRIPTIVE INFORMATION

Alteromonas macleodii is the only validly published species included in the genus at present.

When growing on various complex or defined seawater or NaCl-containing media such as Difco marine 2216 agar (ZoBell, 1941), basal medium agar¹ (BMA), or yeast extract agar² (YEA) (Baumann et al., 1972), the colonies possess an off-white color, smooth texture, circular convex shape, entire edge, and creamy consistency.

All strains grow best on media containing a seawater salts base and fail to grow if Na⁺ ions are absent or if Na⁺ ions are completely replaced by K⁺ ions. *A. macleodii* is relatively halotolerant and able to grow fairly well on media containing 10% NaCl (Ortigosa et al., 1994). Strains are also characteristically mesophilic, growing between 10 and 40°C, with an optimal temperature of 30–35°C. *A. macleodii* and the closely related "*Alteromonas infernus*" are chemoheterotrophs able to utilize a wide range of carbohydrates for growth (Baumann et al., 1984a; Raguénès et al., 1997b). In addition, *A. macleodii* can utilize a range of organic acids, alcohols, and amino acids, but is unable to grow on aromatic compounds (Table BXII.γ.136). *A. macleodii* produces several extracellular enzymes and is able to hydrolyze gelatin, lipids, and starch. A few strains can also attack alginate, but are unable to degrade agar, chitin, or cellulose.

A. macleodii possesses an inducible Entner–Doudoroff pathway (Baumann and Baumann, 1973), which is used for catabolism of D-fructose and D-glucose. Activity for aspartokinase, which is involved in the initial stages of the synthesis of the aspartate family of amino acids, is absent (Baumann and Baumann, 1974).

A. macleodii is not susceptible to predation by *Bdellovibrio* (strains of which were initially isolated by using *Vibrio* species as hosts) (Taylor et al., 1974). *Alteromonas*-like strains have been found to form proteinaceous substances, referred to as marinostatins, which are potent inhibitors of a range of proteases, including subtilisin, papain, ficin, and α-chymotrypsin (Imada et al., 1985a, b, 1986). Several *Alteromonas* strains have been reported to form antibiotic-like substances; however, it is likely that these strains actually belong to the genus *Pseudoalteromonas* (see section on the genus *Pseudoalteromonas*).

The mol% G + C values for the DNA of *A. macleodii* range from 44.9–46.4 (*T_m*). The mol% G + C contents of "*A. macleodii* subsp. *fijiensis*" (Raguénès et al., 1996) and "*A. infernus*" (Raguénès et al., 1997b) have slightly higher values, of 47–48 (*T_m*).

Ubiquinone-8 (Q-8) is the major quinone (97% of total), with traces of Q-9 (2%) and Q-10 (1%) present (Akagawa-Matsushita et al., 1992b). The major whole-cell fatty acids of *A. macleodii* (determined only for strain ATCC 27126) include C_{15:0} (3%), C_{15:1 ω8c} (2%), C_{16:1 ω7c} (37%), C_{16:0} (24%), C_{17:1 ω8c} (7%), C_{17:0} (5%), and C_{18:1 ω7} (9%) (Svetashev et al., 1995). Analysis using neural networks is not able to distinguish the fatty acid pattern of *A. macleodii* from those of some other members of the genus *Pseudoalteromonas* (Bertone et al., 1996). Hydroxy fatty acids comprise only a small amount (~1%) of the whole cell fatty acids. The major polyamines in *A. macleodii* include putrescine, 2-hy-

TABLE BXII.γ.136. Characteristics of *Alteromonas macleodii*^a

Characteristics	Reaction or result for	
	<i>A. macleodii</i>	ATCC 27126 ^T
Accumulation of poly-β-hydroxybutyrate	–	–
Catalase, oxidase	+	+
Hydrogen autotrophy	–	–
Arginine dihydrolase	–	–
Nitrate reduction	–	–
Requirement for Na ⁺ ions for growth	+	+
Growth at:		
4°C	–	–
30°C	+	+
40°C	d	+
45°C	–	–
Lipase, gelatinase, amylase	+	+
Alginase	(–)	+
Chitinase	–	–
Utilization of: ^b		
D-Fructose, D-galactose, D-glucose, cellobiose, lactose, maltose, melibiose, sucrose, trehalose, salicin, glycerol, acetate, propionate, butyrate, isobutyrate, valerate, caproate, caprylate, pelargonate, caprate, pyruvate, DL-glycerate, L-alanine	+	+
D-Gluconate, D-galacturonate, ethanol	(+)	+
L-Tyrosine	(+)	–
N-Acetylglucosamine, D-mannitol, propanol, heptanoate, DL-lactate, glycine	d	+
Propane-1,2-diol, isovalerate, DL-3-hydroxybutyrate, L-arginine, L-aspartate, L-glutamate, L-isoleucine, L-leucine, L-serine, L-valine	d	–
D-Xylose	(–)	+
Mol% G + C of DNA (<i>T_m</i>)	45–46	46

^aSymbols: +, positive for 90% or more strains; (+), positive for 75–89% of strains; d, positive for 26–74% of strains; (–), positive for 11–25% of strains; –, positive for 0–10% of strains.

^bThe following substrates are not utilized: D-ribose, D-arabinose, L-arabinose, D-fucose, L-rhamnose, D-mannose, inulin, D-erythritol, D-sorbitol, *meso*-inositol, adonitol, ethane-1,2-diol, butane-2,3-diol, methanol, isopropanol, isobutanol, saccharate, mucate, D-glucuronate, formate, oxalate, malonate, succinate, adipate, pimelate, suberate, azelate, sebacate, DL-malate, D(–)-tartrate, L(+)–tartrate, *meso*-tartrate, 2-oxoglutarate, fumarate, citrate, aconitate, maleate, levulinate, citraconate, itaconate, mesaconate, D-mandelate, L-mandelate, benzoyl formate, benzoate, *o*-hydroxybenzoate, *m*-hydroxybenzoate, *p*-hydroxybenzoate, phenylacetate, quinate, D-alanine, β-alanine, L-citrulline, L-histidine, L-lysine, L-norleucine, L-orithine, L-phenylalanine, L-proline, L-threonine, D-tryptophan, L-tryptophan, DL-kynurenine, kynurenate, anthranilate, *m*-aminobenzoate, *p*-aminobenzoate, γ-aminobutyrate, δ-aminovalerate, hippurate, methylamine, ethanolamine, benzylamine, putrescine, spermine, histamine, tryptamine, butylamine, 2-aminolamine, pentylamine, betaine, sarcosine, creatine, pantothenate, acetamide, nicotine, nicotinamide, trigollenine, allantoin, adenine, guanine, cytosine, thymine, uracil, and *n*-hexadecane.

droxyputrescine, cadaverine, 5-hydroxy spermidine, and spermidine (Hamana, 1997). *Alteromonas* species form acidic polysaccharides. For example, "*A. macleodii* subsp. *fijiensis*" produces a uronic acid-rich polysaccharide with a viscosity similar to that of xanthan gum (Raguénès et al., 1996). "*A. infernus*" secretes two unusual polysaccharides, the first of which is a water-soluble exopolymer made up of monomers of glucose, galactose, galacturonate, and glucuronate and an insoluble gel-forming exopolymer (Raguénès et al., 1997b). Several other *Alteromonas* strains produce a range of novel acidic exopolysaccharides (Ivanova et al., 1994; Vincent et al., 1994; Nazarenko et al., 1993; Samain et al., 1997).

Little is known about the specific ecology of *Alteromonas*. Stud-

1. BMA consists of (per liter): Tris-HCl, 6.1 g (to give a pH of 7.5); NH₄Cl, 1.0 g; K₂HPO₄·3H₂O, 0.075 g; FeSO₄·7H₂O, 0.028 g; artificial seawater (ASW), 500 ml; distilled water, 500 ml; and agar, 20 g. ASW (MacLeod, 1968) consists of (g/l distilled water): NaCl, 23.4; MgSO₄·7H₂O, 24.6; KCl, 1.5; and CaCl₂·H₂O, 2.9.

2. YEA medium consists of BMA supplemented with 0.5% (w/v) yeast extract.

ies suggest the species is a common inhabitant of waters of the Mediterranean Sea (Ortigosa et al., 1994) and is probably common in subtropical and tropical waters, owing to its mesophilic temperature requirement. Many *Alteromonas* strains have been isolated from tropical oceanic regions. The type strain of *A. macleodii* was isolated from seawater in the Hawaiian archipelago area (Baumann et al., 1972). "*A. infernus*" and "*A. macleodii* subsp. *fijiensis*" were isolated from a field of hydrothermal vents populated by dense stands of the tubeworm *Riftia pachyptila* where the water temperature was 9–18°C (Raguénès et al., 1996, 1997b).

ENRICHMENT AND ISOLATION PROCEDURES

Alteromonas macleodii strains can be isolated directly from seawater. After filtering seawater samples through membrane filters having a pore diameter of 0.22 or 0.45 µm, the filter is placed on BMA agar containing 0.2% (w/v) lactose. After several days of incubation at approximately 25°C, colonies are transferred and purified on the same medium (Gauthier and Breittmayer, 1992). "*Alteromonas infernus*" can be isolated from hydrothermal vent fluid by enrichment in marine 2216 broth buffered with 50 mM MOPS buffer at pH 7.5, with incubation at 25°C for 2 d. Isolation from the enrichment broths then proceeds by serially diluting the enrichment broth onto marine 2216 agar (Raguénès et al., 1997b).

MAINTENANCE PROCEDURES

Alteromonas strains can be maintained at approximately 15°C on marine agar slants and transferred on a monthly basis. Storage at 15°C is preferred, because survival at 4°C seems to be poor. Cryopreservation and storage in liquid nitrogen using glycerol as a cryoprotectant or lyophilization is effective for long-term storage.

DIFFERENTIATION OF THE GENUS *ALTEROMONAS* FROM OTHER GENERA

Differentiation of the *Alteromonas* species from other marine genera appears to be problematic. The most phenotypically similar genus, *Pseudoalteromonas*, is chemotaxonomically and metabolically very similar. Both genera possess similar fatty acid profiles (Svetashev et al., 1995), quinones (Akagawa-Matsushita et al., 1992b), and polyamine profiles, although *Pseudoalteromonas* species lack 2-hydroxyputrescine and 5-hydroxy spermidine (Hamana, 1997). The major traits distinguishing these genera are growth at 4°C and utilization of a variety of carbon compounds (see Key to Some Genera of the Family *Alteromonadaceae* and Table BXII.γ.135 of the chapter describing the family). Beyond these limited phenotypic criteria, only 16S rRNA sequencing can provide a definitive identification. *Alteromonas* differs from the marine genera *Marinobacterium*, *Microbulbifer*, *Marinobacter*, and *Halomonas* primarily by possessing lower mol% G + C values for the DNA. In addition, *Alteromonas* can be distinguished from the genera *Oceanospirillum* and *Marinomonas* based on cellular morphology.

TAXONOMIC COMMENTS

In the last decade or so, following rRNA–DNA hybridization and 5S and 16S rRNA sequence analyses, the genus *Alteromonas* has undergone a reclassification of most of its species and has been reduced to a single validly described species, *A. macleodii*. A 1983 study using rRNA–DNA hybridization was the first to indicate that *A. macleodii* should be classified separately from other species that were originally placed in *Alteromonas* (van Landschoot and De Ley, 1983). The 23S rRNA–DNA hybridization values range from 68.5–71.5 $T_{m(e)}$ °C, which indicate *A. macleodii* is genotypically distinct at the genus level. Subsequent phylogenetic analysis of 16S rRNA sequences has also demonstrated that *A. macleodii* is distinct from the other *Alteromonas* species (Kita-Tsukamoto et al., 1993; Gauthier et al., 1995a). This resulted in nomenclatural restructuring of the genus *Alteromonas*, with *Alteromonas macleodii* being retained as the sole species of the genus because it was the type species of the genus (as stipulated by rule 39b of the Nomenclatural Code (Lapage et al., 1992). Former *Alteromonas* species having a predominantly straight rod morphology, including pigmented and nonpigmented strains, have been classified in the genus *Pseudoalteromonas* (Gauthier et al., 1995a). The bioluminescent species *Alteromonas hanedai* (Jensen et al., 1980) and the oyster species *Alteromonas colwelliana* (Weiner et al., 1988) were renamed *Shewanella hanedai* (MacDonell and Colwell, 1985) and *Shewanella colwelliana* (Coyne et al., 1989), respectively. *Alteromonas vaga* and *Alteromonas communis* have been found to form a distinct branch in the vicinity of the genus *Pseudomonas* on the basis of rRNA–DNA homology (van Landschoot and De Ley, 1983) and 16S rRNA sequences (Gauthier et al., 1992; Kita-Tsukamoto et al., 1993) and have been renamed *Marinomonas vaga* and *Marinomonas communis* (van Landschoot and De Ley, 1983). Methanol and methylamine-utilizing marine strains initially described as "*Alteromonas thalassomethanica*" (Yamamoto et al., 1980) have been reclassified as *Methylophaga thalassica* (Janvier et al., 1985; Urakami and Komagata, 1987).

Overall, the genus *Alteromonas* is most closely related to the genus *Pseudoalteromonas* and to various pigmented psychrophilic isolates from Antarctic sea ice (Bowman et al., 1997a). DNA–DNA hybridization studies indicate *A. macleodii* shares no significant levels of hybridization with phenotypically similar, nonpigmented species of the genus *Pseudoalteromonas* (Akagawa-Matsushita et al., 1993).

Isolates forming unusual polysaccharides ("*A. macleodii* subsp. *fijiensis*" and "*A. infernus*") have been shown to be closely related to *A. macleodii* on the basis of 16S rRNA sequences, forming a tight cluster (sequence divergence ~2%) in the *Gammaproteobacteria* (Raguénès et al., 1996, 1997b). A significant level of DNA hybridization (48%) occurs between "*A. infernus*" and the type strain of *A. macleodii* (ATCC 27126), establishing that "*A. infernus*" is a distinct but closely related genospecies (Raguénès et al., 1997b). The names of these organisms have not yet been validated.

List of species of the genus *Alteromonas*

1. ***Alteromonas macleodii*** Baumann, Baumann, Mandel and Allen 1972, 418^{AL}
mac.leod' i.i. M.L. gen. n. *macleodii* of MacLeod, named after R.A. MacLeod, a Canadian microbiologist.

Cells are straight rods, about 1 × 2–3 µm. Motile by a single polar flagellum.

Colonies on marine agar are off-white, smooth, circular, convex, with an entire edge, and butyrous in consistency.

Physiological and nutritional traits are presented in Table BXII.γ.136.

Isolated from seawater.

The mol% G + C of the DNA is: 45–46 (T_m).BNL-1

Type strain: 107, ATCC 27126, DSM 6062, IAM 12920, NCMB 1963.

GenBank accession number (16S rRNA): X82145.

Species Incertae Sedis

1. “**Alteromonas infernus**” Raguénès, Peres, Ruimy, Pignet, Christen, Loaëc, Rougeaux, Barbier and Guezennec 1997b, 428.

in.fer' nus. Gr. n. *infernus* from infernal regions, deep-sea hydrothermal vents.

Cells are straight rods, $0.6\text{--}0.8 \times 1.4\text{--}2.0\ \mu\text{m}$. Encapsulated and motile by a single polar flagellum. Poly- β -hydroxybutyrate is accumulated. Strictly aerobic chemoheterotroph. Oxidase and catalase positive. Does not denitrify. Grows between 20–35°C, with no growth at 4°C or 45°C. Na^+ ions are required for growth. Does not require growth factors. Hydrolyzes starch, esculin, and gelatin. Positive for β -galactosidase, alkaline phosphatase, C_8 esterase, and leucine arylamidase. Acid phosphatase and C_4 esterase activity is weak.

Utilizes the following carbon substrates: D-xylose, D-fructose, D-galactose, D-gentiobiose, D-glucose, D-turanose, cel-

lobiose, lactose, maltose, melibiose, sucrose, amygdalin, esculin, salicin, D-mannitol, and D-gluconate. The following carbon substrates are not utilized: D-melezitose, trehalose, arbutin, N-acetylglucosamine, and glycogen.

Isolated from seawater from a deep-sea hydrothermal field at 2000 m depth in a rift system of the Guaymas basin.

The mol% G + C of the DNA is: 48.1 (T_m).

Deposited strain: GY785, CNCM I-1628.

2. “**Alteromonas macleodii subsp. fijiensis**” Raguénès, Pignet, Gauthier, Peres, Rougeaux, Christen, Barbier and Guezennec 1996, 72.

Phenotypic characteristics are similar to those for *Alteromonas macleodii* ATCC 27126; however, this subspecies can be distinguished from *Alteromonas macleodii* ATCC 27126 in that it is able to utilize D-ribose, α -methyl-D-glucoside, and raffinose, but not D-xylose, mannitol, N-acetylglucosamine or melezitose.

Isolated from seawater from a deep-sea hydrothermal field near Fiji, Pacific Ocean.

The mol% G + C of the DNA is: 47.6 (T_m).

Deposited strain: ST716, CNCM I-1627.

Genus II. **Alishewanella** Fonnesbech Vogel, Venkateswaran, Christensen, Falsen, Christiansen and Gram 2000, 1140^{VP}

THE EDITORIAL BOARD

A.li.she.wa.nel' la. L. pron. *alius* the other; M.L. fem. n. *Alishewanella* the other *Shewanella*.

Gram-negative, facultatively anaerobic, nonmotile rods. Electron acceptors are O_2 , nitrate, thiosulphate, and TMAO. **Nonfermentative.** One polar flagellum. Catalase and oxidase positive. **Requires NaCl.** Hydrolyzes esculin and gelatin. Does not produce arginine dihydrolase, β -galactosidase, urease, indole, or H_2S .

The mol% G + C of the DNA is: 51.

Type species: **Alishewanella fetalis** Fonnesbech Vogel, Venkateswaran, Christensen, Falsen, Christiansen and Gram 2000, 1141.

FURTHER DESCRIPTIVE INFORMATION

Alishewanella fetalis R2422019 was isolated from a human fetus at autopsy and originally thought to be related to *Shewanella*.

Analysis of 16S rDNA sequences showed that strain R2422019 was related to but distinct from members of the genera *Shewanella*, *Aeromonas*, *Vibrio*, *Photobacterium*, *Erwinia*, *Klebsiella*, and *Shigella*.

DIFFERENTIATION OF THE GENUS *ALISHEWANELLA* FROM OTHER GENERA

Fonnesbech Vogel et al. (2000) provide tables of characteristics that distinguish *Alishewanella fetalis* R2422019 from *Shewanella putrefaciens* and *S. algae* (potential human pathogens having phenotypes similar to R2422019) as well as from the genera *Shewanella*, *Aeromonas*, *Vibrio*, *Photobacterium*, *Erwinia*, *Klebsiella*, and *Shigella*.

List of species of the genus *Alishewanella*

1. **Alishewanella fetalis** Fonnesbech Vogel, Venkateswaran, Christensen, Falsen, Christiansen and Gram 2000, 1141^{VP}
fe.ta' lis. L. adj. *fetalis* pertaining to the fetus, from which the organism was isolated.

Description as for the genus with the following addi-

tional characteristics. Cells $0.5\text{--}1.0 \times 2.0\ \mu\text{m}$. Grows in medium containing up to 8% in NaCl.

The mol% G + C of the DNA is: 51 (HPLC).

Type strain: R2422019, ATCC BAA-284, CCUG 30811, CIP 106648.

GenBank accession number (16S rRNA): AF144407.

Genus III. **Colwellia** Deming, Somers, Straube, Swartz and MacDonell 1988b, 328^{VP} (Effective publication: Deming, Somers, Straube, Swartz and MacDonell 1988a, 159)

JODY W. DEMING AND KAREN JUNGE

Col.wel' li.a. M.L. ending *-ia*; M.L. fem. n. *Colwellia* of Colwell, named in honor of the American microbiologist Professor Rita R. Colwell.

Curved or straight rods $0.4\text{--}1.0 \times 1.5\text{--}5.0\ \mu\text{m}$. Generally motile by polar flagella. Gram negative. Asporogenous. Colonies are off-white, convex, and mucoid, except for the type strain, *Colwellia*

psychrerythraea, which produces red pigment. **Facultatively anaerobic. Chemoorganotrophic.** Able to grow by fermentation of a suitable carbohydrate as indicated by acid (but no gas) produc-

tion. Generally able to utilize amino acids and organic acids, but not many of the sugars or alcohols. Capable of **nitrate reduction** but not iron or trimethylamine *N*-oxide reduction or indole or H₂S production. Generally **chitinolytic**, able to degrade starch, and positive for oxidase, catalase, and alkaline phosphatase. Stimulated by, but not dependent on, growth factors in yeast extract and vitamin solutions. Require **sea salts**, with no growth occurring below 25% sea-salt concentration. **Psychrophilic**. Able to grow on solid media in the temperature range of 0° to <20°C (optimum usually <15°C), with slightly higher maximal and lower minimal temperatures in liquid media. **Sometimes barophilic**, requiring high hydrostatic pressure for growth. Produce docosaheptaenoic acid (C_{22:6 ω3} or DHA) and other **polyunsaturated fatty acids** to help maintain homeoviscosity of cellular membranes under low temperature and high pressure. Cultured strains derived from **permanently cold**, and often deep, marine habitats.

The mol% G + C of the DNA is: 35–46.

Type species: Colwellia psychrerythraea (ex D' Aoust and Kushner 1972) Deming, Somers, Straube, Swartz and MacDonell 1988b, 328 (Effective publication: Deming, Somers, Straube, Swartz and MacDonell 1988a, 159 ("Vibrio psychrerythrus" D' Aoust and Kushner 1972, 342.)

FURTHER DESCRIPTIVE INFORMATION

Cell morphology varies from spherical to curved to straight rod-shaped cells with occasional short filaments. The fine structure appears typical of asporogenous Gram-negative bacteria, with no obviously unique structural features, even in the obligate barophiles (see electron micrographs in Deming et al., 1988a, and Deming and Baross, 2000). *C. rossensis*, however, is unique among *Colwellia* species for its ability to form gas vesicles (Gosink and Staley, 1995). Potentially unique cell-membrane composition is implied by the documented production of polyunsaturated fatty acids.

The most distinctive features of the genus *Colwellia* derive from its habitation of permanently cold marine environments. All characterized species are cold adapted, if not strictly psychrophilic, as defined by Morita (1975) (T_{opt} for growth of 15°C or less, T_{max} of ≤20°C). *Colwellia* (along with *Moritella*) thus appears to be unusual among all genera for encompassing exclusively cold-adapted organisms. Cardinal growth temperatures of individual species tend to be several degrees higher in liquid media than in solid media, causing at least one of the type species, *C. psychrotropica*, to be considered psychrotolerant (T_{max} of 25°C in liquid) (Bowman et al., 1998a). *C. psychrerythraea* strain 34H grows at −5°C in liquid media (Huston, 2003) and swims at subzero temperatures to −10°C (Junge et al., 2003). Two of the known strains, *C. hadaliensis* (Deming et al., 1988a) and *Colwellia* strain MT41 (DeLong et al., 1997), are not only strictly psychrophilic but also obligately barophilic, derived from the deepest portions of the cold Atlantic and Pacific Oceans, respectively. The production of polyunsaturated fatty acids (PUFAs), a feature common to members of the genus so analyzed, is attributed to the need to maintain membrane fluidity for metabolism and growth at the constantly cold temperatures (<4°C) and often high pressures of their native habitats, e.g., polar waters, sea ice, and hadal depths in the cold ocean. Their common growth requirement for sea salts and preference for full-strength sea salt concentration also reflect their native marine habitats.

The type species, *C. psychrerythraea*, is unique in the genus for its production of red pigment during colony formation. All other

species produce chalk-white, off-white, or colorless colonies, typically of convex, circular, raised, and mucoid morphology, when grown on solid media. The obligate barophiles do not grow on standard agar plates at atmospheric pressure, but rather form spherical off-white colonies when grown within a solid medium under high hydrostatic pressure. *C. psychrotropica* produces a diffusible brown pigment during L-tyrosine hydrolysis in liquid media.

All *Colwellia* species are chemoorganotrophic facultative anaerobes able to grow by fermentation of a suitable carbohydrate, as indicated by acid production in Leifson oxidation-fermentation medium. In general, no gas is detected during fermentation. Carbohydrate oxidation and fermentation vary between species, as does the ability to hydrolyze various substrates (Tables BXII.γ.137 and BXII.γ.138). Utilization of amino acids and organic acids is favored over sugars and alcohols. Most species are able to utilize L-glutamate, L-proline, acetate, butyrate, fumarate, pyruvate, succinate, N-acetylglucosamine, oxalacetate, DL-lactate, valerate, caproate, and γ-aminobutyrate, but not lactose, maltose, D-melibiose, L-raffinose, sucrose, trehalose, D-xylose, D-adonitol, D-arabitol, *m*-inositol, D-mannitol, D-sorbitol, L-threonine, L-valine, D-gluconate, D-glucuronate, isovalerate, nonanoate, pimelate, saccharate, α-glycerophosphate, or putrescine. Individual species are distinguished by specific metabolic or hydrolytic traits: e.g., only *C. demingiae* utilizes L-serine; only *C. hornerae* produces acid from L-rhamnose; only *C. rossensis* produces acid from D-galactose, L-arabinose, D-fructose, and D-gluconate; and only *C. psychrotropica* degrades uric acid. Some strains produce extracellular hydrolytic enzymes with unique properties; e.g., *Colwellia* sp. strain 34H produces the most cold-adapted extracellular proteases yet discovered (Huston et al., 2000).

Only *C. hornerae* and *C. maris* are susceptible to vibriostatic agent O/129. Antigenic structure and other antibiotic or drug sensitivities, along with mutants and plasmids, remain to be documented or developed, although *Colwellia*-specific phages have been reported (Borriess et al., 2003). None of the *Colwellia* species are believed to be pathogenic, in keeping with their growth requirement for cold temperatures.

In direct phylogenetic analyses of environmental samples, 16S rRNA sequences specific to *Colwellia* have been detected in cold marine samples from the deep sea (e.g., Moyer et al., 1995), in keeping with what is known about cultured species, but also in coastal waters at temperate latitudes (DeLong et al., 1997). Absent any culturing information, however, the temperature adaptations of such organisms remain unknown.

ENRICHMENT AND ISOLATION PROCEDURES

Members of the genus *Colwellia* are isolated from cold marine samples, using liquid enrichment media containing complex carbon sources, such as peptone, and full-strength sea-salt concentration. The most common medium used is 2216 marine broth (Difco). An initial enrichment period of days to weeks in liquid media, before transfer and cultivation on agar plates, appears to favor the recovery of these psychrophilic strains, although *C. maris* was first enriched on agar medium (Takada et al., 1979). Initial colony formation may require up to 2 months of incubation time at cold temperature. Attention to low-temperature (0° to <4°C) storage of the original sample and of subsequent enrichment incubations is important.

Sea-ice samples are first melted in seawater or brine solutions prior to enrichment procedures to avoid hypotonic shock to salt-requiring strains. Deep-sea samples can be recovered at atmo-

TABLE BXII.γ.137. Characteristics differentiating the species of the genus *Colwellia*^a

Characteristic	1. <i>C. psychrerythraea</i>	2. <i>C. demingiae</i>	3. <i>C. hadaliensis</i>	4. <i>C. hornerae</i>	5. <i>C. maris</i>	6. <i>C. psychrotropica</i>	7. <i>C. rossensis</i>
Growth at 25°C	—	—	—	—	—	+	—
Barophilic growth	—	—	+	—	—	—	—
Prodigiosin-like pigments	d ^b	—	—	—	—	—	—
Motility	+	+	+	+	+	+	—
Gas vesicles	—	—	—	—	—	—	+
Susceptible to vibriostatic agent O/129	—	—	—	+	+	—	—
<i>Hydrolysis of:</i>							
Chitin	+	+	+	—	—	+	+
Starch	+	+	—	+	—	—	+
Uric acid	—	—	—	—	—	+	—
Urea	+	—	—	—	—	+	+
Tween 80	+	—	—	+	+	+	NG
Gelatin	+	—	—	—	+	—	—
Casein	+	+	—	+	—	+	—
DNA	—	—	—	—	+	—	—
<i>Production of acid from:</i>							
D-Glucose	+	—	+	—	+	—	+
Maltose	+	—	—	—	—	—	—
Glycerol	—	—	—	+	—	—	+
Mol% G + C of DNA	35–40	37	46	39	39	42	38

^aSymbols: see standard definitions; blank space, no data available; NG, no growth occurred on test medium.

^bType strain is positive.

spheric pressure, but barophilic *Colwellia* strains are subsequently enriched and isolated under hydrostatic pressures simulating those encountered *in situ*. Methodologies for cultivation of bacteria, in either liquid or solid media, under high hydrostatic pressure are described in detail by Deming (2002).

MAINTENANCE PROCEDURES

Colwellia spp. are routinely maintained by periodic subcultivation on 2216 marine agar or broth (Difco) at low temperatures: <4°C for strict psychrophiles, but also up to 10°C. Obligate barophiles must be subcultivated at low temperatures under high hydrostatic pressure (Deming, 2002). Cryopreservation is the preferred form of long-term storage, freezing pelleted cultures in glycerol (15% glycerol is effective at –20°C to –80°C; 30% glycerol at –80°C) or using lyophilization. Cryopreservation is advised even for short-term storage of the type strain, *C. psychrerythraea*, since it is not recovered reliably from plates, and for obligate barophiles, which have been problematic to store (the availability of recoverable *C. hadaliensis* is currently uncertain). All *Colwellia* cultures should be shipped frozen and packed on ice to maximize recoverability.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Special characteristics of *Colwellia* include growth requirements at low temperatures and sometimes at high pressures and the production of PUFA. Standard incubation devices, from refrigerators to water baths, are used to achieve and maintain low temperatures, but special equipment is required for testing growth and other parameters under high pressure, as described by Deming (2002). Many standard tests for physiological and nutritional features have not yet been developed for high-pressure applications, making molecular assessments essential to determining phylogenetic status. Analysis of 16S rRNA sequences proceeds normally for *Colwellia* spp., if attention is paid to low-temperature requirements in first preparing the cultures. Whole-cell fatty acid analyses rely on standard GC and GC-MS procedures (Bowman et al., 1998a).

DIFFERENTIATION OF THE GENUS *COLWELLIA* FROM OTHER GENERA

The genus *Colwellia* is differentiated from other closely related genera by the combination of being cold adapted, facultatively anaerobic, and able to produce chitinase. Nearest neighbors at the genus level, *Idiomarina*, *Alteromonas*, and *Pseudoalteromonas*, are not usually adapted to growth at low temperatures, are strictly aerobic, and do not typically produce chitinase (some species of *Alteromonas* and *Pseudoalteromonas* do produce the enzyme). The production of PUFA also appears to be an important distinguishing trait for *Colwellia*. Use of 16S rRNA sequence data, however, remains the essential means to distinguish *Colwellia* definitively from other genera. The main features for distinguishing one *Colwellia* species from another are provided in Table BXII.γ.138.

TAXONOMIC COMMENTS

A phylogenetic treatment based on 16S rRNA sequence data from organisms in culture indicates that the closest relatives to members of the genus *Colwellia* are species of the newly described genus *Idiomarina* (Ivanova et al., 2000b), as represented by the deep-sea isolate *I. abyssalis* in Fig. BXII.γ.152. Sequence similarities between *I. abyssalis* and *Colwellia* spp. range from 85.2% (*Colwellia* sp. strain MT41) to 89.6% (type species *C. psychrerythraea*). *Colwellia* also has close phylogenetic affinities, as evidenced by similar degrees of sequence similarity, to members of the genera *Alteromonas* and *Pseudoalteromonas*, as described previously by Bowman et al. (1998a) and depicted in Fig. BXII.γ.152.

Analysis of 16S rRNA sequences for *Colwellia* species in culture (Fig. BXII.γ.152) indicates sequence similarities that range from 92.4% to 98.7%. The low end of the range derives from a comparison of the deep-sea obligate barophile, *Colwellia* sp. strain MT41, with sea-ice isolate *C. demingiae*. Strain MT41 in general has the lowest sequence similarity values to the other described *Colwellia* species (92.4–94.6%), with the exception of *C. maris* (96.3%). Only 5S (not 16S) rRNA sequence data are available for the other obligate barophile, *C. hadaliensis*, precluding comparisons of the two with each other or with other species. A

TABLE BXII.γ.138. Other characteristics of the species of the genus *Cokwellia*^a

Characteristic	1. <i>C. psycherythraea</i>	2. <i>C. demingiae</i>	3. <i>C. hadaliensis</i>	4. <i>C. hornerae</i>	5. <i>C. maris</i>	6. <i>C. psychrotropica</i>	7. <i>C. rossensis</i>
Temperature range for growth in liquid media, °C	0–19	0–18	2–10	0–23	0–22	0–25	0–15
Single polar flagellum	+	—	+	—	+	—	+
Gas vesicles	—	—	—	—	—	—	—
Tolerance of 200% sea salts	d ^b	—	—	+	—	+	—
Growth on 0–25% sea salts	—	—	—	—	— ^c	—	—
Growth on 400% sea salts	—	—	—	—	—	—	—
Growth on NaCl	2.75%	—	—	—	3–6.5%	—	—
Susceptibility to vibriostatic agent O/129	d ^b	—	—	+	+	—	—
Growth factor requirement (yeast extract)	—	—	—	+	—	—	—
Catalase, nitrate reduction	+	+	—	+	+	+	+
Oxidase	+	+	+	+	+	+	+
Alkaline phosphatase	+	+	+	+	—	+	+
Acid phosphatase, trypsin	—	—	+	—	—	—	—
Fe(III) reduction, trimethylamine N-oxide reduction	—	—	—	—	—	—	—
Indole production	—	—	—	—	—	—	—
H ₂ S production	—	—	—	—	—	—	—
<i>Hydrolysis of:</i>							
Alginic acid	—	—	—	—	—	—	—
Casein	+	+	—	+	—	+	—
Chitin	+ ^e	+ ^e	+	—	—	+ ^e	+ ^e
Dextran	—	—	—	—	—	—	—
DNA	—	—	—	—	+	—	—
Esculin	+	+	—	+	—	—	NG
Gelatin	d ^b	—	—	—	+	—	—
ONPG	—	—	—	—	—	—	—
Starch	+	+	—	+	—	—	+
Tween 80	+	—	—	+	+	+	NG
Tyrosine	d ^d	—	—	—	—	+	—
Urea	d ^b	—	—	—	—	+	+
Uric acid	—	—	—	—	—	+	—
Arginine dihydrolase, lysine decarboxylase, ornithine decarboxylase	—	—	—	—	—	—	—
<i>Production of acid from:</i>							
N-Acetylglucosamine	+ ^e	+ ^e	—	—	—	+ ^e	+ ^e
L-Arabinose, D-fructose, D-mannose, melibiose, sucrose	—	—	—	—	—	—	—
D-Arabinose, inositol, mannitol, sorbitol, trehalose, xylose, lactose	—	—	—	—	—	—	—
Cellobiose	d ^b	—	—	—	—	—	—
D-Galactose	—	—	—	—	—	—	+
D-Glucose	+ ^e	—	+	—	—	—	+ ^e
Maltose	+ ^e	—	—	—	—	—	—
L-Rhamnose	—	—	—	+ ^e	—	—	—
Glycerol	—	—	—	+	—	—	+
<i>Utilization of:</i>							
L-Proline, acetate, butyrate, fumarate, L-glutamate, succinate, pyruvate	+	+	—	+	—	+	+
Oxalacetate	+	+	—	+	—	+	—
Valerate, caproate, γ-aminobutyrate	—	+	—	+	—	+	+
N-Acetylglucosamine	+	+	—	—	—	+	+
3-Hydroxybutyrate	+	—	—	—	—	+	+
Citrate	—	+	—	+	—	—	+
Glycogen	+	d ^d	—	—	—	—	+
D,L-Lactate	d ^b	+	—	+	—	+	—
D-Glucose	d ^b	—	—	—	+	—	+
Propionate	d ^d	+	—	+	—	—	—
L-Asparagine	—	+	—	—	—	+	+
L-Alanine, L-aspartate	—	d ^b	—	—	—	+	+
Hydroxy-L-proline	—	d ^b	—	+	—	—	—
L-Phenylalanine	—	d ^b	—	—	—	—	—
L-Serine	—	+	—	—	—	—	—
Azelate	—	d ^b	—	+	—	—	—
Caprylate	—	d ^b	—	—	—	—	—
Glutarate	—	+	—	+	—	—	—
Heptanoate	—	db	—	+	—	—	—
Isobutyrate	—	db	—	—	—	+	—
Malonate	—	db	—	—	—	—	+
D-Mannose, D-raffinose	—	—	—	—	—	—	—
D-Melibiose	—	—	—	—	—	—	—
Glycerol	—	—	—	+	—	—	+
L-Malate	—	—	—	—	—	+	+

(continued)

TABLE BXII.γ.138. (cont.)

Characteristic	1. <i>C. psychrerythraea</i>	2. <i>C. demingiae</i>	3. <i>C. hadaliensis</i>	4. <i>C. hornerae</i>	5. <i>C. maris</i>	6. <i>C. psychrotropica</i>	7. <i>C. rossensis</i>
2-Oxoglutarate	—	—		—		+	—
L-Arabinose, D-fructose, D-gluconate	—	—		—	—	—	+
L-Raffinose, trehalose, D-adonitol, D-arabitol, <i>m</i> -inositol, D-mannitol, D-sorbitol, L-threonine, L-valine, aconitate, adipate, D-gluconate, D-glucuronate, nonanoate, saccharate, α-glycerophosphate, putrescine, isovalerate, pimelate	—	—		—		—	—
Lactose, maltose, sucrose, D-xylose	—	—		—	—	—	—
<i>Fatty acid composition (% of total):</i>							
C _{14:0}	5.1–7.8	7.6–8.0		3		0.8	4.6
C _{14:1 ω7c}	5.1–7.3	9.1–9.3		2.8		2	2.8
C _{15:1 ω8c}	0–2.3	1.9–2.6		20.3		4.2	4.1
C _{15:1 ω6c}	0–0.4	—		1.1		0.1	—
C _{15:0}	1.7–11.0	0.9–1.4		14.3		2.7	2.9
C _{16:0 iso}	0–0.2	—		10.3		—	—
C _{16:1 ω9c}	6.2–8.8	9.5–11.8		2	20	—	1.8
C _{16:1 ω7c}	31.4–36.3	37.5–37.8		15.4	6	56.8	43.4
C _{16:1 ω9t}					20		
C _{16:0}	26.8–33.2	21.9–23.6		13.5	18	21.9	27.1
C _{17:1 ω8c}	0–1.3	tr ^c		5.6		4.5	0.5
C _{17:1 ω6c}	0–0.9	—		1.9		0.9	—
C _{17:0}	0–1.3	tr		2.5	5	1.5	0.1
C _{18:1 ω11c}					6		
C _{18:1 ω9c}	0–1.7	0.2		1.4		0.3	0.8
C _{18:1 ω7c}	0.3–2.1	1.3–1.4		—		1.9	4.2
C _{18:0}	0.1–2.4	0.2–0.5		2		0.4	tr
C _{20:5 ω3}	0–1.5	—		—		0.1	tr
C _{22:6 ω3}	5.5–8.0	1.7–2.2		2.1		0.7	6
Isoprenoid quinone					ubiquinone-8 (Q-8)		

^aSymbols: see standard definitions; blank space, no data available; NG, no growth occurred on test medium; tr, trace fatty acid component making up 0.1% or less of total fatty acid content.

^bReaction differs among strains; type strain is positive.

^cNo growth occurs without NaCl.

^dReaction differs among strains; type strain is negative.

^eAcid (but no gas) was formed from this substrates in Leifson oxidation/fermentation medium.

possible close relationship between phylogeny and obligate barophily within the genus thus remains elusive (DeLong et al., 1997). The high end of the sequence similarity range for *Colwellia* species (98.7%) derives from comparison of the type strain *C. psychrerythraea* with *Colwellia* sp. strain 34H, suggesting the latter is also *C. psychrerythraea*. Confirmation of this suggestion comes from DNA–DNA hybridization tests (Huston, 2003). The availability of the whole genome sequence for strain 34H from The Institute for Genomic Research* thus well represents the genus *Colwellia* and enables a first-order genetic analysis of psychrophily and other traits specific to *Colwellia* or unique to the type species. Existing DNA–DNA hybridization tests show clear separation at the species level, with homologies of 12–35% among the type species (Table BXII.γ.139).

The list of *Colwellia* type species has expanded from the first two, described in 1988, to seven only in the last few years. This expansion reflects renewed ecological and biotechnological interests in cold-adapted organisms and their enzymes (Bowman et al., 1998a; Deming and Baross, 2000, 2002). Characterizations of numerous additional strains and species can be expected in

the near future, including *Colwellia* sp. strain 34H and possibly strain MT41. Current culture collections of psychrophilic bacteria likely contain many undescribed *Colwellia* spp. Recent acquisition of 16S rRNA sequence data from our collection reveals at least three species, awaiting further taxonomic characterization (Junge et al., 2002).

ACKNOWLEDGMENTS

We gratefully acknowledge John Bowman for helpful communications and his substantial contributions to the discovery and characterization of *Colwellia* species.

FURTHER READING

- Bowman, J.P., J.J. Gosink, S.A. McCammon, T.E. Lewis, D.S. Nichols, P.D. Nichols, J.H. Skerratt, J.T. Staley and T.A. McMeekin. 1998. *Colwellia demingiae* sp. nov., *Colwellia hornerae* sp. nov., *Colwellia rossensis* sp. nov., and *Colwellia psychrotropica* sp. nov.: psychrophilic Antarctic species with the ability to synthesize docosahexaenoic acid (22:6ω3). Int. J. Syst. Bacteriol. 48: 1171–1180.
- Deming, J.W., L.K. Somers, W.L. Staube, D.G. Swartz and M.T. MacDonell. 1988. Isolation of an obligately barophilic bacterium and description of a new genus, *Colwellia* gen. nov. Syst. Appl. Microbiol. 10: 152–160.
- Yumoto, I., K. Kawasaki, H. Iwata, H. Matsuyama and H. Okuyama. 1998. Assignment of *Vibrio* sp. strain ABE-1 to *Colwellia maris* sp. nov., a new psychrophilic bacterium. Int. J. Syst. Bacteriol. 48: 1357–1362.

*Editorial Note: At the time this volume went to press, the website for the Institute for Genomic Research was <http://www.tigr.org>.

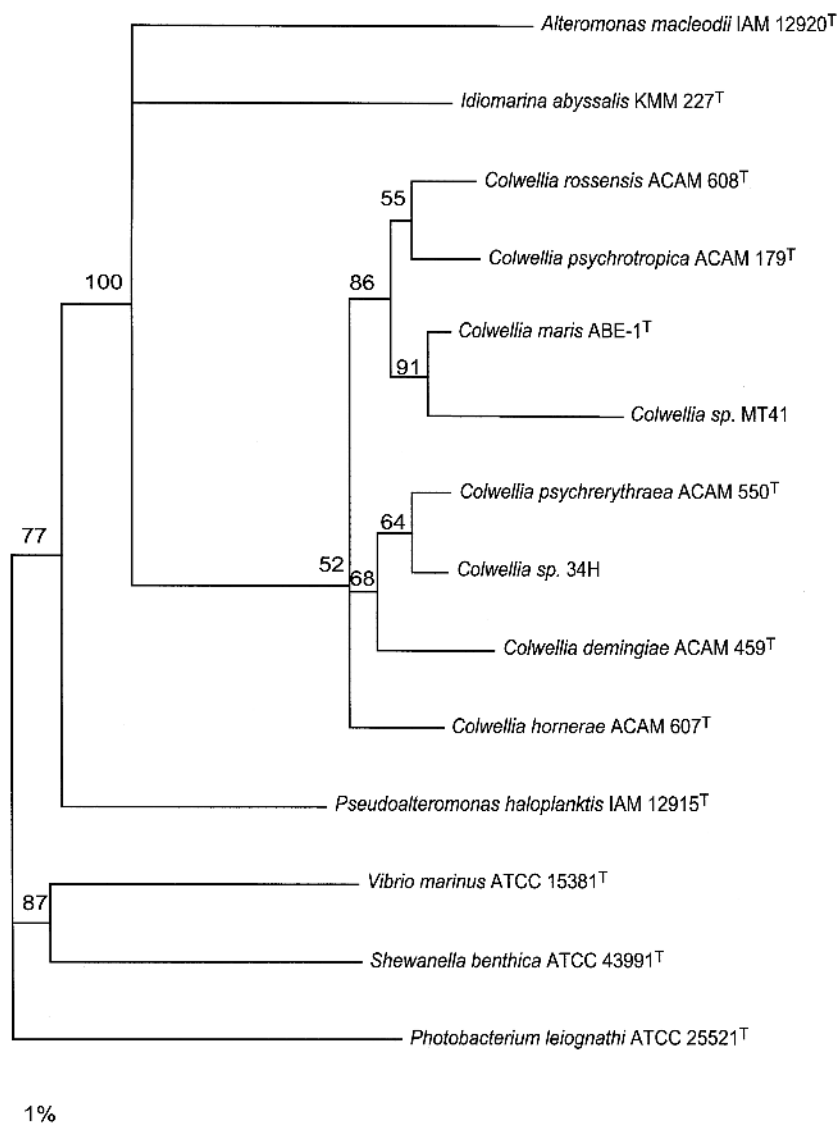


FIGURE BXII.γ.152. Phylogenetic tree based on 16S rRNA sequence comparisons showing the positions of members of the genus *Colwellia* within the *Gammaproteobacteria*. Phylogenetic analysis was conducted with reference sequences obtained from GenBank. Preliminary alignments were made using the Sequence Align program of the Ribosomal Database Project Web site (RDP) and requesting that common gaps be preserved. Complete sequence alignments were prepared with the sequence alignment editor SeqApp (D.G. Gilbert, 11 July 1994, posting date, Seq-App program, version 1.9a169, <http://ftp.bio.indiana.edu/molbio/seqapp>) and by manual comparison to secondary structures provided by RDP. For further phylogenetic analysis, aligned sequences corresponding to *E. coli* positions 30–1452 were used. Phylogenetic relationships were inferred using the PAUP program (4.0b8, D.L. Swofford, Smithsonian Institution, 2001). Tree topology was inferred from neighbor-joining analysis determined with the Kimura two-parameter model. *Photobacterium leiognathi* was used as an outgroup. Results from neighbor-joining bootstrap analysis (1000 replicates) are indicated by numbers (as percentages) near each node. Only nodes supported in greater than 50% of the bootstrap replicates are indicated. The scale bar corresponds to 1 fixed mutation per 100 sequence positions. The following 16S rDNA sequences (with their GenBank accession numbers in parentheses) were obtained for the phylogenetic analysis: *Alteromonas macleodii* (X82145), *Colwellia demingiae* (U85845), *Colwellia hornerae* (U85847), *Colwellia maris* (AB002630), *Colwellia psychrerythraea* (AF001375), *Colwellia psychrotropica* (U85846), *Colwellia rossensis* (U14581), *Colwellia* sp. MT41 (U91595), *Colwellia* sp. 34H (AF396670), *Idiomarina abyssalis* (AF05270), *Photobacterium leiognathi* (X74686), *Pseudoalteromonas haloplanktis* (X67024), *Shewanella benthica* (U91594), and *Vibrio marinus* (X82142).

List of species of the genus *Colwellia*

1. *Colwellia psychrerythraea* (ex D'Aoust and Kushner 1972)
Deming, Somers, Straube, Swartz and MacDonell 1988b,
328^{VP} (Effective publication: Deming, Somers, Straube,

Swartz and MacDonell 1988a, 159 ("Vibrio psychroerythrus"
D'Aoust and Kushner 1972, 342.)
psy.chre.ryth'rae.a. Gr. adj. *psychros* cold; L. adj. *erythraeus*, -a,
-um red.

TABLE BXII.γ.139. DNA–DNA homologies among the species of the genus *Colwellia*^a

Percentage DNA–DNA hybridization	1. <i>C. psychrerythraea</i>	2. <i>C. demingiae</i>	3. <i>C. hadaliensis</i>	4. <i>C. hornerae</i>	5. <i>C. maris</i>	6. <i>C. psychrotropica</i>	7. <i>C. rossensis</i>
1. <i>C. psychrerythraea</i>	100	35		12	22	12	13
2. <i>C. demingiae</i>	35	100		21			21
3. <i>C. hadaliensis</i>							
4. <i>C. hornerae</i>	12	21		100		18	22
5. <i>C. maris</i>	22						
6. <i>C. psychrotropica</i>	12			18		100	21
7. <i>C. rossensis</i>	13	21		22		24	100

^aBlank space, not determined.

The characteristics are as described for the genus and as listed in Tables BXII.γ.137 and BXII.γ.138, with the following additional information. Cell length, 2.5–3.5 μm; width, 0.5 μm. Gas vesicles not formed. Motile by means of a single, long polar flagellum. Colonies are white at first appearance (7–8 d) becoming red within 10–12 d, opaque, raised, 3–5 mm in diameter, with entire edge. Psychrophilic. Growth in liquid media occurs at 0–19°C, but not at temperatures higher than 20°C when cell lysis occurs. Requires 2.75% NaCl for growth. Growth factors are not required. Resistant to vibriostatic agent O/129 (100 μg/ml), although growth is slowed by it. Acid (with no gas) is formed fermentatively and oxidatively from glucose and maltose, chitin, and *N*-acetylglucosamine. Casein, gelatin, esculin, chitin, starch, urea, and Tween 80 are hydrolyzed. The following compounds are utilized as sole sources of carbon and energy: D-glucose, glycogen, *N*-acetylglucosamine, 3-hydroxybutyrate, acetate, butyrate, succinate, fumarate, pyruvate, DL-lactate, oxalacetate, L-glutamate, and L-proline. Some strains can also utilize propionate. Isolated from flounder eggs, Norway, and fast sea ice, Antarctica.

The mol% G + C of the DNA is: 35–40 (*T_m*).

Type strain: ATCC 27364, DSM 8813.

GenBank accession number (16S rRNA): AF001375.

2. ***Colwellia demingiae*** Bowman, Gosink, McCammon, Lewis, Nichols, Nichols, Skerrat, Staley, and McMeekin 1998a, 1178^{VP}

de.ming' i.ae. L. adj. *demingiae* in honor of Jody W. Deming, an American microbiologist who has expanded the knowledge of deep-sea bacteria.

Individual cells are straight to curved rods, with spherical and short filamentous cells occasionally occurring. Cell length, 1.5–4.5 μm; width, 0.4–0.6 μm. Gas vesicles not formed. Motile. Colonies are off-white, have a mucoid consistency, raised elevation, and convex circular shape with entire to lobate edges. Psychrophilic. In liquid media *T*_{opt} is about 10–12°C and *T*_{max} is about 18°C. Requires sea salts for growth. Growth factors are not required. Resistant to vibriostatic agent O/129 (100 μg/ml). Acid (with no gas) is formed fermentatively and oxidatively from chitin and *N*-acetylglucosamine. Casein, esculin, chitin, and starch are hydrolyzed. The following compounds are utilized as sole sources of carbon and energy: *N*-acetylglucosamine, acetate, propionate, butyrate, valerate, caproate, succinate, glutarate, citrate, fumarate, pyruvate, DL-lactate, oxalacetate, L-asparagine, L-glutamate, L-proline, L-serine, and γ-amino-

butyrate. Some strains can also utilize glycogen, isobutyrate, heptanoate, caprylate, malonate, azelate, L-alanine, L-aspartate, L-phenylalanine, and hydroxy-L-proline. Isolated from fast sea ice of the Prydz Bay coast, Antarctica.

The mol% G + C of the DNA is: 37 (*T_m*).

Type strain: ACAM 459.

GenBank accession number (16S rRNA): U85845.

3. ***Colwellia hadaliensis*** Deming, Somers, Straube, Swartz and MacDonell 1988b, 328^{VP} (Effective publication: Deming, Somers, Straube, Swartz and MacDonell 1988a, 159.)
ha'dal.i.en'sis. M.L. adj. ending *-iensis*; Eng. adj. *hadal* used in oceanographic terminology; of, or pertaining to, the greatest depths of the ocean; derived from Gr. n. *Hades* god of the underworld.

Individual cells are curved rods. Cell length, 3.0–5.0 μm; width, 0.8 μm. Gas vesicles not formed. Motile by means of polar flagella. Colonies are off-white but form only within solid media under hydrostatic pressure. Psychrophilic. Growth occurs in liquid media at 2–10°C, but not at temperatures of 15°C or higher. Obligately barophilic. Requires hydrostatic pressures of 300–1020 atm for growth at 2°C and 500–1020 atm at 10°C. Cell lysis occurs at <200 atm and 2°C and at <400 atm and 10°C; absolute upper pressure limits for growth and survival unknown. Acid is formed fermentatively and oxidatively from glucose. Chitin is hydrolyzed. Associated with cold (<4°C), hadal (depth of 6000–11,000 m) marine waters and sinking particles.

The mol% G + C of the DNA is: 46 (*T_m*).

Type strain: BNL-1.

Additional Remarks: The 16S rDNA sequence is not available for *C. hadaliensis*, but see Deming et al. (1988a) for 5S rRNA sequence.

4. ***Colwellia hornerae*** Bowman, Gosink, McCammon, Lewis, Nichols, Nichols, Skerrat, Staley and McMeekin 1998a, 1178^{VP}

horn.ner' ae. L. adj. *hornerae* in honor of Rita Horner, an American biologist who pioneered studies on sea-ice microbiota.

Individual cells are straight to curved rods, with spherical cells occasionally occurring. Cell length, 1.5–3.0 μm; width, 0.4–0.8 μm. Gas vesicles not formed. Motile. Colonies are off-white, have a mucoid consistency, raised elevation, and convex circular shape with entire to lobate edges. Psychrophilic. In liquid media *T*_{opt} is about 12°C and *T*_{max} is about 23°C. Requires sea salts for growth. Growth occurs in the

presence of sea salts at three times normal concentration and is sensitive to vibriostatic agent O/129 (100 µg/ml). Growth factors are not required. Acid (with no gas) is formed fermentatively and oxidatively from L-rhamnose. Acid is formed oxidatively from glycerol. Tween 80, casein, esculin, and starch are hydrolyzed. The following compounds are utilized as sole sources of carbon and energy: glycerol, acetate, propionate, butyrate, valerate, caproate, heptanoate, succinate, glutarate, azelate, citrate, fumarate, pyruvate, DL-lactate, oxaloacetate, L-glutamate, L-proline, hydroxy-L-proline, and γ-aminobutyrate. Isolated from fast sea ice of the Prydz Bay coast, Antarctica.

The mol% G + C of the DNA is: 39 (T_m).

Type strain: ACAM 607, CIP 105821.

GenBank accession number (16S rRNA): U85847.

5. **Colwellia maris** Yumoto, Kawasaki, Iwata, Matsuyama and Okuyama 1998, 1361^{VP}

mar'is. L. gen. n. *maris* of the sea.

Individual cells are curved rods. Cell length, 2–4 µm; width, 0.6–1.0 µm. Gas vesicles not formed. Motile by means of a single polar flagellum. Colonies are colorless. Catalase and oxidase reactions are positive. Psychrophilic. Growth occurs in liquid media at 0–22°C, with T_{opt} of 15°C. Growth factors not required. Growth occurs in media supplemented with 3% or 4% NaCl, but not in the absence of NaCl or at a salinity higher than 6.5%. Sensitive to vibriostatic agent O/129 (10 and 150 µg/ml). Acids are produced from D-glucose under aerobic, but not anaerobic, conditions. No acids are produced from L-arabinose, D-fructose, maltose, D-mannose, melibiose, and sucrose either aerobically or anaerobically. Gelatin, DNA and Tweens 20, 40, 60, and 80 are hydrolyzed, but not casein, chitin, starch, or alginic acid. D-glucose is utilized as a sole source of carbon and energy. Does not utilize D-mannose, raffinose, D-xylose, L-arabinose, D-fructose, glycerol, lactose, maltose, melibiose, or sucrose. The major isoprenoid quinone is Q-8. The whole-cell fatty acids contain saturated and monounsaturated fatty acids with 10–18 C atoms; saturated and monounsaturated C₁₆ fatty acids are predominant in cells grown at 15°C. The strain contains a unique trans-unsaturated fatty acid (9-trans-hexadecanoic acid, C_{16:1 ω9t}). Isolated from ice-impacted seawater in the Okhotsku Sea, Japan.

The mol% G + C of the DNA is: 39 (HPLC).

Type strain: ABE-1, CIP 106458, JCM 10085.

GenBank accession number (16S rRNA): AB002630.

6. **Colwellia psychrotropica** Bowman, Gosink, McCammon, Lewis, Nichols, Nichols, Skerrat, Staley and McMeekin 1998a, 1179^{VP}

psy.chro.tro.pi'ca. Gr. adj. *psychros* cold; Gr. n. *tropica* circle; M.L. fem. adj. *psychrotropica* having an affinity for cold.

Individual cells are straight to curved rods, with spherical cells occasionally occurring. Cell length, 1.5–3.0 µm; width, 0.4–0.8 µm. Gas vesicles not formed. Motile. Colonies are off-white, have a mucoid consistency, raised elevation, and convex circular shape with entire to lobate edges. Psychrophilic. In liquid media T_{opt} is about 18°C and T_{max} is about 25°C. Requires sea salts for growth. Growth factors not required. Growth occurs in the presence of sea salts at three times normal concentration and is sensitive to vibriostatic agent O/129 (100 µg/ml). Acid (with no gas) is formed fermentatively and oxidatively from chitin and N-acetylglucosamine. Urea, uric acid, Tween 80, casein, chitin, and L-tyrosine are hydrolyzed. The following compounds are utilized as sole sources of carbon and energy: N-acetylglucosamine, acetate, butyrate, valerate, caproate, succinate, 2-oxoglutarate, 3-hydroxybutyrate, L-malate, fumarate, pyruvate, DL-lactate, oxaloacetate, L-alanine, L-aspartate, L-asparagine, L-glutamate, L-proline, and γ-aminobutyrate. Isolated from the pycnocline of an Antarctic marine-salinity meromictic lake (Burton Lake).

The mol% G + C of the DNA is: 42 (T_m).

Type strain: ACAM 179.

GenBank accession number (16S rRNA): U85846.

7. **Colwellia rossensis** Bowman, Gosink, McCammon, Lewis, Nichols, Nichols, Skerrat, Staley and McMeekin 1998a, 1179^{VP}

ross.en'sis. M.L. fem. adj. *rossensis* for the Ross Sea, Antarctica.

Individual cells are straight to curved rods, with spherical cells occasionally occurring. Cell length, 1.5–3.0 µm; width, 0.4–0.8 µm. Gas vesicles are formed. Nonmotile. Colonies are chalky-white, have a raised elevation and convex circular shape with entire edges. Psychrophilic. In liquid media T_{opt} is about 10°C and T_{max} is about 15°C. Requires sea salts for growth. Yeast extract required for growth. Resistant to vibriostatic agent O/129 (100 µg/ml). Acid (with no gas) is formed fermentatively and oxidatively from chitin, N-acetylglucosamine, and D-glucose. Also produces acid oxidatively from D-galactose and glycerol. Urea, chitin, and starch are hydrolyzed. The following compounds are utilized as sole sources of carbon and energy: glycogen, L-arabinose, N-acetylglucosamine, D-fructose, D-glucose, glycerol, acetate, butyrate, valerate, caproate, malonate, succinate, citrate, 3-hydroxybutyrate, L-malate, fumarate, pyruvate, L-alanine, L-aspartate, L-asparagine, L-glutamate, L-proline, and γ-aminobutyrate. Isolated from the sea/ice interface of fast sea ice McMurdo Sound, Antarctica (Gosink and Staley, 1995).

The mol% G + C of the DNA is: 38 (T_m).

Type strain: S51-W(gv)1, ACAM 608.

GenBank accession number (16S rRNA): U14581.

Genus IV. Ferrimonas Rosselló-Mora, Ludwig, Kämpfer, Amann and Schleifer 1996, 362^{VP}
(Effective publication: Rosselló-Mora, Ludwig, Kämpfer, Amann and Schleifer 1995a, 200)

RAMON A. ROSSELLÓ-MORA

Fer'ri.mo.nas. L. neut. n. *ferrum* iron; Gr. n. *monas* unit; M.L. fem. n. *Ferrimonas* iron (III)-reducing cell.

Straight rods 0.3–0.5 × 1.2–1.5 µm with rounded ends. Occur singly, occasionally with some pairs or short chains. No resting stages are known. Definite capsules are not evident. Gram neg-

ative. **Motile by means of a single polar flagellum.** No intracellular inclusions have been observed. **Chemoorganotroph, having a strictly respiratory metabolism with oxygen, nitrate, Fe(III)-oxy-**

hydroxide, Fe(III)-citrate, and MnO₂ used as electron acceptors.

Lactate is oxidized to CO₂ but not to acetate. Sodium dependent; a minimum of 0.5% NaCl is required in the medium. Catalase, oxidase, phenylalanine deaminase, DNase, and lipase (Tween 80 and Tween 20) positive. Gelatinase, urease, amylase, Simmons citrate, arginine dihydrolase, and lysine decarboxylase negative. Growth at 42°C. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 54 (*T_m*).

Type species: Ferrimonas balearica Rosselló-Mora, Ludwig, Kämpfer, Amann and Schleifer 1996, 362 (Effective publication: Rosselló-Mora, Ludwig, Kämpfer, Amann and Schleifer 1995a, 200.)

FURTHER DESCRIPTIVE INFORMATION

Fresh isolates may not form colonies on PYG¹ agar medium, but after several subcultivations in enrichment medium colony-forming units may appear. Colonies on TSI-agar² medium produce a black iron precipitate. Aerobic growth on solid medium is often brown and mucous. *Ferrimonas* undergoes autolysis within 5 d under aerobic conditions. Reduction of Fe(III)-oxyhydroxide with lactate as sole electron donor yields magnetite under anaerobic conditions. The range of NaCl tolerance is 0.5–7.5%. *Ferrimonas* does not exhibit growth at 5°C and 44°C but at 42°C. The pH range for growth is 6–9.

ENRICHMENT AND ISOLATION PROCEDURES

Only one strain of *Ferrimonas* has been isolated; it was simultaneously obtained from the same sample by using two different electron donors. Inocula were obtained from the upper few centimeters of a marine sediment of the Palma de Mallorca harbor (Spain). Sediments were inoculated in amorphous Fe(III)-oxyhydroxide medium³ containing either 0.4% (w/v) sodium acetate or 0.25% (w/v) sodium tartrate as sole electron donors, and in both instances 2% NaCl was added to the medium. The anaerobic atmosphere was composed of 97% N₂ and 3% H₂. Incubations were done at 20°C without shaking and in the dark. Enrichments yielded magnetite after 1 month of incubation. After three subcultivations in enrichment medium the isolation of the organism was carried out by streaking the enrichment culture

on Fe(III)-citrate agar. Pure culture was obtained by continuous restreaking under anaerobic incubations for at least 2 weeks.

MAINTENANCE PROCEDURES

Aerobic cultures should be reinoculated often due to the rapid autolysis of this bacterium; it does not survive more than 5 days on either solid or liquid complex medium incubated aerobically. However, anaerobic growth on Fe(III)-oxyhydroxide lactate agar shakes can be maintained for several months. The organism survives for several years frozen at –80°C in nutrient PYG broth containing 15% glycerol. It also survives freeze-drying. However, in cultures maintained by both methods viability may be low due to the tendency of the cells to lyse.

DIFFERENTIATION OF THE GENUS *FERRIMONAS* FROM OTHER GENERA

Ferrimonas can be distinguished from other strictly respiratory Gram-negative genera of the *Gammaproteobacteria* able to undergo Fe(III) reduction (i.e. *Shewanella putrefaciens*, *S. baltica*, and *S. algae*) as well as from *Alteromonas* species by the following characters: its ability to denitrify, to grow at 42°C and its phenylalanine deaminase activity, its inability to grow in medium containing 0% NaCl, its lack of gelatinase and urease, and its negative Simmons citrate reaction.

An oligonucleotide probe specific for *Ferrimonas* has been designed (5'-ACCCCCCTCTCAAGGACT-3' complementary to the position 654-672 in *E. coli* 16S rRNA numbering; Brosius et al., 1981). A comparison with 5300 complete or almost complete accessible 16S rRNA sequences present in the RDP database (Maidak et al., 1997) did not reveal any identical complementary sequences other than that of the target organism. *T_m*, derived from the thermal denaturation curve, is 53.2°C. Optimal stringency conditions for dot blot hybridization with extracted nucleic acids are at 62°C. This probe is also optimized for whole cell *in situ* hybridization. Specificity of the probe is obtained by using 35% formamide with the standard protocol (Manz et al., 1992).

TAXONOMIC COMMENTS

The genus *Ferrimonas* is phylogenetically affiliated with the *Gammaproteobacteria* as illustrated in Fig. BXII.γ.153. The genus presently consists of a single species, *F. balearica*; to date, no similar strains have been reported.

ACKNOWLEDGMENTS

K.-H. Schleifer and W. Ludwig are gratefully acknowledged for critically reviewing this chapter.

FURTHER READING

Rosselló-Mora, R.A., W. Ludwig, P. Kämpfer, R. Amann and K.-H. Schleifer. 1995. *Ferrimonas balearica* gen. nov., sp. nov., a new marine facultative Fe(III)-reducing bacterium. Syst. Appl. Microbiol. 18: 196–202.

1. PYG broth: 1% peptone, 0.5% yeast extract, 0.5% glucose, and 0.8% NaCl.

2. TSI-agar: Triple Sugar Iron agar, Difco.

3. Anaerobic basal mineral medium of Lovley and Phillips (1986): (in grams per liter deionized water) NaHCO₃, 2.5; CaCl₂·2H₂O, 1.0; KCl, 0.1, NH₄Cl, 1.5; NaH₂PO₄, 0.6; NaCl, 20.0; MgCl₂·6H₂O, 5.3; MgSO₄·7H₂O, 0.1; MnCl₂·4H₂O, 0.005; NaMoO₄·2H₂O, 0.001; and yeast extract, 0.05. Media is adjusted to pH 7. Electron acceptors were used at 250 mmol/l Fe(III)-oxyhydroxide (for preparation see Lovley and Phillips, 1986) or 20 mmol/l Fe(III)-citrate.

List of species of the genus *Ferrimonas*

1. *Ferrimonas balearica* Rosselló-Mora, Ludwig, Kämpfer, Amann and Schleifer 1996, 362^{VP} (Effective publication: Rosselló-Mora, Ludwig, Kämpfer, Amann and Schleifer 1995a, 200.)
ba.le.a'ri.ca. L. fem. adj. *balearica* of the Balearic Islands where the organism was isolated.

The description is the same as that given for the genus.

The mol% G + C of the DNA is: 54 (*T_m*).

Type strain: PAT, DSM 9799.

GenBank accession number (16S rRNA): X93021.

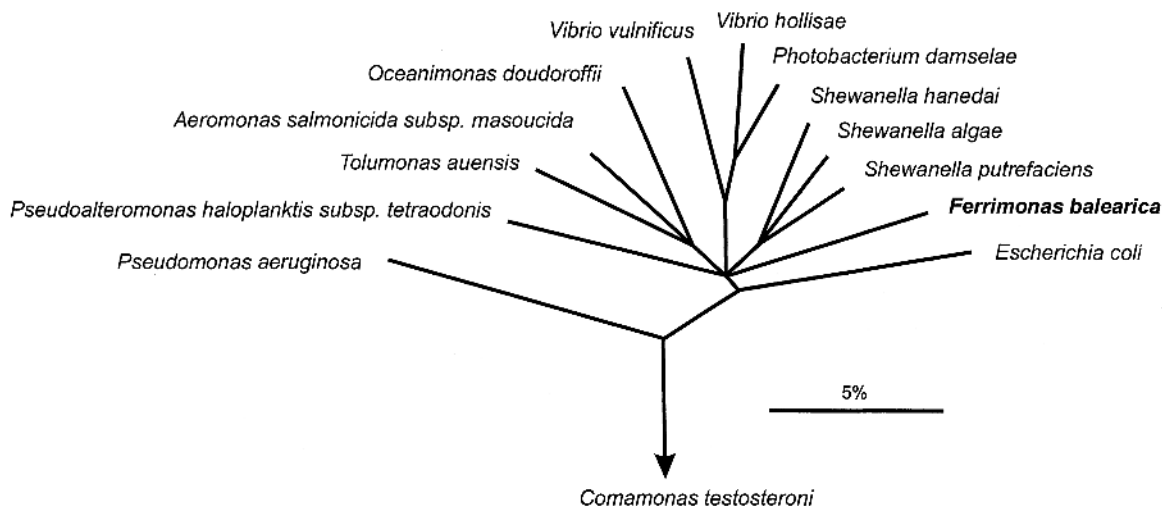


FIGURE BXII.γ.153. 16S rRNA based tree reflecting the phylogenetic affiliation of *Ferrimonas balearica* and a selection of *Proteobacteria*. The tree is based on the results of a distance matrix analysis including only sequence positions that share at least 50% of all available complete or almost complete 16S rRNA sequences from bacteria of this class. Tree topology was evaluated and corrected according to the results of distance matrix, maximum parsimony, and maximum likelihood analyses of various data sets. Accession numbers of shown sequences in the tree are the following: AB021371, M11224, M34133, X74680, X74700, X74707, X76333, X80725, X81623, X82132, X82139, X92889, X93021, and Z95486. Multifurcations indicate topologies that could not be unambiguously resolved. The bar indicates 5% estimated sequence divergence.

Genus V. *Glaciecola* Bowman, McCammon, Brown and McMeekin 1998b, 1220^{VP}

JOHN P. BOWMAN

Gla.ci.e' co.la. L. fem. n. *glacies* ice; L. gen. n. *incola* an inhabitant; M.L. fem. n. *Glaciecola* inhabitant of ice.

Slender rod-shaped cells, 0.2–0.4 × 1.0–5.0 μm. May be slightly curved, curved, or spiral. Gram negative. **Motile.** Nonsporeforming. **Aerobic**, having a strictly respiratory type of metabolism with oxygen as the only known terminal electron acceptor. Chemoheterotrophic. **Psychrophilic**, growing best at temperatures well below room temperature. In liquid media, the maximal growth temperature for most strains is at about 25–26°C; on agar media, no growth occurs above 20°C. **Seawater is required for growth.** **Oxidase and catalase positive.** Major fatty acids are hexadecanoic acid (C_{16:1 ω7c}), hexadecanoic acid (C_{16:0}), and octadecanoic acid (C_{18:1 ω7c}). Belongs to the *Gammaproteobacteria*. Habitat: Antarctic sea-ice diatom assemblages.

The mol% G + C of the DNA is: 40–46.

Type species: *Glaciecola punicea* Bowman, McCammon, Brown and McMeekin 1998b, 1220.

FURTHER DESCRIPTIVE INFORMATION

Glaciecola cells have a generally rod-shaped morphology (0.2–0.4 × 1.0–5.0 μm) and are motile. At the optimal growth temperature, ~15°C, cells are very slender and appear spiral or curved. At supraoptimal (>20°C) and suboptimal (<10°C) temperatures, the cells appear as straight rods. Short filaments also occur, especially in stationary growth cultures. All strains are motile, although information on the type and position of flagella present is still lacking.

The colonies of *G. punicea* are circular and convex in shape, with lobate edges and a butyrous consistency. After 14 d incubation, colony diameter is about 3–5 mm. *G. pallidula* strains form even-edged colonies that have a mucoid consistency and are 3–5 mm after 14 d incubation. Various *Glaciecola* strains are pigmented with unknown compounds that are not extractable with either polar or nonpolar solvents. In the presence of a strong base (20% KOH), the pigment is bleached white. Pigmentation is most obvious in strains of *G. punicea*, which appear musk pink to magenta in color. By comparison, *G. pallidula* colonies have a pale pink coloration.

Strains grow well on marine media, such as Marine 2216 agar (Difco Laboratories). All strains require seawater salts for optimal growth. No growth occurs if NaCl is added alone to the medium; Mg²⁺ and Ca²⁺ ions are also required for growth. Strains of *G. pallidula* are nutritionally non-exacting and will grow in a defined mineral salts medium, such as that used for testing carbon and energy sources of *Shewanella*, *Alteromonas*, and *Pseudoalteromonas* (Baumann et al., 1984a). Most strains of *G. punicea* require yeast extract for growth, and growth of this species can be achieved in a defined mineral salts medium supplemented with 0.02% yeast extract. In any case, growth in minimal media by most *Glaciecola* strains is relatively slow and scant, but it can be boosted somewhat by the addition of a vitamin solution, such as that formulated by Balch et al. (1979).

All strains are psychrophilic. In liquid media, the optimal growth temperature is about 15°C and the maximal growth temperature for most strains is about 25–26°C. Strains will grow at –2°C in liquid marine 2216 media. On agar media, cardinal growth temperature points are reduced by up to 5°C, with no growth occurring at temperatures above 20°C. A detailed study of the salt and temperature optima of *Glaciecola punicea* ACAM 611^T has been performed by Nichols et al. (1999).

All *Glaciecola* strains are strictly aerobic chemoheterotrophs and strongly produce catalase and cytochrome *c* oxidase. None of the strains grows anaerobically by either fermentation or by anaerobic respiration. Oxidative acid production from carbohydrates is weak and slow on Leifson's oxidation–fermentation medium (Leifson, 1963) and is restricted to a small range of carbohydrates (Table BXII.γ.140). Up to 10 d incubation at 15°C is required before acid production is detectable. The hydrolytic capacity of *Glaciecola* strains is relatively limited. Some strains of *G. punicea* can degrade uric acid (slowly) and esculin, whereas most strains of *G. pallidula* can degrade Tween 80 and produce an amylase. All strains so far tested are able to form alkaline phosphatase, α-galactosidase and β-galactosidase. *G. punicea* strains also produce a 6-phospho-β-galactosidase. Strains are able to utilize a limited range of organic acids, amino acids, and other compounds as sole carbon sources (Table BXII.γ.140). The met-

abolic pathways and nutrient transport mechanisms in *Glaciecola* strains have yet to be studied.

G. punicea strains possess a DNA base composition of 44–46 mol% G + C—significantly higher than that of *G. pallidula* strains, which have an average value of 40 mol% G + C.

Whole cell fatty acid profiles of *Glaciecola* species are very similar to one another. The major fatty acids include hexadecanoic acid (C_{16:1} ω_{7c}; 52–62%), octadecanoic acid (C_{18:1} ω_{7c}; 12–21%), hexadecanoic acid (C_{16:0}; 9–12%), and heptadecanoic acid (C_{17:1} ω_{8c}; 4–5%). Strains also contain low levels of saturated iso-branched-chain fatty acids (C_{13:0} iso, C_{14:0} iso, C_{15:0} iso, C_{16:0} iso, and C_{17:0} iso; total 2.5–3.3%). Polyunsaturated fatty acids and hydroxylated fatty acids (usually associated with the cell envelope) are either absent or are present only at trace levels. The fatty acids present are similar to those of *Alteromonas macleodii* (Svetashev et al., 1995); however, monounsaturated fatty acids predominate in *Glaciecola* strains, whereas saturated fatty acids are the most significant components in *Alteromonas* strains.

So far, *Glaciecola* strains have only been isolated from sea-ice algal assemblages. None have been isolated from under-ice seawater below the same sea-ice collection points. The genus is present at both poles; strains have been isolated from both Antarctic and Arctic (Barrow, Alaska) sea ice (Staley and Gosink, 1999).

ENRICHMENT AND ISOLATION PROCEDURES

Glaciecola strains have been obtained only from algal assemblages present within coastally attached sea ice (fast sea ice). Algal assemblages in fast ice usually occur perennially as a bottom assemblage. To obtain this bottom assemblage, care must be taken to avoid losing the layer when drilling out the ice core. The ice samples are melted in seawater at 4°C or less to avoid hypotonic shock to the bacteria present. The melted sample is then added to Marine 2216 liquid media for 1–2 d at 2°C before plating onto Marine 2216 agar. This enrichment phase encourages the growth of isolates that will not grow readily when directly plated onto a solid agar surface, such as *Glaciecola* spp. Incubation of marine agar plates proceeds for at least 4 weeks at 2°C. The unusual pigments produced by *Glaciecola* strains, particularly those of *G. punicea*, make them distinctive on primary isolation plates and easy to select for further purification. Some strains (all so far undescribed) that are related to the genera *Psychrobacter* and *Marinobacter* can also appear pale pink and so can be potentially confused with *G. pallidula* strains.

MAINTENANCE PROCEDURES

Glaciecola strains can be stored as active cultures on Marine 2216 agar plates or slants at 2°C for at least 12 months. The medium should contain an antifungal agent (nystatin or cycloheximide) to prevent contamination. Strains can also be cryopreserved. For cryopreservation, a heavy suspension of cells from a fresh culture is made in Marine 2216 liquid media supplemented with 30% glycerol. The suspension is then frozen at –20°C and then stored at –70°C. The frozen suspension is then checked within 2 weeks of preparation and every 6 months thereafter. Multiple small tubes of frozen suspension should be prepared to avoid loss of viability from repeated freezing and thawing.

DIFFERENTIATION OF THE GENUS *GLACIECOLA* FROM OTHER GENERA

The sea-ice habitat, distinctive pigmentation and psychrophilic properties of the genus *Glaciecola* make them readily identifiable. The genera that are most closely related to *Glaciecola* include

TABLE BXII.γ.140. Characteristics differentiating the species of the genus *Glaciecola*^a

Characteristic ^{b,c}	1. <i>G. punicea</i>	2. <i>G. pallidula</i>
Pigmentation	Pink-red	Pale pink
Yeast extract requirement	d	–
β-Galacto(6-phosphate)sidase	+	–
Acid from maltose	–	(+)
Hydrolysis of:		
Urate	d	–
Esculin	d	–
Starch	–	d
Tween 80	d	+
Utilization of:		
α-Glycerophosphate	d	+
Succinate, L-malate, fumarate, L-proline	+	–
Glycerol, glycogen, acetate, pyruvate, DL-lactate, L-glutamate	–	+
L-Tyrosine	d	–
Butyrate	–	d
Mol% G + C of the DNA	44–46	40

^aSymbols: +, test positive for all strains; d, variable (10–90% of strains) results between strains; –, strains negative for test; (+), the test exhibited a weak and delayed reaction.

^bAll *Glaciecola* strains tested are positive for the following tests: growth on Marine 2216, R2A (Oxoid Pty. Ltd.) + 2.5% NaCl, and ZoBell's marine medium (5 g bacteriological peptone, 2 g yeast extract dissolved in 1000 ml seawater); growth at <15°C on agar media; catalase, oxidase, α-galactosidase, β-galactosidase; utilization of oxaloacetate; and acid production (but weak and delayed) from D-glucose, D-galactose, D-melibiose, and glycerol.

^cAll *Glaciecola* strains tested are negative for the following tests: growth at 25°C or higher on agar media; growth with 3× or 4× strength seawater; tolerance to 5% ox-bile salts; nitrate as a nitrogen source; nitrate reduction, denitrification, and nitrogen fixation; hydrolysis of urea, xanthine, egg yolk, casein, gelatin, elastin, chitin, dextran, and DNA; production of L-phenylalanine deaminase, L-tryptophan deaminase, arginine dihydrolase, glutamate decarboxylase, lysine decarboxylase, ornithine decarboxylase, α-glucosidase, β-glucosidase, α-fucosidase, α-arabinosidase, β-glucuronidase, β-N-acetylglucosaminidase; production of indole from L-tryptophan; production of hydrogen sulfide from thiosulfate or L-cysteine; acid production from L-arabinose, D-mannose, D-fructose, L-rhamnose, D-xylose, D-mannitol, N-acetylglucosamine, sucrose, lactose, cellobiose, trehalose, D-raffinose, dextran, adonitol, L-arabitol, D-sorbitol, and D-inositol; utilization of N-acetylglucosamine, L-arabinose, cellobiose, and D-xylose.

Alteromonas and *Pseudoalteromonas*. *Alteromonas* strains have not been isolated from polar environments and seem to be more adapted to tropical and temperate marine waters, such as the Mediterranean Sea or mid-Pacific Ocean. *Pseudoalteromonas* species are common in southern ocean seawater and have also been isolated from sea ice (mostly strains of *P. antarctica*, *P. nigrifaciens*, and *P. prydzensis*) (Bowman, 1998); however these species are easy to distinguish from *Glaciecola* spp. because they are nonpigmented, able to grow at temperatures up to 30–35°C on agar, and nutritionally versatile, with strong saccharolytic activity.

TAXONOMIC COMMENTS

The 16S rDNA sequence analysis of *Glaciecola* strains indicates that they form a lineage within the order *Alteromonadales*, which

includes other marine genera, such as *Colwellia*, *Shewanella*, *Moritella*, *Ferrimonas*, *Pseudoalteromonas*, and *Alteromonas*. The genus *Alteromonas* is by far the closest related taxon, with a sequence similarity of 0.90–0.92. *Glaciecola punicea* and *Glaciecola pallidula* strains both form individually distinct branches and share a sequence similarity of about 0.93.

The levels of DNA hybridization between strains of the different species have been examined using spectrophotometric-based renaturation rate kinetics (Huss et al., 1983). The levels are equivalent to background levels of reassociation.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *GLACIECOLA*

Characteristics differentiating *G. punicea* and *G. pallidula* are listed in Table BXII.γ.140.

List of species of the genus *Glaciecola*

1. ***Glaciecola punicea*** Bowman, McCammon, Brown and McMeekin 1998b, 1220^{VP}
pu.nice' a. L. fem. adj. *punicea* pink-red, referring to the species pigmentation.

The characteristics are as described for the genus and as listed in Table BXII.γ.140. Other characteristics are as follows. Optimum temperature, 15°C; maximum 20–25°C. Colonies are bright pink-red, circular, and convex, with an entire edge and butyrous consistency. Yeast extract is required by some strains; growth is stimulated by vitamin growth factors. Oxidative acid production from carbohydrates is weak and slow; acid may be formed from D-glucose, D-galactose, D-melibiose, and glycerol. Some strains slowly degrade uric acid, esculin, and L-tyrosine. The following enzymes are produced: α-galactosidase, β-galactosidase, 6-phospho-β-galactosidase, and alkaline phosphatase. The following compounds are used as carbon and energy sources: succinate, L-malate, fumarate, oxaloacetate, and L-proline. Some strains utilize α-glycerophosphate. The major fatty acids are hexadecaenoic acid (C_{16:1 ω7c}), hexadecanoic acid (C_{16:0}), and octadecaenoic acid (C_{18:1 ω7c}). The type strain was isolated from a sea-ice core from the O'Gorman Rocks area of Prydz Bay, Antarctica.

The mol% G + C of the DNA is: 44–46 (*T_m*).

Type strain: IC067, ACAM 611.

GenBank accession number (16S rRNA): U85853.

2. ***Glaciecola pallidula*** Bowman, McCammon, Brown and McMeekin 1998b, 1221^{VP}
pal.lid' u.la. L. adj. *pallidula* pallidula somewhat pale, referring to the weak pigmentation of the species.

The characteristics are as described for the genus and as listed in Table BXII.γ.140. Other characteristics are as follows. Optimum temperature, 10–15°C; maximum, 18–20°C. Colonies are pale pink, circular, and raised convex, with entire edges and a butyrous–mucoid consistency. Growth is stimulated by vitamin growth factors. Oxidative acid production from carbohydrates is weak and slow; acid may be formed from D-glucose, D-galactose, D-melibiose, and glycerol. Tween 80 is hydrolyzed. Produce the following enzymes: α-galactosidase, β-galactosidase, and alkaline phosphatase. Some strains slowly degrade starch. Utilize the following compounds as carbon and energy sources: glycogen, glycerol, α-glycerophosphate, acetate, pyruvate, DL-lactate, oxaloacetate, and L-glutamate. Some strains utilize butyrate. Major fatty acids are hexadecaenoic acid (C_{16:1 ω7c}), hexadecanoic acid (C_{16:0}), and octadecaenoic acid (C_{18:1 ω7c}). The type strain was isolated from sea-ice cores from Taynaya Bay, Antarctica.

The mol% G + C of the DNA is: 40 (*T_m*).

Type strain: IC079, ACAM 615.

GenBank accession number (16S rRNA): U85854.

Genus VI. *Idiomarina* Ivanova, Romanenko, Chun, Matte, Matte, Mikhailov, Svetashev, Huq, Mangel and Colwell 2000b, 906^{VP}

THE EDITORIAL BOARD

I.di.o.ma.ri' na. Gr. adj. *idios* original, true; L. fem. adj. *marina* of the sea, marine; M.L. fem. n. *Idiomarina* pertaining to the peculiar, true marine nature of microorganisms from the ocean (seawater).

Chemoorganotrophic Gram-negative obligately aerobic rods. Motile; one polar flagellum. **Require Na⁺ for growth;** no required vitamins or amino acids. Temperature optimum 20–22°C; pH optimum 7.5–8.0. **Oxidase, catalase, gelatinase, lipase, and DNase positive.** Do not hydrolyze starch or agar. Major cellular

fatty acids are branched *iso* forms having odd numbers of carbon atoms.

The mol% G + C of the DNA is: 48.9–50.4.

Type species: ***Idiomarina abyssalis*** Ivanova, Romanenko, Chun, Matte, Matte, Mikhailov, Svetashev, Huq, Mangel and Colwell 2000b, 906.

FURTHER DESCRIPTIVE INFORMATION

The type strains of both *Idiomarina* species utilized alaninamide, L-alanine, L-alanyl-glycine, L-arginine, glycyl-L-glutamic acid, α -ketobutyric acid, α -ketovaleric acid, monomethylsuccinate, and L-tyrosine in the Biolog[®] test system. Neither utilized D-arabinose, glycerol, lactose, L-lysine, maltose, mannitol, D-mannose, melibiose, L-phenylalanine, D-rhamnose, or sucrose (Ivanova et al., 2000b).

The type strains of both *Idiomarina* species were resistant to ampicillin, benzylpenicillin, kanamycin, lincomycin, oleandomycin, oxacillin, tetracycline, and vancomycin; both were susceptible to erythromycin (Ivanova et al., 2000b).

Analysis of 16S rDNA sequences of *I. abyssalis* and *I. zobellii* placed them together in a clade in the *Gammaproteobacteria* with *Alteromonas*, *Pseudoalteromonas*, and *Colwellia* species (Ivanova et al., 2000b).

ENRICHMENT AND ISOLATION PROCEDURES

Strains were isolated from seawater that was collected at 4000–5000 m, spread on seawater agar amended as described by Ivanova et al. (2000b), and incubated at atmospheric pressure.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *IDIOMARINA*

I. abyssalis KMM 227 grows in the presence of NaCl up to 15% at 4°C; *I. zobellii* KMM 231, up to 10%. *I. zobellii* KMM 231 possesses fimbriae and produces chitinase; *I. abyssalis* KMM 227 does not.

MAINTENANCE PROCEDURES

Strains can be maintained at 4°C on semisolid medium B covered with mineral oil (Ivanova et al., 2000b) or at –80°C suspended in Marine Broth containing 30% (v/v) glycerol (Ivanova et al., 2000b).

DIFFERENTIATION OF THE GENUS *IDIOMARINA* FROM OTHER GENERA

Characteristics that differentiate the genus *Idiomarina* from other members of the family *Alteromonadaceae* (*Alteromonas*, *Pseudoalteromonas*, and *Colwellia*) are given by Ivanova and Mikhailov (2001a); characteristics that differentiate *Idiomarina* from those three genera and from members of the genera *Halomonas*, *Marinomonas*, and *Oceanospirillum* are given by Ivanova et al. (2000b). A comparison of fatty acid profiles of members of the genera *Idiomarina*, *Marinomonas*, *Alteromonas*, *Pseudoalteromonas*, and *Glaucicola* was presented by Ivanova et al. (2000c).

List of species of the genus *Idiomarina*

1. ***Idiomarina abyssalis*** Ivanova, Romanenko, Chun, Matte, Matte, Mikhailov, Svetashev, Huq, Mangel and Colwell 2000b, 906^{VP}
a.bys.sal' is. L. fem. masc. adj. *abyssalis* deep sea; M.L. fem. adj. *abyssalis* of abyssal depths of the ocean (1000–6000 m) from which the organism was isolated.

Description as for the genus with the following additional characteristics. Grows in the presence of 0.6–15% NaCl. Produces proteinase.

The mol% G + C of the DNA is: 50.4 (T_m).

Type strain: KMM 227, ATCC BAA-312.

GenBank accession number (16S rRNA): AF052740.

2. ***Idiomarina zobellii*** Ivanova, Romanenko, Chun, Matte, Matte, Mikhailov, Svetashev, Huq, Mangel and Colwell 2000b, 906^{VP}
zo.bell' i.i. M.L. fem. adj. *zobelli* of Zobell; named after C.E. Zobell, a pioneer marine microbiologist.

Description as for the genus with the following additional characteristics. Grows in the presence of 1–10% NaCl. Produces chitinase.

The mol% G + C of the DNA is: 48 (T_m).

Type strain: KMM 231, ATCC BAA-313.

GenBank accession number (16S rRNA): AF052741.

Genus VII. *Marinobacter* Gauthier, Lafay, Christen, Fernandez, Acquaviva, Bonin and Bertrand 1992, 574^{VP}

JOHN P. BOWMAN AND THOMAS A. McMEEKIN

Mar.i' no.bac.ter. L. adj. *marinus* of the sea; Gr. n. *bacterion* rod or staff; M.L. masc. n. *Marinobacter* rod of the sea.

Rod-shaped cells $0.6\text{--}0.8 \times 1.6\text{--}2.0$ μm . Form surface blebs when grown on *n*-eicosane and other hydrocarbons. **Motile by a single polar flagellum.** **Gram negative.** Nonpigmented colonies. Aerobic with a strictly respiratory type of metabolism. Oxidase and catalase positive. **Can grow anaerobically by denitrification** coupled to the oxidation of a suitable donor carbon substrate. Chemoheterotrophic. Constitutive arginine dihydrolase system is absent. **Na⁺ is required for growth.** **Halotolerant**, growing in 0.08–3.5 M NaCl. **Mesophilic** growing at temperatures 10–45°C. Organic acids, alcohols, amino acids, and hydrocarbons are used as carbon and energy sources. **Carbohydrates are not utilized.** Polysaccharides are not degraded. Type species produces a nondi-

alyzable bioemulsifier when grown on hydrocarbons and is resistant to vibriostatic agent O/129. The major polyamine is spermidine. Isolated from oil brine. **Member of the *Gammaproteobacteria*.**

The mol% G + C of the DNA is: 53 (T_m).

Type species: ***Marinobacter hydrocarbonoclasticus*** Gauthier, Lafay, Christen, Fernandez, Acquaviva, Bonin and Bertrand 1992, 574.

FURTHER DESCRIPTIVE INFORMATION

Marinobacter hydrocarbonoclasticus is represented by a single strain (ATCC 49840). The cells do not accumulate poly- β -hydroxybu-

tyrate (PHB) inclusions or form spores or cysts. When cultivated on synthetic salt (SM) medium¹ containing the C₂₀ alkane eicosane as a sole carbon source (SM-alkane medium), cells are covered by bleb-like extensions of the cell wall. Flagellation is lost when the strain is grown at Na⁺ concentrations well above and below the optimal level - i.e., >1.0 M and <0.2 M NaCl.

On marine agar and SM-peptone (0.5% w/v) agar young colonies are white, circular, and convex, possessing entire edges and a creamy consistency. Colonies usually reach 2–4 mm after 7 d of incubation at 32°C. Older colonies develop a pinkish pigment probably due to accumulation of cytochromes. Diffusible and fluorescent pigments are not produced. The cells are not bioluminescent.

M. hydrocarbonoclasticus has an absolute requirement for Na⁺ ions with an optimal growth level of 0.6 M NaCl. The requirement for Na⁺ cannot be reversed by replacement of K⁺ and Li⁺ ions. The strain also exhibits considerable halotolerance and grows at NaCl levels of 0.08–3.5 M NaCl. When suspended in a solution containing 0.5 M NaCl and in the absence of Mg²⁺ ions, cells undergo severe hypotonic shock with the suspension losing about 60% of its optical density. When suspended or washed in the presence of Mg²⁺, cell walls are considerably stabilized and show only limited, if any, hypotonic shock even if subsequently suspended in distilled water. Stability appears to be due to linking of Mg²⁺ ions between the anionic groups of membrane components (Rayman and MacLeod, 1975).

The optimal growth temperature is 32°C; range 10–45°C. The optimal pH is 7.0–7.5; range 6.0–9.5. The strain is nutritionally nonexacting and does not require growth factors. Nitrate and ammonia ions serve as nitrogen sources.

Cytochrome *c* oxidase and catalase are readily detectable in cell masses; however, cytochrome *aa3* oxidase and cytochrome P-450 appear to be completely absent. Cells can grow anaerobically with nitrate as electron acceptor and a suitable carbon source/electron donor such as citrate, acetate, or succinate. A variety of organic acids and hydrocarbons, such as eicosane and hexadecane, and a few amino acids can be used as carbon and energy sources. Lipase and lecithinase occur but the cells do not hydrolyze polysaccharides, gelatin, or DNA. A constitutive arginine dihydrolase system is absent.

The cells form a nondialyzable bioemulsifier when grown on hydrocarbons as sole sources of carbon and energy (Fernandez-Linares et al., 1996a). When grown on eicosane at various NaCl concentrations, ectoine is the major compatible solute in cells; however, ectoine is readily replaced by glycine betaine (Fernandez-Linares et al., 1996b). Glycine betaine reduces the growth rate at high Na⁺ ion concentrations, possibly by interfering with eicosane metabolism as a consequence of decreased ectoine synthesis (Gauthier et al., 1997). *M. hydrocarbonoclasticus* has been shown to degrade 6,10,14-trimethylpentadecan-2-one, a common isoprenoid ketone found in marine sediment (Rontani et al., 1997).

Spermidine is the predominant polyamine constituent (Hamana, 1997), while the major phospholipids of cells include phosphatidylethanolamine, phosphatidylglycerol, and diphosphatidylglycerol (Gerin and Goutx, 1993).

Little is known about the broader ecology of the genus. The ability to degrade alkane hydrocarbons and broad halotolerance suggests the organisms are commonly associated with oil brines. A gradient diffusion chamber containing increasing levels of NaCl and *p*-toluate resulted in the enrichment of *M. hydrocarbonoclasticus*-like strains (identified as *Pseudomonas nautica*) (Emerson and Breznak, 1997). *M. aquaeolei* was isolated from the head of an offshore oil/gas platform (Nguyen et al., 1999). Several isolates from sea ice and seawater (Bowman et al., 1997c) have been found to be closely related to *M. hydrocarbonoclasticus* and include a variety of psychrophilic and psychrotrophic strains including a strain possessing gas vesicles (Gosink and Staley, 1995).

ENRICHMENT AND ISOLATION PROCEDURES

M. hydrocarbonoclasticus can be isolated by enriching oil brine, seawater, and oil brine contaminated soil samples with SM-alkane liquid broth, which contains a long-chain alkane such as eicosane or hexadecane as the sole carbon and energy source. Enrichment flasks are shaken for several days at approximately 35°C. Growth from these cultures is serially diluted onto SM-alkane agar or marine agar for further purification. Subsequent testing of the traits presented in Table BXII.γ.141 should follow in order to confirm identity of the isolates.

M. aquaeolei VT8 was obtained from an enrichment culture incubated in a complex medium containing 10% NaCl (Nguyen et al., 1999).

MAINTENANCE PROCEDURES

M. hydrocarbonoclasticus can be maintained at –80°C in Marine 2216 broth containing a cryoprotectant agent such as 20–30% glycerol or dimethylsulfoxide. Strains can also be freeze-dried using 20% skim milk as a cryoprotectant. *M. aquaeolei* can be stored in 25% glycerol, 5% NaCl at –70°C (Nguyen et al., 1999).

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Denitrification in *M. hydrocarbonoclasticus* can be determined by measuring accumulation of nitrous oxide during anaerobic growth with nitrate. The testing medium is SM supplemented with 0.1% acetate, 0.1% succinate, and 4 mM nitrate, and is prepared anaerobically using 100% nitrogen. Denitrification is halted by the addition of acetylene (10 kPa). Nitrous oxide levels are measured by gas chromatography (Bonin et al., 1987).

DIFFERENTIATION OF THE GENUS *MARINOBACTER* FROM OTHER GENERA

Marinobacter is most likely to be confused with nonfermentative Gram-negative, rod-shaped halophilic marine genera including *Halomonas*, *Marinomonas*, *Marinobacterium*, *Microbulbifer*, *Alteromonas*, *Pseudoalteromonas*, and *Shewanella*. The most useful traits available for differentiation of *Marinobacter* strains from these genera include the inability to utilize carbohydrates, or to degrade polysaccharides or gelatinase, the ability to denitrify, and broad halotolerance. Traits useful for differentiation of the genus from some other Gram-negative marine bacteria are presented in Table BXII.γ.141.

TAXONOMIC COMMENTS

Only limited chemotaxonomic data are available for *M. hydrocarbonoclasticus*. However, the phenotypic traits of *Marinobacter*

1. SM medium consists of (per liter): 30 g NaCl, 12.3 g Tris, 3.7 g NH₄Cl, 6.2 g MgSO₄·7H₂O, 1.5 g CaCl₂, and 0.75 g KCl. The pH of the medium is adjusted with 10 M HCl to 7.5. For solid media, agar (15 g/liter) can be added. SM-alkane medium consists of SM medium supplemented with 0.1% (w/v) alkane (eicosane, hexadecane etc.). SM-peptone medium consists of SM medium supplemented with 0.5% peptone.

are not particularly distinct and it possesses some similarity to species of several families of the *Gammaproteobacteria* including *Halomonadaceae* and *Alteromonadaceae*.

A phylogenetic analysis of *M. hydrocarbonoclasticus* ATCC 49840 using 16S rDNA sequence data indicated that its closest named relative is *Microbulbifer hydrolyticus*, a polysaccharide-degrading marine species (González et al., 1997). Spröer et al. (1998) later showed that *Pseudomonas nautica* is also very closely related to ATCC 49840 and proposed the transfer of *P. nautica* to *M. hydrocarbonoclasticus* (see below). The 6,10,14-trimethylpentadecan-2-one-degrading strain CAB and an isolate from an oil well (strain VT8) were also very closely related to ATCC 49840; Nguyen et al. (1999) have described strain VT8 as *Marinobacter aquaeolei*. Other strains whose 16S rDNA sequences were closely related to *Marinobacter* include a variety of isolates from Antarctic sea ice and under-ice seawater (Gosink and Staley, 1995; Bowman et al., 1997c) (Fig. BXII.γ.154). These strains share the inability to catabolize carbohydrates with *M. hydrocarbonoclasticus* but have other traits different from the current definition of *Marinobacter*: they are unable to reduce nitrate and are psychrophilic and stenohaline. These differences could be sufficient to support the description of a separate genus, closely related to *Marinobacter*.

16S rDNA sequence data (>99% similarity) provided compelling evidence that the species *Pseudomonas nautica* (Baumann et al., 1983b) is a subjective synonym of *M. hydrocarbonoclasticus*. In addition, Spröer et al. (1998) reported 100.8% hybridization of DNA from the type strains of *P. nautica* and *M. hydrocarbonoclasticus* in DNA-DNA hybridization experiments. Finally, the phenotypic similarities of *P. nautica* and *M. hydrocarbonoclasticus* (Table BXII.γ.142), including the ability to utilize a range of hydrocarbon alkanes (Monpert, 1996; Husain et al., 1997a, b) and active denitrification, provide a strong argument to support the transfer of *P. nautica* to *M. hydrocarbonoclasticus*. One initial discrepancy was a difference in the reported DNA base compositions. Reported values for the mol% G + C of the DNA of *P. nautica* were 57.8–61.1 (determined by buoyant density centrifugation; Baumann et al., 1972); these values were higher than the value of 52.7 reported for *M. hydrocarbonoclasticus* (determined by thermal denaturation; Gauthier et al., 1992). However, Spröer et al. (1998) analyzed DNA from the type strain of each species by HPLC and obtained values of 57.7 mol% G + C for *P. nautica* and 57.3% for *M. hydrocarbonoclasticus*. The mol% G + C of *M. aquaeolei* is 55.7 ± 0.1 (determined by HPLC; Nguyen et al., 1999).

TABLE BXII.γ.141. Phenotypic characteristics differentiating the genus *Marinobacter* from other marine bacteria

Characteristic	<i>Marinobacter</i>	<i>Marinobacterium</i>	<i>Microbulbifer</i>	<i>Marinomonas</i>	<i>Oceanospirillum</i>
Morphology	Rod	Rod	Rod	Curved rods	Curved, helical
Na ⁺ requirement	+	+	+	+	+
Gelatinase activity	—	—	+	—	D
Denitrification	+	—	—	—	—
Carbohydrate utilization	—	+	+	+	—
Mol% G + C	52.7 ^b ; 57.3 ^c	55	58	45–50	45–51

^aSymbols: +, positive; D, test results vary between species; — negative.

^bGauthier et al., 1992.

^cSpröer et al., 1998.

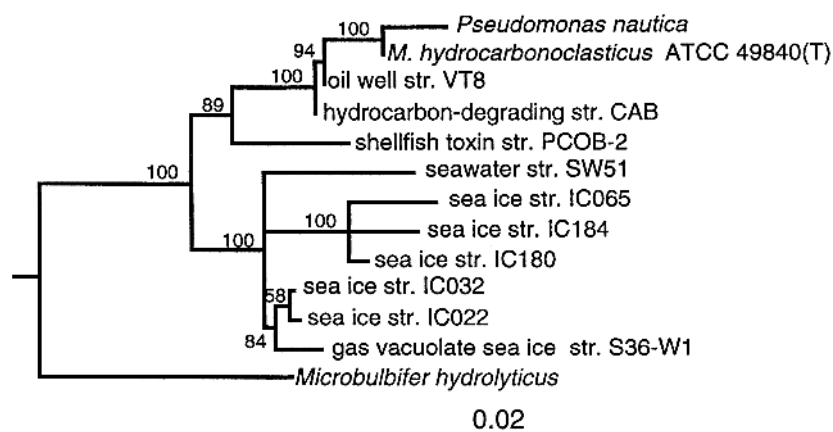


FIGURE BXII.γ.154. Phylogenetic tree of *Marinobacter hydrocarbonoclasticus* ATCC 49840, *Pseudomonas nautica* ATCC 27132, and close relatives based on nearly complete 16S rRNA sequences. The tree was determined using maximum likelihood distances and clustered by neighbor-joining. Strain VT8 has been described as *M. aquaeolei* (Nguyen et al., 1999; GenBank accession no. AJ000726). Various other strains on the tree remain unclassified, including hydrocarbon-degrading strain CAB (Rontani et al., 1997; U61848), shellfish toxin strain PCOB-2 (Maas et al., unpublished; AJ000647); under-ice seawater and sea ice strains SW51, IC065, IC180, IC184, IC032 and IC022 (Bowman et al., 1997c; AF001374, U85863-7); and a gas vacuolate sea ice strain S36-W1 (Gosink and Staley, 1995; U14584). Numbers at branch nodes are bootstrap values (calculated from 250 replicate analyses). Only bootstrap values over 50% are shown. Bar equals a distance of 0.02.

TABLE BXII.γ.142. Phenotypic characteristics of *Marinobacter hydrocarbonoclasticus* and *Pseudomonas nautica*^{a,b,c}

Characteristic	<i>Marinobacter hydrocarbonoclasticus</i>	<i>Pseudomonas nautica</i>
Morphology	Rods	Rods
Flagella	Single polar or none	Single polar or none
Accumulation of poly-β-hydroxybutyrate	—	—
Optimal temperature for growth	32°C	35°C
Growth at 4°C	—	—
Growth at 45°C	+	d
NaCl requirement	+	+
Tolerates 20% NaCl	+	ND
Catalase, oxidase, denitrification	+	+
Hydrogen autotrophy	—	—
Arginine dihydrolase, lysine decarboxylase, ornithine decarboxylase	—	—
Gelatin liquefaction	—	—
Starch hydrolysis	—	—
Lipase (Tween 80 hydrolysis)	+	+
Lecithinase (egg yolk reaction)	+	+
Urease activity	—	—
Acid from glucose	—	—
Utilization of:		
Acetate, butyrate, caproate, succinate, fumarate, DL-lactate, <i>p</i> -hydroxybenzoate, L-proline, hexadecane, eicosane	+	+
Adipate, citrate, L-glutamate	+	d
DL-Alanine, benzoate	—	d
Propionate, isobutyrate, valerate, heptanoate, caprylate, pelargonate, caprate, malonate, pyruvate, DL-malate, DL-3-hydroxybutyrate, L-histidine	ND	+
Isovalerate, sebacate, propyleneglycol, ethanol, propanol, butanol, isobutanol	ND	d
Tetradecane, heneicosane, pristane, phenyldecane, phenanthrene	+	ND

^aSymbols: +, 80–100% of strains are positive; d, 20–80% of strains are positive; —, 0–20% of strains are positive; ND, no data.

^bPhenotypic data from Gauthier et al. (1992) and Baumann et al. (1983b). *M. hydrocarbonoclasticus* and *P. nautica* do not utilize DL-arabinose, D-ribose, D-fructose, D-glucose, D-mannose, lactose, sucrose, cellobiose, D-mannitol, D-sorbitol, glycerol, D-gluconate, N-acetylglucosamine, glycolate, 2-oxoglutarate, *m*-hydroxybenzoate, *o*-hydroxybenzoate, DL-mandelate, propionate, L-arginine, L-asparagine, L-aspartate, glycine, L-lysine, L-methionine, L-ornithine, L-serine, DL-tryptophan, and sarcosine.

^cIn addition most strains of *P. nautica* (<20% positive) do not utilize D-xylose, D-fucose, L-rhamnose, D-galactose, trehalose, maltose, melibiose, saccharate, mucate, D-galacturonate, D-glucuronate, salicin, inulin, cellulose, agar, adonitol, *i*-erythritol, *m*-inositol, ethyleneglycol, methanol, isopropanol, geraniol, formate, oxalate, glutarate, pimelate, suberate, azelate, L(+)-tartrate, D(–)-tartrate, meso-tartrate, glycerate, aconitate, maleate, levulinate, citraconate, itaconate, mesaconate, benzoyl formate, phenylacetate, phenylethanediol, naphthalene, quinate, β-alanine, L-citrulline, L-isoleucine, L-leucine, L-norleucine, L-phenylalanine, L-threonine, L-tyrosine, L-valine, kynurenate, kynurenine, trigollenine, anthranilate, *m*-aminobenzoate, *p*-aminobenzoate, α-aminobutyrate, γ-aminobutyrate, δ-aminovalerate, methylamine, ethanalamine, benzylamine, tryptamine, butylamine, 1-amilamine, 2-amilamine, putrescine, spermine, betaine, creatine, histamine, hippurate, acetamide, pantothenate, nicotinate, nicotinamide.

List of species of the genus *Marinobacter*

- 1. *Marinobacter hydrocarbonoclasticus*** Gauthier, Lafay, Christen, Fernandez, Acquaviva, Bonin and Bertrand 1992, 574^{VP} *hy' dro. car. bo. no. clas. ti. cus.* M.L. part. adj. *hydrocarbonoclasticus* hydrocarbonoclastic, hydrocarbon dismantler.

Phenotypic characteristics are described in the text and presented in Tables BXII.γ.141 and BXII.γ.142. Isolated from seawater of the Mediterranean Sea, near a petroleum refinery outlet, Gulf of Fos, France.

The mol% G + C of the DNA is: 53 (*T_m*); 57.3 (HPLC).

Type strain: SP.17, ATCC 49840, DSM 8798.

GenBank accession number (16S rRNA): X67022 (ATCC 49840).

- 2. *Marinobacter aquaeolei*** Nguyen, Denner, Dang, Wanner and Stan-Lotter 1999, 373^{VP} *a. quae. o' le. i.* L. n. *aqua* water; L. n. *oleum* oil; M.L. gen. n. *aquaeolei* from water of oil, isolated from an oil-field brine.

Gram-negative motile rods (0.4–0.5 × 1.4–1.6 μm). Single polar flagellum. Nonsporeforming. Growth at 13–50°C, optimum 30°C; pH 5–10, optimum 7.3; and NaCl concentrations 0–20%; optimum 5%. Catalase and oxidase positive. Grows aerobically. Also grows anaerobically on acetate, citrate, and succinate when nitrate is present. Carbon and energy sources include acetate, L-alanine, butyrate, citrate, fumarate, L-glutamate, L-glutamine, L-isoleucine, DL-lactate, L-leucine, L-proline, and succinate. Degrades *n*-hexadecane, pristane, and some compounds in crude oil. Isolated from the head of an offshore oil/gas platform in southern Vietnam.

The mol% G + C of the DNA is: 55.7 ± 0.1 (HPLC).

Type strain: VT8, ATCC 700491, DSM 11845.

GenBank accession number (16S rRNA): AJ000726.

TABLE BXII-γ.143. Differential characteristics of *Marinobacterium* and some aerobic, rod-shaped, marine bacteria that require NaCl to grow^a

Characteristic	<i>Marinobacterium</i>	<i>Alteromonas</i> ^b <i>Pseudalteromonas</i> ^d and <i>Shewanella</i> ^d	<i>Halomonas marina</i>	<i>Halomonas</i> spp. ^e	<i>Marinobacter</i> <i>hydrocarbonoclasticus</i>	<i>Marinomonas</i> ^f	<i>Mesophilobacter</i>	<i>Methylobacterium</i> <i>pelagicum</i> ^g	<i>Methylobacterium</i> <i>marina</i>	<i>Oceanospirillum</i> <i>jannaschii</i> ^h	<i>Oceanospirillum</i> <i>krugi</i>	<i>Pseudomonas</i> <i>doudoroffii</i> ⁱ	<i>Pseudomonas</i> <i>stuartii</i>	<i>Pseudomonas</i> <i>stutzeri</i> ^j
<i>Cell shape:</i>														
Straight rod	+	+	+	+	+	D	+	+	+	+	+	+	+	+
Curved rod	-	- ^l	-	-	-	D ^m	-	-	+	-	-	-	-	-
Motility	+	+	+	+	+	+	-	+	+	+	+	+	+	+
<i>Flagellar arrangement:</i>														
Polar	+	+	+	-	+	+	nd	+	+	+	+	+	+	+
Lateral	-	-	-	+	-	-	nd	-	-	-	-	-	-	-
Cell diameter, μm	0.5–0.7	0.5–1.5	0.8–1.1	0.8–1.1	0.3–0.5	0.7–1.5	0.5–0.6	0.8–1.2	0.2	1.0–1.4	0.8–1.2	0.7–1.2	0.6–0.8	0.5–0.7
Poly-β-hydroxybutyrate accumulation	-	-	+	+	-	-	-	-	nd	+	+	+	+	-
<i>Utilization of:</i>														
Dl-Malate	+	-	+	D	d	-	+	-	-	d	+	+	d	+
α-Ketoglutarate	nd	- ^o	+	D	-	+	nd	-	-	nd	+	+	+	+
β-Hydroxybutyrate	+	-	nd	nd	d	+	nd	-	-	-	+	+	+	+
Methane	nd	-	-	-	-	-	nd	+	+	-	-	-	-	-
Methanol	+	-	-	-	-	-	nd	+	D	-	-	-	-	-
Methylamine	+	-	-	-	-	-	nd	-	nd	+	-	-	-	-
NO ₃ ⁻ reduced to NO ₂ ⁻	-	- ^p	-	D	d	-	+	nd	nd	+	+	d	+	+
Oxidase	+	+	-	D	+	D	+	nd	nd	+	+	+	+	+
Lipase	-	+	nd	nd	+	-	nd	nd	nd	-	+	-	-	-
Gelatinase	-	+	-	-	d	-	+	+	+	-	+	-	-	-
D-Glucose	+	+	+	+	-	+	+	-	-	-	+	-	-	+
D-Fructose	+	D	+	D	-	+	+	-	+	-	+	+	-	+
Saccharate	nd	nd	-	D	-	+	nd	nd	nd	-	-	-	-	+
Glycolate	nd	nd	+	D	-	+	+	nd	nd	-	+	-	-	-
Mannitol	nd	nd	+	D	-	+	+	nd	nd	-	+	-	-	-
Isobutanol	nd	nd	-	D	d	-	nd	nd	nd	+	-	-	+	+
Quinate	nd	nd	-	D	-	+	nd	nd	nd	-	+	-	-	-
δ-Aminovalerate	nd	nd	-	D	-	-	nd	nd	nd	+	+	-	d	-
L-Tyrosine	nd	D	+	D	-	-	nd	nd	nd	+	-	+	+	+
Sarcosine	nd	nd	d	D	-	+	nd	nd	nd	+	-	+	-	-

^aFor symbols, see standard definitions; nd, not determined.^bThe species considered is *A. macleodii*.^cThe species considered are *P. haloplanktis*, *P. espejiana*, *P. undina*, *P. rubra*, *P. lateroviolacea*, *P. citrea*, *P. aurantia*, *P. nigrifaciens*, and *P. denitrificans*, all of which were previously included in the genus *Alteromonas*.^dThe species considered are *S. hanedai* and *S. cobwelliana*, which were previously included in the genus *Alteromonas*.^eThe species considered are *H. aquamarina*, *H. cupida*, *H. halophila*, *H. pacifica*, and *H. venusta*, all of which were previously included in the genus *Deleya*.^fThe species considered are *M. communis* and *M. vaga*, which were formerly assigned to the genus *Alteromonas*.^g*Methylobacterium pelagicum* was previously included in the genera *Methylobacter* and *Methylobacter*.^hThe two species considered here are straight rods. All other species of *Oceanospirillum* are helical.ⁱMarine pseudomonads differ from terrestrial species by requiring over 75 mM NaCl for optimum growth rate and cell yield.^jRefers to strains previously classified as *P. pelagetum* but shown to belong to *P. stutzeri* by DNA hybridization experiments.^kCoccioid cells occur in old cultures.^lOnly one species, *P. undina*, is a curved rod. Long, filamentous helices, often exceeding 20 μm, occur when *S. cobwelliana* grows on surfaces, in nutrient-poor media, or during late growth phases.^m*M. communis* is curved; *M. vaga* is straight.ⁿSome cells of *P. stutzeri* (*P. pelagetum*) may be slightly curved.^o*S. cobwelliana* utilizes α-ketoglutarate.^p*S. hanedai*, *S. cobwelliana*, and some strains of *P. haloplanktis* are positive. *P. denitrificans* carries out denitrification with gas formation.^q*S. cobwelliana* and some strains of *S. hanedai* are unable to use D-glucose.

Genus VIII. Marinobacterium González, Mayer, Moran, Hodson and Whitman 1997, 375^{VP}

JOSÉ M. GONZÁLEZ

Ma.ri.no.bac.te'ri.um. L. adj. *marinus* of the sea; Gr. n. *bakterion* a small rod; M.L. neut. n. *Marinobacterium* small marine rod.

Rod-shaped cells, 0.5–0.7 × 1.6–2.3 µm. Motile by means of a single polar flagellum. Gram negative. Nonsporeforming and noncystforming. **Have a strictly aerobic type of metabolism with O₂ as the terminal electron acceptor.** Denitrification does not occur. Nonpigmented. **Oxidase and catalase positive. Sea salt-based medium is required for growth.** Chemoorganotrophic, growing on various sugars, fatty acids, aromatic compounds, and amino acids. C₁ compounds are utilized, such as methanol or formate. Vitamins, in the form of 0.005% yeast extract or a complex vitamin solution, are required for optimal growth.

The mol% G + C of the DNA is: 54.9.

Type species: Marinobacterium georgiense González, Mayer, Moran, Hodson and Whitman 1997, 375.

FURTHER DESCRIPTIVE INFORMATION

When the type strain is observed by electron microscopy and negative staining, the cells exhibit numerous blebs and vesicles on their surface, and an S layer composed of repeating units covers the surfaces of the blebs and vesicles. Poly-β-hydroxybutyrate is not accumulated.

Temperature range for growth, 4–41°C; optimum, 37°C. The NaCl range for growth is 0.01–2 M; optimum, 0.1–0.5 M.

The type strain contains the following major fatty acids: C_{16:0}, C_{10:0} 3OH, C_{10:0}, C_{12:0}, C_{18:0}, C_{18:1} ω7c, C_{18:1} ω9c, C_{18:1} ω12c, C_{16:1} ω7c, and C₁₅ iso 2OH.

ENRICHMENT AND ISOLATION PROCEDURES

The type strain was isolated from a marine enrichment community growing on the high molecular weight fraction of a black liquor sample from pulp mill effluent, which is rich in lignin, lignin byproducts, and other plant polymers in a smaller proportion (González et al., 1997). The original inoculum was from

a salt marsh on the coast of Georgia, U.S.A. The enrichment medium consists of filter-sterilized seawater containing 5 µM NH₄NO₃, 1 µM KH₂PO₄, and the liquor fraction at a concentration of 20 mg/l. This medium is inoculated with cloudy, brown-green seawater from the salt marsh. Flasks are incubated aerobically with shaking. After enrichment, the organism is isolated on YTSS¹ agar plates incubated at room temperature.

MAINTENANCE PROCEDURES

The type strain is maintained in YTSS broth with 15% glycerol and 15% DMSO at –70°C. Lyophilized cultures are also used.

DIFFERENTIATION OF THE GENUS *MARINOBACTERIUM* FROM OTHER GENERA

Table BXII.γ.143 (see previous page) lists characteristics differentiating *Marinobacterium* from other aerobic marine bacteria. The NaCl requirement for growth excludes *Marinobacterium* from the family *Halomonadaceae* and the genus *Marinobacter*. The mol% G + C is different from values of members of the genera *Alcaligenes*, *Oceanospirillum*, *Marinomonas*, and *Microbulbifer*.

TAXONOMIC COMMENTS

Based on 16S rRNA sequence analysis, *Marinobacterium* is related to *Pseudomonas*, *Oceanospirillum*, *Marinomonas*, *Microbulbifer*, *Halomonas*, and *Marinobacter*. However, the percent similarity with any of these genera is less than 90%, indicating that *Marinobacterium* is a distinct genus.

1. YTSS agar (g/l distilled water): yeast extract (Difco Laboratories, Detroit, MI, USA), 4.0; tryptone (Difco), 2.5; sea salts (Sigma Chemical Co., St. Louis, MO), 20; and agar, 18. YTSS broth is the same except that agar is omitted.

List of species of the genus Marinobacterium

1. ***Marinobacterium georgiense*** González, Mayer, Moran, Hodson and Whitman 1997, 375^{VP}
geor.gi.en'se. L. adj. *georgiense* from Georgia, U.S.A., referring to the place where it was first isolated.

The description is as given for the genus and as given in Table BXII.γ.143. In addition, fimbriae are present and small capsules are produced.

Colonies on Marine Agar 2216 (Difco Laboratories, Detroit, Mich.) are translucent.

Growth occurs between pH 5.5 and 9.5; optimum, 7.5. The NaCl range for growth is 0.01–2.0 M, optimum, 0.1–0.5 M.

Growth occurs on the C₁ compounds methanol and formate, organic acids, monosaccharides, disaccharides, amino acids, and aromatic compounds. Included among the aromatic compounds are lignin-related compounds, such as *p*-coumarate, cinnamate, ferulate, and vanillate. Phenol, *p*-hydroxybenzoate, and benzoate are also metabolized.

Cellulose, chitin, gelatin, starch, agar, and xylan are not hydrolyzed.

The mol% G + C of the DNA is: 54.9 (HPLC).

Type strain: KW-40, ATCC 700074, CIP 105236, DSM 11526.

GenBank accession number (16S rRNA): U58339.

Genus IX. Microbulbifer González, Mayer, Moran, Hodson and Whitman 1997, 375^{VP}

JOSÉ M. GONZÁLEZ

Mi.cro.bul'bi.fer. Gr. adj. *micro* small; L. masc. n. *bulbus* bulb; suff. *-fer* carrying, bearing; L. masc. n. *Microbulbifer* small bearer of bulbs.

Rod-shaped cells, 0.3–0.5 × 1.1–1.7 µm. Gram negative. Capsules are produced. Nonsporeforming and noncystforming. **Nonmotile.** Poly-β-hydroxybutyrate is not produced. Colonies are non-

pigmented. **Possess a strictly aerobic type of metabolism with oxygen as the terminal electron acceptor.** Denitrification does not occur. **Oxidase and catalase positive. A sea salt-based medium**

is required for growth. Chemoorganotrophic, growing on sugars, fatty acids, aromatic compounds, and amino acids. Cellulose, xylan, chitin, gelatin, starch, and Tween 80 are hydrolyzed. Related to but outside the group of true *Pseudomonas* in the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 57.6.

Type species: ***Microbulbifer hydrolyticus*** González, Mayer, Moran, Hodson and Whitman 1997, 375.

FURTHER DESCRIPTIVE INFORMATION

The cell envelope has numerous surface vesicles derived from the outer membrane.

Microbulbifer contains the following major fatty acids: C_{15:0} iso, C_{17:1} iso ω9c, C_{11:0} iso 3OH, C_{11:0} iso, C_{17:0} iso, C_{15:0}, C_{17:1} ω8c, C_{17:0}, C_{18:1} ω7c, C_{18:1} ω9t, C_{18:1} ω12t, C_{16:1} ω7c, and C₁₅ iso 2OH.

ENRICHMENT AND ISOLATION PROCEDURES

The type strain was isolated from a marine enrichment community growing on the high-molecular-weight fraction of a black liquor sample from pulp mill effluent, which is rich in lignin, lignin byproducts, and other plant polymers in a smaller proportion (González et al., 1997). The original inoculum was from a salt marsh on the coast of Georgia. The enrichment medium consists of filter-sterilized seawater containing 5 μM NH₄NO₃, 1 μM KH₂PO₄, and the liquor fraction at a concentration of 20 mg/l. This medium is inoculated with cloudy, brown-green sea water from the salt marsh. Flasks are incubated aerobically with shaking. After enrichment, the organism is isolated on YTSS¹ agar plates incubated at room temperature.

List of species of the genus *Microbulbifer*

1. ***Microbulbifer hydrolyticus*** González, Mayer, Moran, Hodson and Whitman 1997, 375^{VP}

hy.dro.ly'ti.cus. Gr. n. *hydor* water; Gr. adj. *lyticus* dissolving, splitting; M.L. adj. *hydrolyticus* splitting with water, referring to the hydrolytic activity of the bacterium.

Cells are rod shaped, 0.3–0.5 × 1.1–1.7 μm. Cells appear as single cells or short chains. In medium with xylan or viscous medium, they form long chains. They do not possess flagella. An S layer is present. Capsules are produced in liquid media. Polyhydroxybutyrate is not produced.

Colonies on marine agar 2216 are circular, convex with entire margins, and cream colored.

Growth occurs between pH 6.5 and 8.5; optimum, 7.5. The temperature range for growth is 10–41°C; optimum,

MAINTENANCE PROCEDURES

The type strain is maintained in YTSS broth with 15% glycerol and 15% DMSO at –70°C. Lyophilized cultures are also used.

DIFFERENTIATION OF THE GENUS *MICROBULBIFER* FROM OTHER GENERA

The NaCl requirement for growth excludes *Microbulbifer* from the family *Halomonadaceae* and the genus *Marinobacter*. The mol% G + C content of the DNA is different from that of members of the genera *Oceanospirillum*, *Marinomonas*, and *Marinobacter*. *Microbulbifer* is able to use a limited range of carbohydrates, in contrast to most *Pseudomonas* species. The fatty acid profile also differs from those of other bacteria. Among other characteristics, the fatty analysis reveals a low resemblance with *Marinobacterium*. Although the characteristics of *Microbulbifer* are very similar to those of *Alteromonas*, the mol% G + C is outside the range of this genus.

TAXONOMIC COMMENTS

Based on its 16S rRNA sequence, besides *Pseudomonas*, other closely related bacteria are the marine genera *Oceanospirillum*, *Marinomonas*, *Marinobacterium*, *Halomonas*, and *Marinobacter*.

1. YTSS agar (g/l distilled water): yeast extract (Difco Laboratories, Detroit, MI, USA), 4.0; tryptone (Difco), 2.5; sea salts (Sigma Chemical Co., St. Louis, MO), 20; and agar, 18. YTSS broth is the same except that agar is omitted.

37°C. Sea salt-based medium is required for growth. The NaCl range for growth is 0.1–1.0 M; optimum, 0.1–0.5 M. Vitamins are required for optimal growth.

Growth occurs on organic acids, amino acids, and a few monosaccharides, disaccharides, and aromatic compounds (glucose, xylose, N-acetyl-D-glucosamine, cellobiose, vanillate, and ferulate). Cellulose, xylan, chitin, gelatin, starch, and Tween 80 are hydrolyzed. This species has been found by oligonucleotide probes to be numerically dominant in a pulp mill effluent enrichment community containing only bacteria.

The mol% G + C of the DNA is: 57.6 (HPLC).

Type strain: IRE-31, ATCC 700072; CIP 105235, DSM 11525.

GenBank accession number (16S rRNA): U58338.

Genus X. *Moritella* Urakawa, Kita-Tsukamoto, Steven, Ohwada and Colwell 1999b, 341^{VP}
(Effective publication: Urakawa, Kita-Tsukamoto, Steven, Ohwada and Colwell 1998, 376)

THE EDITORIAL BOARD

Mo.ri.tel'la. M.L. dim. *-ella* ending; M.L. fem. n. *Moritella* named after Richard Y. Morita to honor his work in marine microbiology.

Gram-negative, chemoorganotrophic, facultatively anaerobic motile rods. Polar flagella. **Halophilic.** Catalase and oxidase positive. Voges–Proskauer negative. Negative for production of indole and H₂S. Do not produce arginine dihydrolase or ornithine decar-

boxylase. Produce acid from D-glucose. Utilize N-acetyl glucosamine.

The mol% G + C of the DNA is: 40–45.

Type species: ***Moritella marina*** (Baumann, Furniss and Lee 1984c) Urakawa, Kita-Tsukamoto, Steven, Ohwada and Colwell

1999b, 341 (Effective publication: Urakawa, Kita-Tsukamoto, Steven, Ohwada and Colwell 1998, 376) (*Vibrio marinus* Baumann, Furniss and Lee 1984c, 356.)

FURTHER DESCRIPTIVE INFORMATION

The genus *Moritella* includes four species; all inhabitants of marine environments. *M. marina* was isolated from fish skin, seawater, and marine sediment (Colwell and Morita, 1964; Urakawa et al., 1998). *Moritella japonica*, which is barotolerant, was obtained from sediments from the Japan Trench at a depth of 6,356 m (Nogi et al., 1998a). *M. yayanosii*, which requires pressures of 50 MPa or greater, was obtained from sediments from the Mariana Trench at a depth of 10,898 m (Nogi and Kato, 1999a).

M. viscosa is an etiologic agent of the disease "winter ulcer" in Atlantic salmon (*Salmo salar*) and may cause skin disease in other species of fish (Lunder et al., 2000).

Analysis of 16S rDNA sequences placed the genus *Moritella* in the *Gammaproteobacteria* (Urakawa et al., 1998).

ENRICHMENT AND ISOLATION PROCEDURES

Sampling and isolation procedures for *Vibrio marinus* (*Moritella marina*) from seawater are described by Colwell and Morita (1964).

Moritella japonica and *Moritella yayanosii* were isolated by diluting bottom mud sediments in Marine Broth 2216[™] (Difco)

and incubating at 4–15°C either at atmospheric pressure or in a pressurized vessel (Kato et al., 1995; Nogi et al., 1998a; Nogi and Kato, 1999a).

Moritella viscosa can be isolated from tissue samples on blood agar made 0.5–2.0% in NaCl as described by Lunder et al. (1995).

MAINTENANCE PROCEDURES

Media and conditions for maintenance of *Vibrio marinus* (*Moritella marina*) are described by Colwell and Morita (1964).

Moritella japonica and *Moritella yayanosii* can be maintained on Marine Agar 2216[™] (Difco) with incubation at 4–15°C either at atmospheric pressure or in a pressurized vessel as appropriate (Kato et al., 1995; Nogi et al., 1998a; Nogi and Kato, 1999a).

Moritella viscosa strains can be maintained at 4°C on Difco brain–heart infusion agar made 2% in NaCl or on Difco Marine Agar 2216[™]; long-term storage can be achieved at –80°C in brain–heart infusion broth made 2% in NaCl and 30% in glycerol or in Marine Agar 2216[™] made 10% in glycerol (Benediktsdóttir et al., 2000; Lunder et al., 2000).

DIFFERENTIATION OF THE GENUS *MORITELLA* FROM OTHER GENERA

Tables of phenotypic characteristics that differentiate *Moritella* from species of the genera *Shewanella* and *Vibrio* are available (Nogi et al., 1998a; Nogi and Kato, 1999a; Lunder et al., 2000).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *MORITELLA*

Nogi et al. (1998a), Nogi and Kato (1999a), and Lunder et al. (2000), provide tables of phenotypic characteristics that differentiate *Moritella* species from each other.

List of species of the genus *Moritella*

1. ***Moritella marina*** (Baumann, Furniss and Lee 1984c) Urakawa, Kita-Tsukamoto, Steven, Ohwada and Colwell 1999b, 341^{VP} (Effective publication: Urakawa, Kita-Tsukamoto, Steven, Ohwada and Colwell 1998, 376) (*Vibrio marinus* Baumann, Furniss and Lee 1984c, 356)

ma.ri'na. L. fem. adj. *marina* marine, of the sea.

Description as for the genus with the following additional characteristics. Nitrate reduced to nitrite. Optimum growth temperature 18°C. Major fatty acids C_{14:0}, C_{16:0}, C_{16:1}, C_{22:6}.

The mol% G + C of the DNA is: 42.2 (*T_m*).

Type strain: ATCC 15381, CIP 102861, NCCB 79030, NCIMB 1144.

GenBank accession number (16S rRNA): AB038033, X82142.

2. ***Moritella japonica*** Nogi, Kato and Horikoshi 1999, 341^{VP} (Effective publication: Nogi, Kato and Horikoshi 1998a, 294.)

ja.pon.i'ca. L. gen. n. *japonica* named after the Japan Trench, where this strain originated.

Description as for the genus with the following additional characteristics. Psychrophilic (growth optimum 10–15°C) and halophilic (growth optimum 3% NaCl). Able to grow both at atmospheric pressure and at pressures up to 70 MPa. Nitrate reduced to nitrite. Lysine decarboxylase produced. Acid produced from glycerol. Major fatty acids C_{14:0}, C_{16:0}, C_{16:1}, C_{22:6}.

The mol% G + C of the DNA is: 45 (HPLC).

Type strain: DSK1, CIP 106291, JCM 10249.

GenBank accession number (16S rRNA): D21224.

3. ***Moritella viscosa*** (Lunder, Sørsum, Holstad, Steigerwalt, Mowinckel and Brenner 2000) Benediktsdóttir, Verdonck, Sproer, Helgasón and Swings 2000, 487^{VP} (*Vibrio viscosus* Lunder, Sørsum, Holstad, Steigerwalt, Mowinckel and Brenner 2000, 447.)

vis.co'sa. L. fem. adj. *viscosa* viscous, because of its thread-forming, adherent colonies.

Description as for the genus with the following additional characteristics. Requires 1–4% NaCl. Lysine decarboxylase produced. Hydrolysis of casein, DNA, gelatin, lecithin, starch (19 of 20 strains), Tween 80, and urea. Acid produced from D-galactose.

The mol% G + C of the DNA is: 42.5 (*T_m*).

Type strain: NVI 88/478, ATCC BAA-105, NCIMB 13584.

GenBank accession number (16S rRNA): Y17574.

4. ***Moritella yayanosii*** Nogi and Kato 1999b, 1325^{VP} (Effective publication: Nogi and Kato 1999a, 76.)

ya.ya.nos.i'i. M.L. gen. n. *yayanosii* in honor of American deep-sea biologist Aristides Yayanos.

Description as for the genus with the following additional characteristics. Cannot grow at pressures less than 50 MPa; able to grow at pressures up to 100 MPa. Requires NaCl. No reduction of nitrate or nitrite. Major fatty acids C_{14:0}, C_{16:0}, C_{16:1}, C_{22:6}.

The mol% G + C of the DNA is: 44.6 (HPLC).

Type strain: DB21MT-5, JCM 10263.

GenBank accession number (16S rRNA): AB008797.

Genus XI. *Pseudoalteromonas* Gauthier, Gauthier and Christen 1995a, 759^{VP}

JOHN P. BOWMAN AND THOMAS A. MCMEEKIN

Pseu.do.al.te.ro.mo' nas. Gr. adj. *pseudes* false; M.L. n. *Alteromonas* the genus *Alteromonas*; M.L. n. *Pseudoalteromonas* false *Alteromonas*.

Straight rods, approximately $1 \times 2-3 \mu\text{m}$, occurring singly or in pairs. Do not form spores or cysts. Do not accumulate poly- β -hydroxybutyrate. Nonbioluminescent. **Gram negative. Motile by means of a single polar flagellum**, either unsheathed or sheathed. Colonies are nonpigmented or pigmented. **Chemoheterotrophic. Aerobic, possessing a strictly respiratory type of metabolism** with oxygen as the terminal electron acceptor. No growth occurs anaerobically. A constitutive arginine dihydrolase system is usually absent. One species (*P. denitrificans*) is able to denitrify. Utilize a range of carbon substrates, including carbohydrates, alcohols, organic acids, and amino acids. Form gelatinase and lipase. Ammonia serves as a nitrogen source. Amino acids may be required by some strains for growth. **Requires Na^+ ions for growth.** Mostly psychrotolerant, with growth occurring between 0 and 40°C. Several species produce autotoxic antibiotic compounds. Isolated from seawater, sediment, sea ice, surfaces of stones, marine algae, marine invertebrates, and salted foods. The major quinone is ubiquinone-8 (Q-8). Major fatty acids are $\text{C}_{16:0}$, $\text{C}_{16:1 \omega 7\text{C}}$, $\text{C}_{17:1 \omega 8\text{C}}$, and $\text{C}_{18:1 \omega 7\text{C}}$. Member of the order *Alteromonadales*, family *Alteromonadaceae*, in the *Gamma*proteobacteria.

The mol% G + C of the DNA is: 38–48 (T_m).

Type species: *Pseudoalteromonas haloplanktis* (Zobell and Upham 1944) Gauthier, Gauthier and Christen 1995a, 759 (*Alteromonas haloplanktis* (Zobell and Upham 1944) Reichelt and Baumann 1973a, 438; "*Vibrio haloplanktis*" Zobell and Upham 1944, 261.)

FURTHER DESCRIPTIVE INFORMATION

Motility Sheathed flagella occur on cells of *P. denitrificans* (Enger et al., 1987), *P. luteoviolacea* (Novick and Tyler, 1985), and *P. piscicida* (Hansen et al., 1963, 1965). Cells occasionally form three or more polar flagella. Older cultures often exhibit significantly reduced levels of motile cells, and filamentous cells and involution bodies often occur.

Cultural characteristics Colonies on Difco Marine 2216 agar are typically circular and convex, possessing an entire edge and a creamy or mucoid consistency. Colonies are nonpigmented or pigmented with noncarotenoid type pigments. When growing on tyrosine-containing media, some species form black-brown, melanin-like pigments (Table BXII.7.144). In some instances, such as in the type strain of *P. nigrifaciens* (ATCC 27126), melanin production occurs constitutively on complex media. *P. citrea*, *P. aurantia*, and *P. piscicida* form yellow or orange pigments. Spectral and preliminary chemical analyses of the pigments of *P. piscicida* suggest the pigments are pteridine derivatives (Weeks et al., 1962). Both *P. rubra* and *P. denitrificans* form bright red prodigiosin-type pigments. When *P. denitrificans* is grown on protamine or L-glutamine, the color of cultures changes from red to bright purple or blue (Enger et al. 1987). Strains of *P. luteoviolacea* form violet colonies owing to accumulation of violacein and can simultaneously form yellow pigments, giving colonies an unusual and characteristic appearance (Gauthier, 1982).

Pseudoalteromonas are either psychrotrophic or mesophilic. No psychrophilic strains, however, have been isolated, even from the cold-water and sea-ice environments of the Antarctic (Bozal et al., 1997; Bowman, 1998).

Antibiotic production Most pigmented and a few nonpigmented *Pseudoalteromonas* strains have the ability to form antibiotic substances. The red pigment of *P. denitrificans*, identified as cycloprodigiosin hydrochloride, has been shown to suppress T-cell proliferation and has been proposed as a immunosuppressant therapeutic agent. Cycloprodigiosin hydrochloride apparently selectively uncouples proton translocation from the ATPase reaction (Kawauchi et al., 1997). Strains of *P. citrea*, *P. aurantia*, *P. rubra*, and *P. luteoviolacea* form autotoxic, high-molecular-weight polyanionic substances (70–100 kDa in size), which can inhibit Gram-negative and Gram-positive bacteria (Gauthier, 1976b; Gauthier and Flatau, 1976; Ballester et al., 1977). Analysis of the agent produced by *P. rubra* indicates it is an outer membrane glycoprotein of complex composition (Ballester et al., 1977). *P. citrea* produces two different proteinaceous components, as determined by electrophoretic profiles (Gauthier, 1977). The mode of action of these agents involves inducing an increase in the rate of oxygen consumption in susceptible bacteria, leading to lethal cell damage due to accumulation of oxygen free radicals (Gauthier, 1976b). In addition to this, *P. luteoviolacea* can form a variety of low-molecular-weight brominated compounds, including pentabromopseudilin, which are bacteriolytic (Gauthier and Flatau, 1976; Sakata et al., 1982, 1986; Hanefeld et al., 1994). *P. haloplanktis* forms an unusually bioactive iron siderophore, bisucaberin, which is able to block DNA synthesis in tumor cells and induce macrophage-mediated cytotoxicity (Kameyama et al., 1987; Takahashi et al., 1987). Siderophore production by *Pseudoalteromonas* strains is thought to give an advantage in the marine environment, where iron often is very scarce (Reid and Butler, 1991; Reid et al., 1993). The ability of *Pseudoalteromonas* to produce antibiotics may have some unexpected advantages; for example, some strains of *P. haloplanktis* produce a heat-labile, proteinaceous compound able to inhibit various pathogenic *Vibrio* species, which has been proposed as a probiotic agent to protect reared scallops and oyster larvae (Riquelme et al., 1996).

The exact scope, ecological role, mechanisms, and ecological impact of antibiotic production by *Pseudoalteromonas* are under increasing investigation. Obviously, antibiotic production would help *Pseudoalteromonas* species effectively compete for potentially scarce nutrients by inhibiting other marine bacteria. Another possible role is that they act as chemical barriers to predation; for example, various marine biota have been observed to possess coats of pigmented, aerobic, rod-shaped bacteria, which produce bioactive substances (Fenical, 1993). *Pseudoalteromonas* strains, however, also have an aggressive role in the marine food chain and are able to kill various types of toxic "red-tide" dinoflagellates and raphidophytes (such as *Gymnodinium* and *Chatonella* species) using uncharacterized, proteinaceous, secreted substances (Imai et al., 1995; Lovejoy et al., 1998). *P. piscicida* has been claimed to produce a neuromuscular toxin, which is able to kill a variety of fish and crab species (Bein, 1954; Meyers et al., 1959; Hansen et al., 1965). In addition, *P. haloplanktis* subsp. *tetraodonis* can apparently form tetrodotoxin, an exceptionally potent neurotoxin that has been isolated from the puffer fish (*Fugu poecilonotus*), from which this subspecies was first described (Yasumoto

TABLE BXII.γ.144. Characteristics differentiating the species of the genus *Pseudoalteromonas*^a

Characteristics	<i>P. haloplanktis</i>	<i>P. antarctica</i>	<i>P. atlantica</i>	<i>P. aurantia</i>	<i>P. carrageenovora</i>	<i>P. citrea</i>	<i>P. denitrificans</i>	<i>P. espejana</i>	<i>P. luteoviolacea</i>	<i>P. nigrifaciens</i>	<i>P. piscicida</i>	<i>P. prydzensis</i>	<i>P. rubra</i>	<i>P. tetradonidis</i>	<i>P. undina</i>
Sheathed polar flagellum	—	—	—	—	—	—	+	—	d	—	+	—	—	—	—
Pigments:															
Yellow-orange	—	—	—	+	—	d	—	—	d	—	+	—	—	—	—
Red (prodigiosin)	—	—	—	—	—	—	+	—	—	—	—	—	+	—	—
Violet (violacein)	—	—	—	—	—	—	—	—	+	—	—	—	—	—	—
Melanin-like	+	—	+	+	d	d	nd	—	d	+	+	—	nd	—	—
NO ₃ →NO ₂	d	—	d	—	—	—	+	—	—	—	—	—	—	+	—
Denitrification	—	—	—	—	—	—	+	—	—	—	—	—	—	—	—
Catalase	+	+	+	+	+	d	+	+	—	+	d	+	—	+	+
Agarase	—	—	+	—	—	d	—	—	—	—	—	—	—	—	—
Chitinase	d	—	—	—	—	—	+	—	—	—	+	+	—	+	+
Alginase	—	—	+	+	+	+	+	+	+	—	+	+	+	+	—
Amylase	d	d	+	—	+	+	+	+	nd	—	—	—	nd	+	d
Growth at:															
4°C	—	+	+	+	+	d	+	—	d	d	+	+	—	+	d
35°C	+	—	+	—	+	d	—	d	+	d	+	—	+	+	—
Tolerates 9% NaCl	+	+	+	—	+	—	—	+	—	+	+	+	—	nd	+
Amino acid requirement	d	—	—	+	—	d	+	+	+	—	+	—	+	+	+
Antibiotic production	d	nd	nd	+	nd	d	+	—	+	nd	+	nd	+	+	—
Utilization of:															
D-Ribose	—	—	—	—	—	+	nd	—	—	+	—	nd	—	—	—
D-Fructose	d	—	+	+	+	+	—	d	—	+	d	—	—	—	—
D-Galactose	d	+	+	—	+	—	—	+	—	+	d	d	—	—	—
D-Mannose	+	d	+	+	—	d	nd	d	—	d	+	+	+	—	—
Cellobiose	—	d	+	—	+	d	—	d	—	d	—	d	—	—	—
Lactose	—	—	+	—	+	d	—	+	—	d	—	—	—	—	—
Maltose	+	+	+	d	+	—	+	+	+	+	+	+	—	+	+
Melibiose	—	—	+	—	+	nd	nd	+	—	+ ^c	—	—	nd	—	—
Sucrose	+	—	+	—	+	d	—	+	—	+	+	+	—	+	+
Trehalose	d	—	+	+	—	d	nd	+	+	+	+	+	+	+	d
Mannitol	d	+	+	—	+	d	—	+	—	+	—	+	—	—	—
D-Gluconate	—	d	—	—	—	d	—	—	—	d	—	+	—	—	—
Glycerol	—	+	d	—	+	d	—	—	—	d	—	+	—	—	—
Acetate	+	—	+	—	+	d	+	+	—	d	+	+	—	+	+
Succinate	+	+	d	—	+	+	—	—	—	d	+	+	—	+	+
Pyruvate	+	nd	nd	—	nd	+	+	+	d	+	+	+	—	+	+
Fumarate	+	nd	d	—	+	d	—	—	—	d	+	+	—	+	+
DL-Malate	—	—	—	—	+	—	nd	—	—	—	+	+	—	+	—
DL-Lactate	—	—	—	—	—	d	—	—	d	+	—	—	—	—	—
Citrate	+	d	+	—	+	d	—	+	—	—	+	+	—	—	—
2-Oxoglutarate	—	nd	—	—	+	—	nd	—	—	—	—	—	—	—	—
Aconitate	+	nd	+	—	—	+	nd	+	nd	+	—	+	nd	—	—
p-Hydroxybenzoate	—	—	—	—	+	+	—	—	—	+	—	nd	nd	—	—
DL-α-Alanine	+	+	—	—	—	d	+	+	+	+	—	—	nd	+	+
L-Glutamate	d	+	—	—	—	+	+	—	—	+	—	+	nd	+	—
L-Ornithine	—	nd	—	—	—	+	—	d	—	+	—	—	nd	—	d
L-Proline	d	+	—	—	—	+	+	+	d	+	d	+	nd	+	d
Mol% G + C (<i>T_m</i> , Bd)	42–44	41–42	41–42	38–43	39–40	39–45	36–37	43–44	40–43	39–41	44–45	38–39	46–48	41	43–44

^aFor symbols see standard definitions; nd, not determined.

et al., 1986; Simidu et al., 1987, 1990; Gallacher and Birkbeck, 1993). *Pseudoalteromonas* spp. are generally not pathogenic; however, *P. haloplanktis* may cause disease in oysters (Colwell and Sparks, 1967) and the red-spot disease of the seaweed *Laminaria japonica* (Ezura et al., 1988a, b; Sawabe et al., unpublished).

Sodium ion requirement All *Pseudoalteromonas* species require Na⁺ ions for growth; the minimal levels vary somewhat, but are usually in the range of 20 mM. Studies have shown that the Na⁺ requirement cannot be replaced by supplying K⁺ ions

but can be partially reduced in some strains by supplying seawater levels of Ca²⁺, Na²⁺, and Mg²⁺ (Thompson et al., 1970; Thompson and MacLeod, 1973). Na⁺ is necessary for the activity of cellular uptake systems in *Pseudoalteromonas*, including the transport of amino acids, tricarboxylic acid intermediates, galactose, orthophosphate, and K⁺ (MacLeod, 1968; Fein and MacLeod, 1975; Droniuk et al., 1987; MacLeod and MacLeod, 1992). Na⁺ ions are also necessary for maintaining the integrity of the cell wall (Forsberg et al., 1970). Most nonpigmented *Pseudoaltero-*

monas species demonstrate some degree of halotolerance and can grow at NaCl concentrations of 9–12%. Strains of *P. frydzensis* can even tolerate NaCl levels of 15–20% (Bowman, 1998). Pigmented species appear to be more stenohaline in nature, with the exception of *P. piscicida*.

Metabolism *Pseudoalteromonas* species are strictly aerobic chemoheterotrophs using a respiratory metabolism with oxygen as the electron acceptor. *P. denitrificans* is able to grow microaerobically by denitrification (Enger et al., 1987), while some strains of *P. haloplanktis* and *P. atlantica* can reduce nitrate to nitrite. A constitutive arginine dihydrolase system is generally absent. Glucose and fructose are assimilated by an inducible Entner–Doudoroff pathway (Sawyer et al., 1977a). All strains possess cytochrome *c* oxidase activity. Catalase activity is usually present, though it may be weakly expressed or absent in certain pigmented species. *P. luteoviolacea* and *P. rubra* possess peroxidase (H_2O_2 oxidoreductase) instead of catalase (Gauthier, 1976a, 1982). The following tests have been found to be negative for all tested species: lysine decarboxylase, ornithine decarboxylase, tryptophan deaminase, phenylalanine deaminase, indole production from tryptophan, fermentation of D-glucose, methyl red test, anaerobic growth with nitrate as the electron acceptor, and sensitivity to vibriostat agent O/129.

Nutrition *Pseudoalteromonas* strains grow well on marine 2216 agar and similar seawater media. Though most strains can use ammonia as a sole nitrogen source, some species, in particular the pigmented species, require amino acids for growth. For these strains, growth in defined media can be achieved by the addition of individual amino acids (Baumann et al., 1984a, c) or by addition of low levels of yeast extract or casein hydrolysate (0.1–0.5% w/v). Media useful for growing *Pseudoalteromonas* species are given in the section on the genus *Alteromonas*.

Pseudoalteromonas species can utilize a range of organic compounds as sole carbon and energy sources, including carbohydrates, alcohols, and amino acids. Pigmented species in general tend to be less nutritionally versatile than the nonpigmented species. *Pseudoalteromonas* species also have the ability to degrade a range of macromolecules. Most species can degrade gelatin, casein, starch, and DNA, and can form a lipase. Several species possess chitinolytic and alginolytic activity. *P. atlantica* possesses a type I β -agarase, the gene for which has been cloned, sequenced, and overexpressed in *Escherichia coli* (Belas, 1989). *P. citrea* may also form agarases (Ivanova et al., 1998), and *P. carragenovora* possesses the unusual ability to degrade κ -carrageenan (Yaphe and Baxter, 1955) by means of a β -1,3-1,4-glucanase-like κ -carrageenase (Barbeyron et al., 1994). The amylase from *P. haloplanktis* strain A23 is cold adapted and thermolabile (Feller et al., 1992). *P. undina* synthesizes an efficient organophosphorus acid anhydrolase (Cheng et al., 1993).

Genetic characteristics The genome structure of *P. haloplanktis* is complex. Analysis by restriction digestion and pulsed-field gel electrophoresis has detected the presence of two genetic units, one with a length of 2.7 Mb and a second replicon with a length of 0.8 Mb (Lanoil et al., 1996).

Lipids, quinones, polyamines The major fatty acids in *Pseudoalteromonas* include $C_{15:0}$ (2–9%), $C_{15:1\omega8c}$ (1–7%), $C_{16:0}$ (18–34%), $C_{16:1\omega7c}$ (35–50%), $C_{17:1\omega8c}$ (1–13%), and $C_{18:1\omega7c}$ (1–9%) (Monteoliva-Sanchez and Ramos-Cormenzana, 1987; Matsui et al., 1991; Svetashev et al., 1995; Bozal et al., 1997; Bowman, 1998). So far, fatty acid characterization has been done only for some

of the nonpigmented species. In any case, the profiles are so similar to that of *Alteromonas macleodii* that neural network analysis cannot tell the difference (Bertone et al., 1996). The major isoprenoid quinone is ubiquinone-8 (Q-8, 84–100%), with smaller amounts of Q-6 (0–9%), Q-7 (0–6%), and Q-9 (0–15%) present (Akagawa-Matsushita et al., 1992b). The major cellular polyamines in *Pseudoalteromonas* include putrescine, cadaverine, and spermidine (Hamana, 1997). Some species of *Pseudoalteromonas* synthesize unusual acidic exopolysaccharides (Gorshkova et al., 1993, 1997; Bozal et al., 1994).

Ecology Due to the relatively limited scope of individual studies and often inadequate identification of isolates, it is difficult to obtain a precise idea of how widely distributed each *Pseudoalteromonas* species is. However, strong evidence indicates the genus is common and widespread in the marine ecosystem, with strains isolated from coastal and estuarine seawater, organic detritus, deep water, benthic sediments, sea ice, surfaces of marine fauna and flora, aquaculture systems, and salterns (Kakimoto et al., 1980; Gavrilovic et al., 1982; Sakata et al., 1982; Austin, 1982; Kameyama et al., 1987; Lemos et al., 1985; Nair and Simidu, 1987; Nissen, 1987; Oliver, 1987; Ezura et al., 1988a; del Moral et al., 1988; Nair et al., 1988; Noble et al., 1990; Gauthier and Breittmayer, 1992; Bozal et al., 1994, 1997; Ortigosa et al., 1994; Uchida et al., 1997; Yoshikawa et al., 1997; Bowman, 1998). In addition, strains of *Pseudoalteromonas nigrifaciens* have also been isolated from various salted food products, including butter, from which it was originally described (White, 1940). *P. haloplanktis* seems to have a very wide distribution, having been isolated from all major oceans, save for the polar regions (Gauthier and Breittmayer, 1992). *P. antarctica* appears to be very common in the Antarctic marine environment and is one of the most common isolates from seawater and sea ice (Bozal et al., 1997; Bowman et al., 1997a; Bowman, 1998). Pigmented species are often very sensitive to UV irradiation, thus affecting potential isolation sites (Gauthier and Breittmayer, 1992; Enger et al., 1987).

ENRICHMENT AND ISOLATION PROCEDURES

The isolation of specific *Pseudoalteromonas* species is dependent on recognition of certain obvious phenotypic properties, especially pigmentation (Gauthier and Breittmayer, 1992). In most instances, special enrichment or selection conditions for *Pseudoalteromonas* species are not available, and the organisms usually can only be directly isolated from suitable source material onto marine agar or BMA supplemented with 0.1–0.2% concentration of a particular carbon source that is utilized by the species in question. Certain species have been isolated by enrichment, including *P. denitrificans*, which was isolated from a medium containing L-arginine and L-glutamine as sole carbon and nitrogen sources (Enger et al., 1987). Antibiotic production of colonies on isolation plates can be used to screen for antibiotic-producing *Pseudoalteromonas* species by means of the replica plating procedure (Smibert and Krieg, 1994), as described by Gauthier and Breittmayer (1992). For more details on methods useful in isolating *Pseudoalteromonas* species, see Gauthier and Breittmayer (1992).

MAINTENANCE PROCEDURES

Most nonpigmented strains can be stored on marine agar plates or slants at 2–5°C for several months before subculturing is necessary. Pigmented strains, however, often produce autotoxic antibiotic compounds and so need to be transferred at least once

per week. *Pseudoalteromonas* strains can be successfully stored for several years as dilute suspensions in sterile, unsupplemented seawater at 7–8°C (Enger et al., 1987). They can also be preserved by freeze-drying with 20% skim milk as a cryoprotectant or by freezing in liquid nitrogen or in a –80°C freezer with 20–30% glycerol as a cryoprotectant.

DIFFERENTIATION OF THE GENUS *PSEUDOALTEROMONAS* FROM OTHER GENERA

Characteristics differentiating the genus *Pseudoalteromonas* from other members of the family *Alteromonadaceae* are shown in Table BXII.γ.135 of the chapter describing the family. Nonpigmented *Pseudoalteromonas* species are most likely to be confused with the genus *Alteromonas*, with which they share many phenotypic and chemotaxonomic similarities. In broad terms, *Pseudoalteromonas* species can be distinguished by their ability to grow at 4°C and inability to utilize a number of carbon substrates. These few phenotypic traits seem inadequate, but until a wider knowledge of the phenotypic differences between these genera are available, the only other effective methods for the conclusive identification are 16S rDNA sequencing, DNA–DNA hybridization, and rRNA–DNA hybridization.

TAXONOMIC COMMENTS

Pseudoalteromonas was created to accommodate most of the former *Alteromonas* species that comprise a phylogenetic clade distinct from *Alteromonas macleodii* (van Landschoot and De Ley, 1983; Kita-Tsukamoto et al., 1993; Gauthier et al., 1995a). *Pseudoalteromonas piscicida*, previously classified as “*Pseudomonas piscicida*” (Buck et al., 1963), was also included in the new genus (Gauthier et al., 1995a).

More specifically, both rRNA–DNA hybridization and 16S rDNA sequencing studies (Kita-Tsukamoto et al., 1993; Gauthier et al., 1995a) indicate *Pseudoalteromonas* species comprise a relatively tight clade adjacent to the genus *Alteromonas*. Fig. BXII.γ.155 shows a phylogenetic tree based on 16S rDNA sequences for *Pseudoalteromonas*. One interesting feature is the high

sequence similarity shared by the nonpigmented species; indeed, six species possess sequences that diverge less than 2%—well within the potential estimated range of sequence divergence for a single species (Stackebrandt and Goebel, 1994). *P. prydzensis* is an outlier of this group. The pigmented species show a greater divergence in sequence than do the nonpigmented species, except for *P. citrea* and *P. aurantia*, which are very closely related.

DNA–DNA (Akagawa-Matsushita et al., 1993) and rDNA–DNA hybridization (van Landschoot and De Ley, 1983) results confirm the very close relatedness of the nonpigmented strains. Overall $T_{m(e)}$ values range from 74.0–77.5°C, while DNA–DNA hybridization values range from 30–60%. *P. piscicida* has a $T_{m(e)}$ of 71.5°C, which corresponds with its more distinct phylogenetic position. On the basis of DNA–DNA hybridization data, Ivanova et al. (1998) have indicated that the species *Alteromonas distincta* (Romanenko et al., 1995b) and *Alteromonas elyakovii* (Ivanova et al., 1996b) are closely related to *P. nigrifaciens* (sharing 40–50% DNA hybridization) while *Alteromonas fuliginea* (Romanenko et al., 1994) is a junior subjective synonym of *P. citrea*.

Several strains are novel, yet to be described, species, based on their phylogenetic positions (Fig. BXII.γ.155). Various sea-ice and under-ice seawater strains (strains IC006, IC013, MB6-03, MB6-05, MB8-02, SW08, and SW29) are distributed throughout the nonpigmented *Pseudoalteromonas* clade; however, based on phenotypic data and DNA hybridization, they are closely related to *P. antarctica* (Bowman, 1998; unpublished data).

Several melanin-producing strains isolated from marine fauna from the eastern Russian-Pacific area have been described as *Alteromonas* species. These include *A. fuliginea* (Romanenko et al., 1994), *Alteromonas distincta* (Romanenko et al., 1995b), and *A. elyakovii* (Ivanova et al., 1996b). Subsequent analysis by DNA–DNA hybridization now indicates that these species have a closer allegiance to the genus *Pseudoalteromonas*, with *A. fuliginea* being a junior subjective synonym of the species *P. citrea* (Ivanova et al., 1998). Because these species do not have their generic positions officially established yet, they have been listed as *Species Incertae Sedis*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *PSEUDOALTEROMONAS*

Characteristics distinguishing *Pseudoalteromonas* species are presented in Table BXII.γ.144. Considerable variation in carbon-source utilization patterns is evident within some species. This may be due to differences in testing procedures, because carbon source results sometimes differ significantly among different studies; examples include the results for the type strains of *P.*

haloplanktis subsp. *tetraodonis* (Simidu et al., 1990; Akagawa-Matsushita et al., 1992b) and *P. nigrifaciens* (Baumann et al., 1984a; Akagawa-Matsushita et al., 1992b; Ivanova et al., 1996a). The carbon source data shown in Tables BXII.γ.144 and BXII.γ.145 should therefore be treated with some degree of caution.

List of species of the genus *Pseudoalteromonas*

1. ***Pseudoalteromonas haloplanktis*** (Zobell and Upham 1944) Gauthier, Gauthier and Christen 1995a, 759^{VP} (*Alteromonas haloplanktis* (Zobell and Upham 1944) Reichelt and Baumann 1973a, 438; *Vibrio haloplanktis* Zobell and Upham 1944, 261.)
ha.lo.plank' tis. Gr. n. *halos* sea; Gr. adj. *planktos* wandering; M.L. adj. *haloplanktis* sea-wandering.

Cells are rod-shaped, 0.8–1.2 × 1.8–3.0 μm. Colonies are nonpigmented, smooth, and convex, with an entire edge. Sodium ions are required for growth. Growth factors (amino acids) are usually not required for growth. Grows between 10°C and 35°C, with optimum growth occurring at approximately 30°C. Some strains may grow at 4°C. Other

characteristics are shown in the genus description and in Tables BXII.γ.144 and BXII.γ.145. Strains utilize D-mannose, citrate, and aconitate, but do not hydrolyze alginate or utilize L-malate, isovalerate, caproate, or L-lysine.

Isolated from seawater.

The mol% G + C of the DNA is: 41–45 (T_m).

Type strain: 545, ATCC 14393, DSM 6060, NCIMB 2084.

GenBank accession number (16S rRNA): X67024; X82143.

2. ***Pseudoalteromonas antarctica*** Bozal, Tudela, Rosselló-Mora, Lalucat and Guinea 1997, 350.^{VP}
ant.ar' ti.ca. L. adj. *antarcticus* southern (in this instance pertaining to the Antarctic environment, where the organism was isolated).

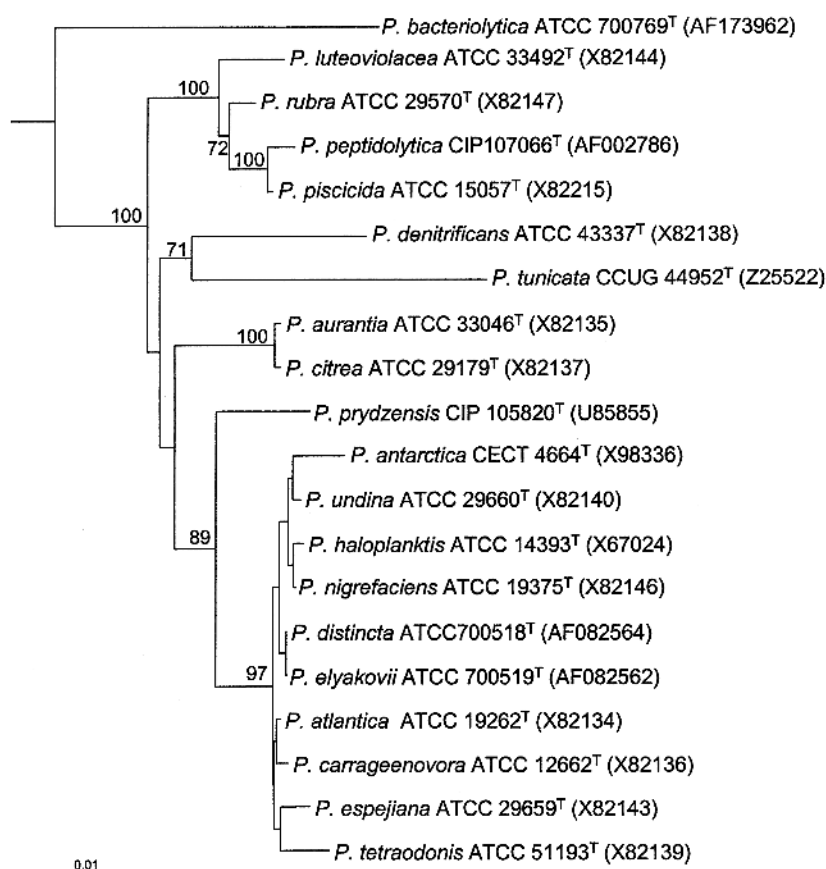


FIGURE BXII.γ.155. Phylogenetic tree of the genus *Pseudoalteromonas* based on nearly complete 16S rRNA sequences, including species validly published as of 2000–2001. The tree was determined using maximum-likelihood distances and clustered by neighbor-joining. Numbers at branch nodes are bootstrap values (calculated from 250 replicate analyses). Only bootstrap values over 50% are shown. *T* denotes a type strain. Bar = distance of 0.01. Numbers in brackets denote GenBank accession codes.

Cells are straight and rod-like, $0.5\text{--}0.9 \times 0.7\text{--}3\text{ }\mu\text{m}$. Colonies are beige, smooth, convex, and mucoid, with entire edges. Require approximately 17 mM NaCl for growth, but do not require amino acids or other growth factors. Grow between 4 and 30°C ; optimum temperature is $25\text{--}30^{\circ}\text{C}$. Other characteristics are as given in the genus description and in Tables BXII.γ.144 and BXII.γ.145. In addition, strains are lecithinase negative, and acid is formed oxidatively from D-glucose, maltose, starch, D-galactose, and D-mannitol, but not from L-arabinose, D-fructose, rhamnose, sucrose, or D-sorbitol. Acid may be formed from D-mannose or glycerol. Can grow on MacConkey agar, but not on Simmons citrate agar. May grow on cetrimide agar. See Bozal et al. (1994, 1997) for additional characteristics.

Isolated from muddy soils and sediments collected from Antarctic coastal areas.

The mol% G + C of the DNA is: 41–42 (T_m).

Type strain: NF3, CECT 4664.

GenBank accession number (16S rRNA): X98336.

3. ***Pseudoalteromonas atlantica*** (Akagawa-Matsushita, Matsuo, Koga and Yamasoto 1992b) Gauthier, Gauthier and Christen 1995a, 760^{VP} (*Alteromonas atlantica* Akagawa-Matsushita, Matsuo, Koga and Yamasoto 1992b, 624.) *at.lan' ti.ca*. L. adj. *atlanticus* from Gr. adj. *Atlantikos* pertaining to the Atlantic Ocean.

Cells are rod-shaped, $0.5\text{--}0.8 \times 1.5\text{--}2.3\text{ }\mu\text{m}$. Colonies are sunken by agarase activity. Colonies are beige to pale orange, smooth, and convex, with an entire edge. May produce a melanin-like pigment on tyrosine-containing media. Sodium ions are required for growth, but strains do not require amino acids or other growth factors. Grow between 5 and 35°C , with optimum growth at about 30°C . Grows between pH 5.5 and 8.5. Other characteristics are as given in the genus description and in Tables BXII.γ.144 and BXII.γ.145. In addition, strains can reduce methylene blue and coagulate litmus milk; gluconate oxidation, 3-ketolactose production from lactose, and Congo red adsorption tests are negative; strains are unable to degrade carrageenan or xanthine; acid is formed from D-glucose and D-mannitol; and no growth occurs on media containing 0.4% phenethyl alcohol.

Isolated from marine macroalgae and seawater.

The mol% G + C of the DNA is: 40.6–41.7 (T_m).

Type strain: ATCC 19262, IAM 12927, LMG 2138, NCIMB 301.

GenBank accession number (16S rRNA): X82134.

4. ***Pseudoalteromonas aurantia*** (Gauthier and Breittmayer 1979) Gauthier, Gauthier and Christen 1995a, 760^{VP} (*Alteromonas aurantia* Gauthier and Breittmayer 1979, 371.)

TABLE BXII.γ.145. Other characteristics of selected species of the genus *Pseudoalteromonas*^a

Characteristics	<i>P. haloplanktis</i>	<i>P. atlantica</i>	<i>P. antarctica</i>	<i>P. carrageenovora</i>	<i>P. espejiana</i>	<i>P. nigrifaciens</i>	<i>P. tetradonis</i>	<i>P. undina</i>
Oxidase, gelatinase, caseinase	+	+	+	+	+	+	+	+
Lipase	+	+	d	+	+	+	+	+
Urease	+	+	d	—	+	+	—	d
β-Galactosidase	—	+	—	—	+	+	—	+
β-Glucosidase (esculin)	nd	+	d	nd	nd	nd	nd	+
Deoxyribonuclease	nd	+	d	nd	nd	nd	nd	+
Phosphatase	nd	+	+	nd	nd	nd	nd	+
H ₂ S production	nd	+	d	nd	nd	nd	nd	+
Voges-Proskauer Test	nd	—	d	nd	nd	nd	nd	d
Decomposition of L-tyrosine	+	+	—	+	+	+	—	+
Utilization of: ^b								
DL-Arabinose	d	—	—	d	—	d	—	—
D-Xylose	—	—	—	—	—	d	—	—
D-Gentiobiose	+	+	—	+	+	—	+	+
D-Glucose	+	+	+	+	+	+	+	+
L-Rhamnose	—	—	—	—	—	d	—	—
D-Raffinose	—	+	—	—	—	—	—	—
Adonitol	—	—	—	—	—	d	nd	—
meso-Inositol	—	—	—	—	—	+	nd	—
D-Sorbitol	—	—	—	—	—	+	—	—
Amygdalin	—	+	d	—	—	—	—	+
Salicin	—	—	—	—	—	—	—	—
α-Methyl-D-mannoside	nd	—	nd	nd	nd	nd	nd	—
D-Glucosamine	—	—	nd	+	—	—	+	—
N-Acetylglucosamine	+	—	d	+	—	—	+	—
Glycogen	—	—	d	—	+	+	—	+
Ethanol	d	nd	nd	+	—	+	nd	nd
n-Propanol	d	nd	nd	+	—	+	nd	nd
n-Butanol	—	nd	nd	+	d	—	nd	nd
Propionate	+	+	—	+	+	+	nd	+
Butyrate	d	—	—	d	+	+	+	d
Isobutyrate	d	—	nd	+	+	+	+	d
Valerate	d	+	—	+	d	+	+	d
Isovalerate	—	—	nd	—	+	—	+	d
Caproate	—	—	nd	—	—	+	+	d
Heptanoate	d	—	nd	—	—	—	—	d
Caprylate	d	+	nd	—	—	d	—	d
Pelargonate	d	+	nd	d	+	—	nd	+
Caprate	+	+	—	+	+	—	nd	+
Oxalate	—	+	nd	—	—	—	nd	+
Malonate	—	—	d	—	—	—	nd	—
Azelate	—	—	nd	—	—	—	nd	—
DL-3-Hydroxybutyrate	d	—	d	d	d	+	+	d
DL-Tartrate	—	—	nd	—	—	—	nd	—
Mesaconate	—	—	nd	—	—	—	—	d
o-Hydroxybenzoate	—	+	nd	—	—	—	—	d
Phenylacetate	—	—	—	—	—	+	nd	—
L-Arginine	d	—	nd	d	d	+	+	—
L-Aspartate	d	—	nd	—	+	—	nd	—
L-Citrulline	—	—	nd	d	+	—	nd	—
L-Cysteine	—	—	nd	nd	nd	nd	nd	—
Glycine	d	—	nd	d	+	—	nd	—
L-Histidine	d	—	—	—	—	d	nd	—
L-Isoleucine	—	—	nd	d	d	—	nd	—
L-Leucine	d	—	d	+	d	—	nd	—
L-Lysine	—	—	—	—	—	+	+	—
L-Methionine	nd	—	nd	nd	nd	nd	nd	—
L-Phenylalanine	—	—	—	d	d	d	—	—
L-Serine	+	—	d	+	+	+	nd	—
L-Threonine	d	—	nd	d	+	+	nd	—
L-Valine	d	—	nd	—	—	—	nd	—
DL-γ-Aminobutyrate	—	—	nd	—	—	—	nd	—
L-Amylamine	—	—	nd	—	—	—	nd	—
Adipate, pimelate	—	—	—	—	—	—	nd	—
Glutarate	—	—	nd	—	—	—	nd	—

(continued)

TABLE BXII.γ.145. (cont.)

Characteristics	<i>P. haloplanktis</i>	<i>P. alantica</i>	<i>P. antarctica</i>	<i>P. carragenovora</i>	<i>P. esojiana</i>	<i>P. nigrifaciens</i>	<i>P. tetradonis</i>	<i>P. undina</i>
Putrescine	—	—	nd	—	—	—	nd	—
<i>m</i> -Hydroxybenzoate	—	—	—	—	—	—	nd	—
DL-Fucose	—	—	—	—	—	—	nd	—
Suberate, sebacate, <i>meso</i> -tartrate, itaconate	—	—	—	—	—	—	nd	—
α-Methyl-D-glucoside, D-turanose, 5-keto-D-gluconate	—	—	nd	—	—	—	—	—
D-Melezitose	—	—	nd	—	—	—	—	—
DL-Tryptophan	—	—	nd	—	—	—	nd	—
Inulin, creatine	—	—	nd	—	—	—	nd	—
<i>l</i> -Erythritol, DL-glycerate, L-norleucine, DL-kynurenine, DL-α-aminobutyrate, DL-α-aminovalerate, DL-δ-aminovalerate, ethanolamine, histamine, tryptamine, spermine, trigollenine, betaine, sarcosine, acetamide, butylamine, benzylamine, anthranilate, <i>m</i> -aminobenzoate, <i>p</i> -aminobenzoate	—	—	nd	—	—	—	nd	—
Glycolate	—	—	nd	—	—	—	nd	—
Citraconate, levulinate, maleate, benzoate, DL-maleate, β-alanine	—	—	nd	—	—	—	nd	—
Arbutin	nd	—	nd	—	—	—	nd	—
D-Glucuronate	—	nd	nd	—	—	—	nd	nd
Saccharate	—	nd	nd	—	—	—	nd	nd
Mucate, methanol, isopropanol, isobutanol, ethylene glycol, propyleneglycol, butylene glycol, formate, quinate, benzoylformate, phenylthanediol, phenol, naphthaene, kynurenate, methylamine, pentylamine, 2-amylamine, hippurate, pantothenate, nicotinate, nicotinamide, allantoin, adenine, cytosine, guanine, thymine, uracil, dodecane	—	nd	nd	—	—	—	nd	nd
DL-Arabitol	nd	—	nd	nd	nd	nd	nd	—
Dulcitol, urea	nd	—	nd	nd	nd	nd	nd	—
D-Lyxose, D-tagatose, sorose, xylitol, β-methyl-D-glucoside, 2-keto-D-gluconate, L-norvaline, ethylamine	nd	—	nd	nd	nd	nd	nd	—
Phthalate, isophthalate, <i>tert</i> -phthalate	nd	—	nd	nd	nd	nd	nd	—
D-Galacturonate	—	nd	nd	—	—	nd	nd	nd

^aFor symbols see standard definitions; nd, not determined.

^bNo information is available concerning ascorbate, oxaloacetate, hydroxy-L-proline, L-asparagine, L-glutamine, protamine, 2-deoxyribose, deoxycholate, galacitol, D-galactosamine, α-glycerophosphate, urate, and glycylglycine.

au.ran' ti.a. M.L. n. *aurantium* orange; M.L. fem. adj. *aurantia* orange-colored.

Cells are rod-shaped, 0.5–1.5 × 1.5–4.0 μm. Colonies on agar media are orange, smooth, and convex, with an entire edge. Sodium ions (optimal levels 0.5–0.6 M) and amino acids are required for growth. Grows between 1 and 8% NaCl. Produces a polyanionic, autotoxic antibiotic on complex media. Grows between 4 and 30°C, with optimum growth occurring at about 25°C. Other characteristics are as shown in the genus description and in Tables BXII.γ.144 and BXII.γ.146. In addition, strains can form phospholipase C, collagenase, leucine aminopeptidase, valine aminopeptidase, and phosphoamidase. Strains may also form α-glucosidase, β-glucosaminidase, and α-chymotrypsin and reduce methylene blue. Tryptophanase, α-galactosidase, β-galactosidase, β-glucuronidase, α-mannosidase, α-fucosidase, and C14 lipase are absent. Trimethylamine *N*-oxide is not reduced. See Gauthier and Breittmayer (1979) for additional characteristics.

Isolated from surface seawater.

The mol% *G* + *C* of the DNA is: 38.8–42.5 (*T_m*).

Type strain: 208, ATCC 33046, DSM 6057, NCIMB 2052.

GenBank accession number (16S rRNA): X82135.

5. **Pseudoalteromonas bacteriolytica** Sawabe, Makino, Tsumi, Nakano, Tajima, Iqbal, Yumoto, Ezura and Christen 1998a, 773^{VP}

bac.te.ri.o.ly'ti.ca. Gr. n. *baktron* rod or staff; Gr. adj. *lytica* dissolving; M.L. fem. adj. *bacteriolytica* bacteria-dissolving.

Gram negative. Cells are rod-shaped, 1.9–2.5 × 0.6–0.9 μm, occurring singly or in pairs. Motile by polar flagella. Does not accumulate poly-β-hydroxybutyrate. Colonies are red-pigmented or non-pigmented, circular, smooth, and convex with entire edges. Requires NaCl and amino acids for growth. Grows from 15–35°C; no growth at 37°C.

Strictly aerobic chemoheterotroph. Oxidase positive. Catalase activity is weak. Produces acid oxidatively from D-glucose. Produces amylase. Digests casein and Tween 80. Does not produce chitinase, alginase, agarase, or κ-carrageenase. Does not reduce nitrate to nitrite. Utilizes D-glucose, D-mannose, D-fructose, sucrose, D-galactose, D-glucosamine, fumarate, succinate, acetate, and propionate. Does not utilize xylose, lactose, melibiose, D-gluconate, D-glucuronate, *N*-acetylglucosamine, D-sorbitol, erythritol, glycerol, citrate, DL-malate, aconitate, α-ketoglutarate, *m*-hydroxybenzoate, putrescine, δ-aminovalerate, γ-aminobutyrate, or L-proline.

TABLE BXII.γ.146. Other characteristics of selected species of the genus *Pseudoalteromonas*^a

Characteristics	<i>P. aurantia</i>	<i>P. citrea</i>	<i>P. denitrificans</i>	<i>P. luteoviolacea</i>	<i>P. piscicida</i>	<i>P. prydzensis</i>
Oxidase, gelatinase, caseinase	+	+	+	+	+	+
Lipase	+	+	+	+	+	+
Urease	—	nd	nd	—	—	d
β-Galactosidase	—	—	—	—	—	—
β-Glucosidase (esculin)	—	nd	nd	d	d	d
Deoxyribonuclease	+	+	+	+	+	+
Phosphatase	+	+	+	+	+	+
H ₂ S production	—	—	nd	—	d	—
Voges-Proskauer Test	—	—	nd	nd	—	—
Decomposition of L-Tyrosine	+	+	+	d	+	—
<i>Utilization of:</i> ^b						
DL-Arabinose	—	—	—	—	—	+
D-Xylose	—	—	nd	—	—	nd
D-Glucose	+	d	+	d	+	+
L-Rhamnose	—	+	nd	—	—	—
D-Raffinose	—	—	nd	—	—	—
Adonitol	—	+	nd	—	—	—
meso-Inositol	—	—	nd	—	—	nd
D-Sorbitol	—	d	—	—	—	—
Amygdalin	—	nd	nd	nd	—	nd
Salicin	—	d	nd	—	—	nd
α-Methyl-D-mannoside	—	nd	nd	nd	+	nd
D-Glucosamine	+	nd	nd	+	+	nd
N-Acetylglucosamine	+	nd	—	+	+	+
Glycogen	—	+	nd	nd	+	+
Ascorbate	d	nd	nd	d	—	nd
Ethanol	nd	+	—	nd	nd	nd
Propionate	nd	+	+	nd	nd	+
n-Butyrate	nd	+	+	nd	nd	+
Isobutyrate	nd	nd	nd	nd	nd	+
n-Valerate	nd	—	nd	nd	nd	d
Isovalerate	nd	nd	nd	nd	nd	d
Caproate	nd	+	nd	nd	nd	—
Heptanoate	nd	nd	nd	nd	nd	—
Caprylate	nd	—	nd	nd	nd	d
Pelargonate	nd	—	—	nd	nd	—
Caprate	—	—	—	—	—	—
Oxalate	—	nd	nd	d	—	nd
Malonate	—	nd	nd	—	—	d
Azelate	nd	nd	nd	nd	nd	d
Oxaloacetate, hydroxy-L-proline	nd	nd	nd	nd	nd	+
DL-3-Hydroxybutyrate	nd	nd	—	—	nd	+
DL-Tartrate	—	nd	nd	d	—	nd
o-Hydroxybenzoate	—	nd	nd	—	—	nd
Phenylacetate	nd	+	nd	nd	nd	nd
L-Arginine	+	+	+	d	+	nd
L-Asparagine	nd	nd	+	+	nd	—
L-Aspartate	—	+	nd	nd	—	—
L-Citrulline	d	nd	—	nd	+	nd
L-Cysteine	—	nd	—	d	—	nd
Glycine	+	nd	nd	nd	—	nd
L-Glutamine, protamine	nd	nd	+	nd	nd	nd
L-Histidine	d	—	nd	d	d	—
L-Isoleucine	—	nd	nd	nd	+	nd
L-Leucine	d	nd	nd	—	+	—
L-Lysine	—	—	nd	d	nd	nd
L-Methionine	—	nd	nd	nd	d	nd
L-Phenylalanine	—	—	nd	d	d	—
L-Serine	—	nd	nd	nd	—	—
L-Threonine	d	d	nd	+	—	—
L-Valine	—	nd	nd	nd	—	nd
γ-Aminobutyrate	—	nd	nd	nd	—	+
L-Amylamine	—	nd	nd	nd	d	nd
Adipate, pimelate	nd	nd	nd	nd	nd	—
Glutarate	nd	—	nd	nd	nd	—
Putrescine	—	nd	—	nd	—	—
m-Hydroxybenzoate	nd	nd	nd	—	nd	nd
DL-Fucose	—	nd	nd	nd	—	nd
α-Methyl-D-glucoside,	—	nd	nd	nd	—	nd
D-Turanose, 5-keto-D-gluconate	nd	nd	nd	nd	nd	nd
DL-Tryptophan	—	nd	—	—	—	nd

(continued)

TABLE BXII.γ.146. (cont.)

Characteristics	<i>P. aurantia</i>	<i>P. citrea</i>	<i>P. denitrificans</i>	<i>P. luteoviolacea</i>	<i>P. piscicida</i>	<i>P. prydzensis</i>
Inulin, creatine	—	nd	nd	—	—	nd
<i>D</i> -Erythritol, <i>D,L</i> -glycerate	nd	nd	nd	—	nd	nd
<i>L</i> -Norleucine, <i>D,L</i> -kynurenine, <i>D,L</i> -α-aminobutyrate, <i>D,L</i> -α-aminovaleate, <i>D,L</i> -δ-aminovaleate, ethanolamine, histamine, tryptamine, spermine, trigollenine, betaine, sarcosine, acetamide, butylamine, benzylamine, anthranilate, <i>m</i> -aminobenzoate, <i>p</i> -aminobenzoate	—	nd	nd	nd	—	nd
Glycolate	nd	nd	—	nd	nd	nd
Arbutin	—	nd	nd	nd	—	nd
<i>D</i> -Glucuronate	nd	nd	nd	—	nd	—
Saccharate	nd	nd	nd	nd	nd	—
<i>D,L</i> -Arabitinol	—	nd	nd	nd	—	—
Dulcitol, urea	nd	nd	nd	—	nd	nd
<i>D</i> -Lyxose, <i>D</i> -tagatose, sorbose, xylitol, β-methyl- <i>D</i> -glucoside, 2-keto- <i>D</i> -gluconate, <i>L</i> -norvaline, ethylamine	—	nd	nd	nd	—	nd
<i>D</i> -Galacturonate	nd	nd	nd	nd	nd	—
2-Keto- <i>D</i> -gluconate	—	nd	nd	nd	—	nd
α-Glycerophosphate, urate	nd	nd	nd	nd	nd	—
Deoxycholate	nd	nd	nd	—	nd	nd
Galacitol, <i>D</i> -galactosamine, glycylglycine	—	nd	nd	nd	—	nd

^aFor symbols see standard definitions; nd, not determined.

^bNo information is available concerning the following substrates: caprate, *D*-gentiobiose, *n*-propanol, *n*-butanol, mesaconate, suberate, sebacate, *meso*-tartrate, itaconate, *D*-melizitose, citraconate, levulinate, maleate, benzoate, *D,L*-maleate, β-alanine, mucate, methanol, isopropanol, isobutanol, ethyleneglycol, propyleneglycol, butyleneglycol, formate, quinate, benzoylformate, phenylethanediol, phenol, naphthalene, kynurenate, methylamine, pentylamine, 2-amilamine, hippurate, pantothenate, nicotinate, nicotinamide, allantoin, adenine, cytosine, guanine, thymine, uracil, dodecane, phthalate, isophthalate, and *tert*-phthalate.

Isolated from, and causative agent of, red spot disease in beds of *Laminaria japonica*.

The mol% G + C of the DNA is: 44–46 (HPLC).

Type strain: IAM 14595.

GenBank accession number (16S rRNA): D89929.

6. **Pseudoalteromonas carrageenovora** (Akagawa-Matsushita, Matsuo, Koga and Yamasoto 1992b) Gauthier, Gauthier and Christen 1995a, 760^{VP} (*Alteromonas carrageenovora* Akagawa-Matsushita, Matsuo, Koga and Yamasoto 1992b, 626.) *car.ra.gee.no'vo.ra*. M.L. n. *carrageenum* carrageen, another name for carrageenan; L. v. *vorare*, to devour; M.L. fem. adj. *carrageenovora* carrageenan decomposing.

Cells are rod-shaped, 0.7–0.8 × 1.9–2.5 μm. Colonies are beige to pale yellow, smooth, and convex, with an entire edge. Produces diffusible melanin-like pigments on tyrosine-containing media. Sodium ions are required for growth, but amino acids and other growth factors are not required. Grow between 5 and 35°C, with optimum growth at about 30°C. Grow between pH 5.5 and 9.0. Other characteristics are as given in the genus description and in Tables BXII.γ.144 and BXII.γ.145. In addition, strains can reduce methylene blue and coagulate litmus milk. Gluconate oxidation, 3-ketolactose production from lactose, and Congo red adsorption tests are negative. Degrade carrageenan, but not xanthine. Acid is formed from *D*-glucose and *D*-mannitol. No growth occurs on media containing 0.4% phenethyl alcohol (Akagawa-Matsushita et al., 1992b).

Isolated from a mixture of seawater and marine macroalgae (Yaphe and Baxter, 1955).

The mol% G + C of the DNA is: 39.5 (*T_m*).

Type strain: ATCC 43555, DSM 6820, IAM 12662, IFO 12985, NCIMB 302.

GenBank accession number (16S rRNA): X82136.

7. **Pseudoalteromonas citrea** (Gauthier 1977) Gauthier, Gauthier and Christen 1995a, 760, emend. Ivanova, Kiprianova,

Mikhailov, Levanova, Garagulya, Gorshkova, Vysotskii, Nicolau, Yumoto, Taguchi and Yoshikawa 1998, 255^{VP} (*Alteromonas citrea* Gauthier 1977, 354; *Alteromonas fuliginea* Romanenko, Lysenko, Mikhailov and Kurika 1995a, 879.) *ci'tre.a*. L. adj. *citreus* lemon-yellow.

Cells are straight rods, 0.5–0.8 × 1.0–1.5 μm. Colonies on agar media are smooth, convex with an entire edge, and either nonpigmented, pigmented yellow-orange, or containing diffusible blue-black or brown, melanin-like pigments. Sodium ions are required for growth; optimal growth occurs with about 2–3% NaCl. May require amino acids for growth. Some strains produce a polyanionic, autotoxic antibiotic on complex media. Grows between 4 and 35°C, with optimal growth at about 30°C. Other characteristics are as shown in the genus description and in Tables BXII.γ.144 and BXII.γ.146. Positive for levan production. Strains may hydrolyze pustulan and laminarin. Strains are resistant to ristomycin, ampicillin, neomycin, kanamycin, streptomycin, benzylpenicillin, lincomycin, and cephalixin, but are sensitive to rifampicin, gentamicin, oxacillin, polymyxin, erythromycin, and ofloxacin.

Isolated from surface seawater, mussels, ascidians, and sponges.

The mol% G + C of the DNA is: 38.9–44.7 (*T_m*).

Type strain: ATCC 29719, DSM 6058, NCIMB 1889.

GenBank accession number (16S rRNA): X82137.

8. **Pseudoalteromonas denitrificans** (Enger, Nygaard, Solberg, Schei, Nielsen and Dundas 1987) Gauthier, Gauthier and Christen 1995a, 760^{VP} (*Alteromonas denitrificans* Enger, Nygaard, Solberg, Schei, Nielsen and Dundas 1987, 421.) *de.ni.tri'fi.cans*. M.L. v. *denitrifico* to denitrify; M.L. part. adj. *denitrificans* denitrifying.

Cells are rod-shaped, 0.5–0.7 × 2–4 μm. Motile by a single, sheathed, polar flagellum; rarely, up to 3 polar flagella may be observed. Cells produce a number of prod-